WireMod Spline Generation Guide for SBND

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1 Introduction & Code Repositories

To produce WireMod splines for SBND, we split the task into two main parts:

1. Generate multi-dimensional histograms with spatial, angular, and hit response variables that have calibration corrections applied.
    - We have an interface for calibration NTuples as well as flat CAF trees.
    - The calibration NTuples allow us to generate splines from through-going muons. These provide a stable dE/dx from the MIP-like cosmic off-beam samples.
    - The flat CAF trees allow us to select protons as a way to probe dE/dx with Wiremod. This population will allow us to study the performance of the WireMod splines extracted from MIP muons when applied to highly ionizing tracks. Additionally, we may be able to add dE/dx as a WireMod dimension.
    - The code repository, `SBNDWireModHists`, can be found here:
        - URL: https://github.com/aantonakis/SBNDWireModHists
2. Extract the data/mc MPV ratios for the hit response variables as a function of the spatial and angular dimensions. We partition the spatial and angular components of the multi-dimensional histograms from step 1 to achieve sufficient track statistics for the MPV extraction. The hit response ratios as a function of the spatial and angular binning define the WireMod splines. These splines typically constitute a family of 2D ratio splines that can leverage interpolation functionaliy from packages such as ROOT.

The spline generation workflow is currently split across two repositories:

- `SBNDWireModSplineGeneration`: Python workflow scripts, ROOT output files, CSV summaries, and diagnostic plots.
    - URL: https://github.com/aantonakis/SBNDWireModSplineGeneration
- `WireMod-Spline-Generation-Guide`: this documentation project.
    - URL: https://github.com/aantonakis/WireMod-Spline-Generation-Guide

The current 2D workflow uses PyROOT for all ROOT I/O and histogram projections, while Matplotlib is used for diagnostic plots. The ROOT environment is loaded through the wrapper script:

```
./scripts/run_with_root.sh <python script> <options>
```

2 WireMod Histogram Generation

2.1 Purpose

The `SBNDWireModHists` repository produces the ROOT `THnSparseD` inputs used by the spline generation workflow. This step reads the selected data or simulation samples, applies the requested calibration corrections and event/track selections, and fills one track-count histogram and one hit-response histogram for each requested wire plane or angular region.

The central Gen2 macro is:

```
macros/multi_dim_tracks_gen2_grid.C
```

For the angle-bin workflow, the corresponding macro is:

```
macros/multi_dim_tracks_gen2_anglebin_grid.C
```

The standard histogram objects are named `hTrackN` and `hHitN`, where N is the wire-plane index. The `hTrackN` histograms are used to count tracks and construct adaptive regions. The `hHitN` histograms store the hit-response axis, for example charge, dQ/dx, or width, together with the selected WireMod coordinates.

2.2 Configuration Files

Histogram axes and selections are controlled by text configuration files under:

`Configs/Gen2`

The configuration file defines the global binning with `Nbins`, `Xmin`, and `Xmax`, while the `dim` field selects which axes are actually stored in the output sparse histograms. For the current angle-bin XYZ workflow, the template configs store:

- `dim: 0 1 2 6` for (x, y, z, Q),
- `dim: 0 1 2 5` for $(x, y, z, dQ/dx)$,
- `dim: 0 1 2 7` for (x, y, z, W).

The angular coordinates are not stored as axes in these reduced output histograms. Instead, they are used as selections. Each generated angle-bin config contains `PlaneIndex`, `UseAbsAngles`, `ThetaXWMin`, `ThetaXWMax`, `ThetaYZMin`, `ThetaYZMax`, `AngleBinRegion`, and `AngleBinTracks`. These fields select a single folded angular region for a single wire plane.

The current templates are:

```
Configs/Gen2/config_xyz_spline_anglebin_template_q.cfg
Configs/Gen2/config_xyz_spline_anglebin_template_dq.cfg
Configs/Gen2/config_xyz_spline_anglebin_template_w.cfg
```

2.3 Generating Angle-Bin Configs

The angle-bin configs are generated from folded 2D angular-bin CSV files produced by the angular binning workflow. The helper script is:

`scripts/generate_gen2_anglebin_configs.py`

It reads the per-plane angular bins, chooses the appropriate observable template, and writes one config per angular region:

```
python scripts/generate_gen2_anglebin_configs.py --dim q
python scripts/generate_gen2_anglebin_configs.py --dim dq
python scripts/generate_gen2_anglebin_configs.py --dim w
```

The generated configs are stored in observable-specific directories:

```
Configs/Gen2/AngleBins_q/PlaneN
Configs/Gen2/AngleBins_dq/PlaneN
Configs/Gen2/AngleBins_w/PlaneN
```

The current folded angular binning gives 27 regions for planes 0, 2, 4, and 5, and 26 regions for planes 1 and 3. The same angular-region definitions are used for the `q`, `dq`, and `w` observable configs.

2.3.1 Switching Data and MC Configs

The generated angle-bin configs contain an `isData` field that controls whether the histogram-generation macro treats the input as data or simulation. After generating the configs, this field can be changed in bulk with:

`scripts/set_config_isdata.py`

For example, to set all `q` angle-bin configs to simulation mode:

```
python scripts/set_config_isdata.py Configs/Gen2/AngleBins_q --value false
```

and to set them back to data mode:

```
python scripts/set_config_isdata.py Configs/Gen2/AngleBins_q --value true
```

The same command can be applied to the other observable-specific config directories, such as `Configs/Gen2/AngleBins_dq` or `Configs/Gen2/AngleBins_w`. To update every angle-bin config directory under `Configs/Gen2` at once, use:

```
python scripts/set_config_isdata.py --all-anglebins --value false
python scripts/set_config_isdata.py --all-anglebins --value true
```

The `--dry-run` option can be added to either command to print the files that would be changed without editing them.

2.4 Submitting Grid Jobs

The plane-level grid wrapper submits the Gen2 angle-bin macro once for each config in a selected plane and observable directory:

`grid/submit_multi_dim_tracks_gen2_anglebin_plane.py`

It loops over the relevant config files and forwards the common grid options to:

`grid/submit_multi_dim_tracks_gen2_anglebin.py`

A representative submission command is:

```
python grid/submit_multi_dim_tracks_gen2_anglebin_plane.py \
  --plane 0 \
  --observable q \
  --sample data \
  --tag gen2 \
  -l input_file_list.txt \
  -nfile 100 \
  -ngrid 20
```

For each config, the wrapper builds an output prefix of the form:

`gen2_<data-or-mc>_<config-name-without-.cfg>`

For example, the first plane-0 charge angular region is labeled:

`gen2_data_config_xyz_spline_anglebin_q_plane0_region001`

The resulting ROOT files are then moved into the spline-generation repository under separate data and MC directories, for example:

```
AngleData/data/plane0
AngleData/mc/plane0
```

3 Simple 2D Spline Generation

3.1 Goal

The first 2D spline production tests use lower-dimensional Gen2 histograms where the first two THnSparse axes are WireMod coordinates or angles, and the third axis is the hit-level response variable. The current samples are:

- `YZQ`: adaptive bins in (y, z), hit response axis Q.
- `YZW`: same (y, z) binning as `YZQ`, hit response axis W.
- `XTXZQ`: adaptive bins in (x, θ_{xw}), hit response axis Q.
- `XTXZW`: same (x, θ_{xw}) binning as `XTXZQ`, hit response axis W.

For each plane, the workflow constructs adaptive 2D regions from the data track-count histogram, extracts data and simulation hit-response distributions in each region, estimates the most probable hit-response scale with an iterative truncated mean, and stores the ratio

$$R_{\mathrm{WireMod}}(a, b) = \frac{\mathrm{MPV}_{\mathrm{data}}(a, b)}{\mathrm{MPV}_{\mathrm{MC}}(a, b)} \tag{1}$$

as a `TGraph2DErrors`. Here (a, b) denotes the two WireMod dimensions used for the file, for example (y, z) or (x, θ_{xw}).

3.2 Input Files

The Gen2 2D input files are stored in:

`/Users/alexanderantonakis/Documents/Codex/2026-06-08/Gen2_2D`

Each file contains one `hTrackN` and one `hHitN` object for each of the six wire planes. The `hTrackN` histograms define the binning statistics. The corresponding `hHitN` histograms provide the hit-response distributions used to extract the data/MC ratio.

3.3 Adaptive Binning

The adaptive binning is produced with `scripts/make_adaptive_binning_2d.py`. The default target is currently 5×10^5 tracks per 2D region. The script projects the 3D `hTrackN` sparse onto axes 0 and 1, converts the projection into a NumPy count matrix, and recursively partitions this plane.

The recursive splitting rule is:

1. Start from the full active 2D plane.
2. Try a quadrant split. If all four quadrants exceed the target count threshold, keep splitting each quadrant.
3. If the quadrant split is not allowed by the threshold, try a binary split along the selected fallback axis.
4. Keep the split only if both children exceed the threshold.
5. Otherwise, keep the current region as a terminal adaptive bin.

For `XTXZ` files, one side of the detector is mostly empty for each plane. The workflow therefore applies an active-x selection before adaptive splitting. The default `auto` behavior uses $x < 0$ for planes 0–2 and $x > 0$ for planes 3–5. This prevents the empty side of the histogram from forcing the algorithm into one overly large bin.

A typical command is:

```
./scripts/run_with_root.sh scripts/make_adaptive_binning_2d.py \
  --input ../Gen2_2D/Wire_Gen2_DATA_YZQ_26Aug2026_Full.root \
  --hist hTrack0 \
  --target 500000 \
  --max-plots 1
```

For the width response files, the same binning is reused from the corresponding charge file: `YZQ` binning is used for `YZW`, and `XTXZQ` binning is used for `XTXZW`.

Table 1: Adaptive 2D region counts from the current Gen2 data track histograms. The `XTXZQ` counts include the active-x side selection.

Plane	`YZQ` regions	`XTXZQ` regions
0	2145	826
1	2129	796
2	1056	913
3	2193	852
4	2140	862
5	1116	912

3.4 Ratio Extraction

The ratio extraction is performed with `scripts/make_2d_ratio_splines.py`. For each adaptive region, the script applies the region as an axis-0/axis-1 selection on the data and MC `hHitN` sparse histograms, projects the response axis, and computes the iterative truncated mean for the data and MC projected distributions.

The truncated-mean settings currently match the earlier 3D spline workflow:

- lower truncation: -2.0 RMS from the current mean,
- upper truncation: $+1.75$ RMS from the current mean,
- convergence tolerance: 10^{-4},
- maximum iterations: 100.

The central spline point is placed at the geometric center of each adaptive region. By default, the script also duplicates points at boundary edges and corners. These edge-augmented points help constrain interpolation near the boundary of the valid 2D phase space. The output ROOT file contains:

- `data_mpv_N`: data truncated-mean MPV graph,
- `mc_mpv_N`: MC truncated-mean MPV graph,
- `splines_N`: data/MC ratio graph.

A typical ratio command is:

```
./scripts/run_with_root.sh scripts/make_2d_ratio_splines.py \
  --data ../Gen2_2D/Wire_Gen2_DATA_YZQ_26Aug2026_Full.root \
  --mc ../Gen2_2D/Wire_Gen2_MC_YZQ_26Aug2026_Full.root \
  --hit-hist hHit0 \
  --idx 0 \
  --binning-label YZQ_26Aug2026
```

For `YZW`, the output label and binning label are different:

```
./scripts/run_with_root.sh scripts/make_2d_ratio_splines.py \
  --data .../Wire_Gen2_DATA_YZW_26Aug2026_Full.root \
  --mc .../Wire_Gen2_MC_YZW_26Aug2026_Full.root \
  --hit-hist hHit0 \
  --idx 0 \
  --output-label YZW_26Aug2026 \
  --binning-label YZQ_26Aug2026
```

3.5 Current Outputs

The current output files are written under:

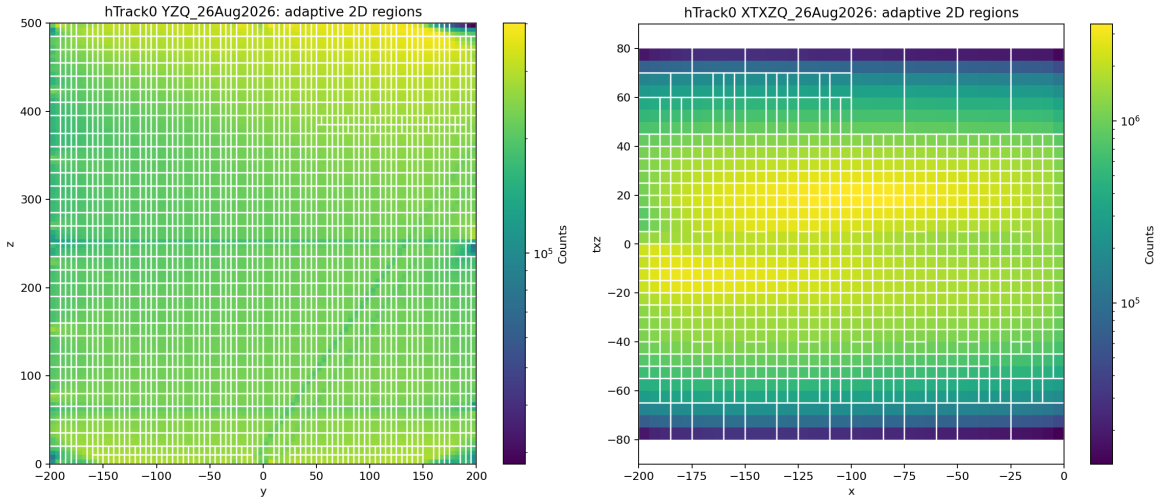
```
plots/adaptive_2d
plots/ratio_splines_2d
```

in the `SBNDWireModSplineGeneration` repository. CSV files store the adaptive regions and MPV ratio summaries, while ROOT files store the final graphs used downstream by WireMod.

Table 2: Current 2D ratio spline point counts. Each entry gives central adaptive-region fits followed by the number of graph points after edge augmentation.

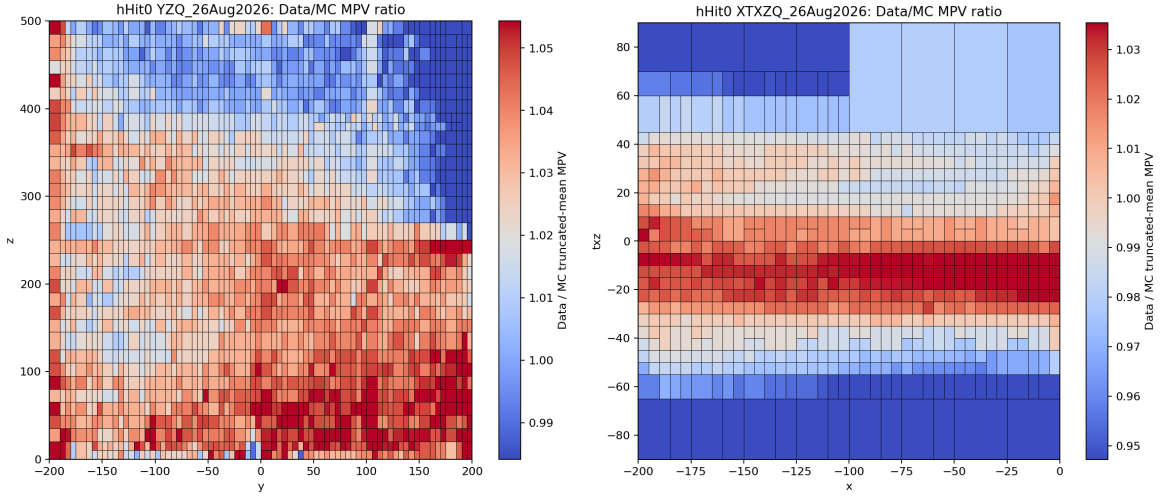
Plane	`YZQ`	`YZW`	`XTXZQ`	`XTXZW`
0	2145 / 2350	2145 / 2350	826 / 897	826 / 897
1	2129 / 2336	2129 / 2336	796 / 864	796 / 864
2	1056 / 1209	1056 / 1209	913 / 996	913 / 996
3	2193 / 2404	2193 / 2404	852 / 920	852 / 920
4	2140 / 2346	2140 / 2346	862 / 930	862 / 930
5	1116 / 1277	1116 / 1277	912 / 990	912 / 990

Figures 1 and 2 show representative outputs for plane 0. The full plane-by-plane diagnostic set is included in Section A.



(a) `YZQ` adaptive regions. (b) `XTXZQ` adaptive regions.

Figure 1: Example adaptive 2D binnings for plane 0. The `XTXZQ` binning uses the active $x < 0$ side for plane 0.



(a) `YZQ` ratio map. (b) `XTXZQ` ratio map.

Figure 2: Example data/MC truncated-mean MPV ratio maps for plane 0.

3.6 Summary Plots

The combined 2D spline ROOT files are also used to make all-plane summary plots for each response observable and coordinate pair. These plots place planes 0–2 in the first row and planes 3–5 in the second row. A fixed color scale from 0.85 to 1.15 is used for all panels so that broad plane-to-plane trends can be compared directly. For the YZ splines, the horizontal axis is z and the vertical axis is y. For the XTXZ splines, the horizontal axis is x and the vertical axis is θ_{xw}.

These summary plots are useful as a common-scale overview of the current 2D results. The appendix figures in Section A provide the complementary per-spline diagnostic view, where the colorbar range is allowed to follow the natural range of each individual 2D spline.

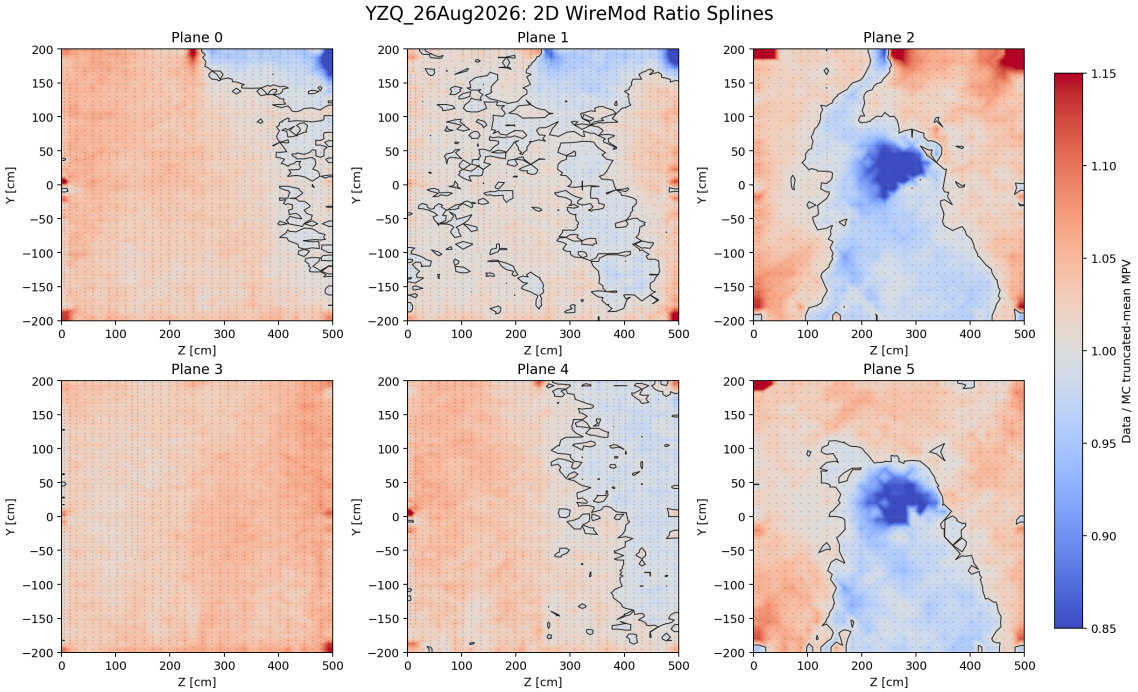


Figure 3: All-plane summary of the `YZQ` data/MC MPV ratio splines. The color scale is fixed from 0.85 to 1.15.

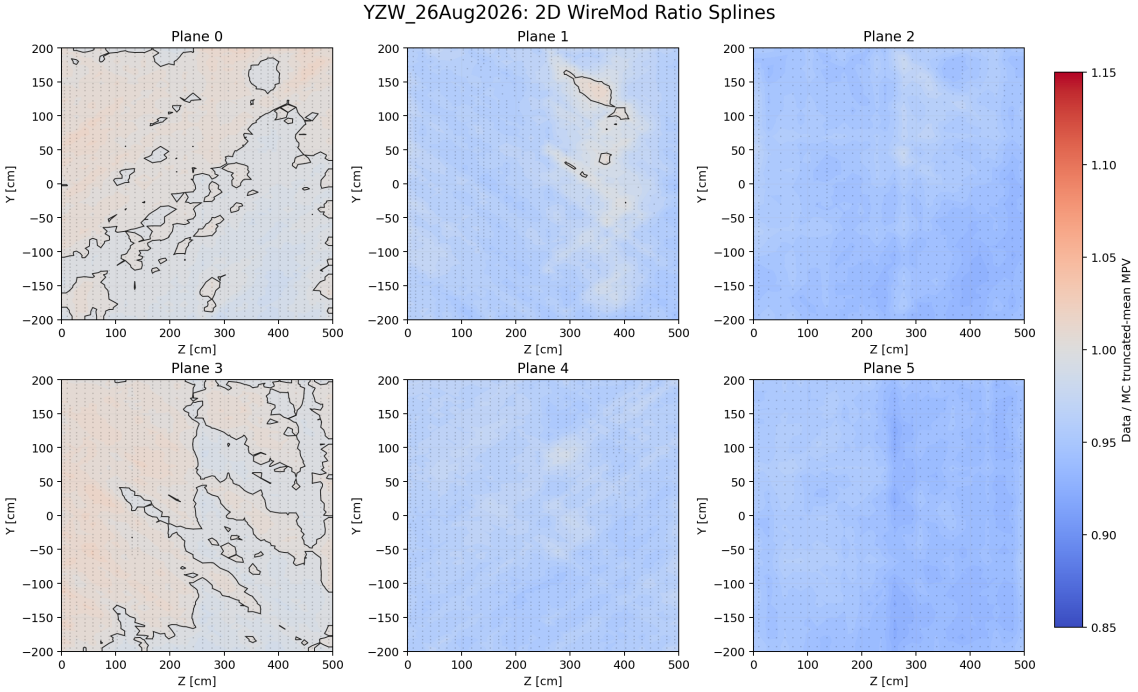


Figure 4: All-plane summary of the `YZW` data/MC MPV ratio splines. The color scale is fixed from 0.85 to 1.15.

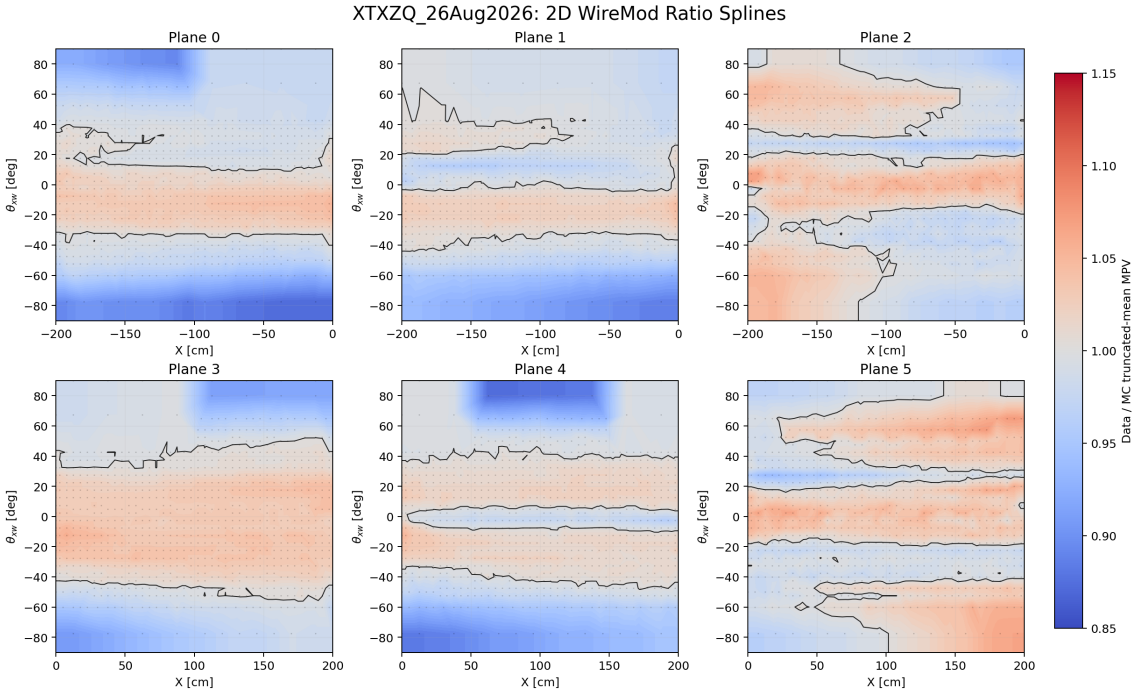


Figure 5: All-plane summary of the `XTXZQ` data/MC MPV ratio splines. The color scale is fixed from 0.85 to 1.15.

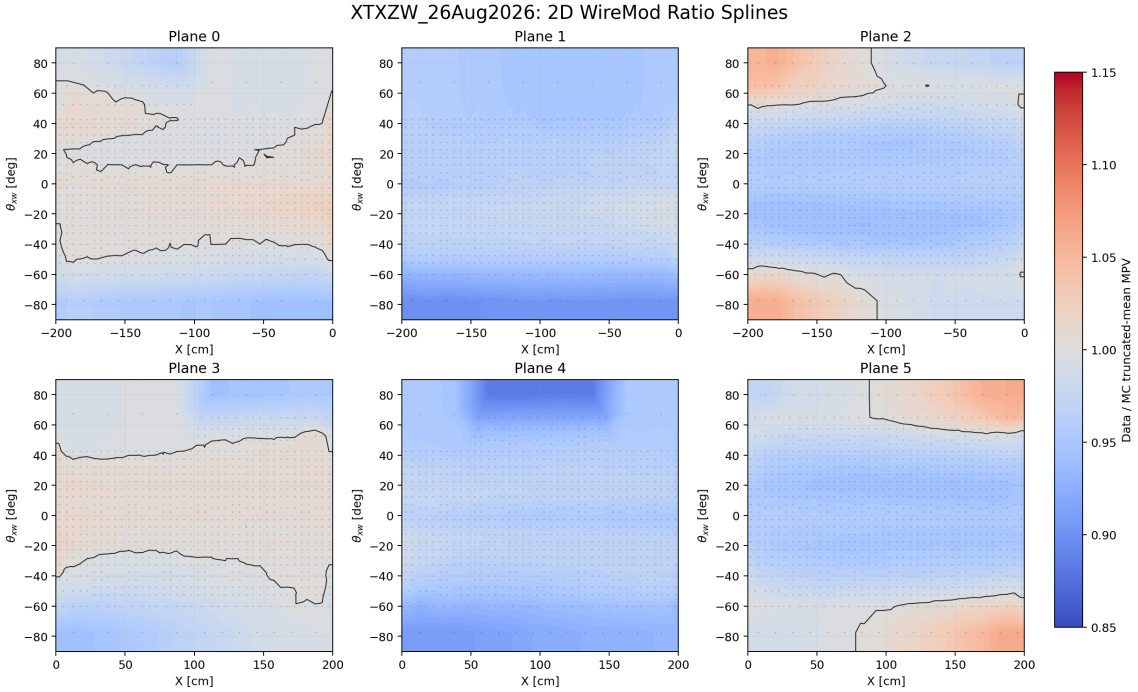


Figure 6: All-plane summary of the `XTXZW` data/MC MPV ratio splines. The color scale is fixed from 0.85 to 1.15.

For the x versus θ_{xw} splines, an alternate symmetric-angle binning was also produced. In this version, the adaptive binning is learned in the $\theta_{xw} > 0$ half-plane and then mirrored to $\theta_{xw} < 0$. This preserves a symmetric angular segmentation while still allowing the positive-angle side to follow the observed track density.

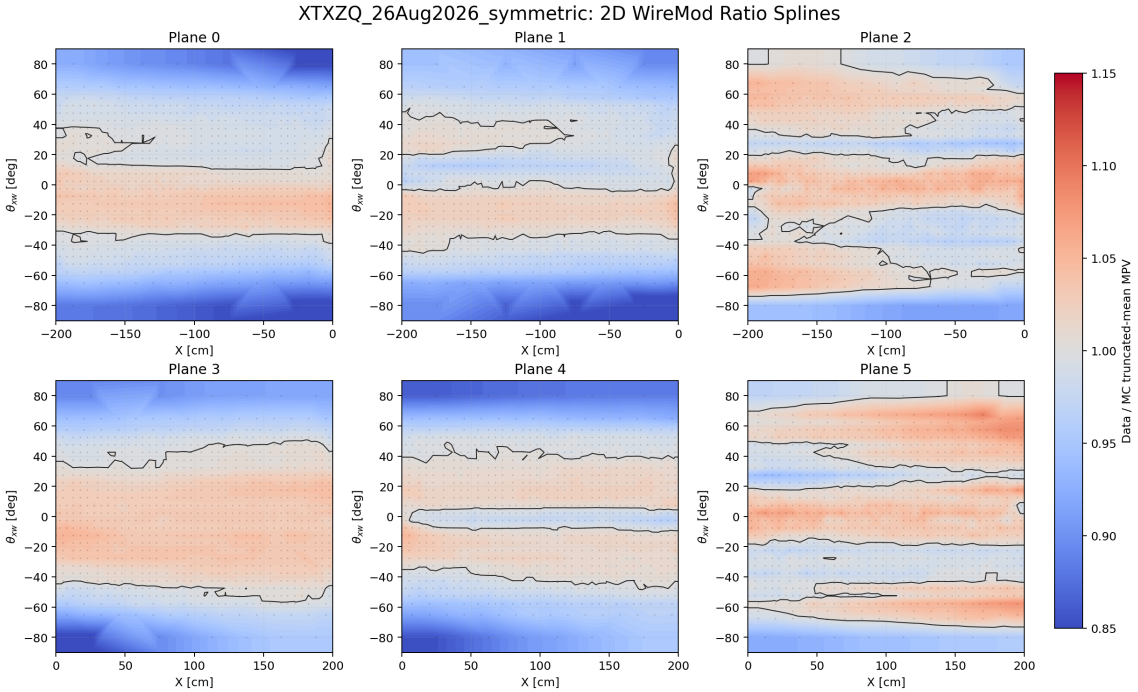


Figure 7: All-plane summary of the symmetric-angle `XTXZQ` data/MC MPV ratio splines. The adaptive binning is learned for $\theta_{xw} > 0$ and mirrored to $\theta_{xw} < 0$. The color scale is fixed from 0.85 to 1.15.

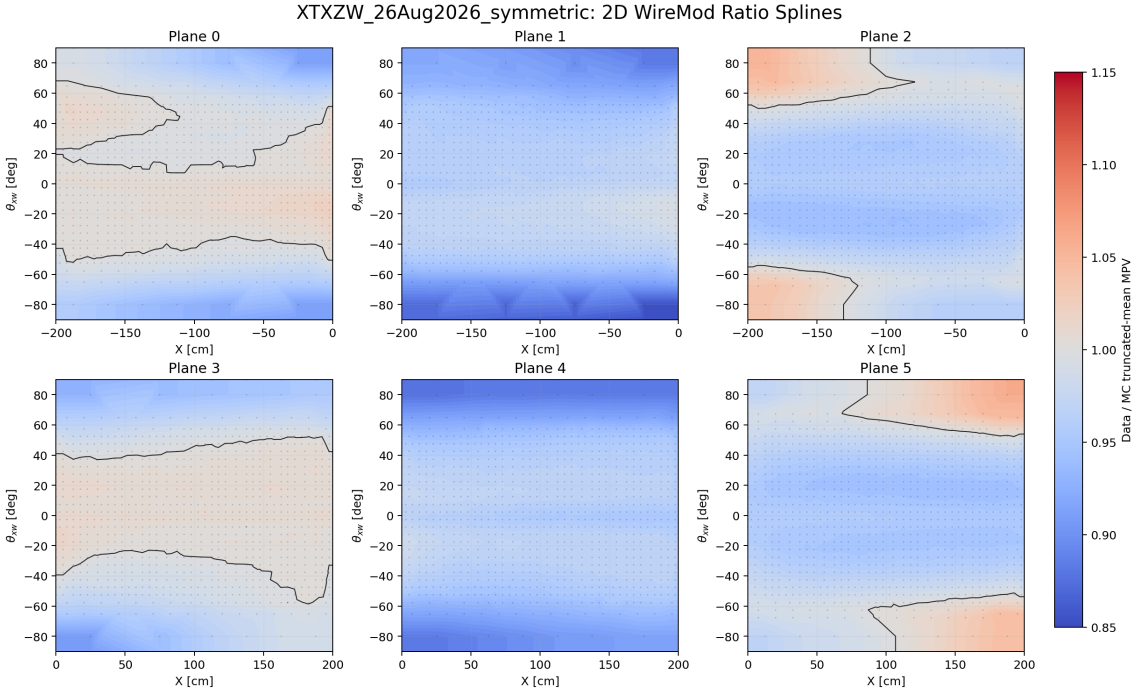


Figure 8: All-plane summary of the symmetric-angle `XTXZW` data/MC MPV ratio splines. The adaptive binning is learned for $\theta_{xw} > 0$ and mirrored to $\theta_{xw} < 0$. The color scale is fixed from 0.85 to 1.15.

4 Higher Dimensional Angle-Binned Spline Generation

4.1 Workflow

The higher-dimensional workflow keeps the angular dependence by first partitioning the two-dimensional angular plane, then producing separate $(x, y, z, \text{response})$ histograms for each angular region. Within a fixed plane and angular region, the workflow proceeds as:

1. Select a folded 2D angular bin in $(|\theta_{xw}|, |\theta_{yz}|)$.
2. Generate data and MC `hTrackN`/`hHitN` histograms in `SBNDWireModHists` using the matching angle-bin config.
3. Use the data `hTrackN` histogram to define adaptive YZ regions independently in each x-bin.
4. In each x-bin and YZ region, project the data and MC `hHitN` histograms onto the response axis.
5. Extract the data and MC truncated-mean MPVs and store the data/MC ratio as a 2D spline in the YZ plane.

The final product for one plane and one angular bin is therefore a family of 2D YZ ratio splines, one spline for each populated x-slice. The full WireMod correction can then be viewed as a set of 2D YZ surfaces indexed by plane, angular bin, x-bin, and response observable.

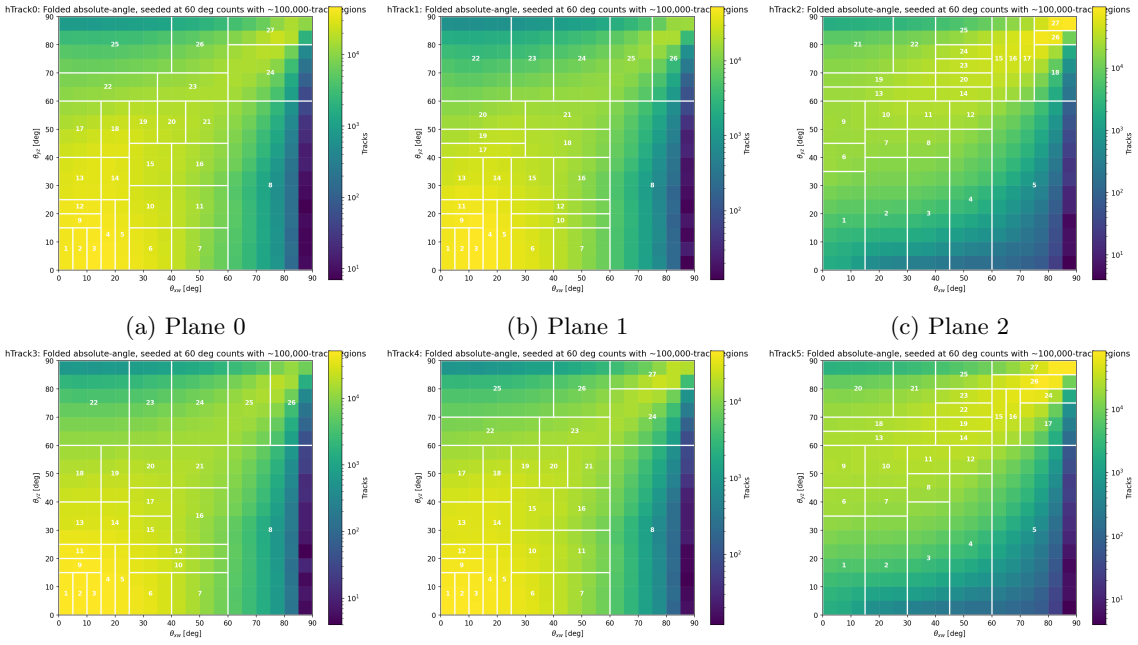
4.2 Angular Binning

The current angular binning uses folded absolute angles, motivated by the approximate symmetry of the hit-response behavior about $\theta = 0$ for both angular variables. The folded angular plane covers:

$$0 \leq |\theta_{xw}| \leq 90^\circ, \qquad 0 \leq |\theta_{yz}| \leq 90^\circ. \tag{2}$$

The binning algorithm first separates the angular space into low-low, low-high, high-low, and high-high regions, with the high-angle boundary at approximately 60°. Adaptive splitting is then run within these regions so that high-angle regions, where stronger gradients are expected, are not swallowed by very large low-angle bins. The current target used for this angular binning is 10^5 tracks per angular region.

Figure 9 shows the folded angular distributions and selected WireMod angular regions for each of the six wire planes.



(a) Plane 0 (b) Plane 1 (c) Plane 2

(d) Plane 3 (e) Plane 4 (f) Plane 5

Figure 9: Folded 2D angular track distributions and selected WireMod angular regions. The angular plane is folded into $(|\theta_{xw}|, |\theta_{yz}|)$ and the adaptive binning prioritizes the high-angle regions above approximately 60°.

4.3 Adaptive YZ Binning in X Slices

The x/YZ adaptive binning is produced with:

`scripts/make_adaptive_binning_xyz.py`

This script expects the reduced four-dimensional sparse output from `SBNDWireModHists`, with axes corresponding to $(x, y, z, \text{response})$. For each physical x-bin, it projects the data `hTrackN` histogram onto the YZ plane. If the x-bin has enough tracks, the YZ plane is adaptively split with the same quadrant-first logic used by the simple 2D workflow.

A representative command is:

```
DATA=AngleData/data/plane0
./scripts/run_with_root.sh scripts/make_adaptive_binning_xyz.py \
  --input $DATA/gen2_data_config_xyz_spline_anglebin_q_plane0_region001_full.root \
  --hist hTrack0 \
  --target 100000 \
  --max-plots 5
```

The output label is inferred from names such as `anglebin_q_plane0_region001`, so each angular-region binning is kept separate. The output files are written to:

`plots/adaptive_xyz`

4.4 XYZ Ratio Spline Extraction

The higher-dimensional ratio extraction is performed with:

`scripts/make_xyz_ratio_splines.py`

For each x-bin and adaptive YZ region, the script projects the data and MC `hHitN` histograms onto the response axis and applies the same iterative truncated-mean MPV extraction used in the simple 2D workflow. The output ROOT file contains one set of graphs per x-bin:

- `data_mpv_N_X`: data MPV graph in the YZ plane,
- `mc_mpv_N_X`: MC MPV graph in the YZ plane,
- `splines_N_X`: data/MC ratio graph in the YZ plane.

Here N is the plane index and X is the x-bin index.

A representative command for charge is:

```
DATA=AngleData/data/plane0
MC=AngleData/mc/plane0
./scripts/run_with_root.sh scripts/make_xyz_ratio_splines.py \
  --data $DATA/gen2_data_config_xyz_spline_anglebin_q_plane0_region001_full.root \
  --mc $MC/gen2_mc_config_xyz_spline_anglebin_q_plane0_region001_full.root \
  --hit-hist hHit0 \
  --idx 0
```

For `dq` or `w`, the same charge-derived adaptive YZ binning can be reused with:

`--binning-observable q`

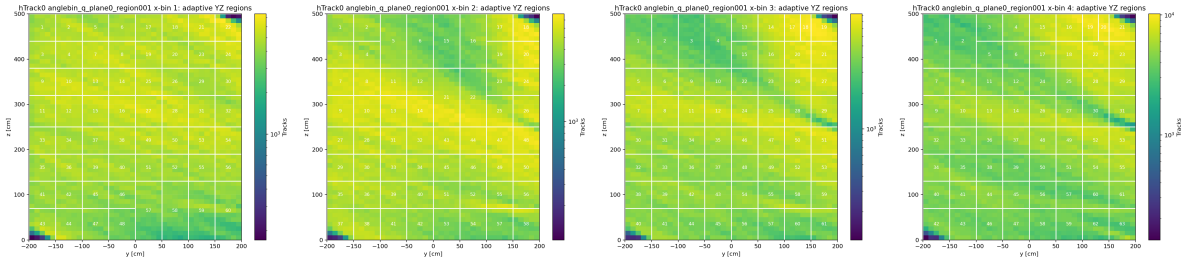
This keeps the spatial binning fixed across response observables for the same plane and angular region.

4.5 Current Higher-Dimensional Tests

The initial end-to-end higher-dimensional test was run for plane 0, angular region 001, and the `q`, `dq`, and `w` observables. The charge workflow has since been completed for all 27 angular regions of plane 0, yielding 10,719 fitted center regions without skipped regions. Plane 2 charge extraction has also been completed for all 27 angular regions using new plane-specific binning, yielding 5,742 fitted center regions without skipped regions. The full x-slice summary figures are collected in Section B. Diagnostic

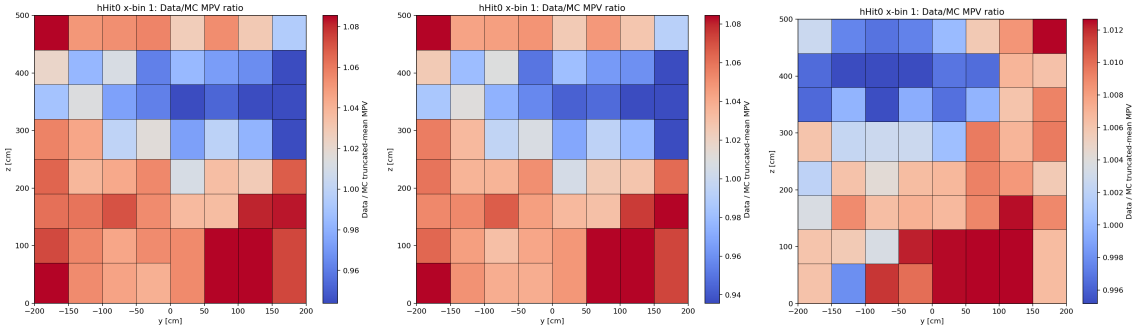
outputs include adaptive YZ region plots for individual x-bins, YZ ratio maps for each x-bin, and optional per-region hit-response plots with data and MC MPV markers.

Figures 10 and 11 show representative plane-0, region-001 outputs. These are useful debugging plots for checking that the YZ binning follows the data track density and that the extracted ratios vary smoothly within each x-slice.



(a) x-bin 1 (b) x-bin 2 (c) x-bin 3 (d) x-bin 4

Figure 10: Example adaptive YZ regions for plane 0, angular region 001, in the first four x-bins.

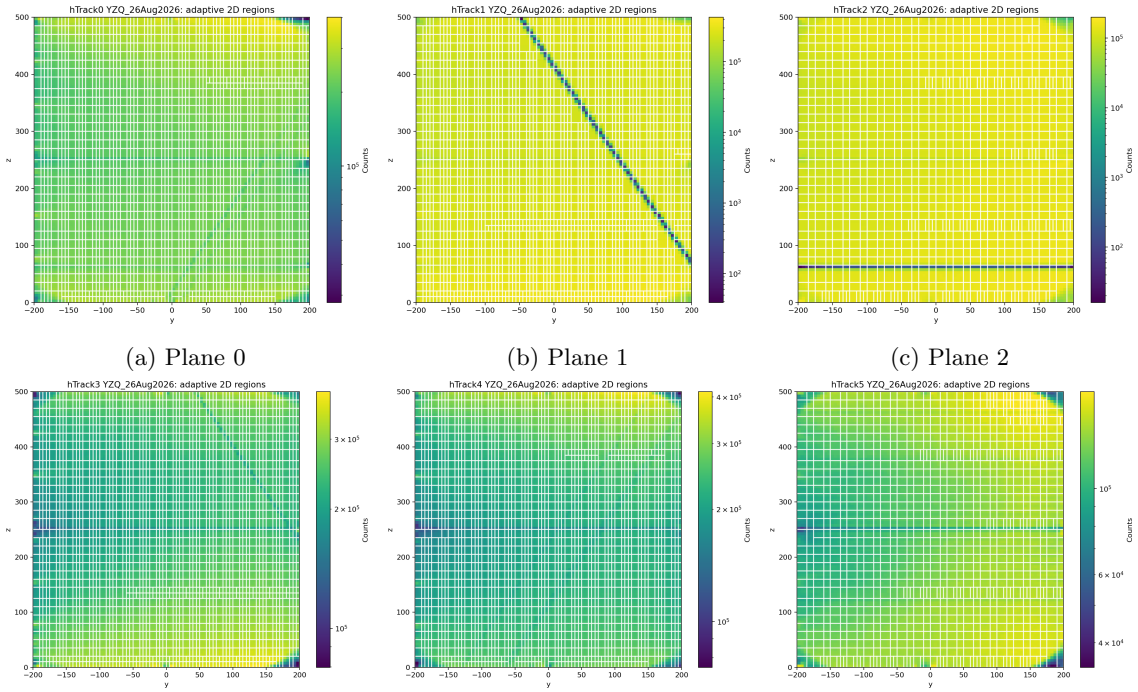


(a) Q, x-bin 1 (b) dQ/dx, x-bin 1 (c) Width, x-bin 1

Figure 11: Example YZ data/MC MPV ratio maps for plane 0, angular region 001, and three response observables.

Appendix A 2D Splines in Gen2

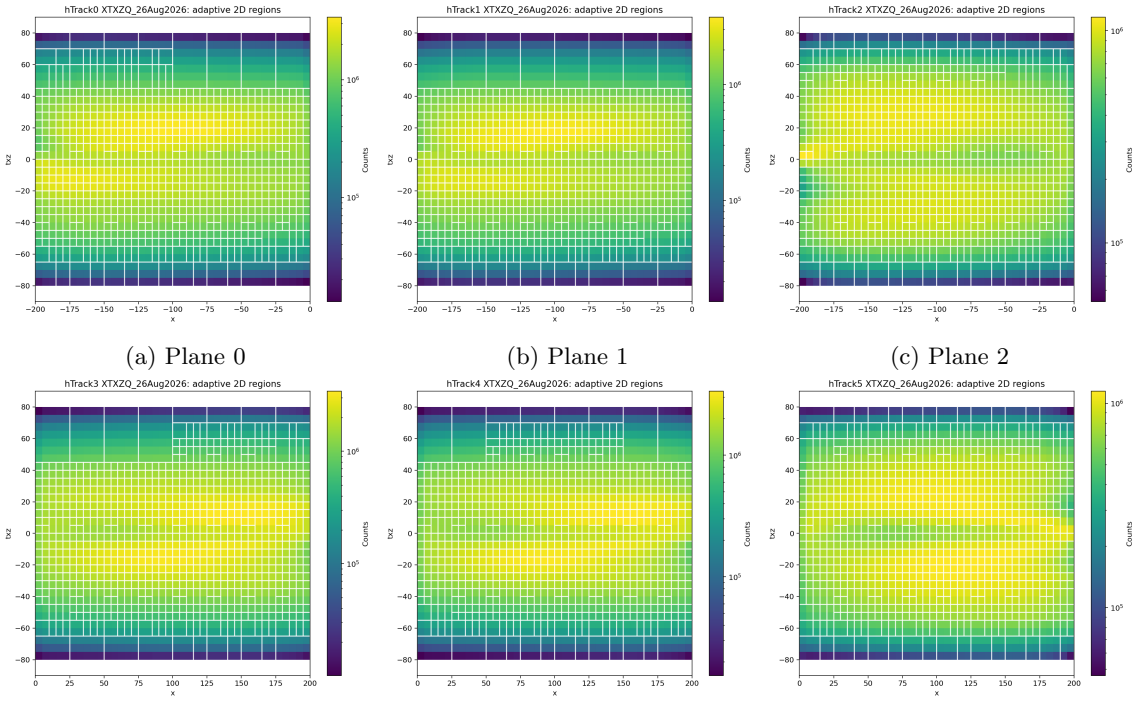
A.1 Adaptive Binning Diagnostics



(a) Plane 0 (b) Plane 1 (c) Plane 2

(d) Plane 3 (e) Plane 4 (f) Plane 5

Figure 12: `YZQ` adaptive 2D regions for all planes.

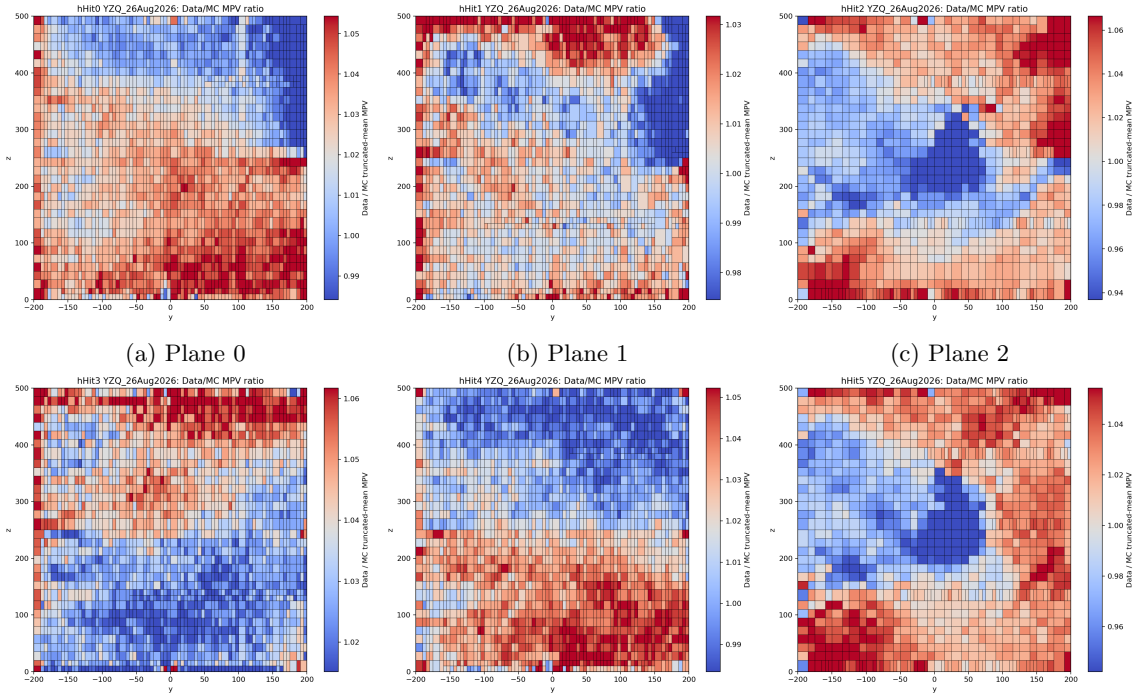


(a) Plane 0 (b) Plane 1 (c) Plane 2

(d) Plane 3 (e) Plane 4 (f) Plane 5

Figure 13: `XTXZQ` adaptive 2D regions for all planes.

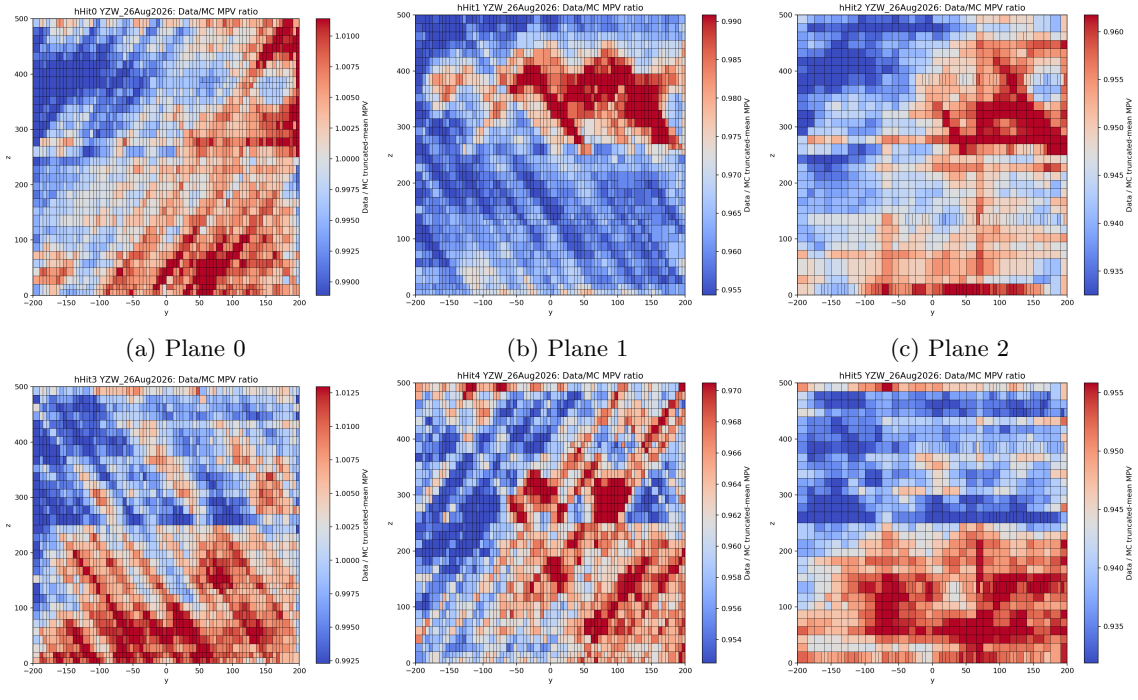
A.2 Ratio Map Diagnostics



(a) Plane 0 (b) Plane 1 (c) Plane 2

(d) Plane 3 (e) Plane 4 (f) Plane 5

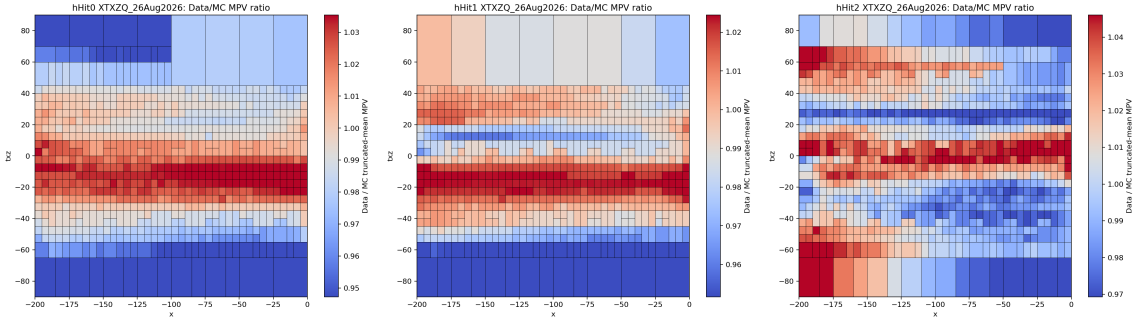
Figure 14: `YZQ` data/MC MPV ratio maps for all planes.



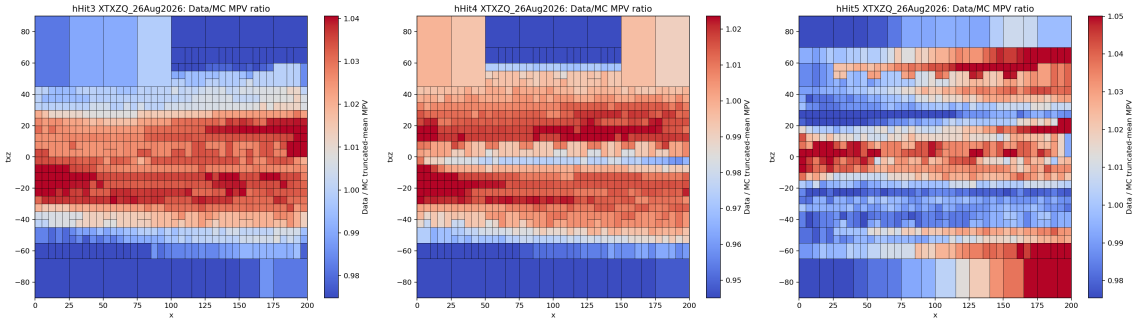
(a) Plane 0 (b) Plane 1 (c) Plane 2

(d) Plane 3 (e) Plane 4 (f) Plane 5

Figure 15: `YZW` data/MC MPV ratio maps for all planes.

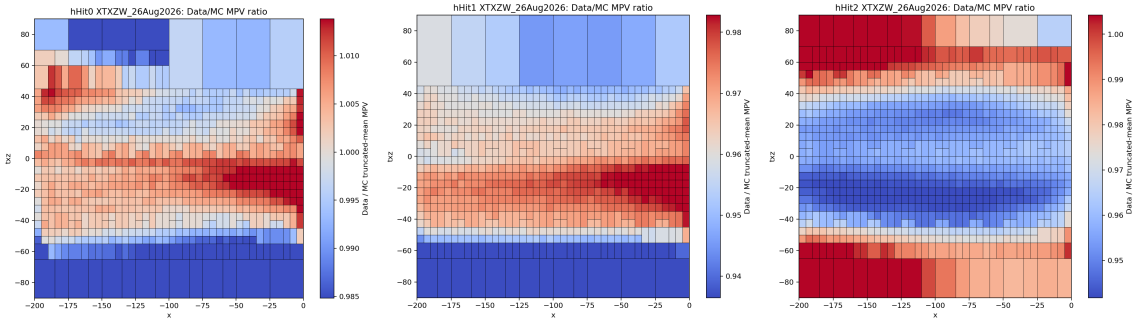


(a) Plane 0 (b) Plane 1 (c) Plane 2

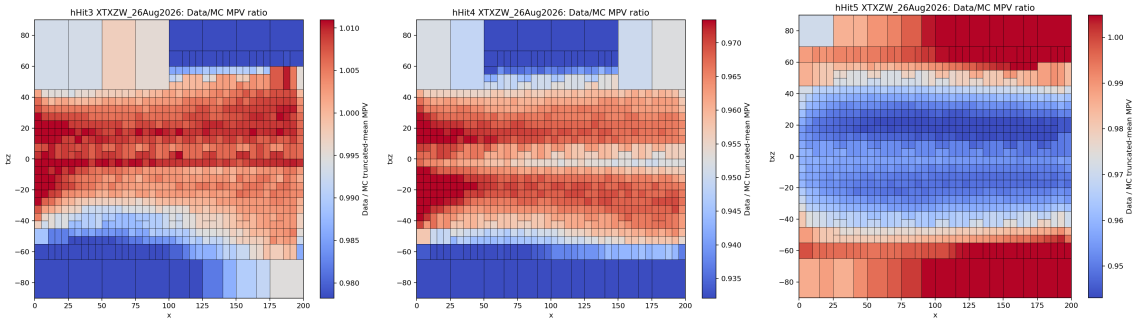


(d) Plane 3 (e) Plane 4 (f) Plane 5

Figure 16: `XTXZQ` data/MC MPV ratio maps for all planes.



(a) Plane 0 (b) Plane 1 (c) Plane 2



(d) Plane 3 (e) Plane 4 (f) Plane 5

Figure 17: `XTXZW` data/MC MPV ratio maps for all planes.

Appendix B Higher-Dimensional Angle-Binned Ratio Splines

This appendix collects the higher-dimensional ratio spline outputs, organized by wire plane. Each angular region has one summary figure containing the YZ ratio surface for every populated x-bin, with the x-bin index and spatial interval shown above each panel. The current results cover all 27 folded two-dimensional angular regions for charge (Q) and dQ/dx in plane 0, and charge in plane 2. Plane 2 uses independently generated adaptive binning from its own data histograms. Plane 1 results will be added when available. The dQ/dx figures follow the charge figures and reuse the same charge-derived adaptive binning. A third set of figures shows $R_Q/R_{dQ/dx}$, where each R is the corresponding data/MC response correction. The two spline surfaces are linearly interpolated separately on the same YZ grid, then divided at each grid location. Graph coordinates and x-bin sets must match. This compares the corrections rather than the raw charge and dQ/dx responses.

The visualization follows `scripts/plot_2d_spline_summaries_linear.py`: duplicate graph coordinates are averaged and the edge-augmented ratio points are linearly interpolated on a 40×40 grid. Z is horizontal and Y is vertical. All panels use the same diverging color scale from 0.85 to 1.15; values outside this interval saturate at the color-scale endpoints. Interpolation is restricted to the convex hull of the input graph points.

The summary plots and plane subsections are generated with:

```
python3 scripts/plot_xyz_spline_summaries_linear.py \
  --observable q --planes 0 \
  --outdir ../WireMod-Spline-Generation-Guide/Figures/WireModXYZ/summary \
  --appendix-output ../WireMod-Spline-Generation-Guide/xyz_spline_appendix.tex
```

Run this command from `SBNDWireModSplineGeneration` in a PyROOT-enabled environment. Additional planes are included automatically when their angle-bin spline outputs and matching binning CSVs are available. To generate the dQ/dx figures after charge within the existing Plane 0 subsection, use:

```
python3 scripts/plot_xyz_spline_summaries_linear.py \
  --observable dq --binning-observable q \
  --planes 0 --omit-plane-heading \
  --outdir ../WireMod-Spline-Generation-Guide/Figures/WireModXYZ/summary \
  --appendix-output ../WireMod-Spline-Generation-Guide/xyz_spline_appendix_dq.tex
```

Plane 2 charge summaries use the same plotting script with `--observable q --planes 2` and a separate appendix fragment, `xyz_spline_appendix_plane2_q.tex`, included after the Plane 0 results.

The Q-to-dQ/dx comparison figures use the same script and figure directory, with these options:

```
--observable q --divide-by dq --planes 0 --omit-plane-heading
--appendix-output \
  ../WireMod-Spline-Generation-Guide/xyz_spline_appendix_q_over_dq.tex
```

B.1 Plane 0

B.1.1 Charge (Q)

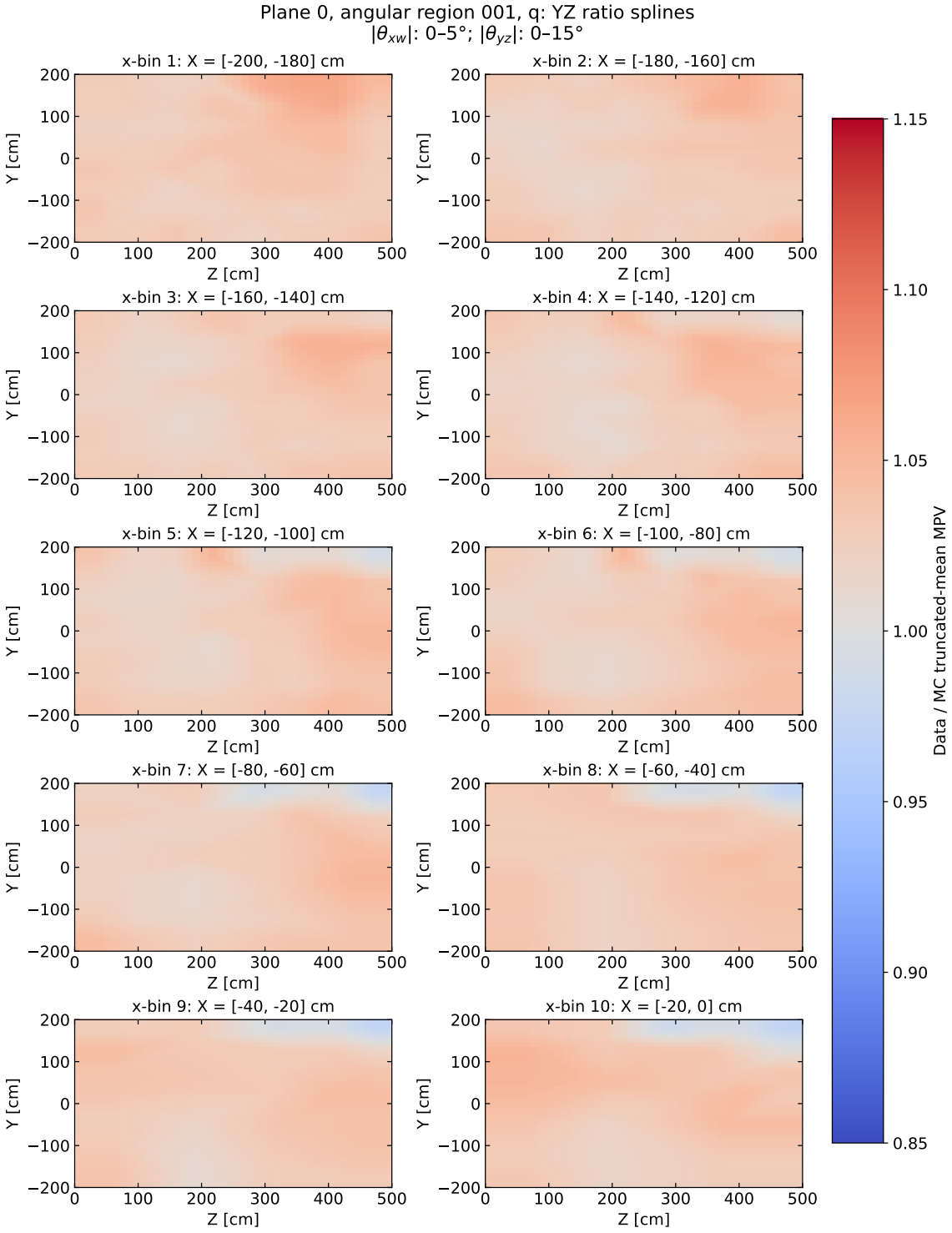


Figure 18: Plane 0, angular region 001: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 5]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

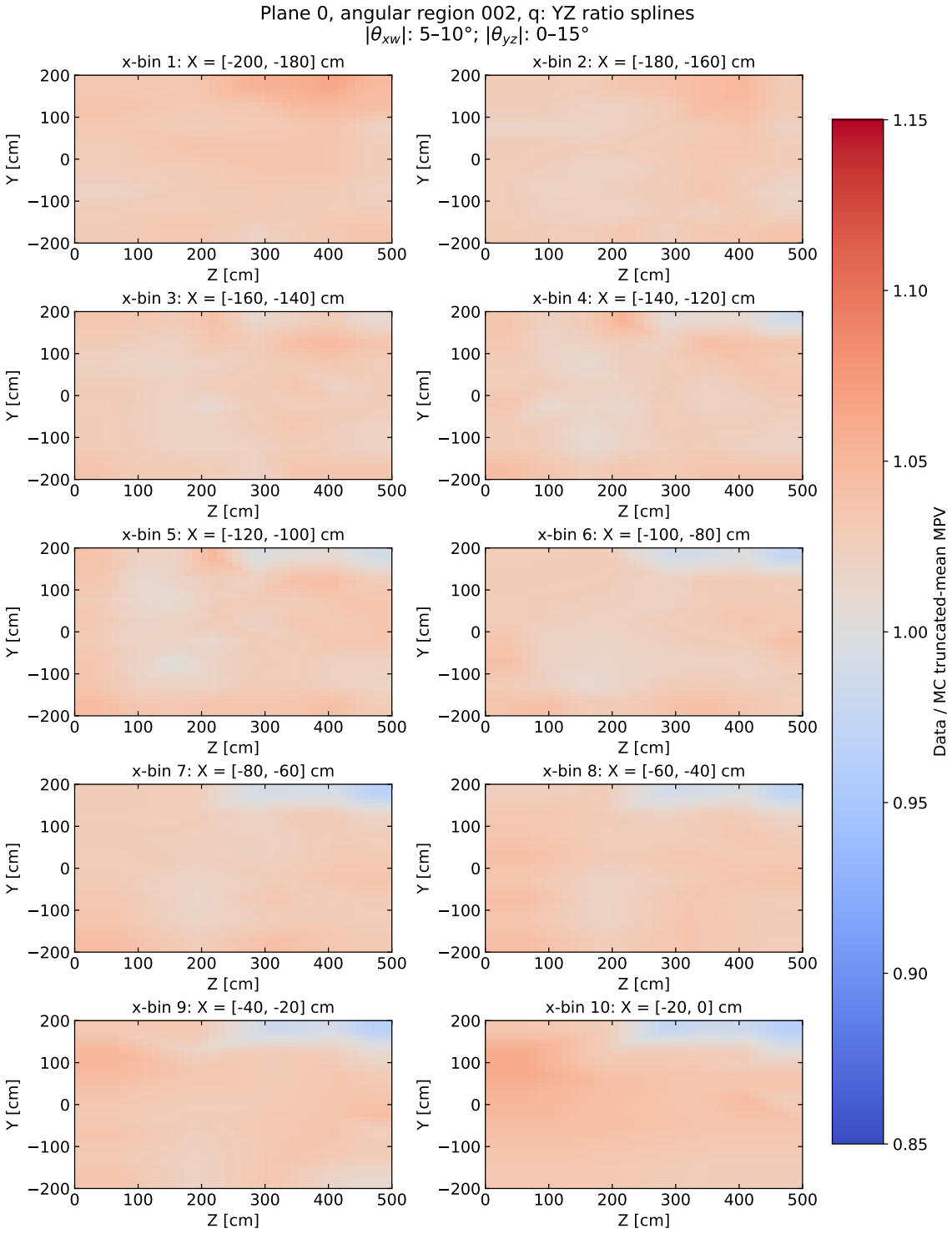


Figure 19: Plane 0, angular region 002: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [5, 10]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

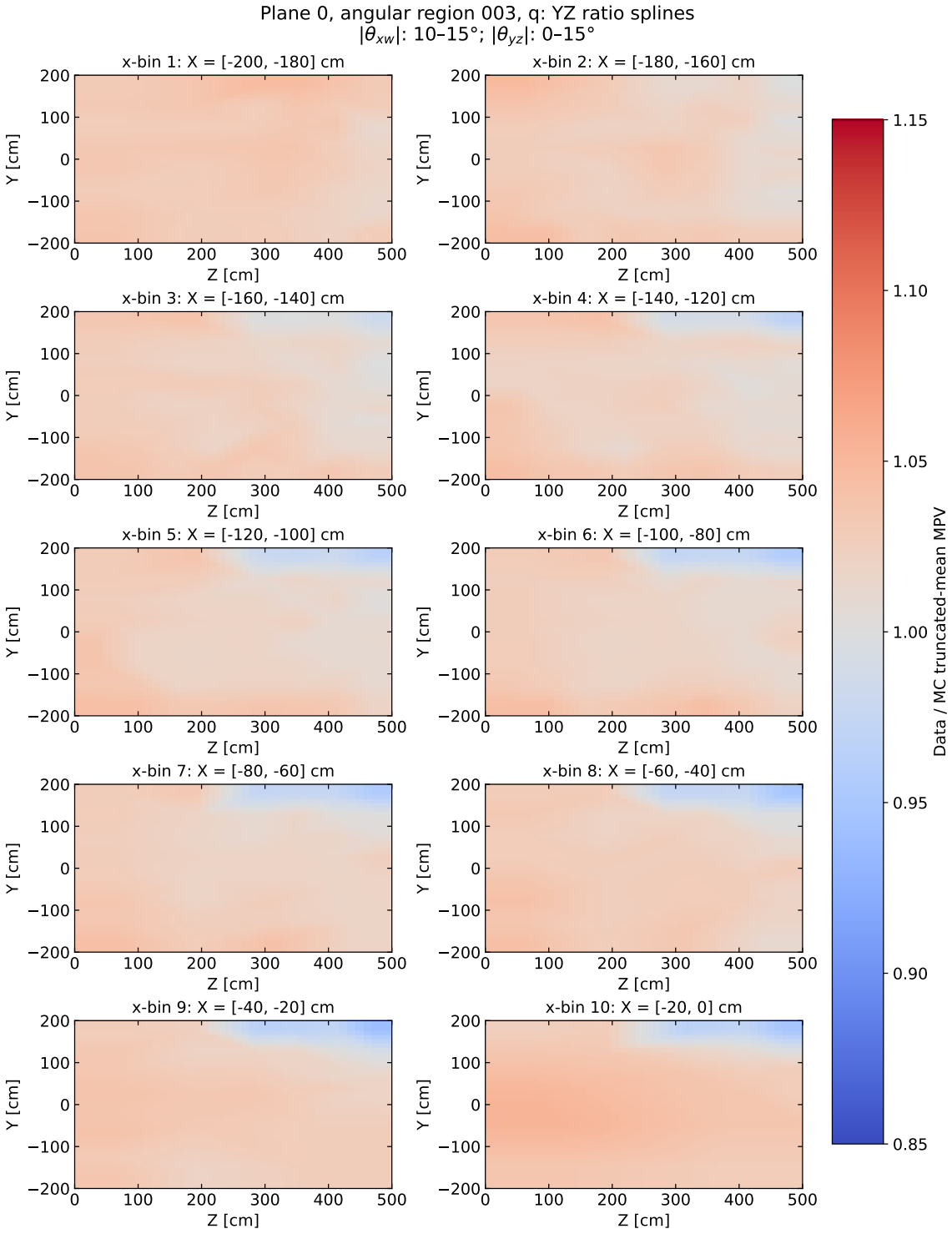


Figure 20: Plane 0, angular region 003: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [10, 15]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

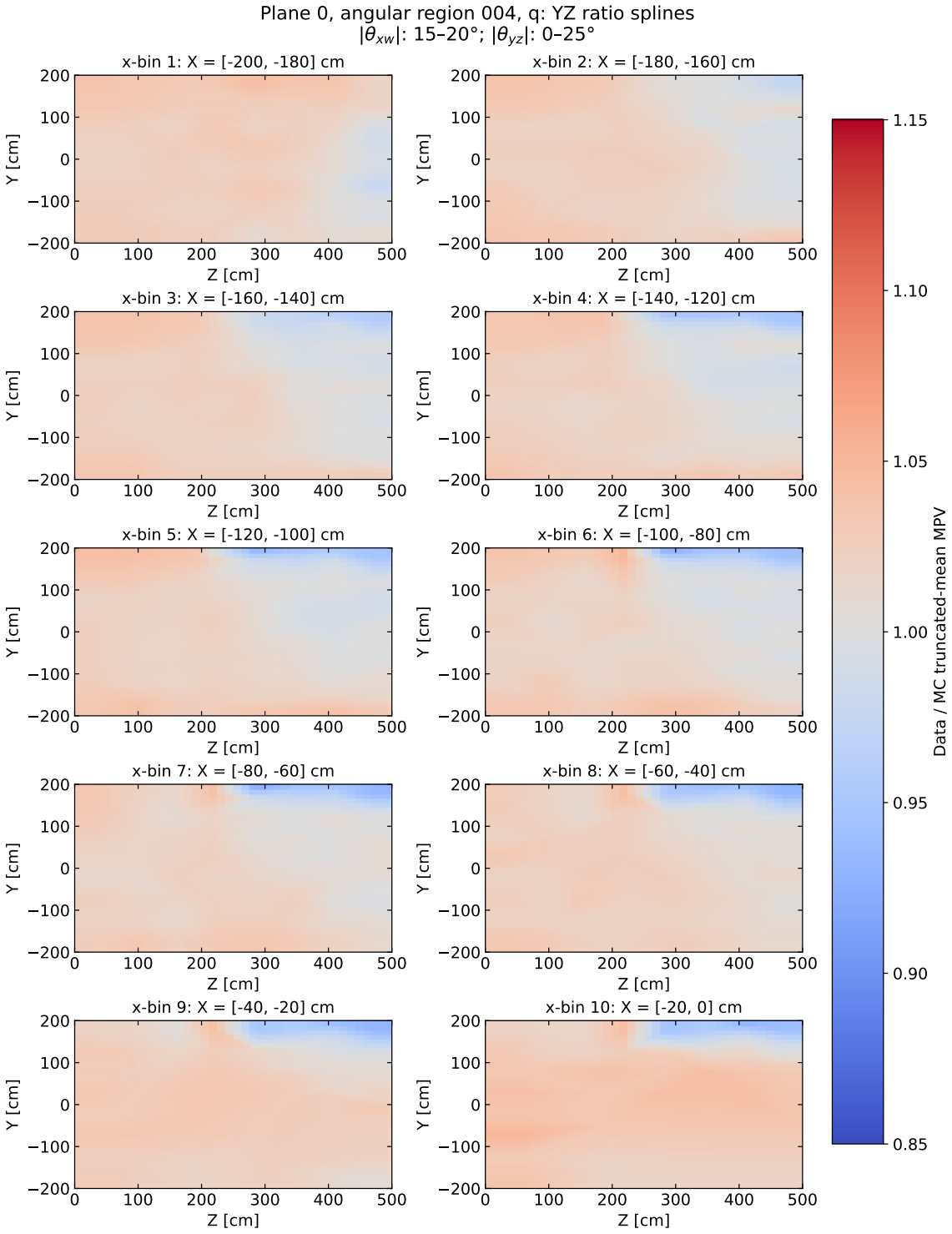


Figure 21: Plane 0, angular region 004: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 20]^\circ$ and $|\theta_{yz}| \in [0, 25]^\circ$.

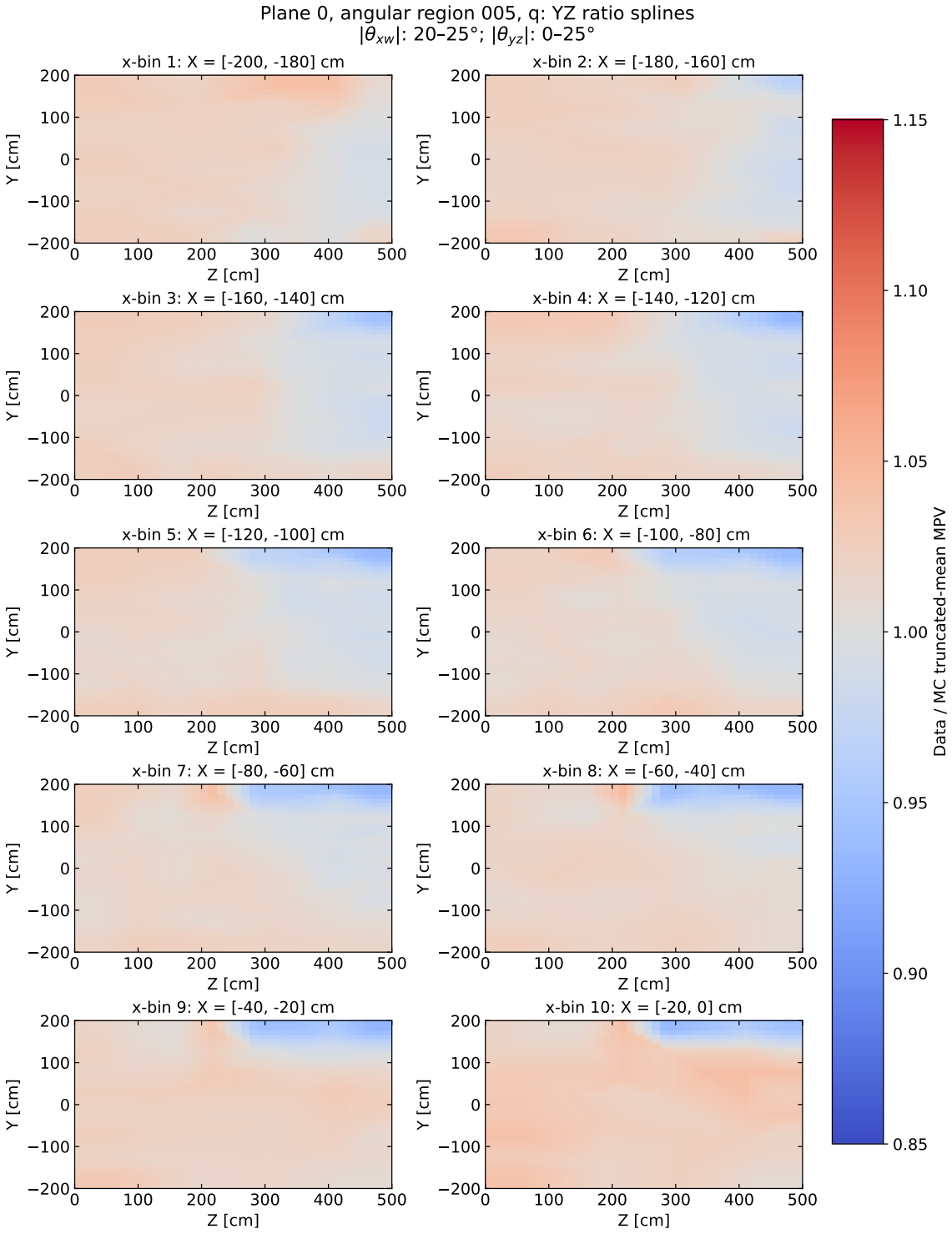


Figure 22: Plane 0, angular region 005: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [20, 25]^\circ$ and $|\theta_{yz}| \in [0, 25]^\circ$.

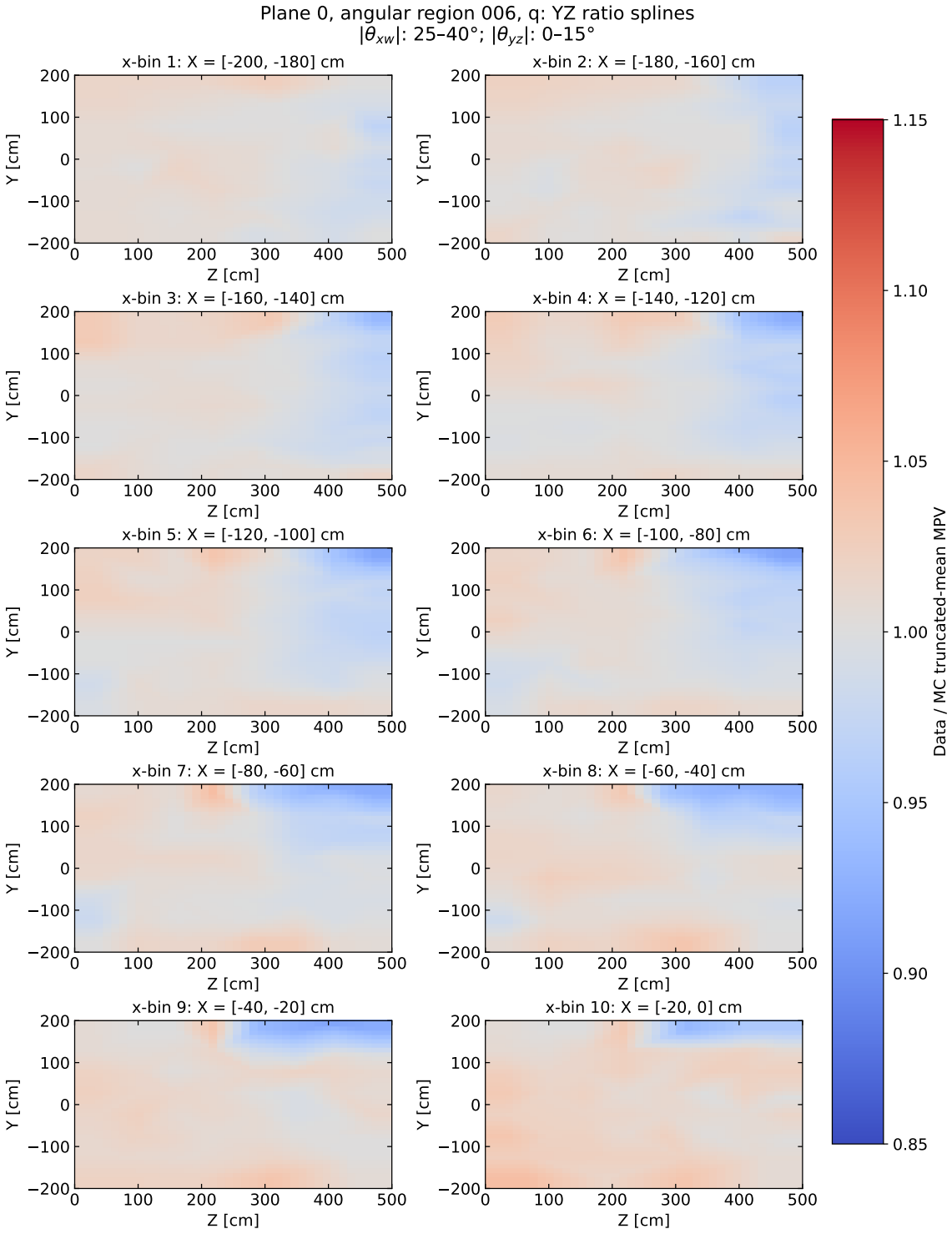


Figure 23: Plane 0, angular region 006: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

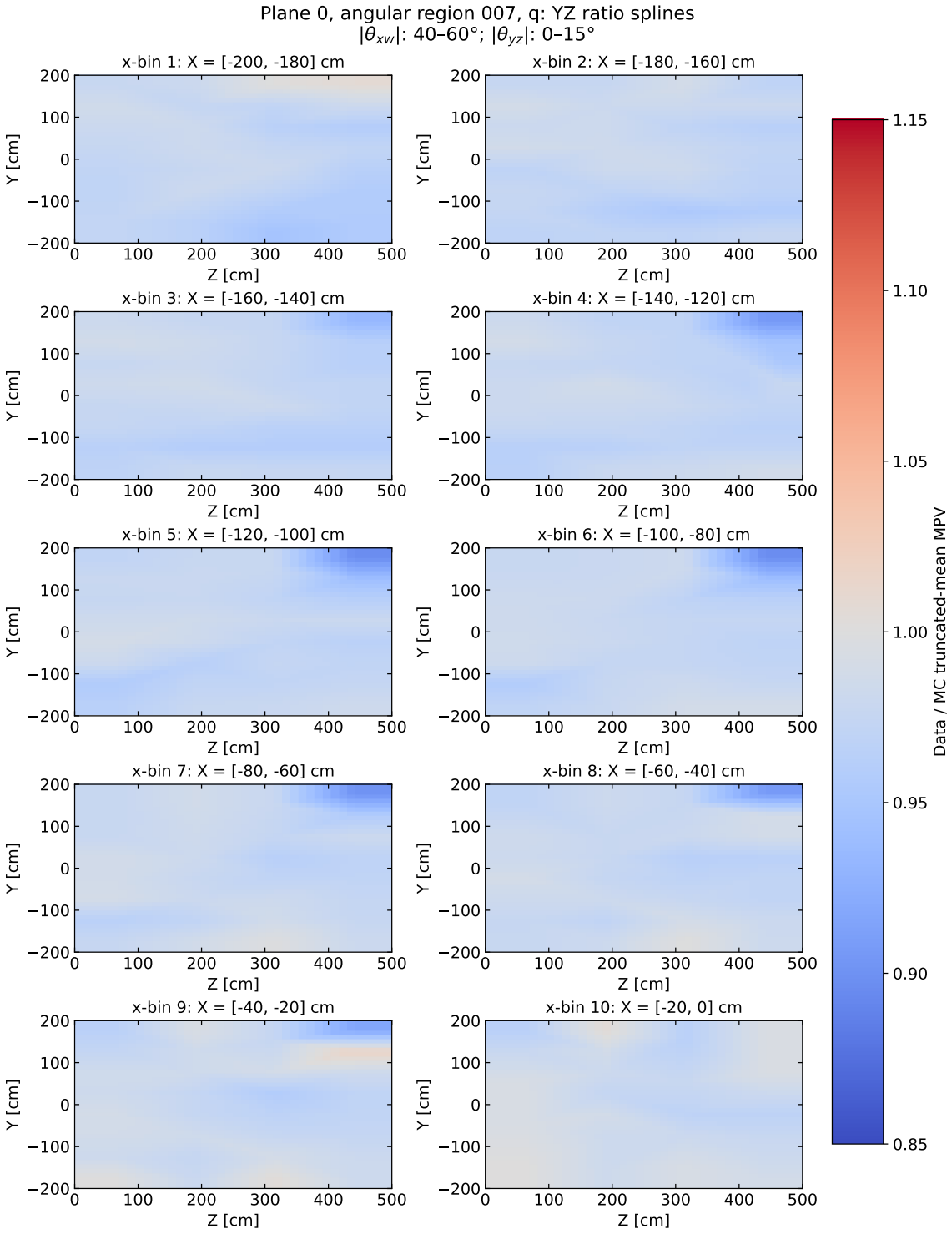


Figure 24: Plane 0, angular region 007: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

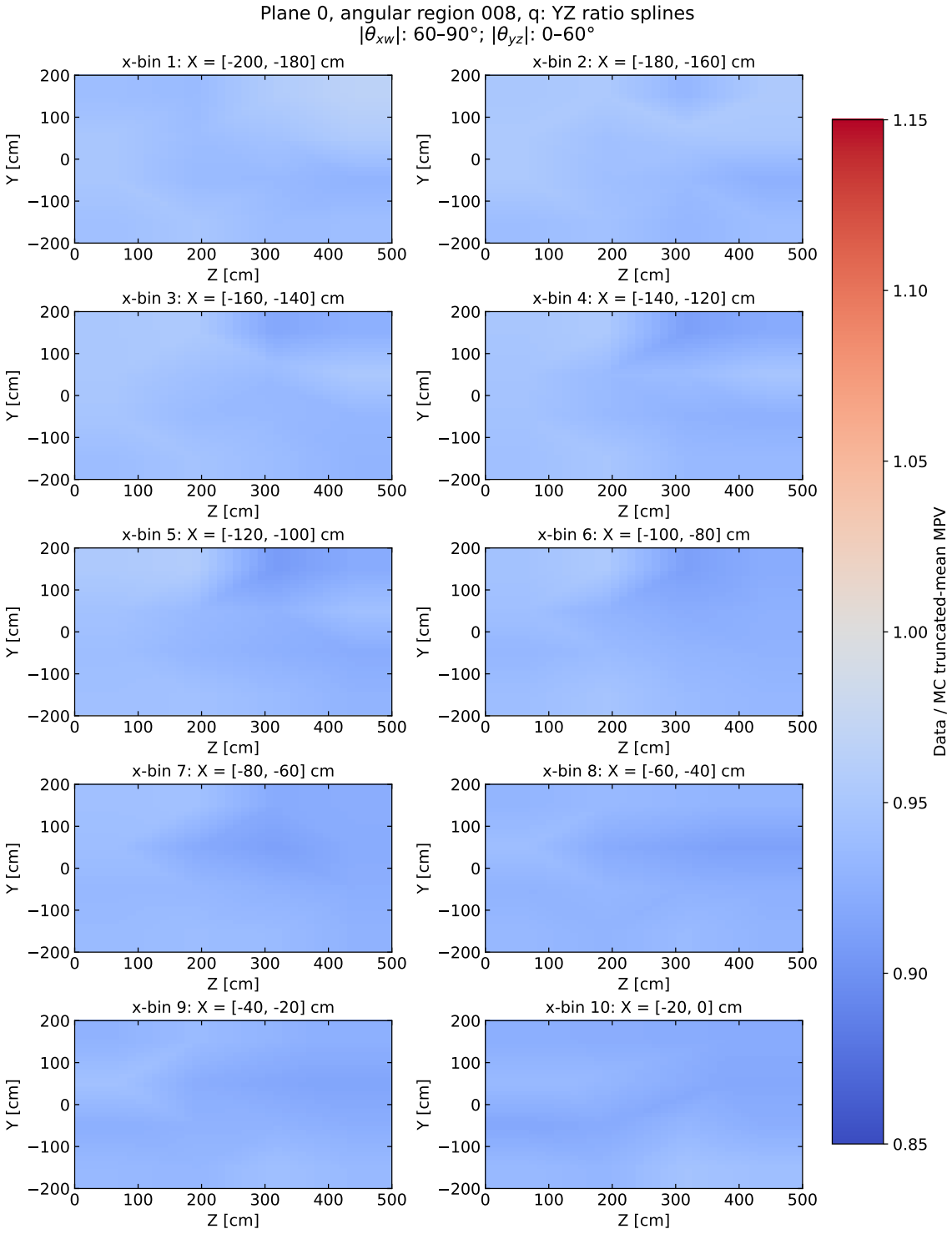


Figure 25: Plane 0, angular region 008: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [0, 60]^\circ$.

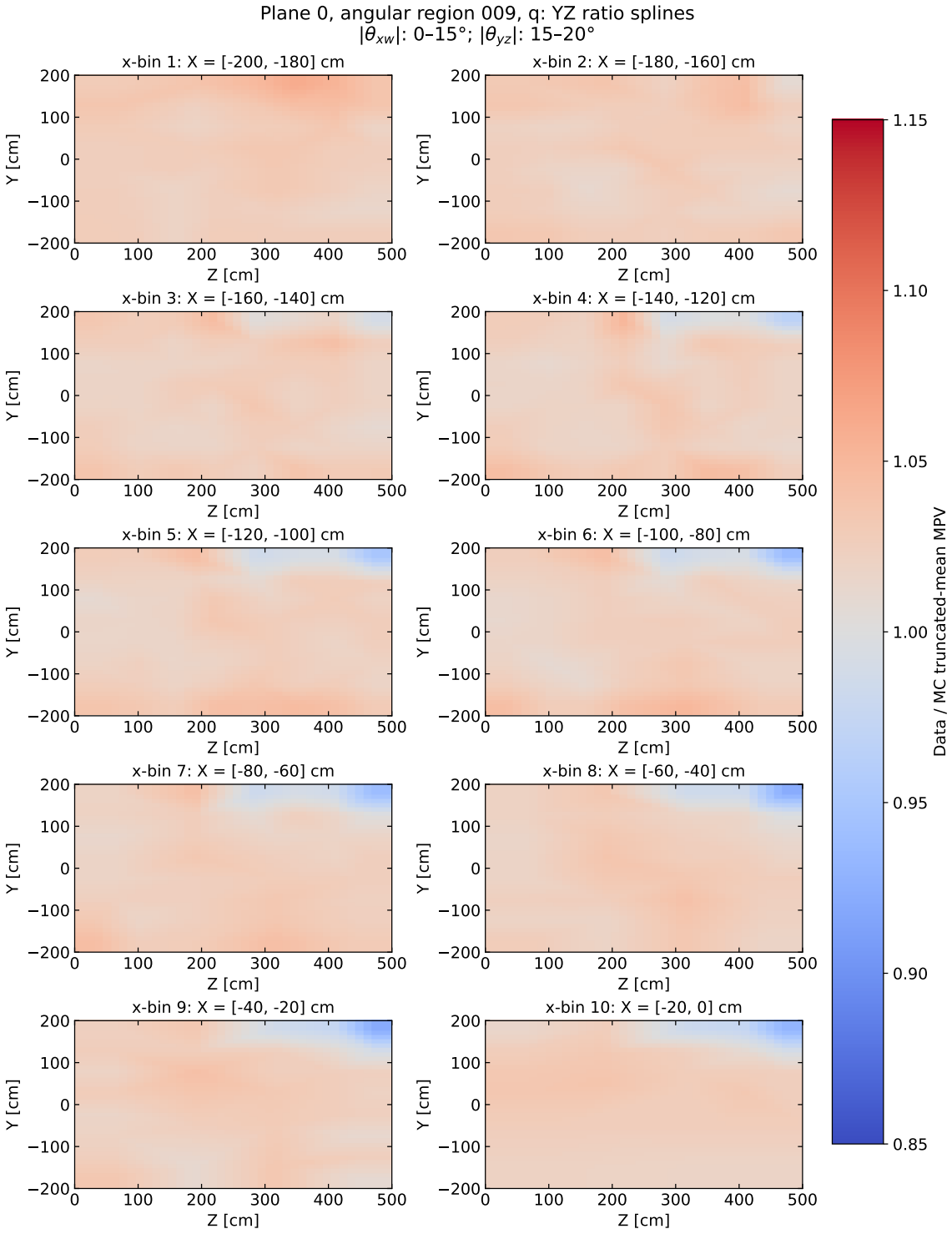


Figure 26: Plane 0, angular region 009: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [15, 20]^\circ$.

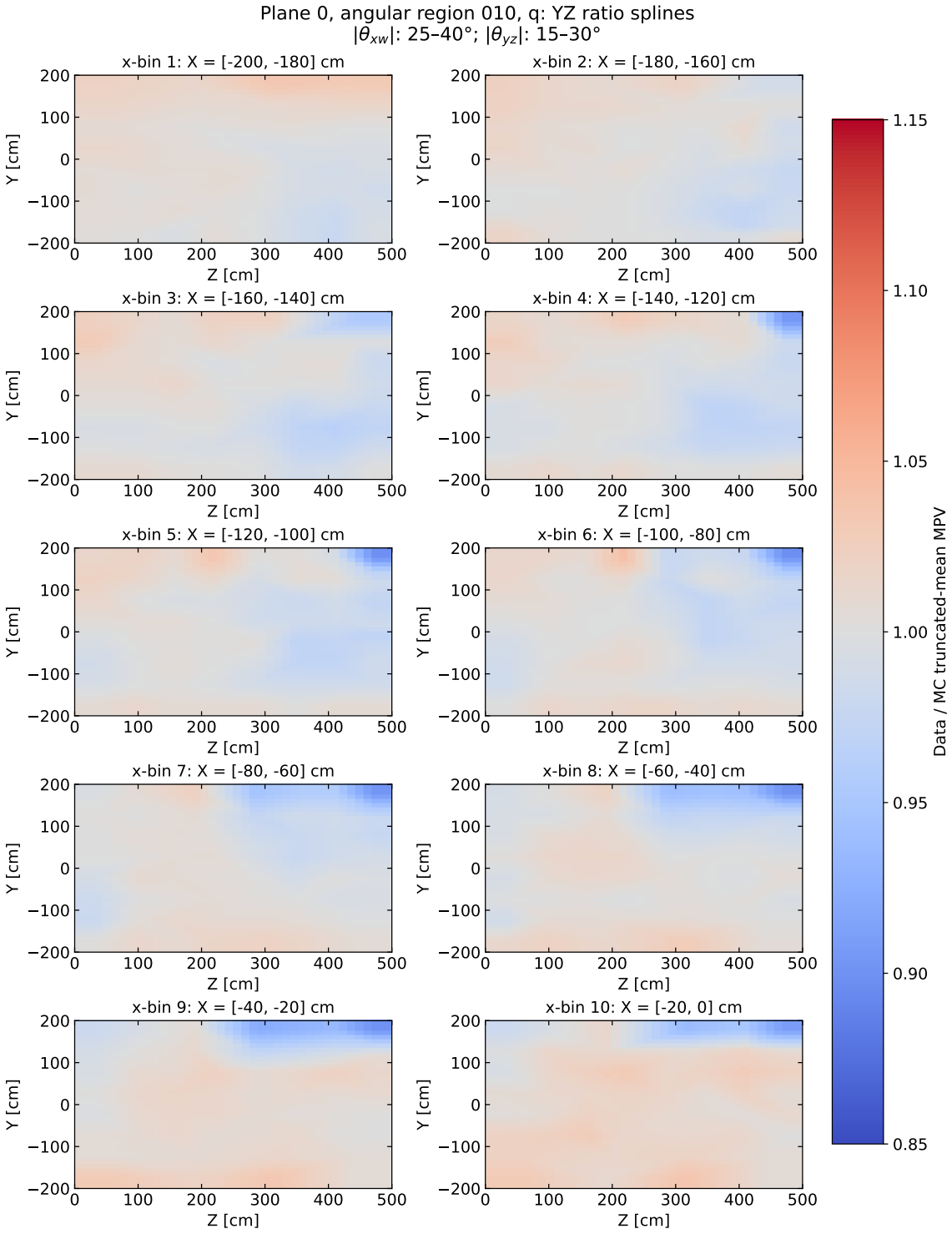


Figure 27: Plane 0, angular region 010: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [15, 30]^\circ$.

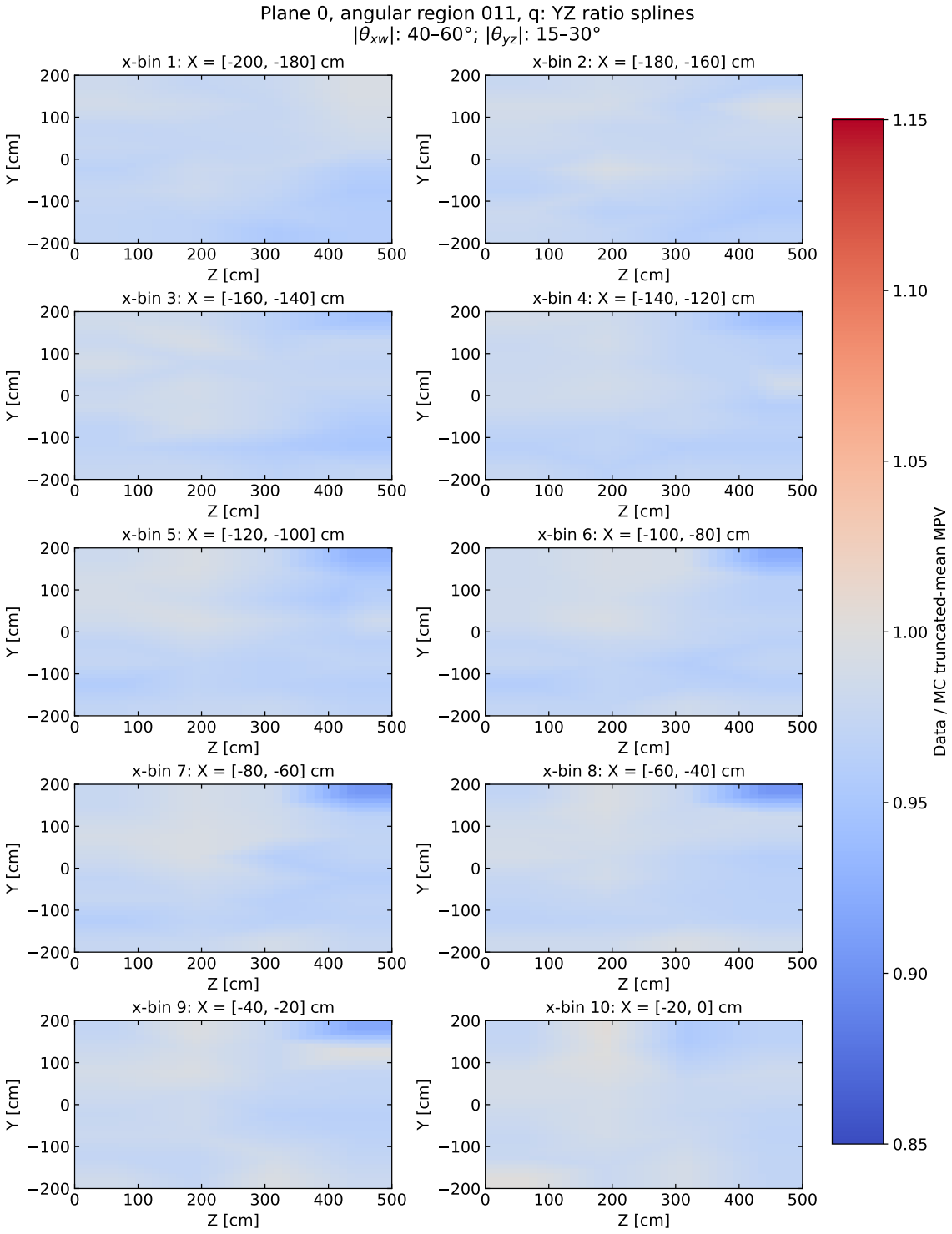


Figure 28: Plane 0, angular region 011: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [15, 30]^\circ$.

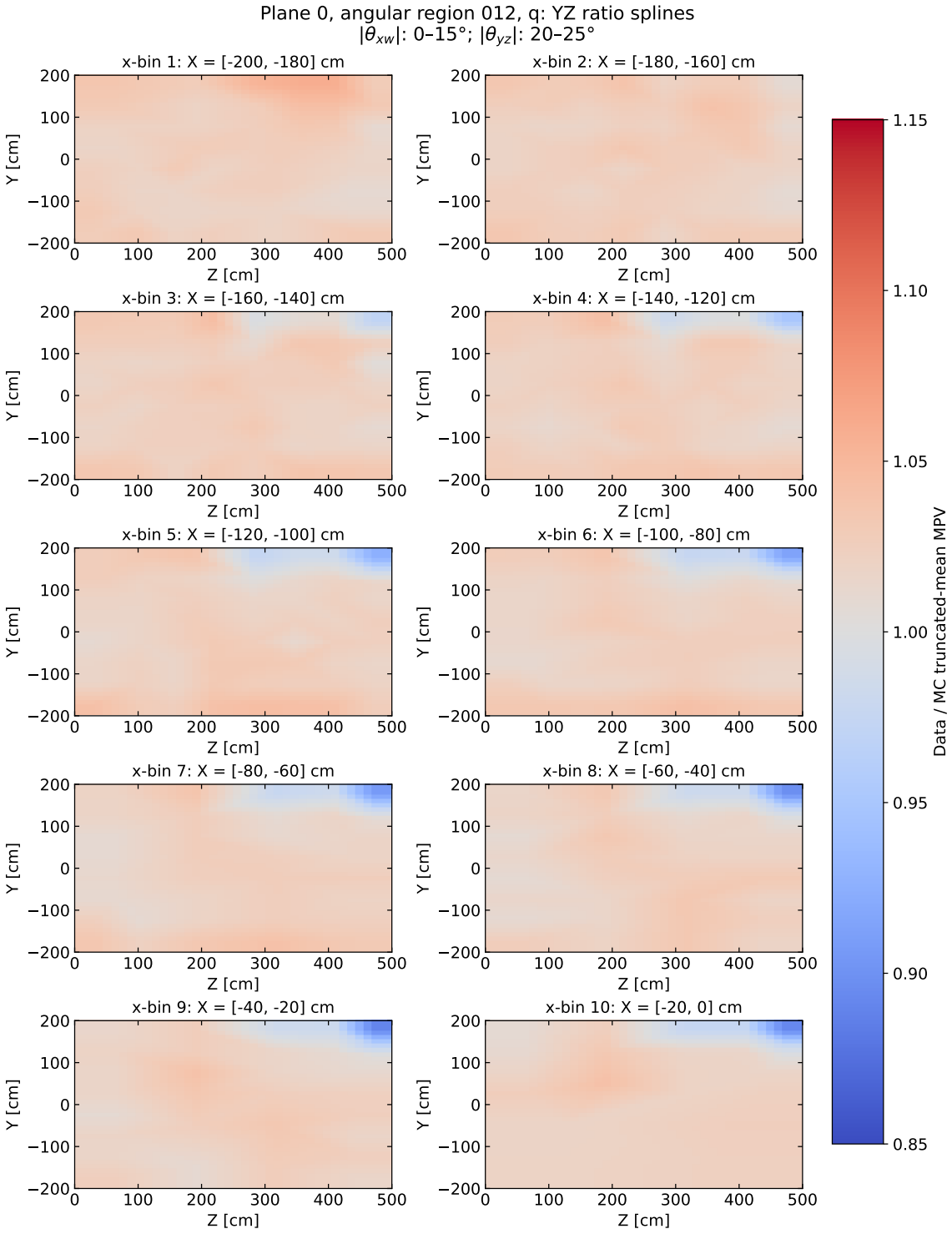


Figure 29: Plane 0, angular region 012: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [20, 25]^\circ$.

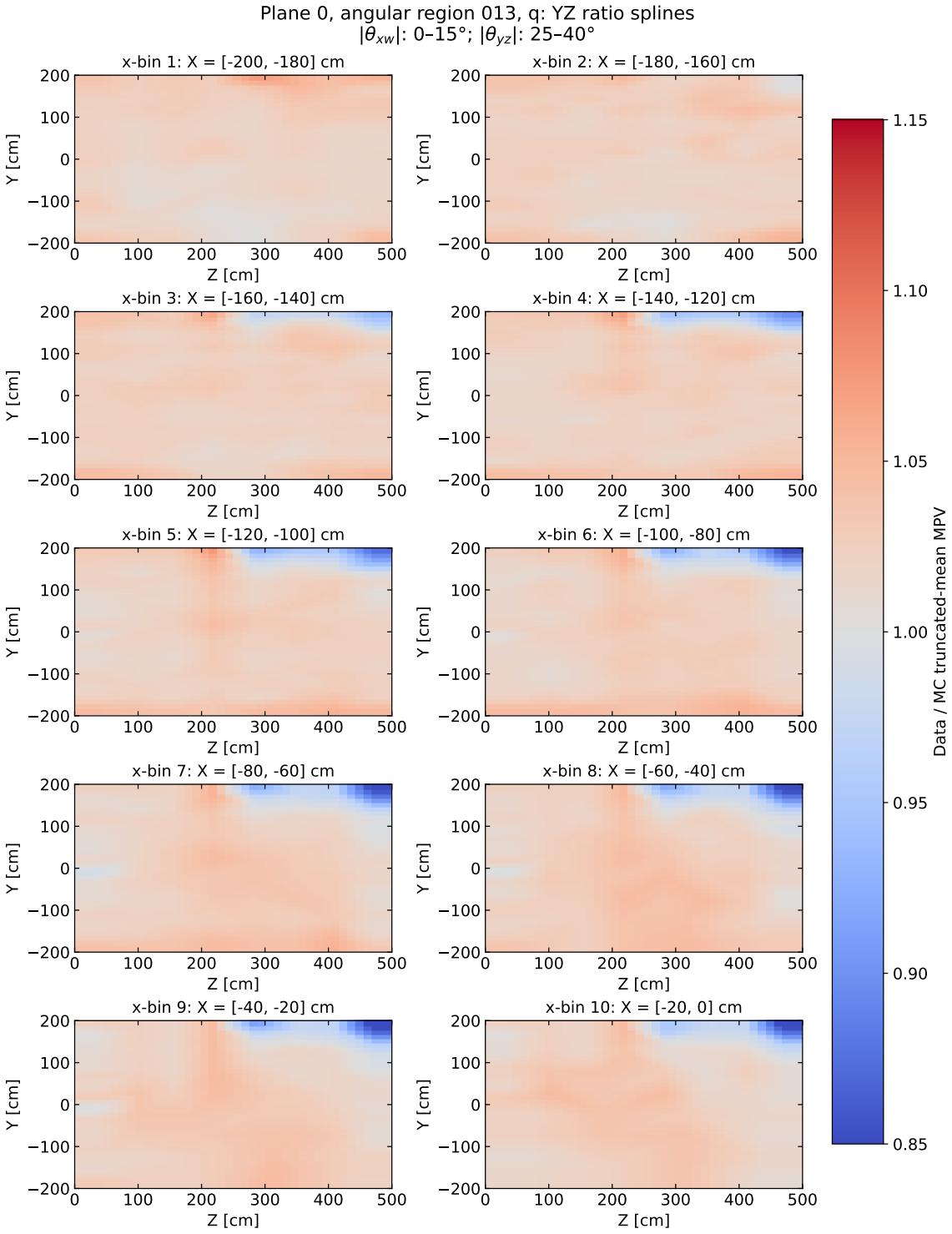


Figure 30: Plane 0, angular region 013: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [25, 40]^\circ$.

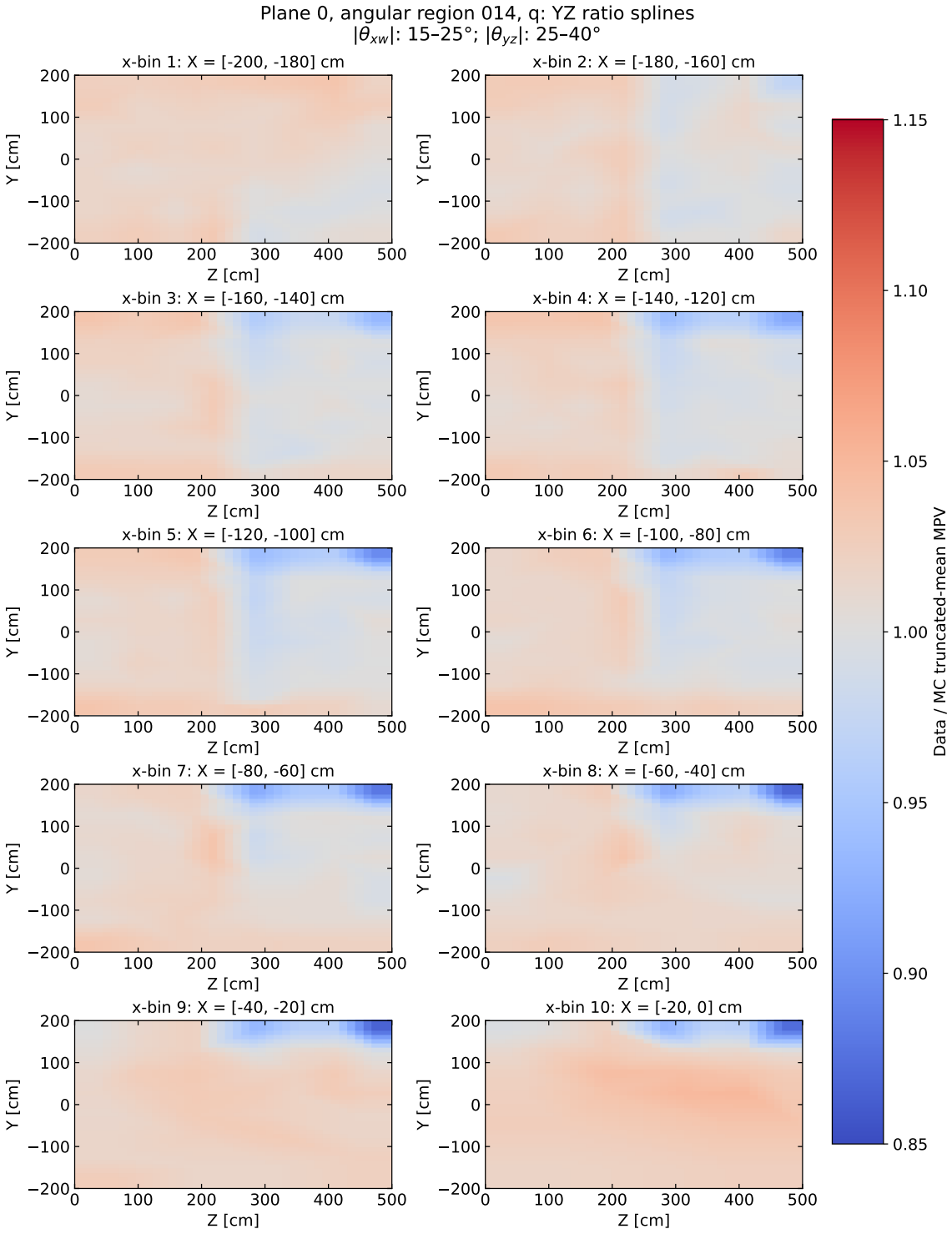


Figure 31: Plane 0, angular region 014: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 25]^\circ$ and $|\theta_{yz}| \in [25, 40]^\circ$.

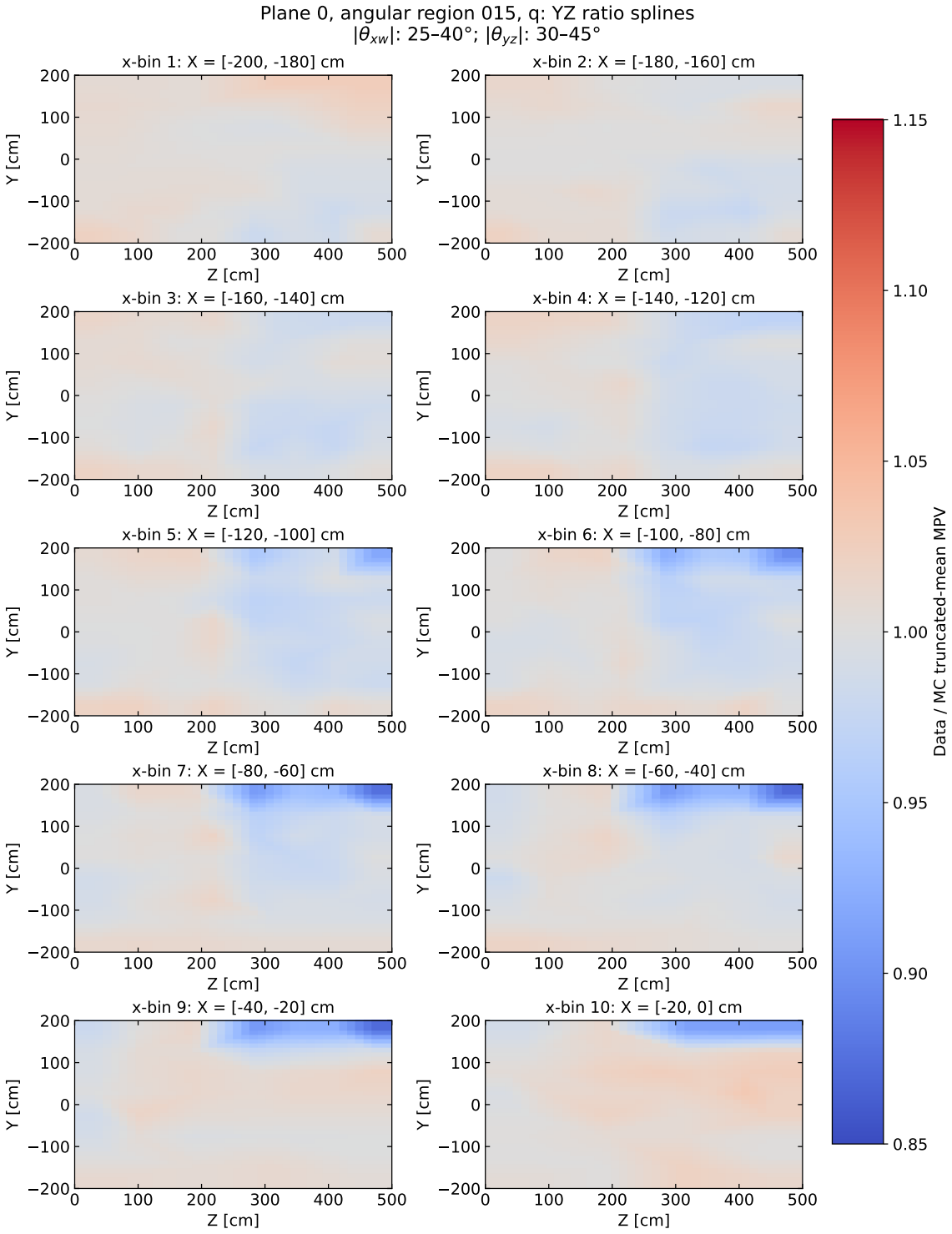


Figure 32: Plane 0, angular region 015: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [30, 45]^\circ$.

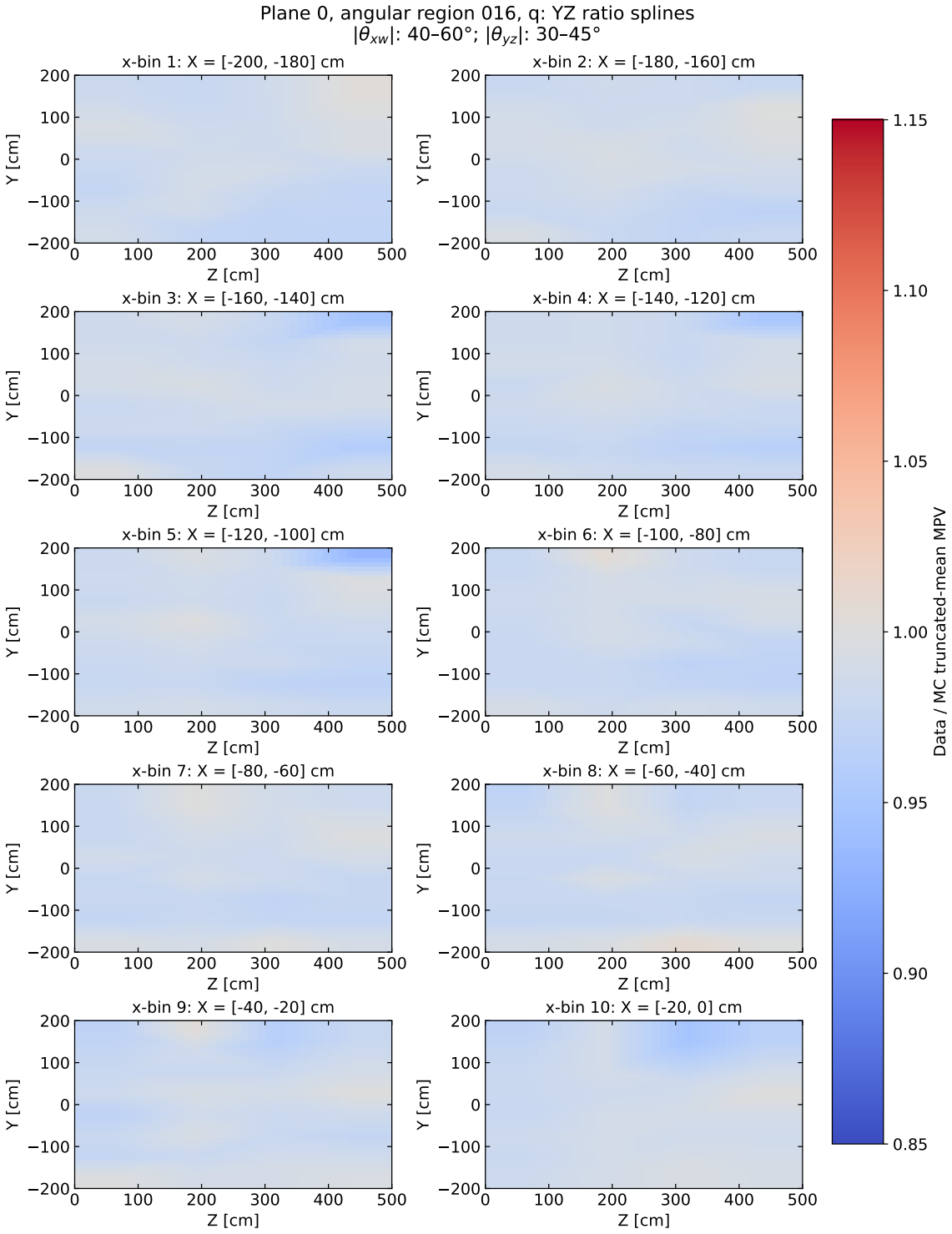


Figure 33: Plane 0, angular region 016: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [30, 45]^\circ$.

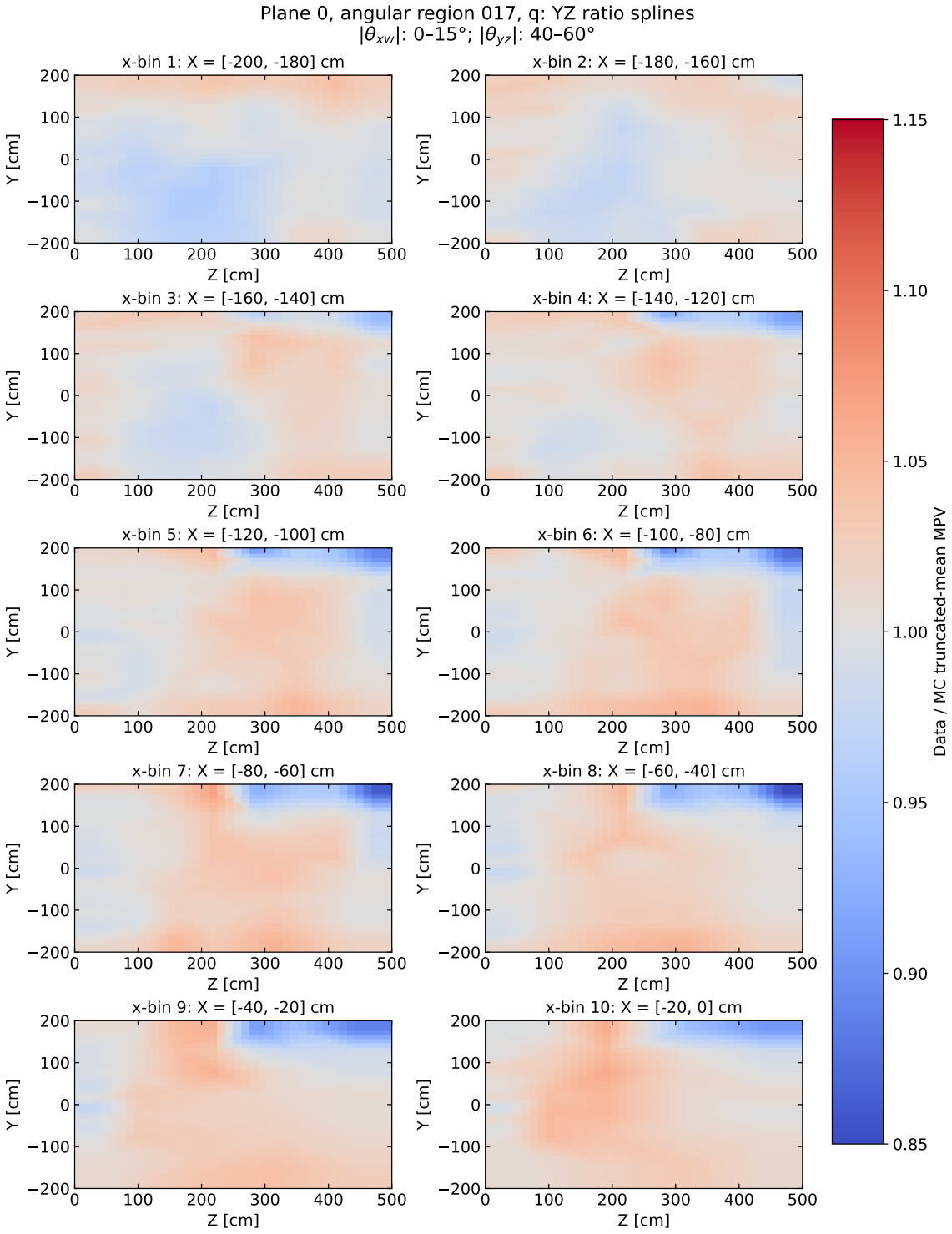


Figure 34: Plane 0, angular region 017: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [40, 60]^\circ$.

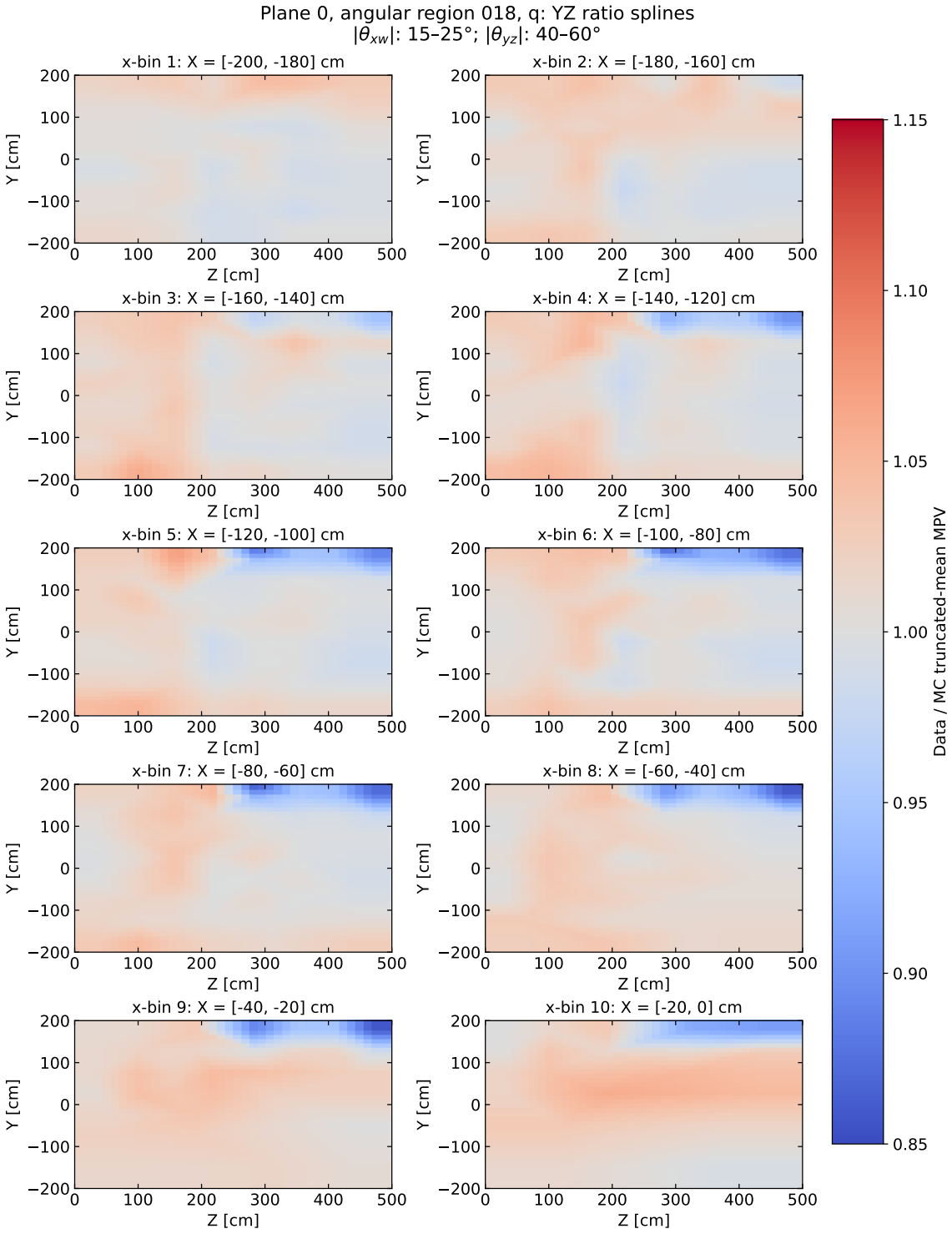


Figure 35: Plane 0, angular region 018: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 25]^\circ$ and $|\theta_{yz}| \in [40, 60]^\circ$.

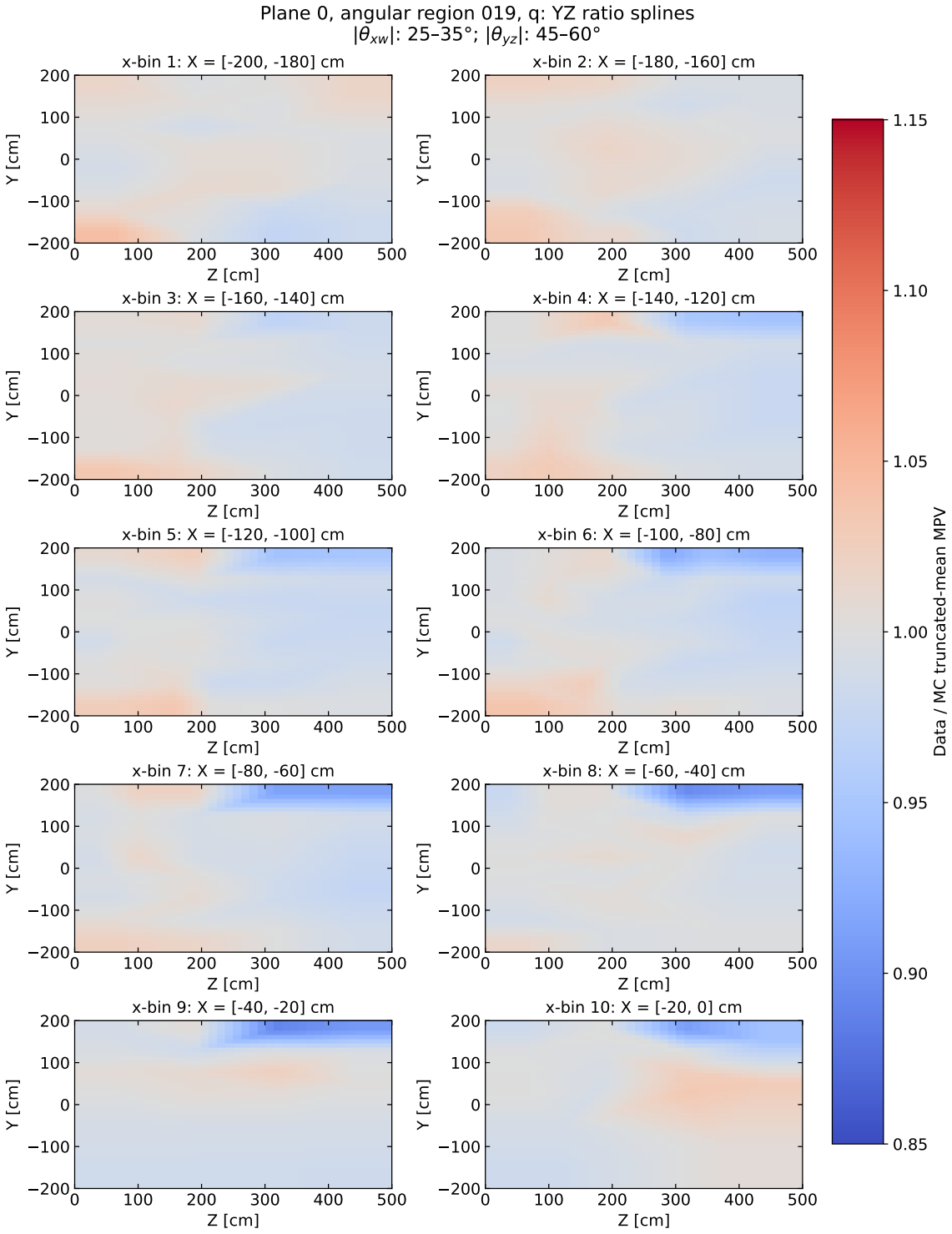


Figure 36: Plane 0, angular region 019: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 35]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

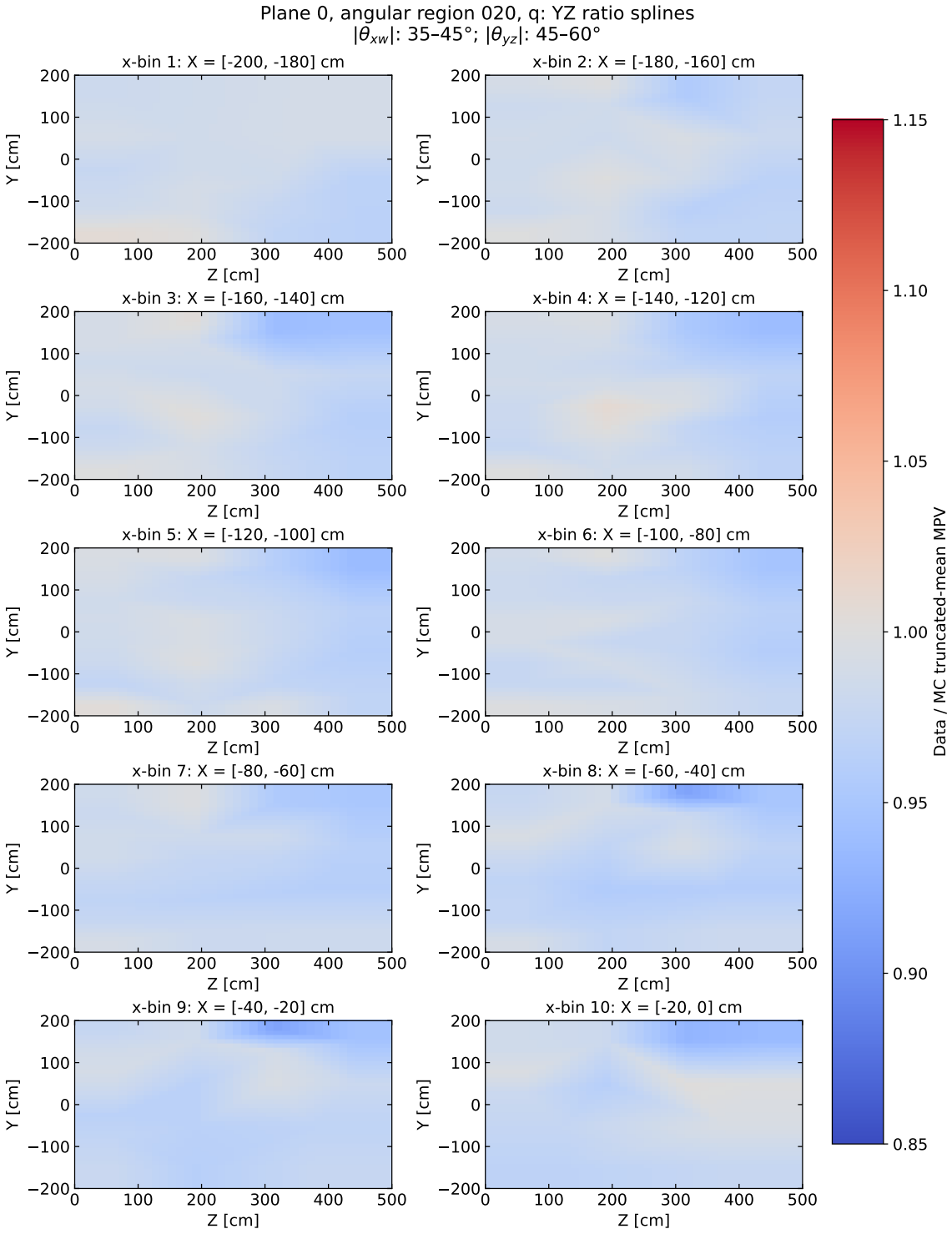


Figure 37: Plane 0, angular region 020: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [35, 45]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

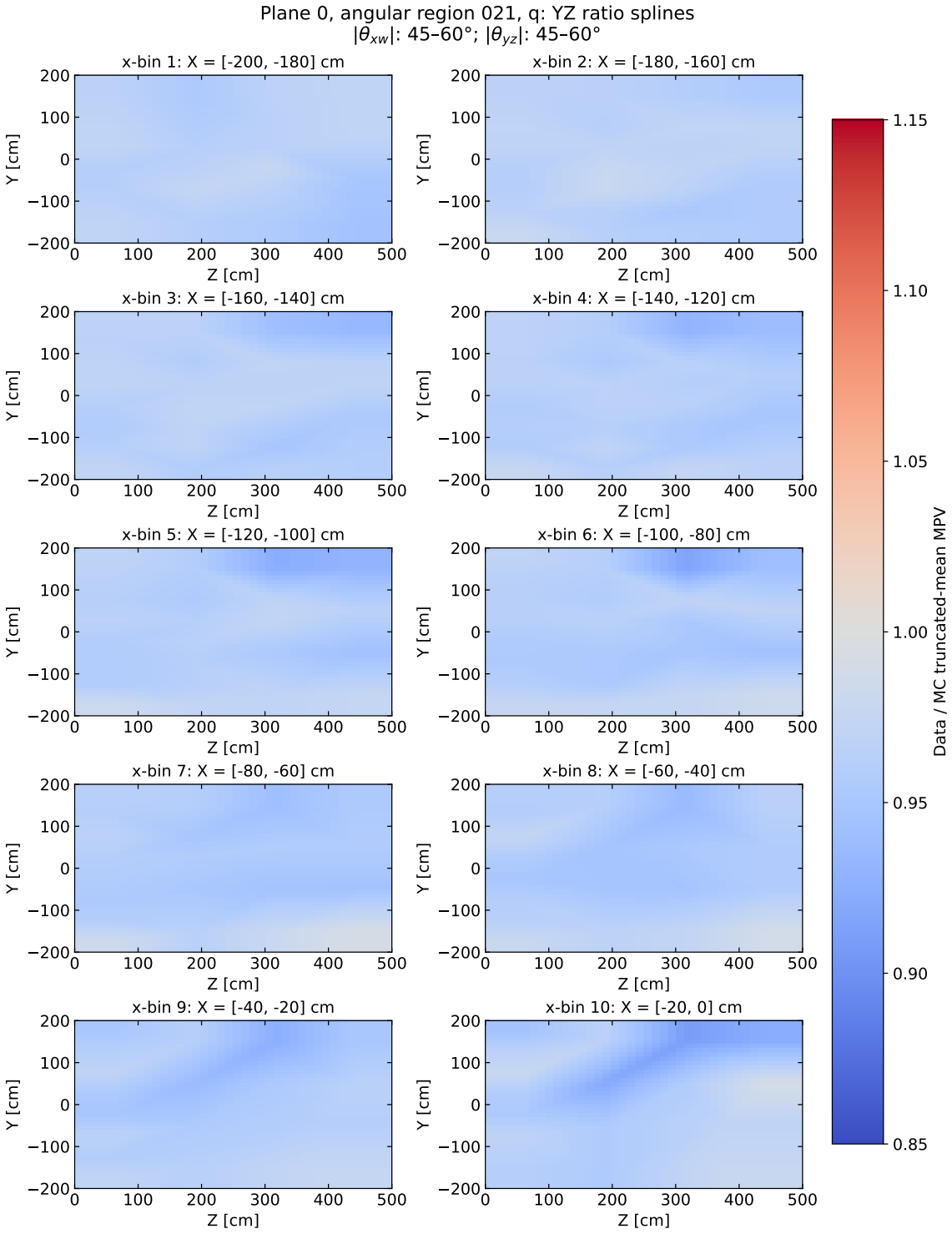


Figure 38: Plane 0, angular region 021: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [45, 60]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

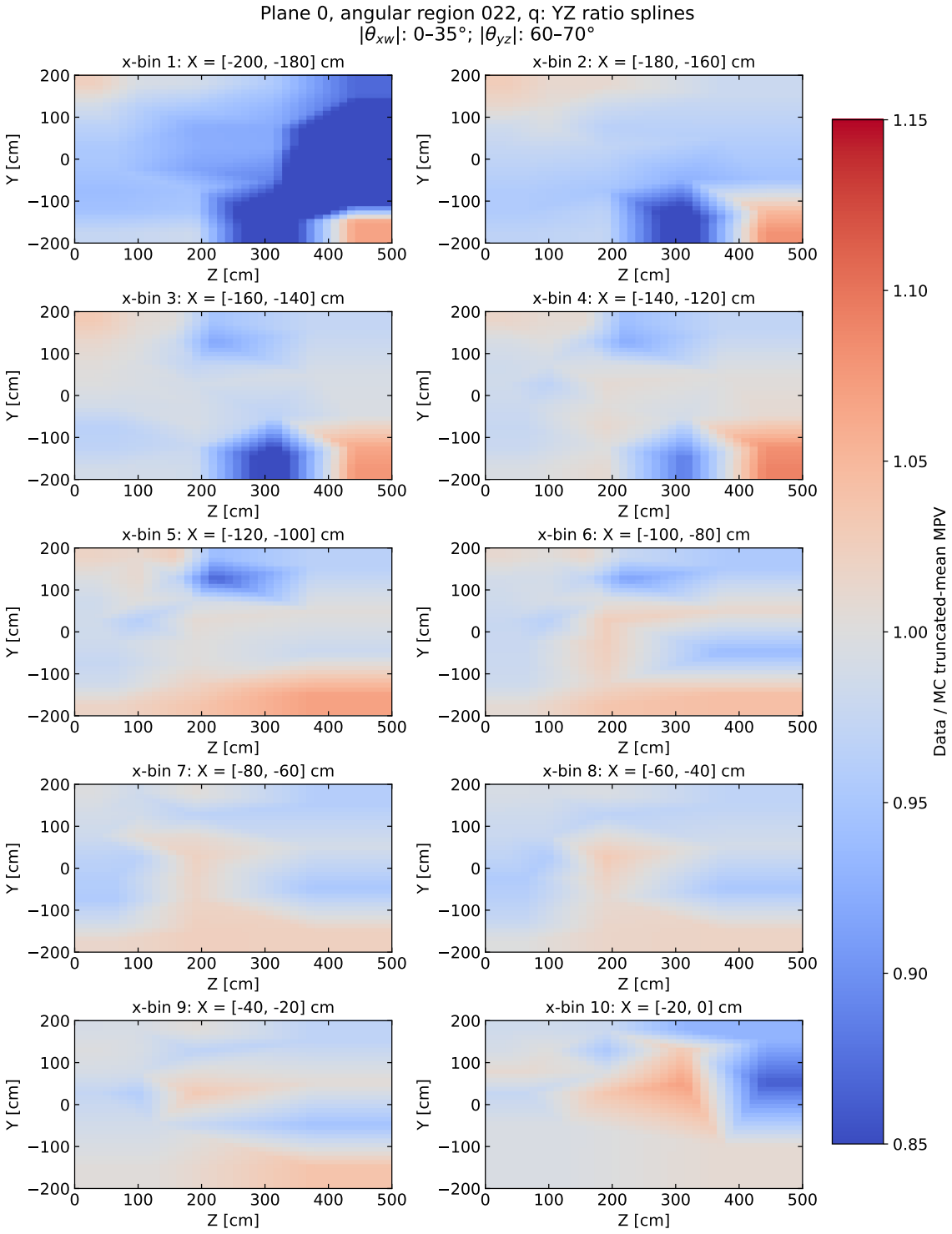


Figure 39: Plane 0, angular region 022: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 35]^\circ$ and $|\theta_{yz}| \in [60, 70]^\circ$.

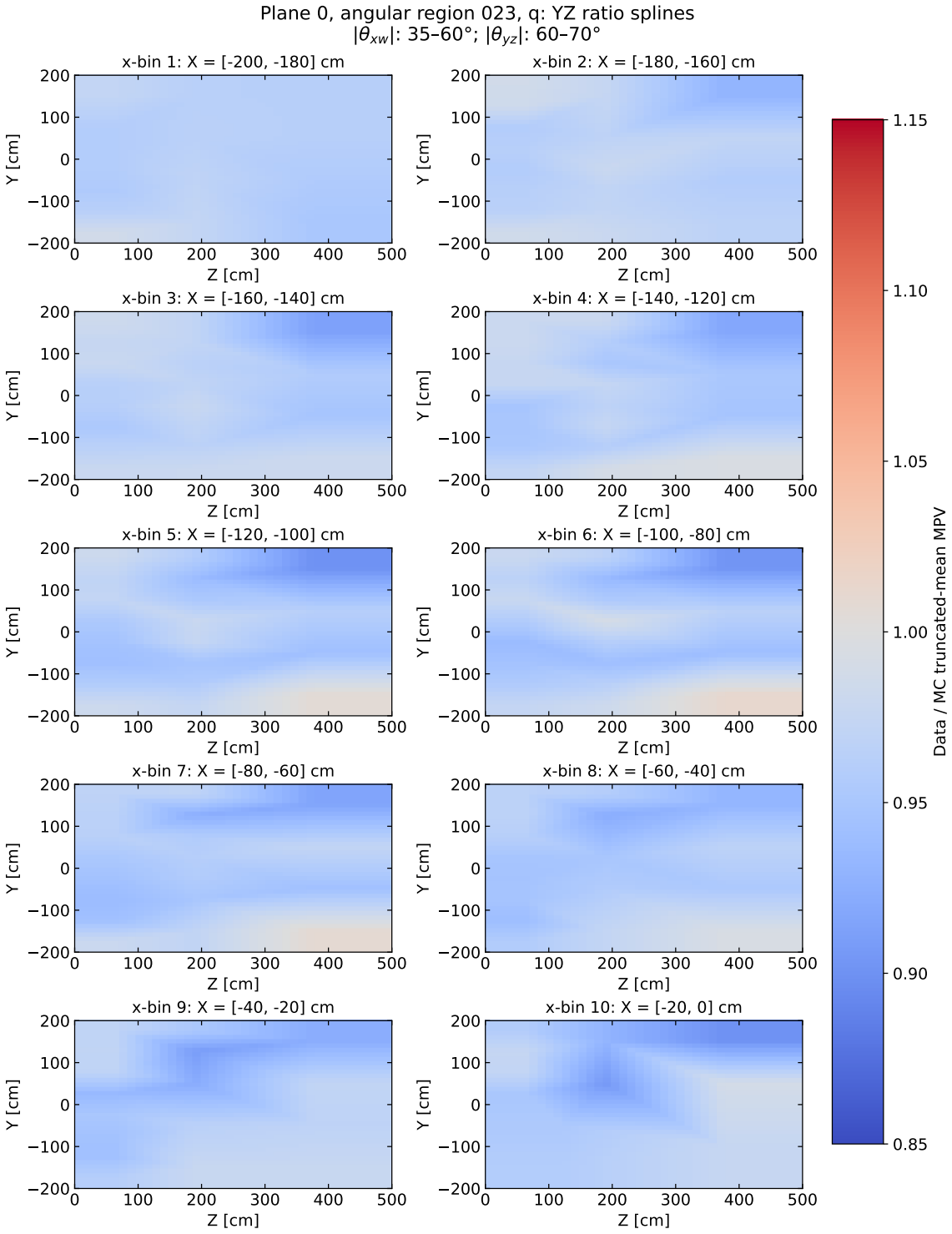


Figure 40: Plane 0, angular region 023: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [35, 60]^\circ$ and $|\theta_{yz}| \in [60, 70]^\circ$.

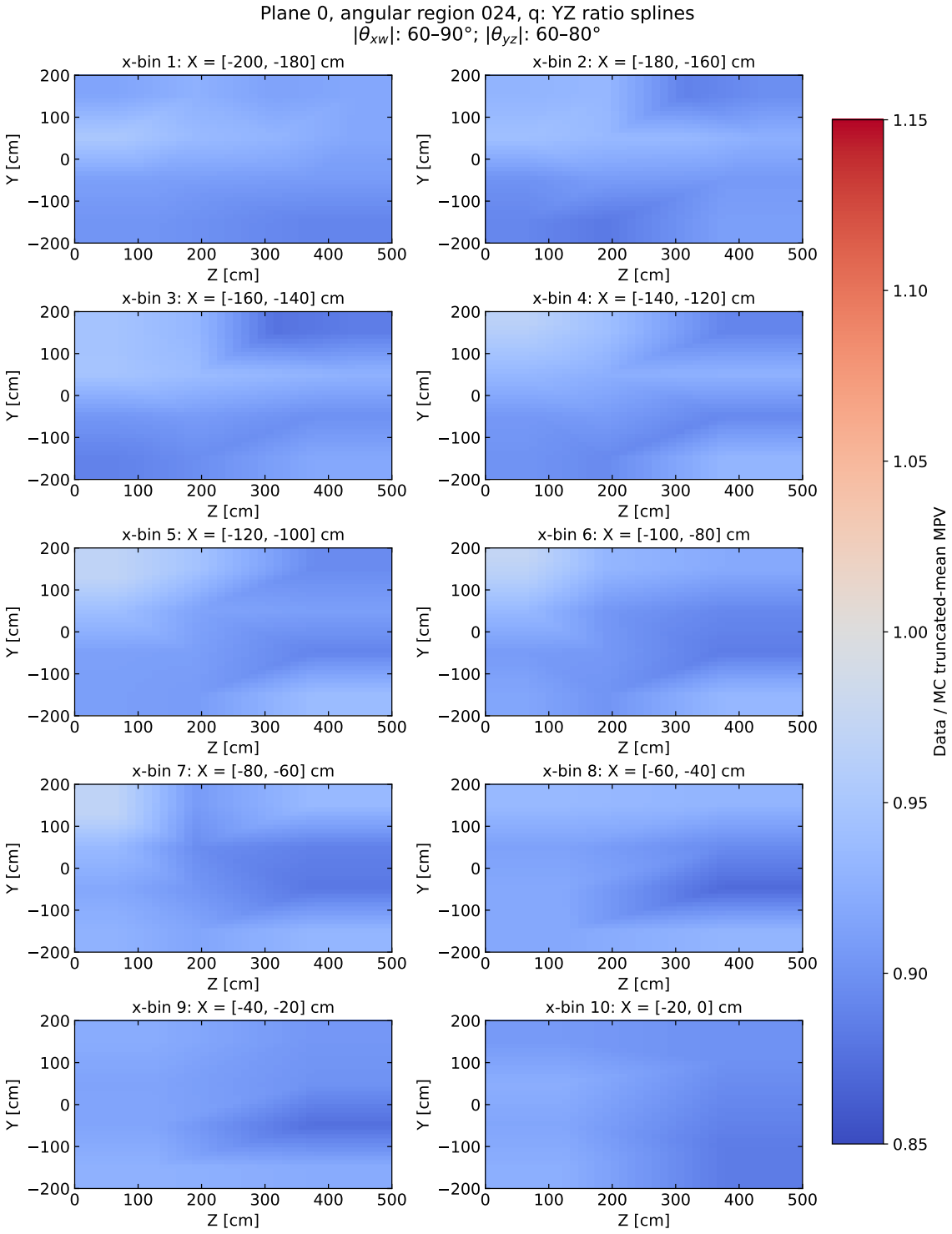


Figure 41: Plane 0, angular region 024: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [60, 80]^\circ$.

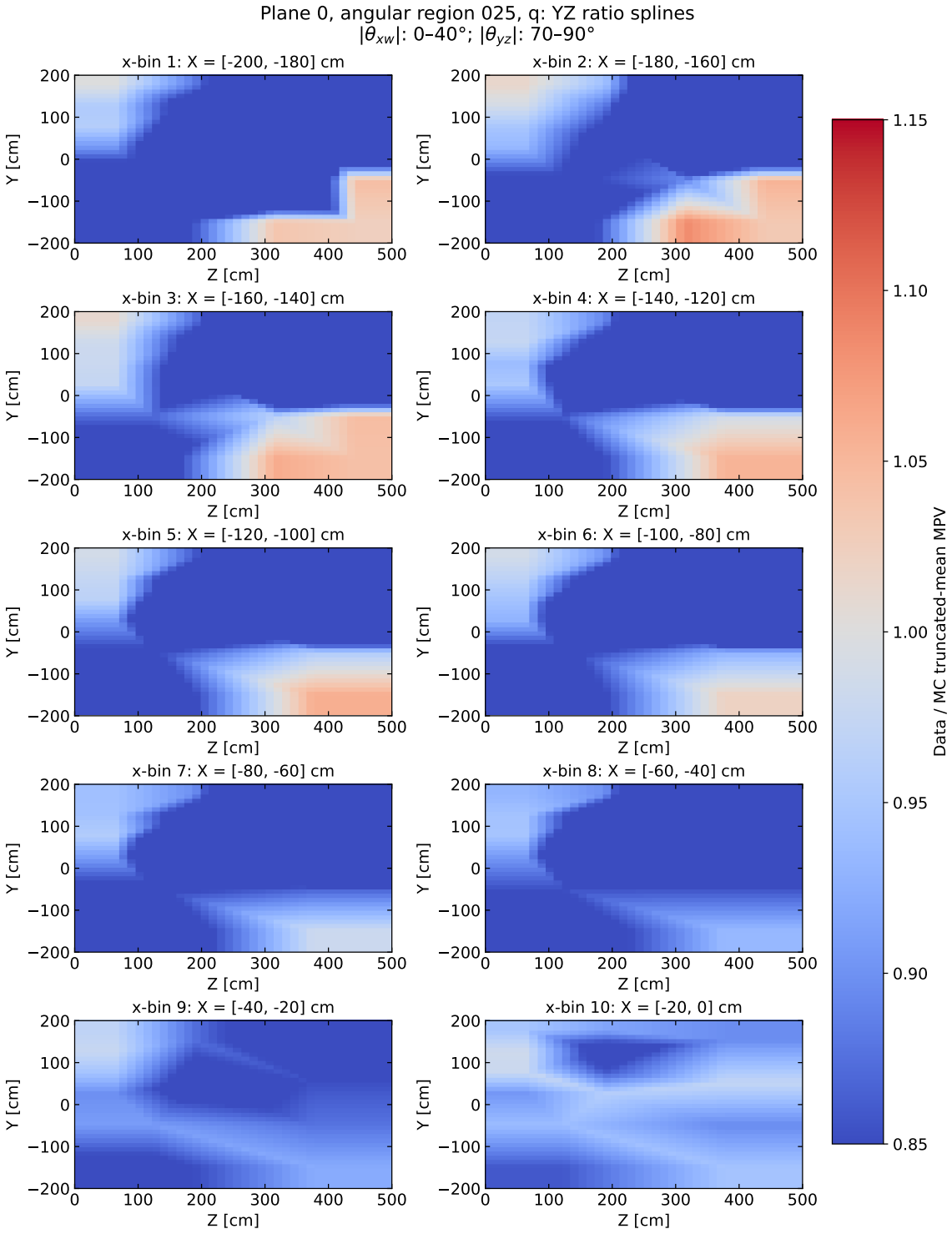


Figure 42: Plane 0, angular region 025: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 40]^\circ$ and $|\theta_{yz}| \in [70, 90]^\circ$.

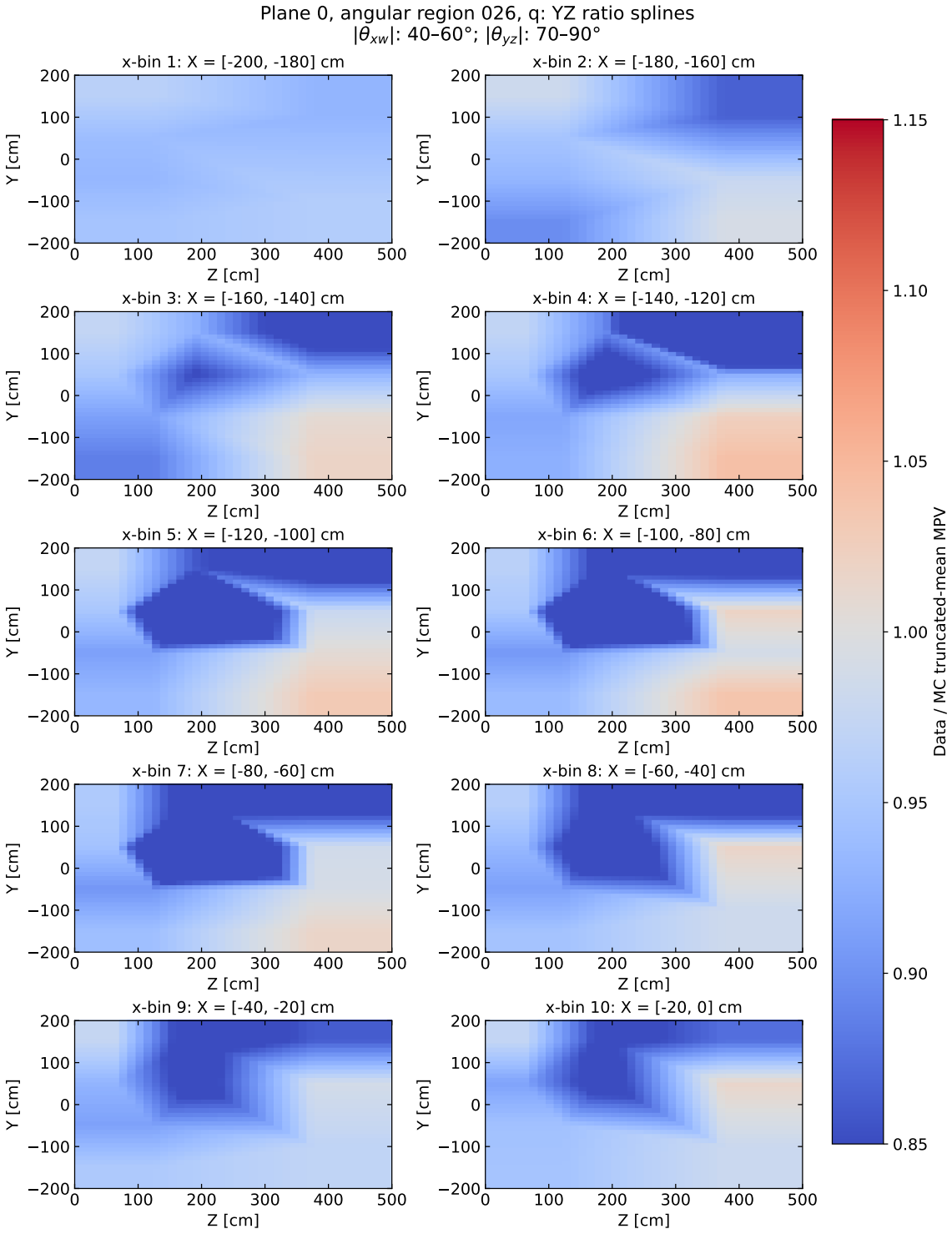


Figure 43: Plane 0, angular region 026: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [70, 90]^\circ$.

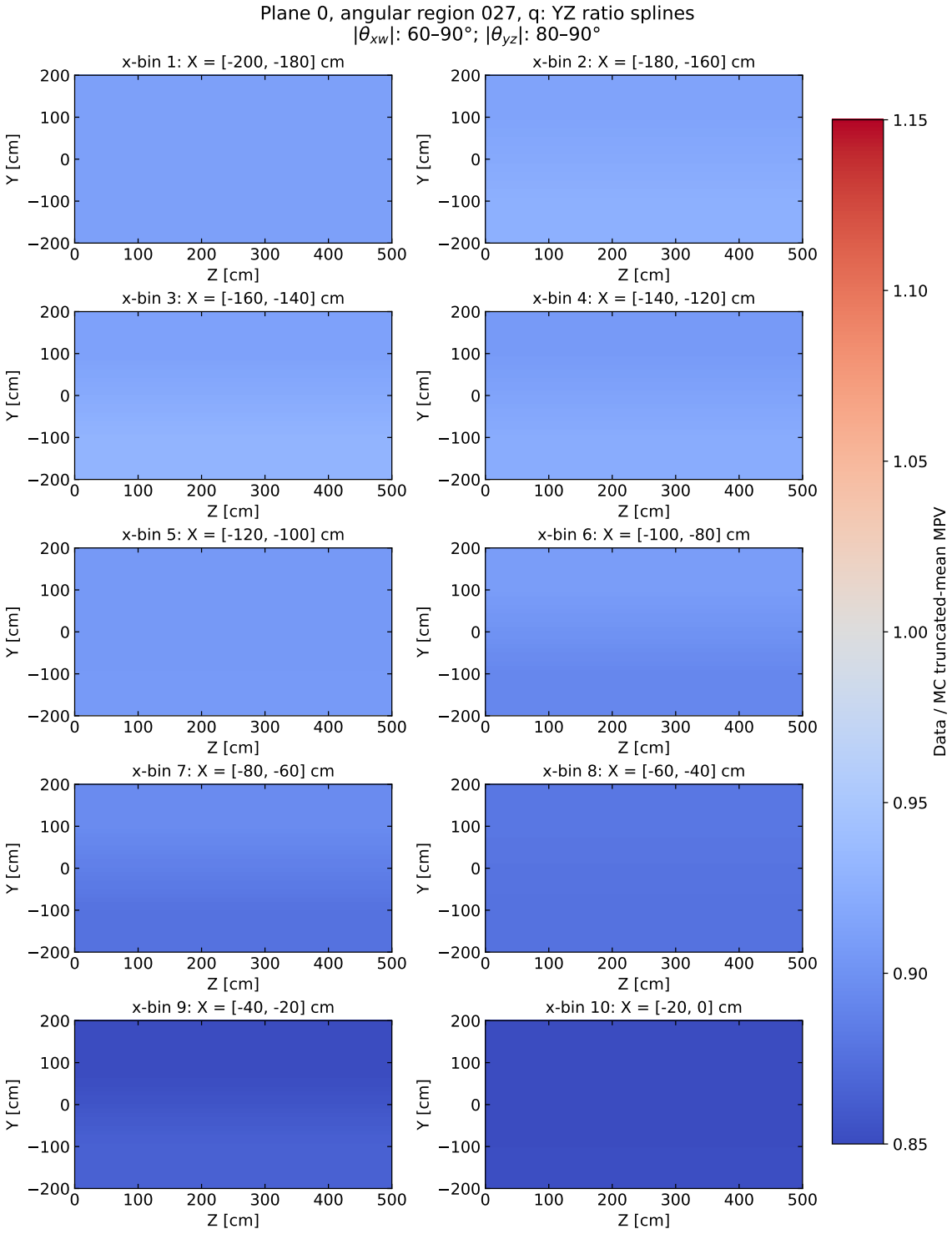


Figure 44: Plane 0, angular region 027: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [80, 90]^\circ$.

B.1.2 dQ/dx

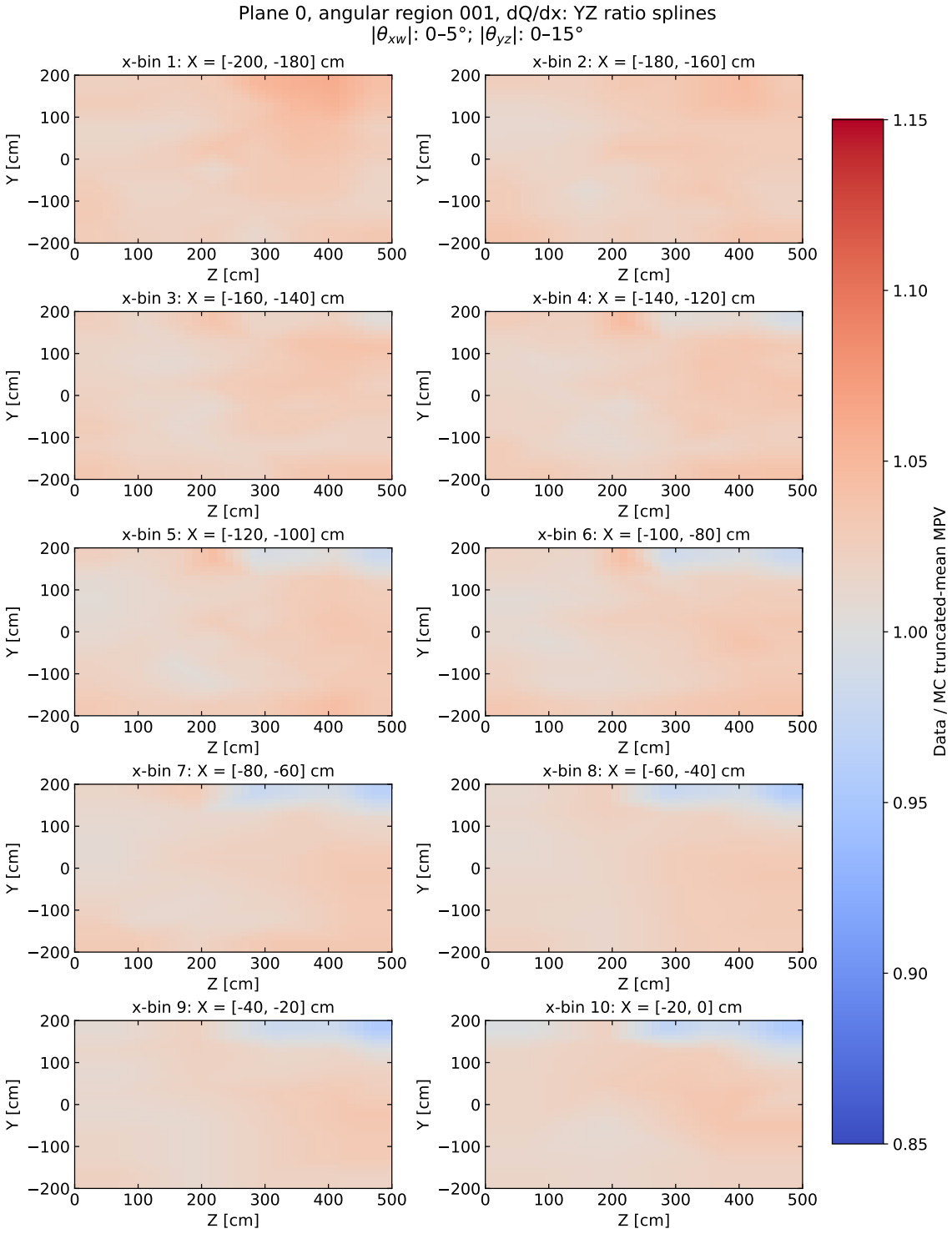


Figure 45: Plane 0, angular region 001: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 5]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

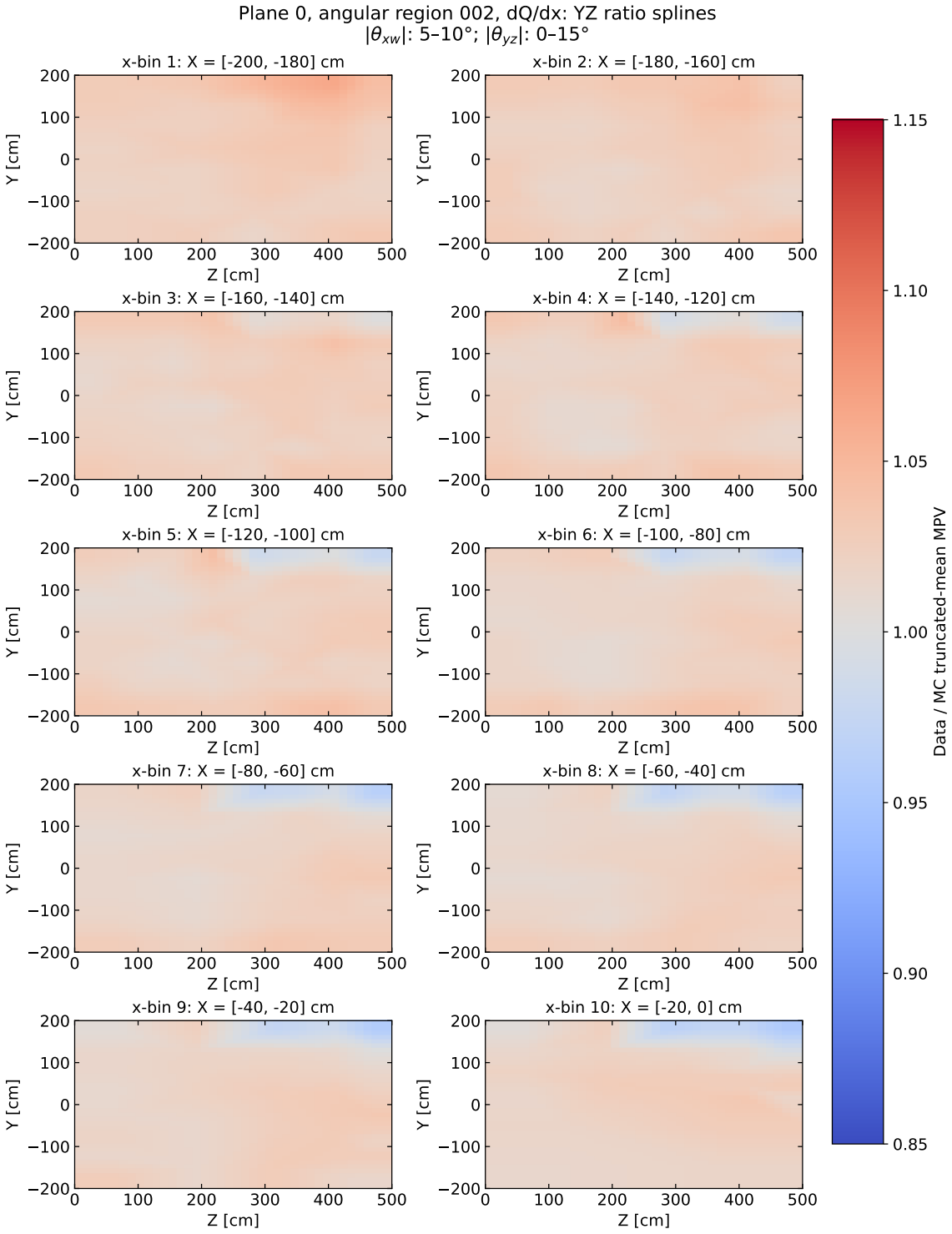


Figure 46: Plane 0, angular region 002: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [5, 10]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

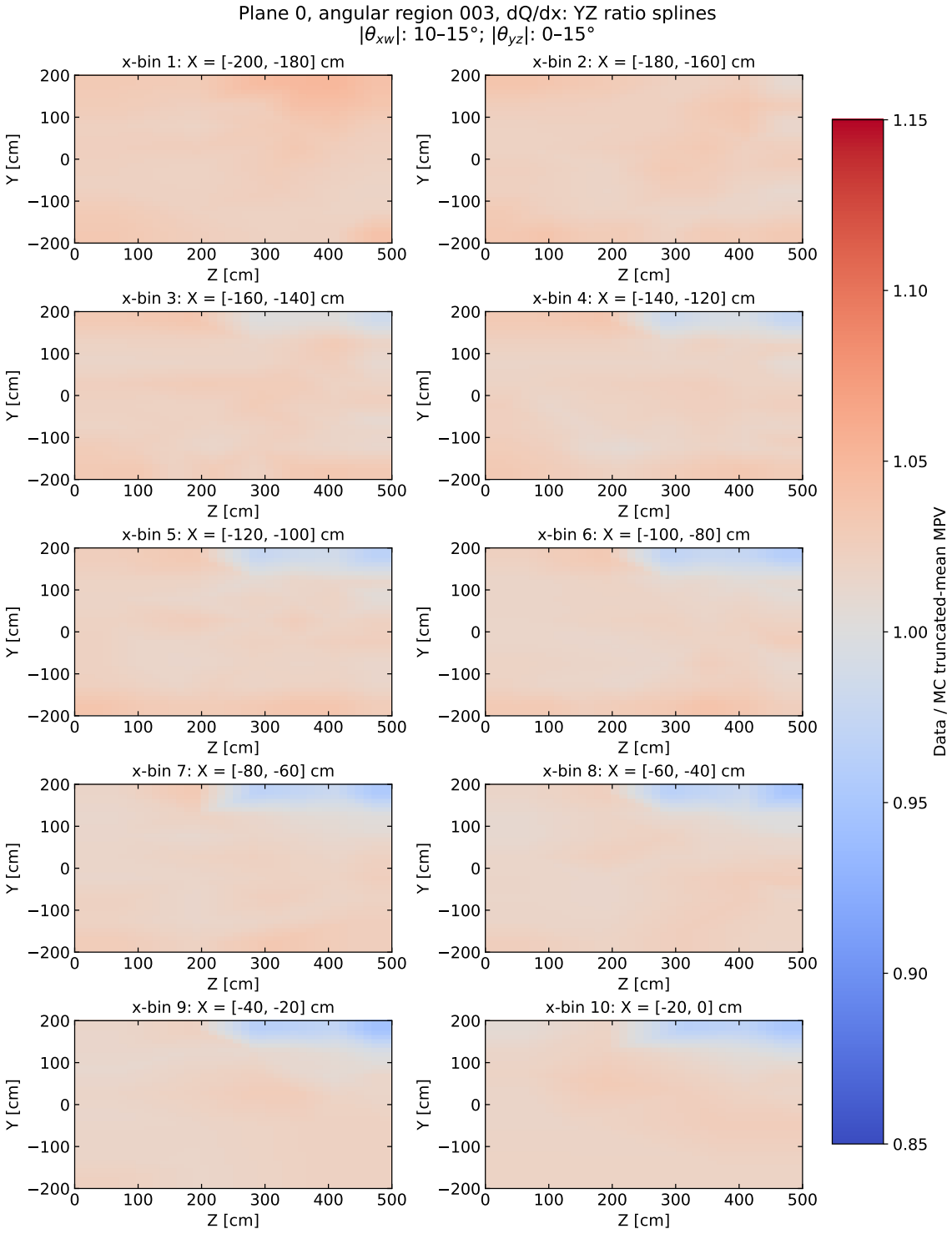


Figure 47: Plane 0, angular region 003: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [10, 15]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

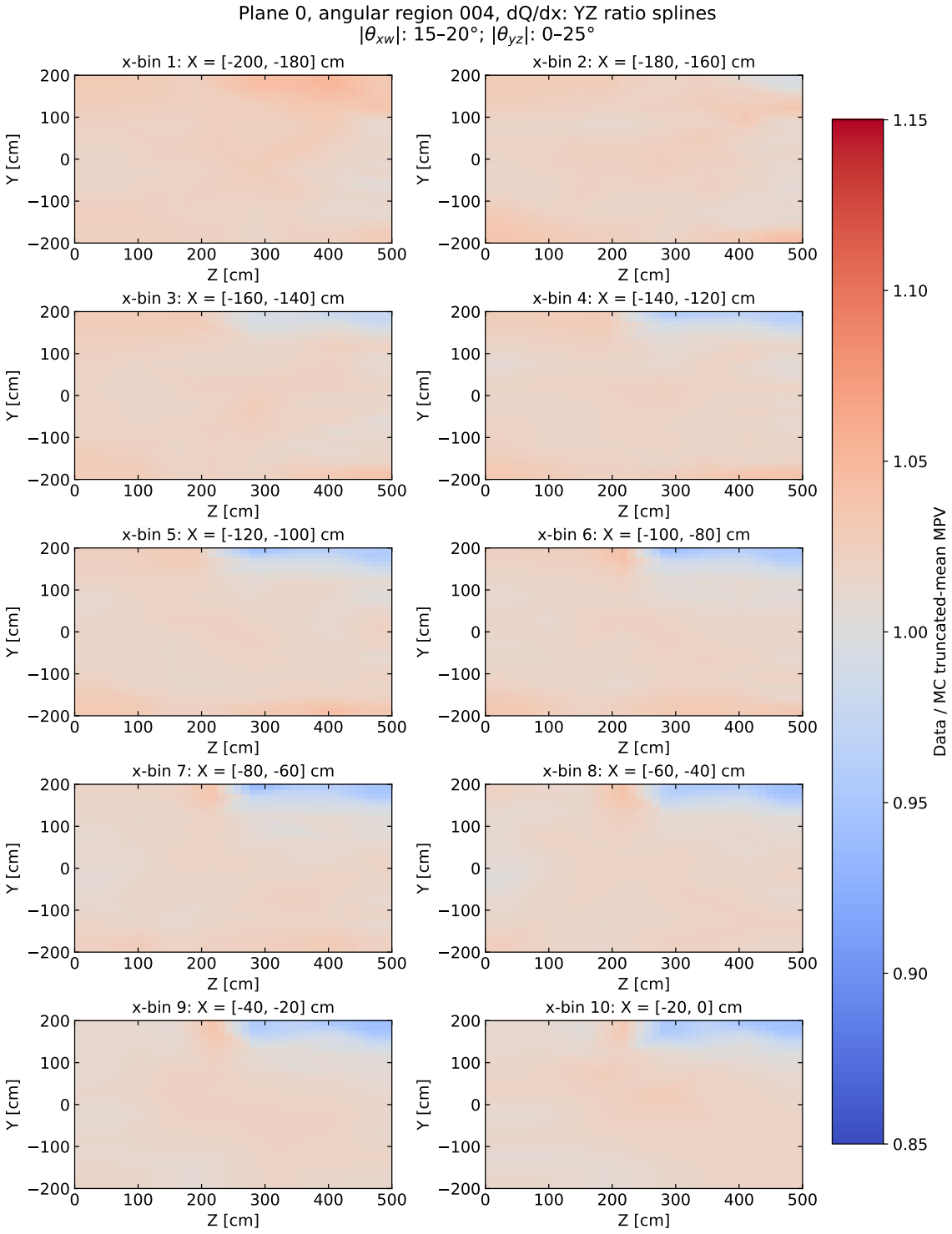


Figure 48: Plane 0, angular region 004: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 20]^\circ$ and $|\theta_{yz}| \in [0, 25]^\circ$.

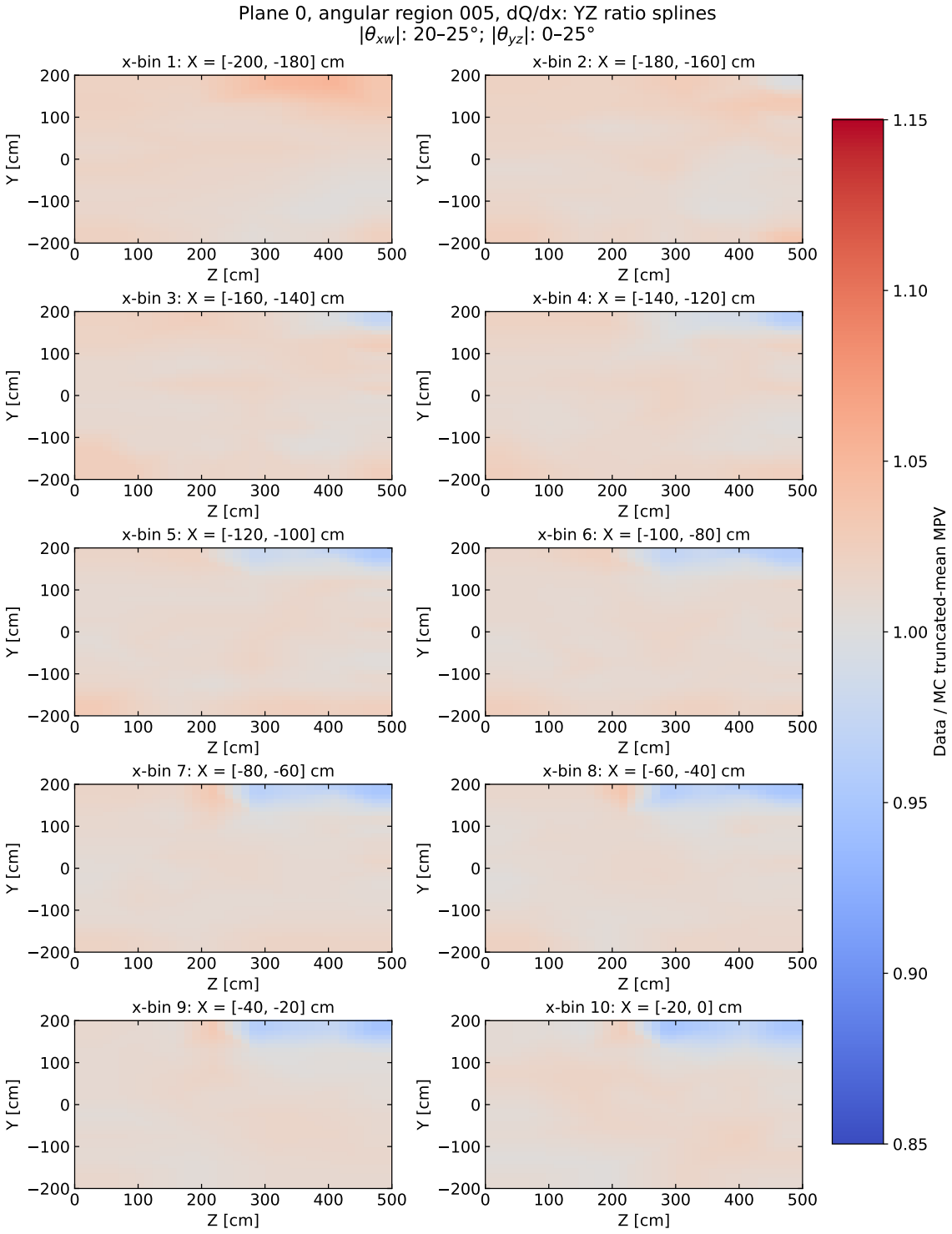


Figure 49: Plane 0, angular region 005: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [20, 25]^\circ$ and $|\theta_{yz}| \in [0, 25]^\circ$.

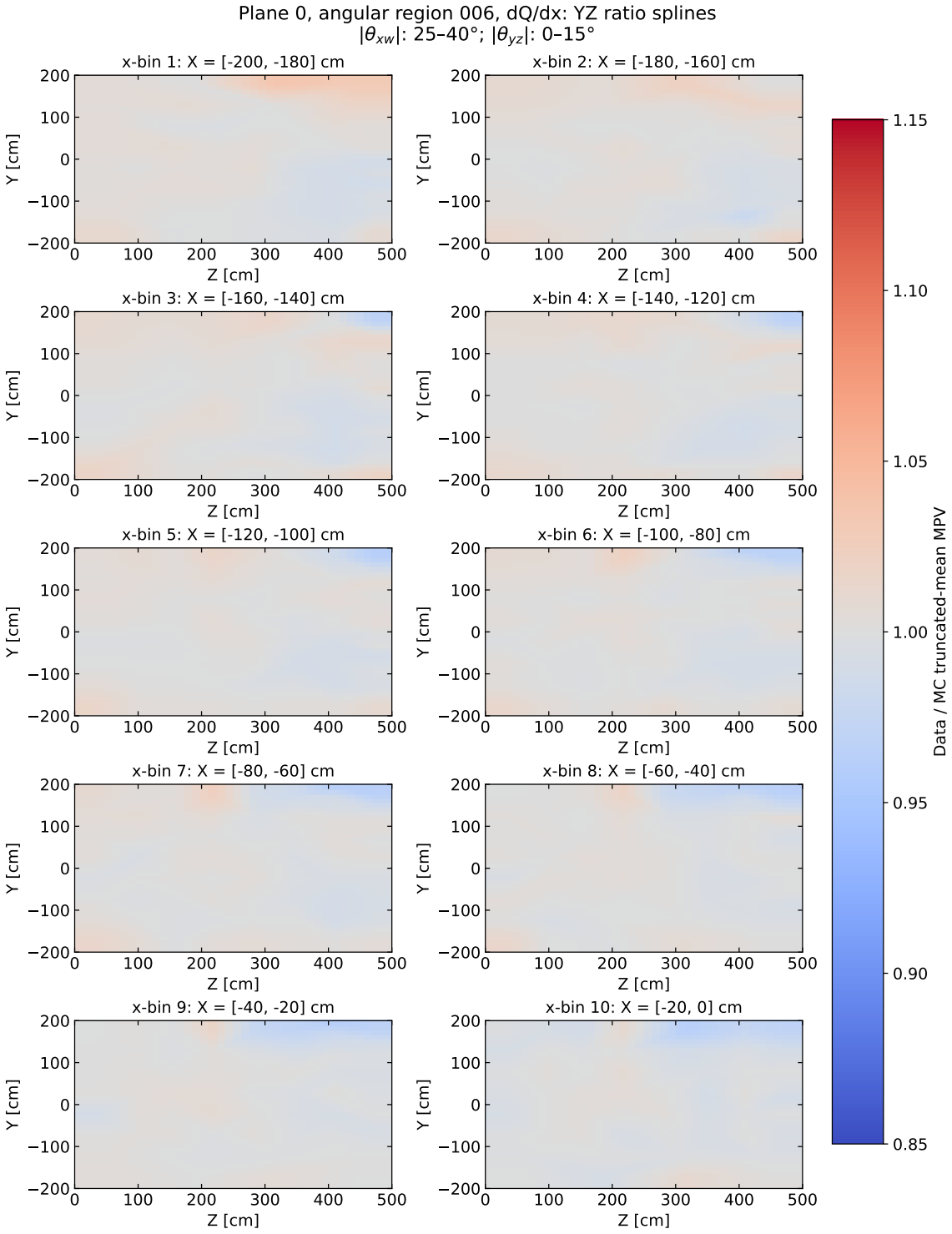


Figure 50: Plane 0, angular region 006: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

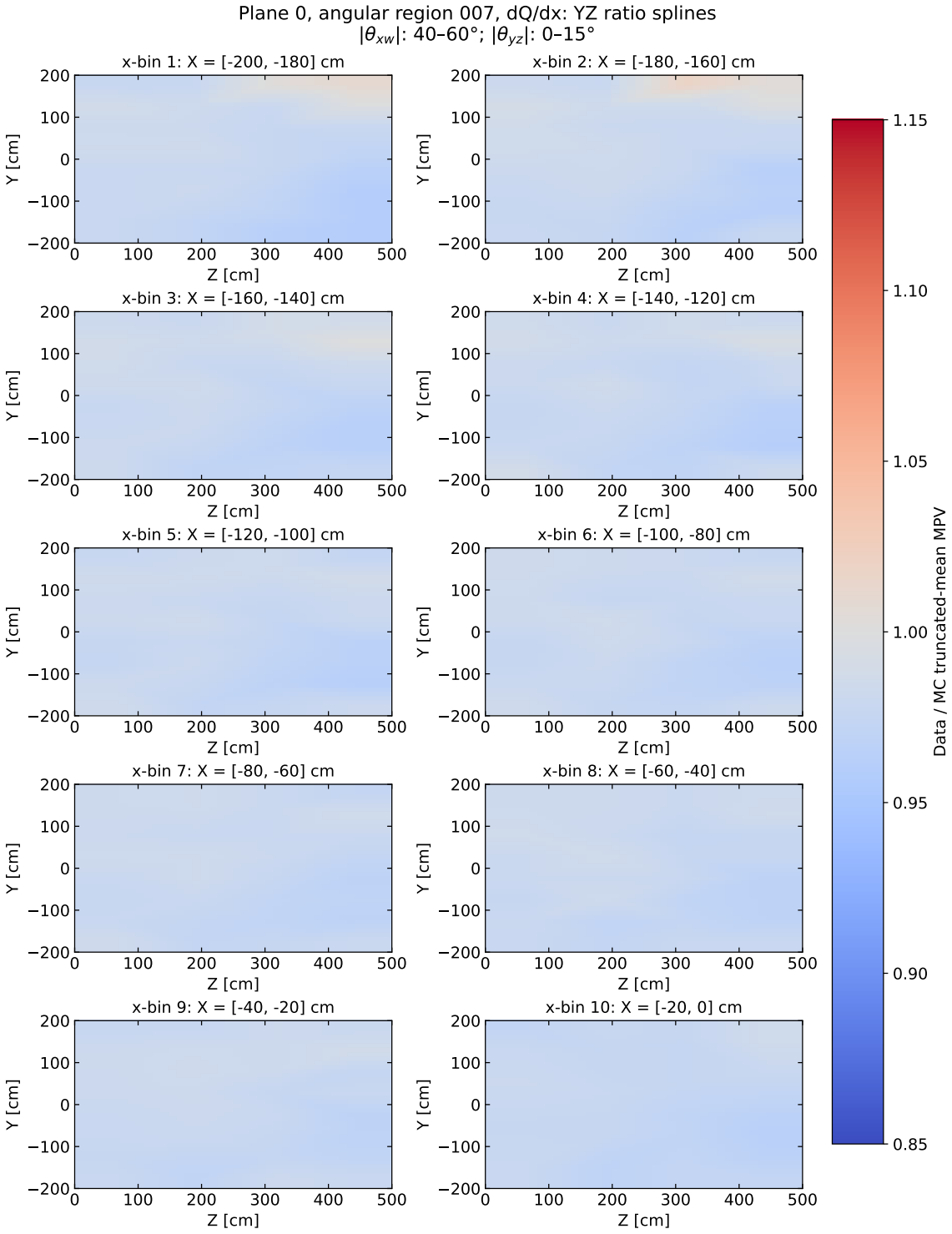


Figure 51: Plane 0, angular region 007: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

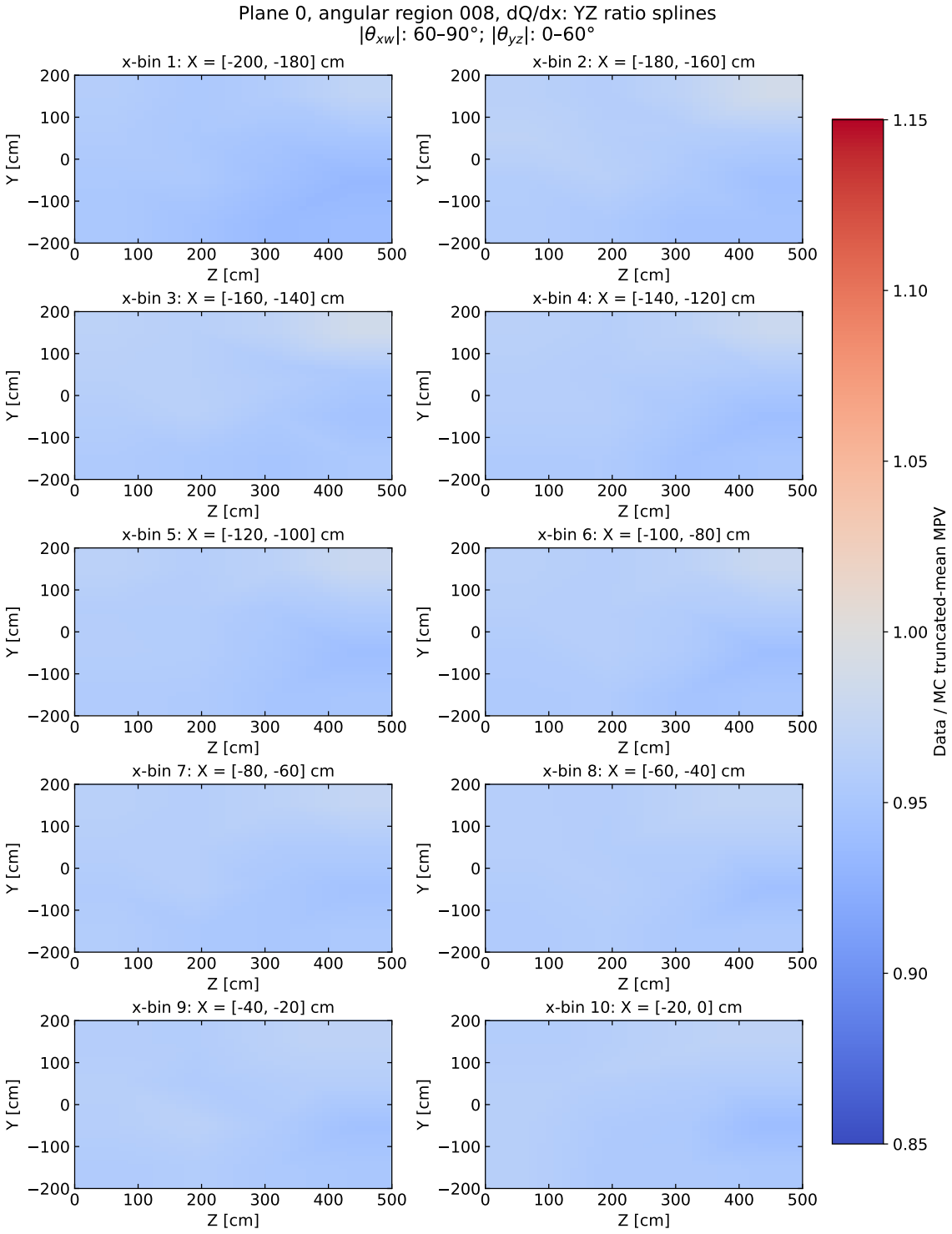


Figure 52: Plane 0, angular region 008: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [0, 60]^\circ$.

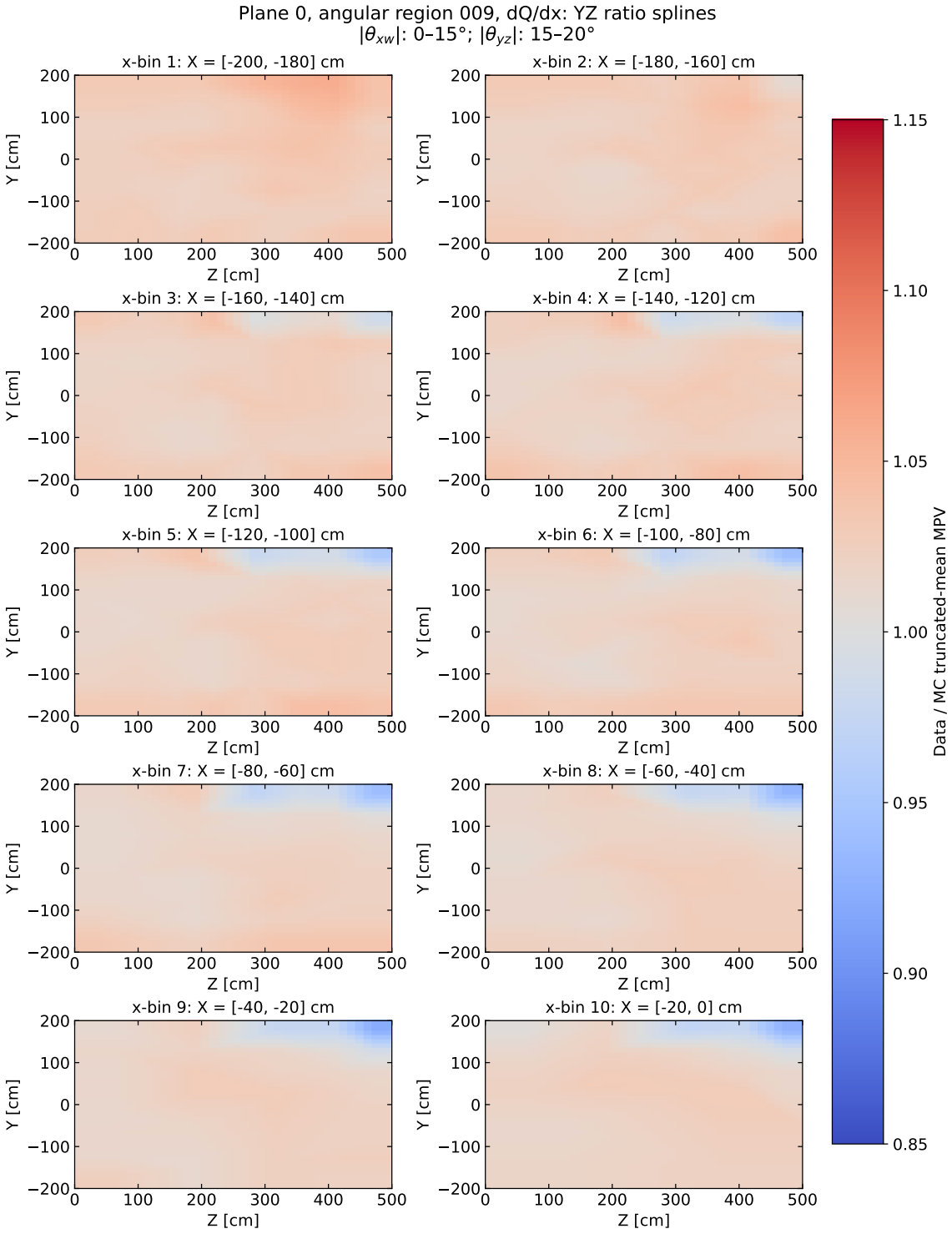


Figure 53: Plane 0, angular region 009: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [15, 20]^\circ$.

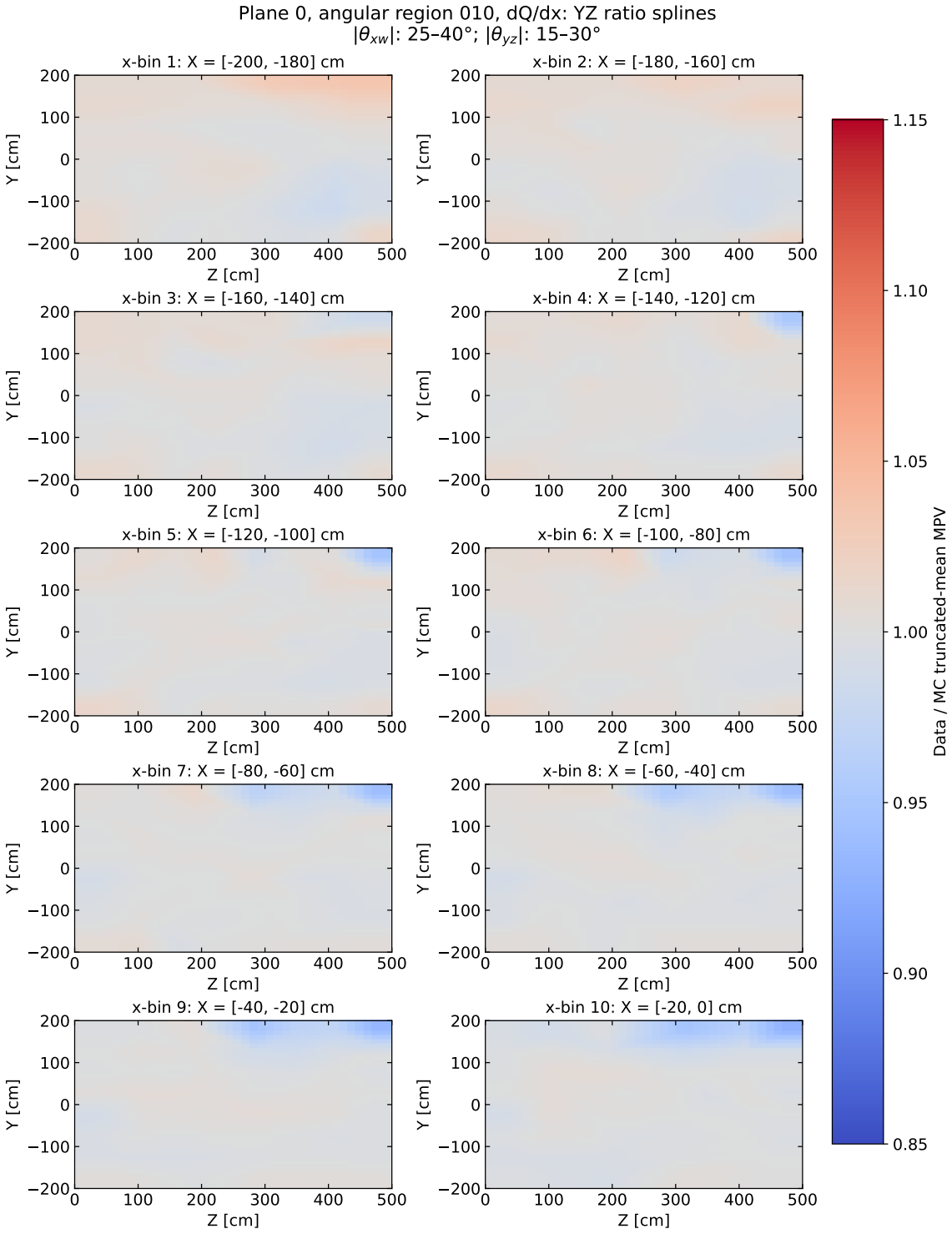


Figure 54: Plane 0, angular region 010: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [15, 30]^\circ$.

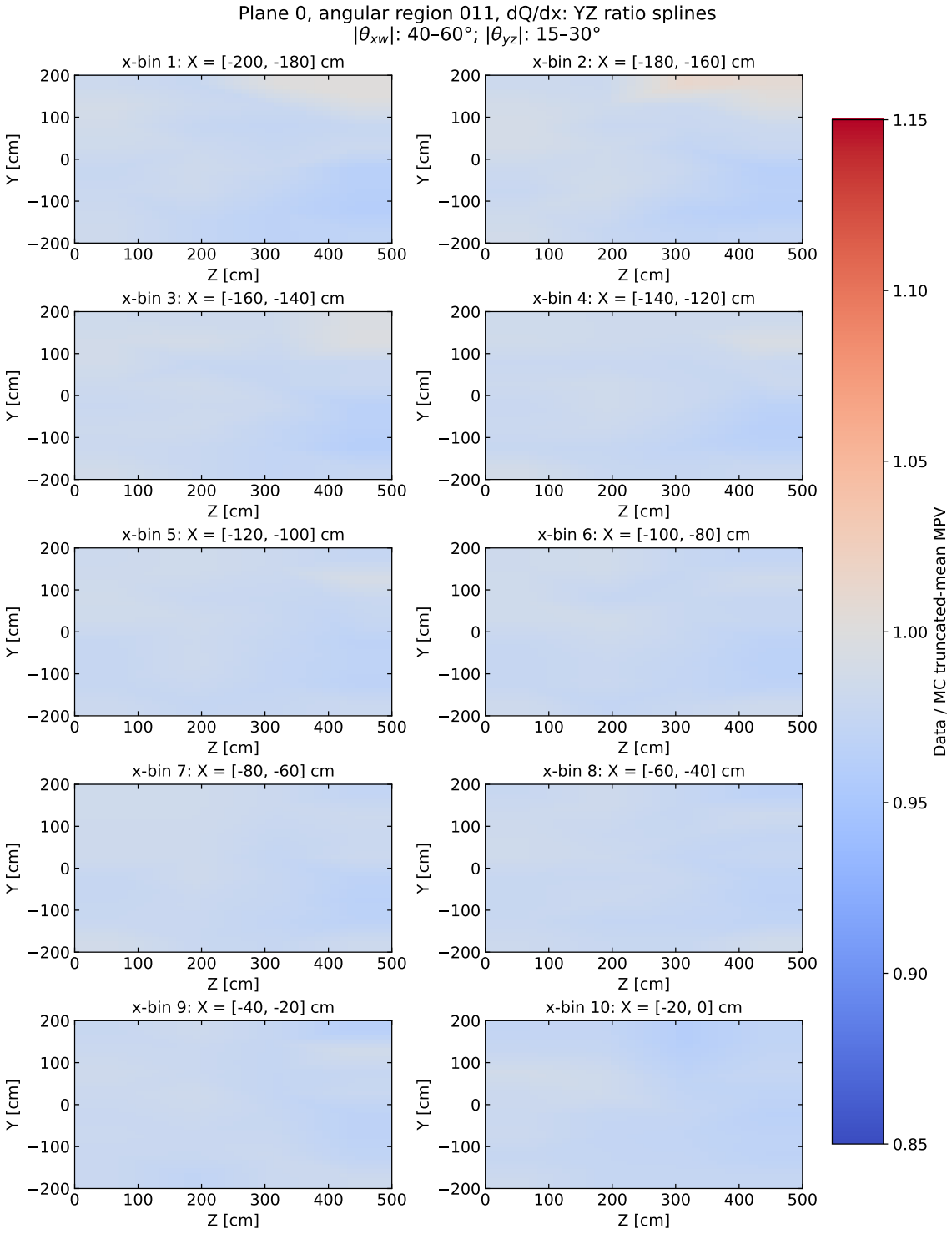


Figure 55: Plane 0, angular region 011: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [15, 30]^\circ$.

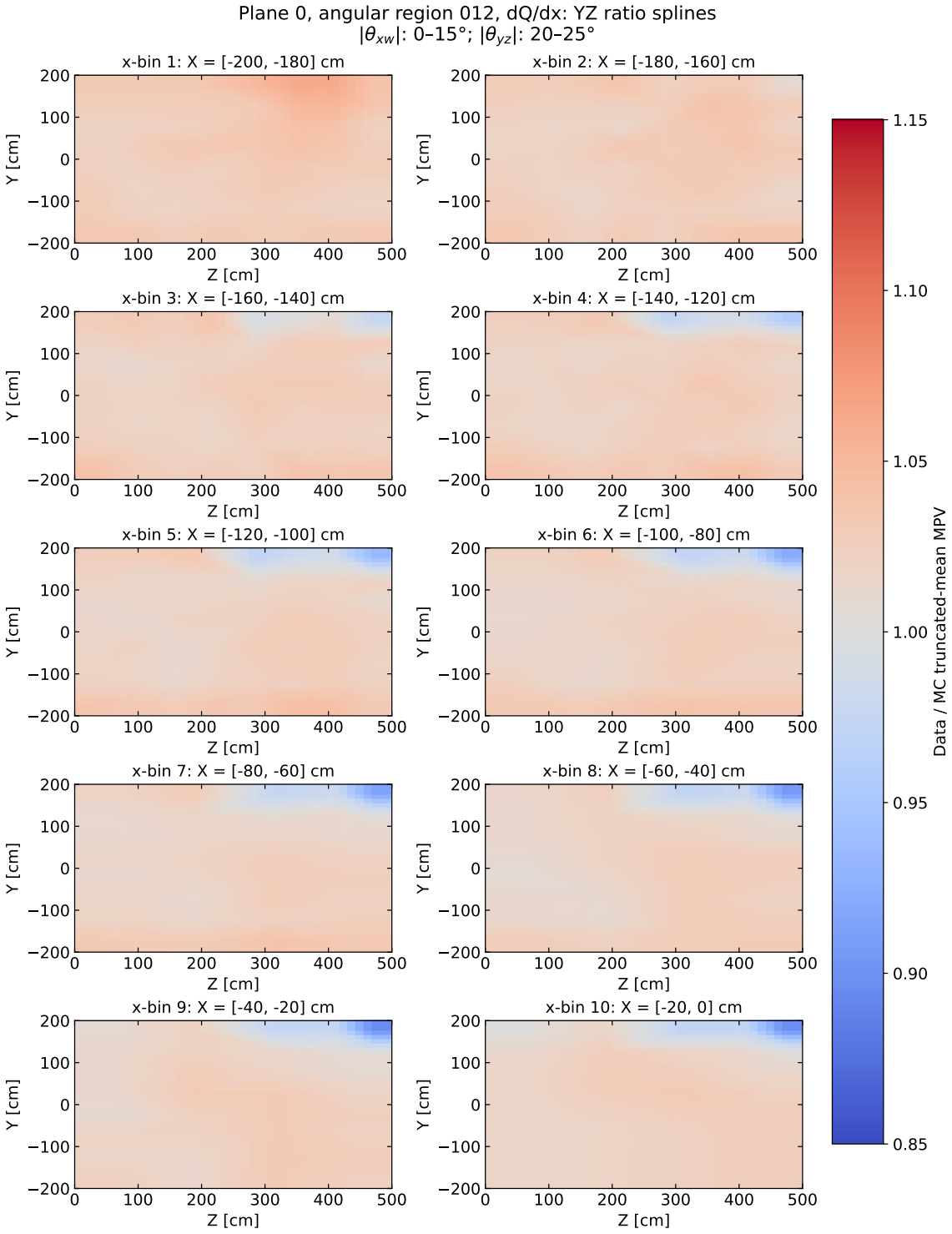


Figure 56: Plane 0, angular region 012: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [20, 25]^\circ$.

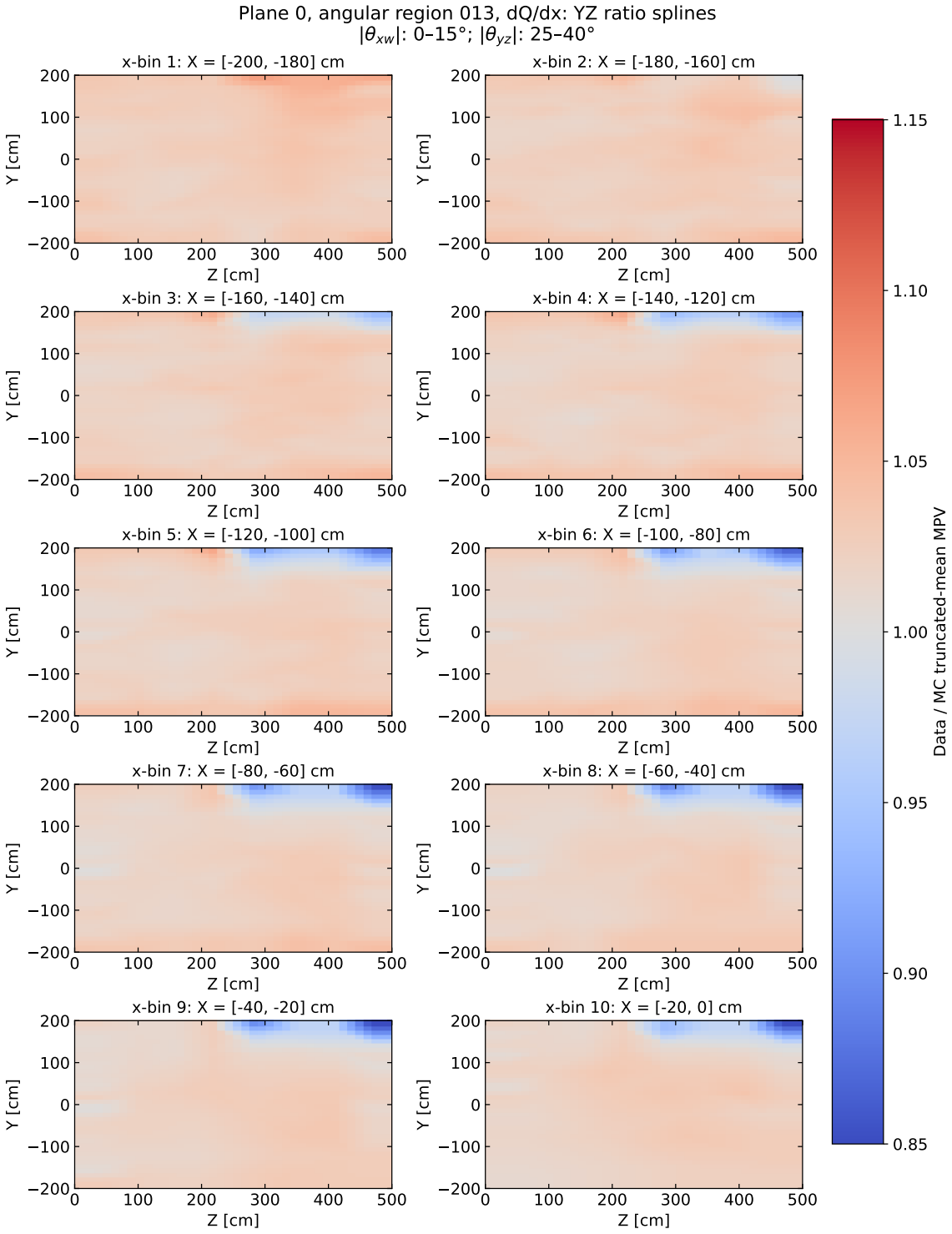


Figure 57: Plane 0, angular region 013: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [25, 40]^\circ$.

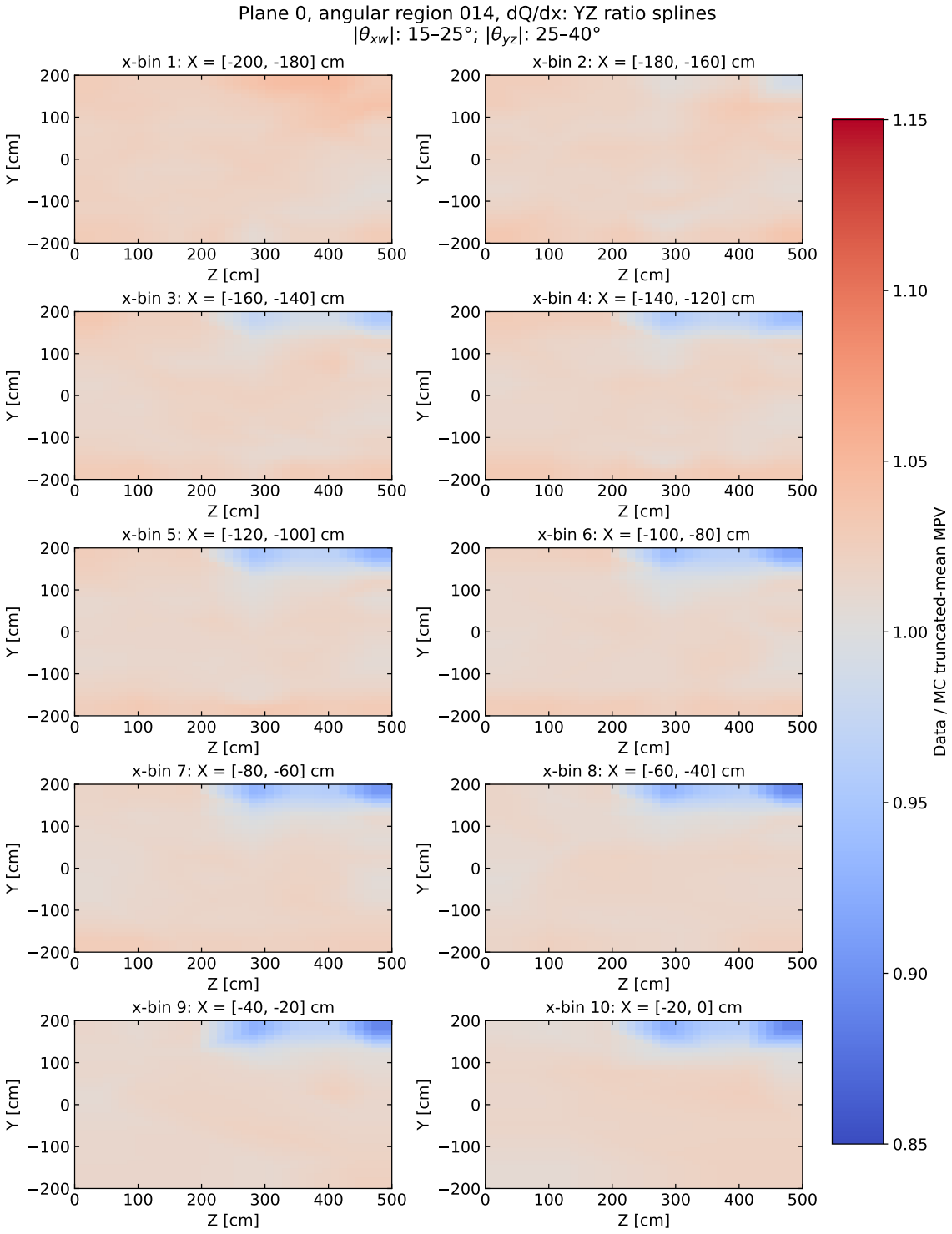


Figure 58: Plane 0, angular region 014: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 25]^\circ$ and $|\theta_{yz}| \in [25, 40]^\circ$.

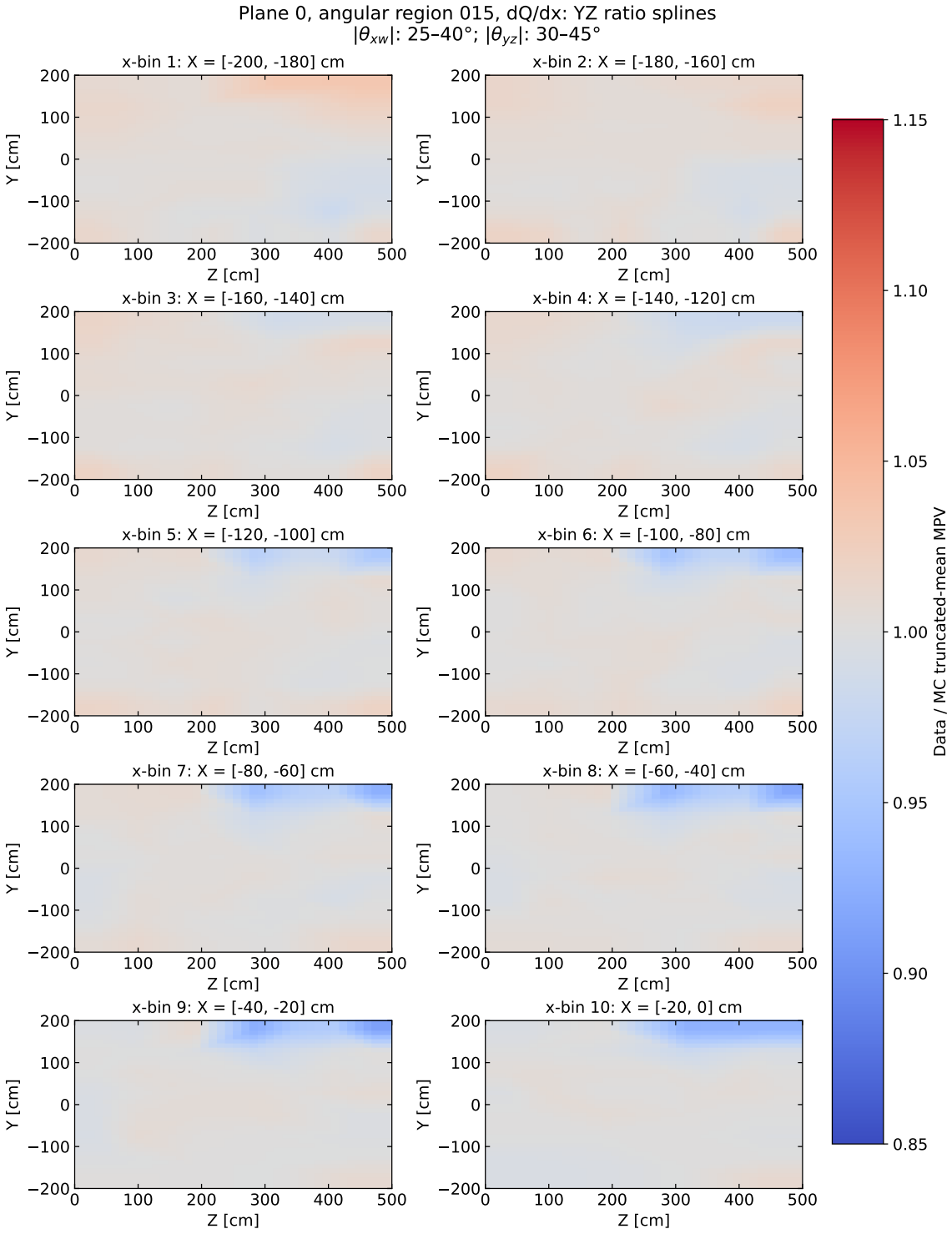


Figure 59: Plane 0, angular region 015: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [30, 45]^\circ$.

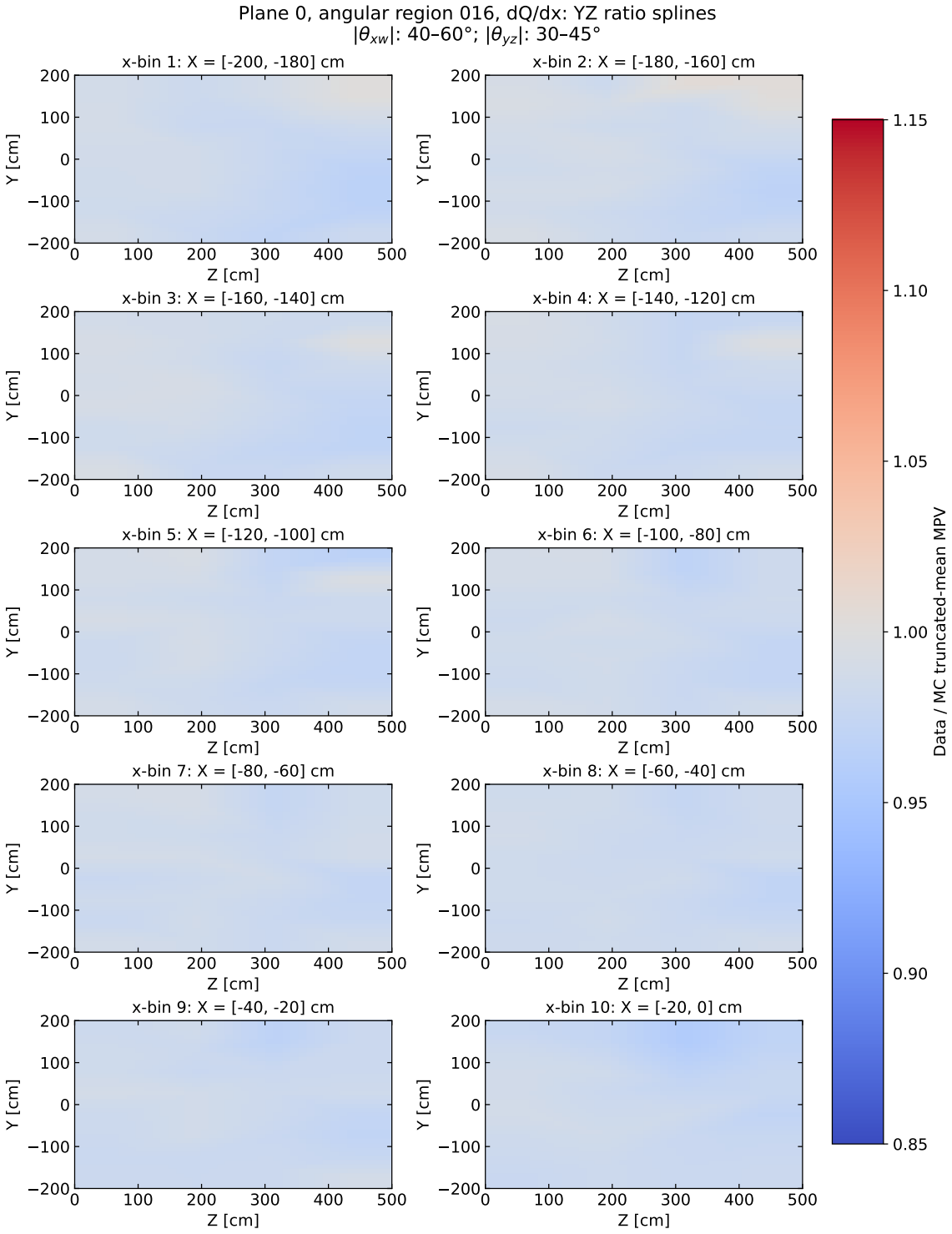


Figure 60: Plane 0, angular region 016: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [30, 45]^\circ$.

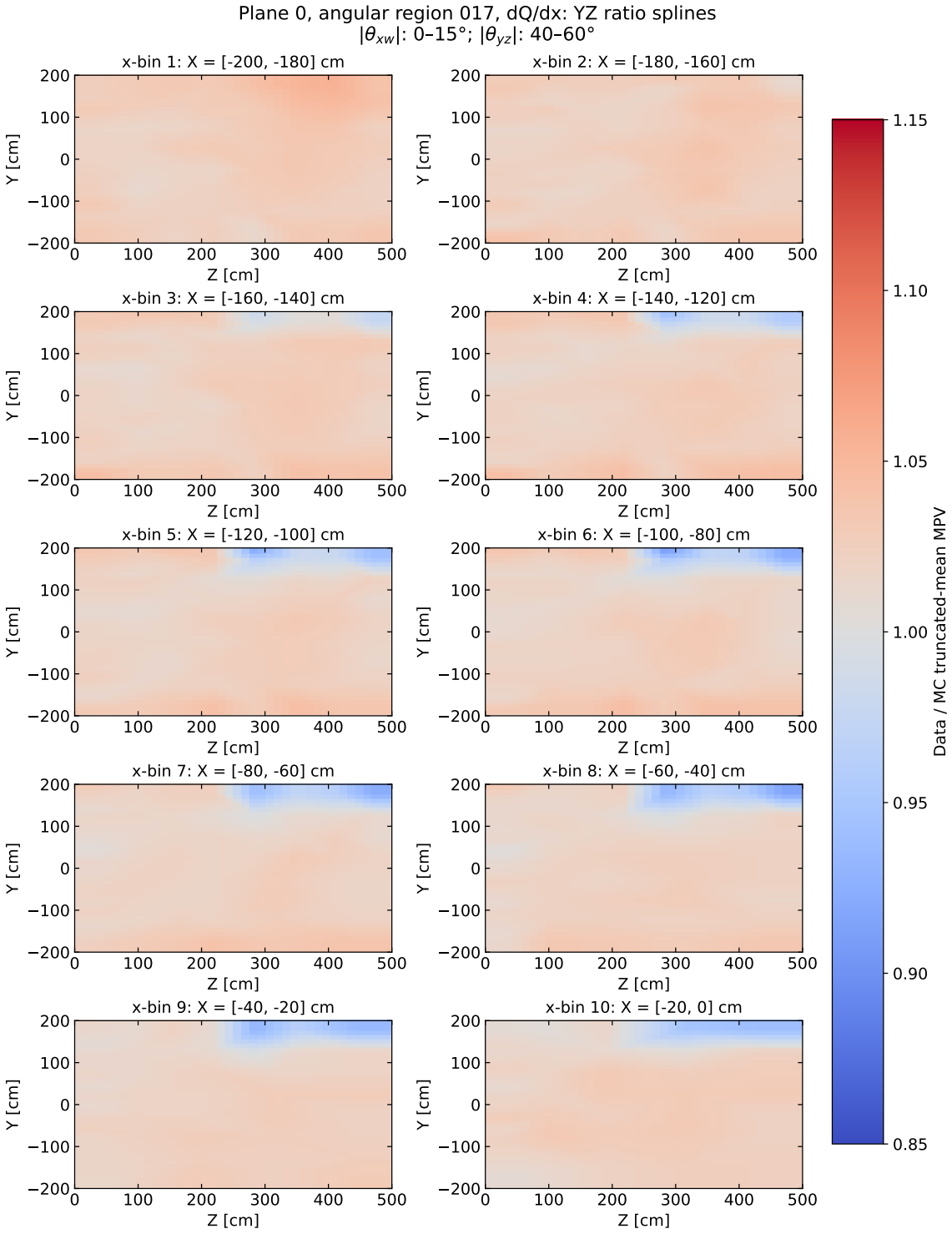


Figure 61: Plane 0, angular region 017: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [40, 60]^\circ$.

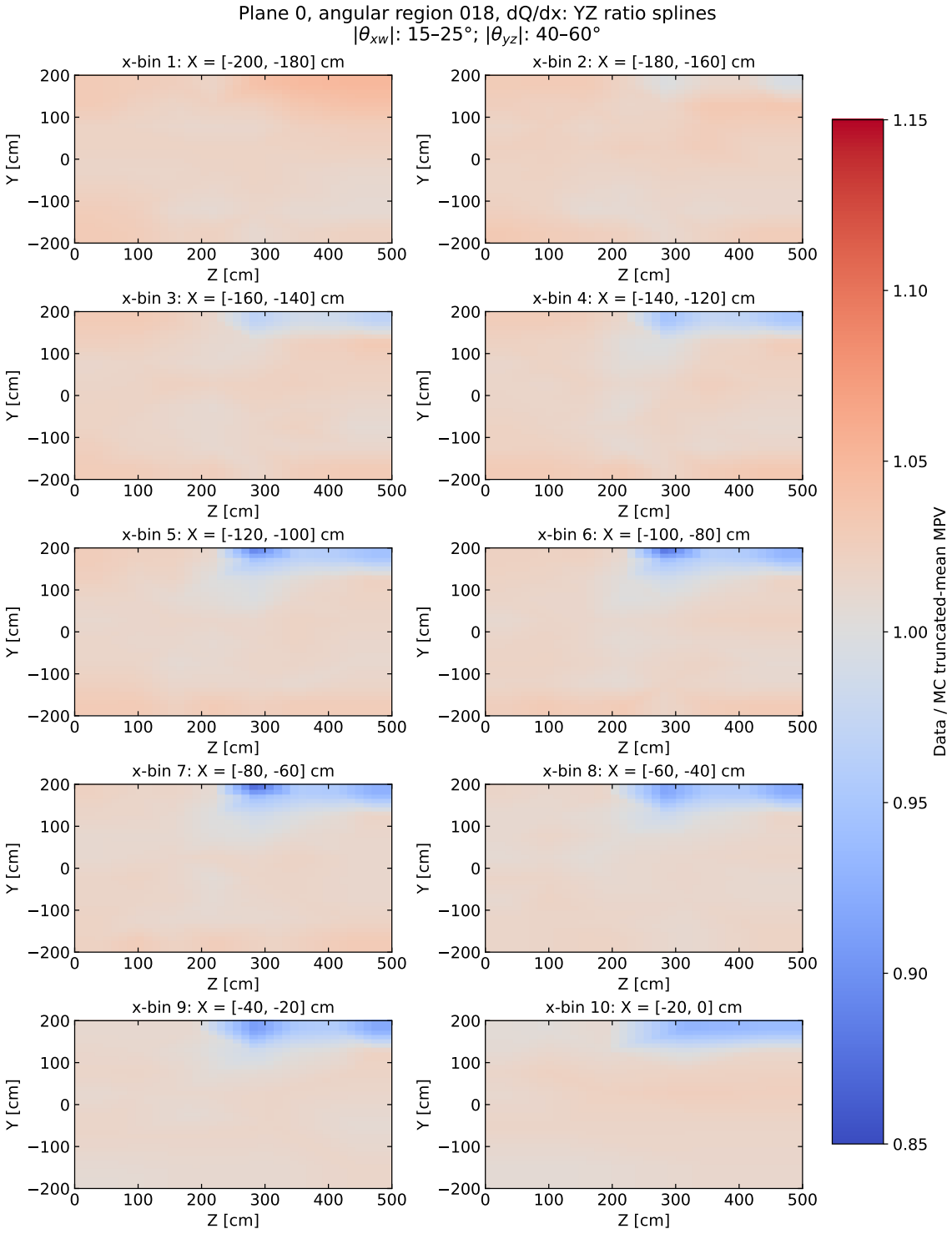


Figure 62: Plane 0, angular region 018: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 25]^\circ$ and $|\theta_{yz}| \in [40, 60]^\circ$.

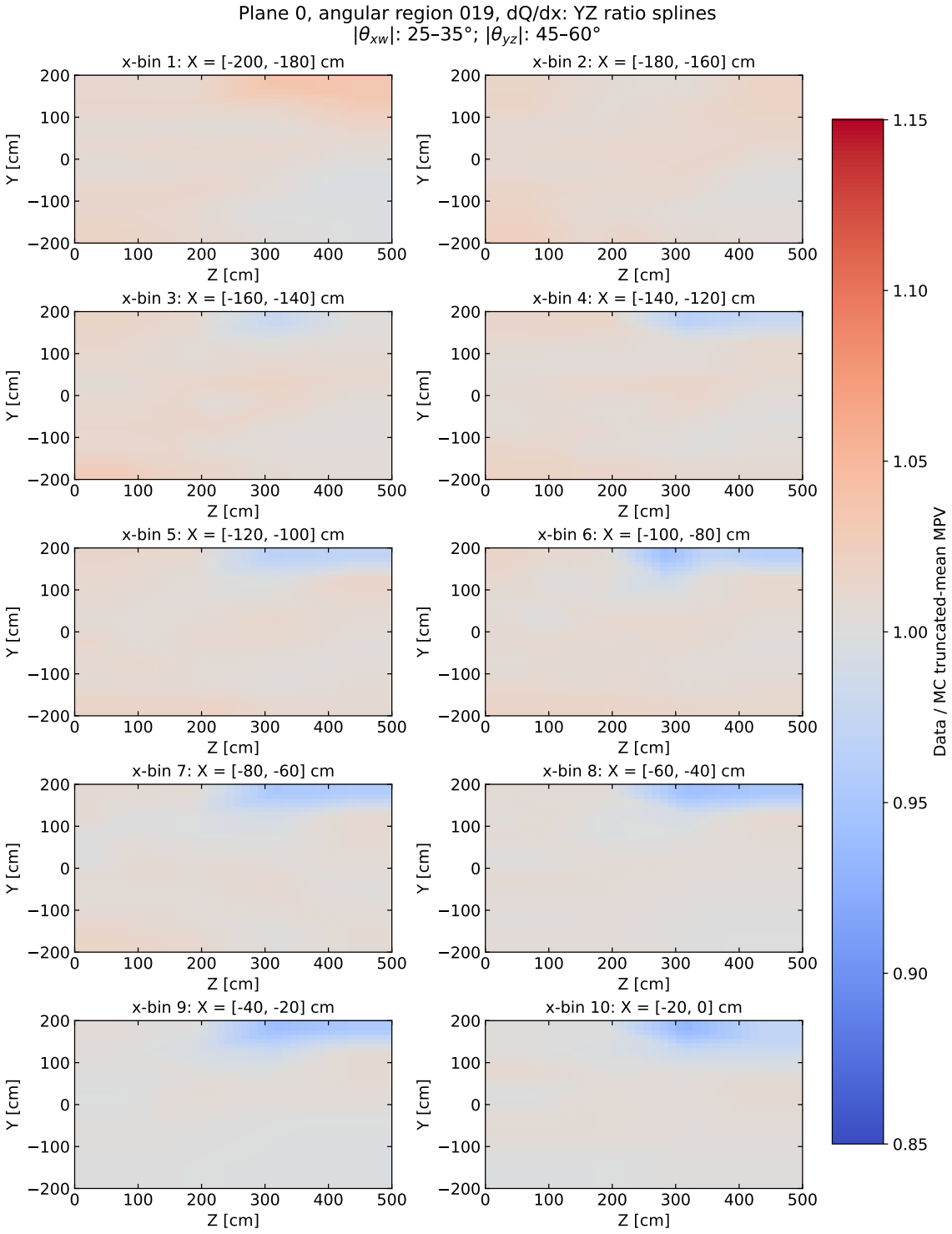


Figure 63: Plane 0, angular region 019: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 35]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

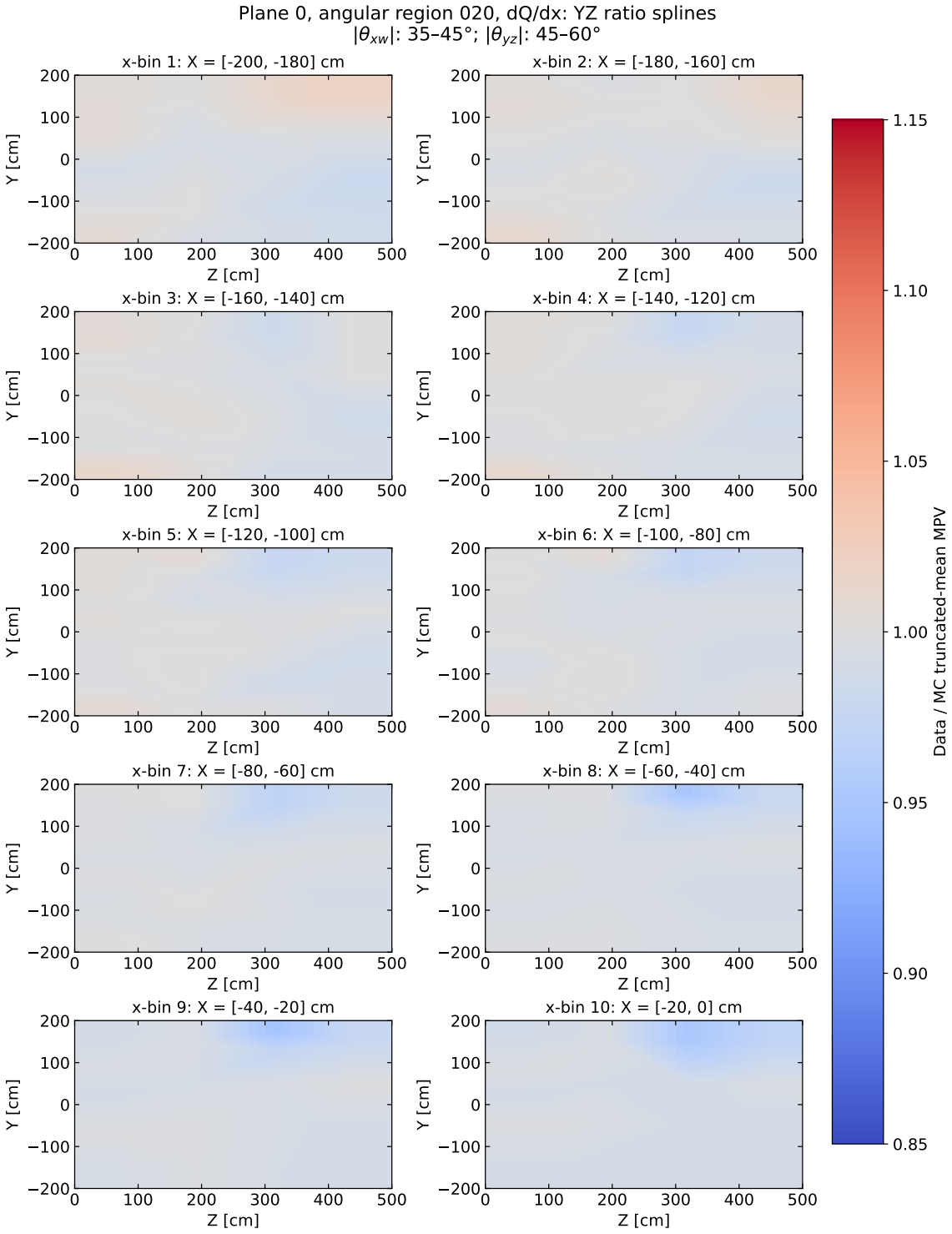


Figure 64: Plane 0, angular region 020: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [35, 45]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

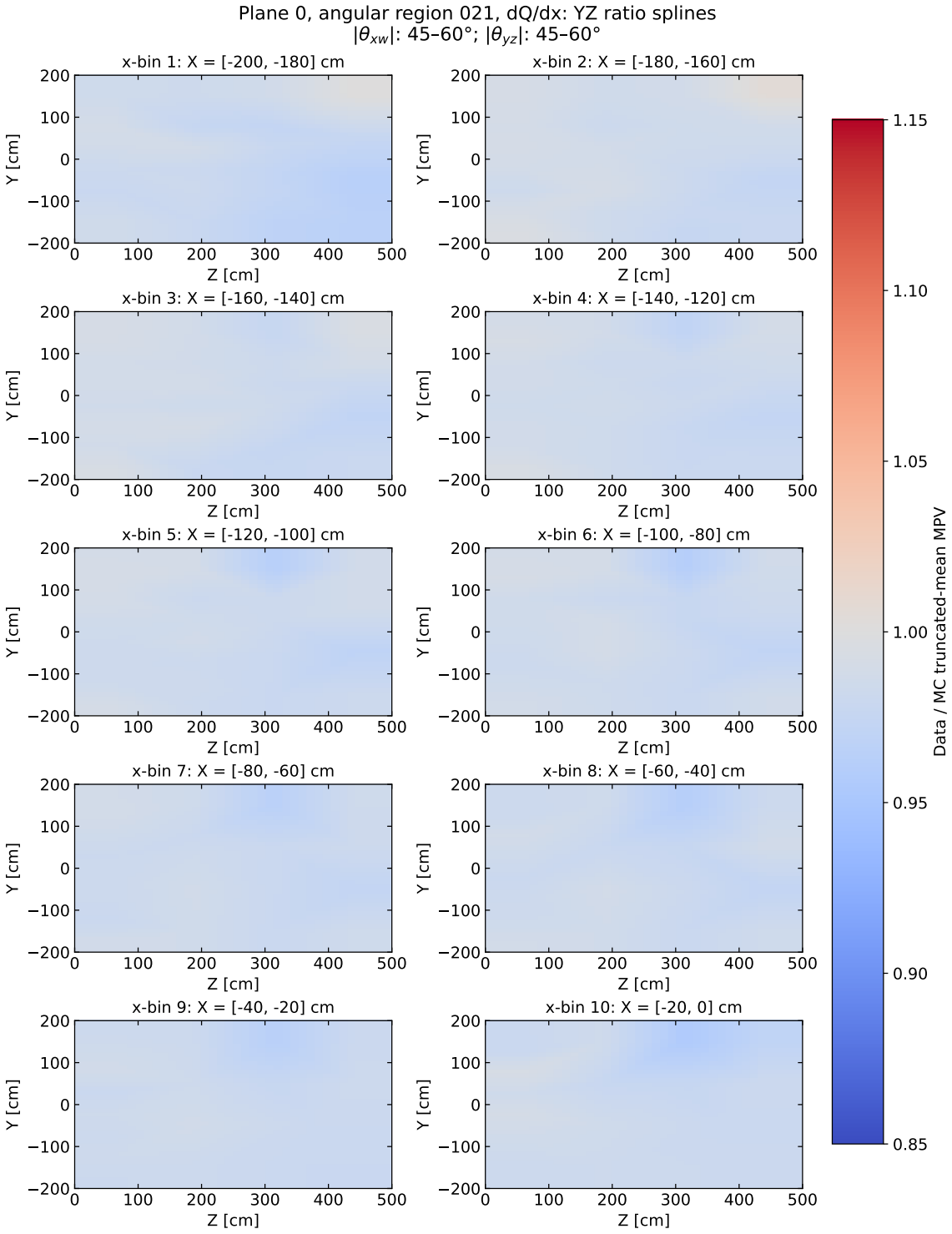


Figure 65: Plane 0, angular region 021: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [45, 60]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

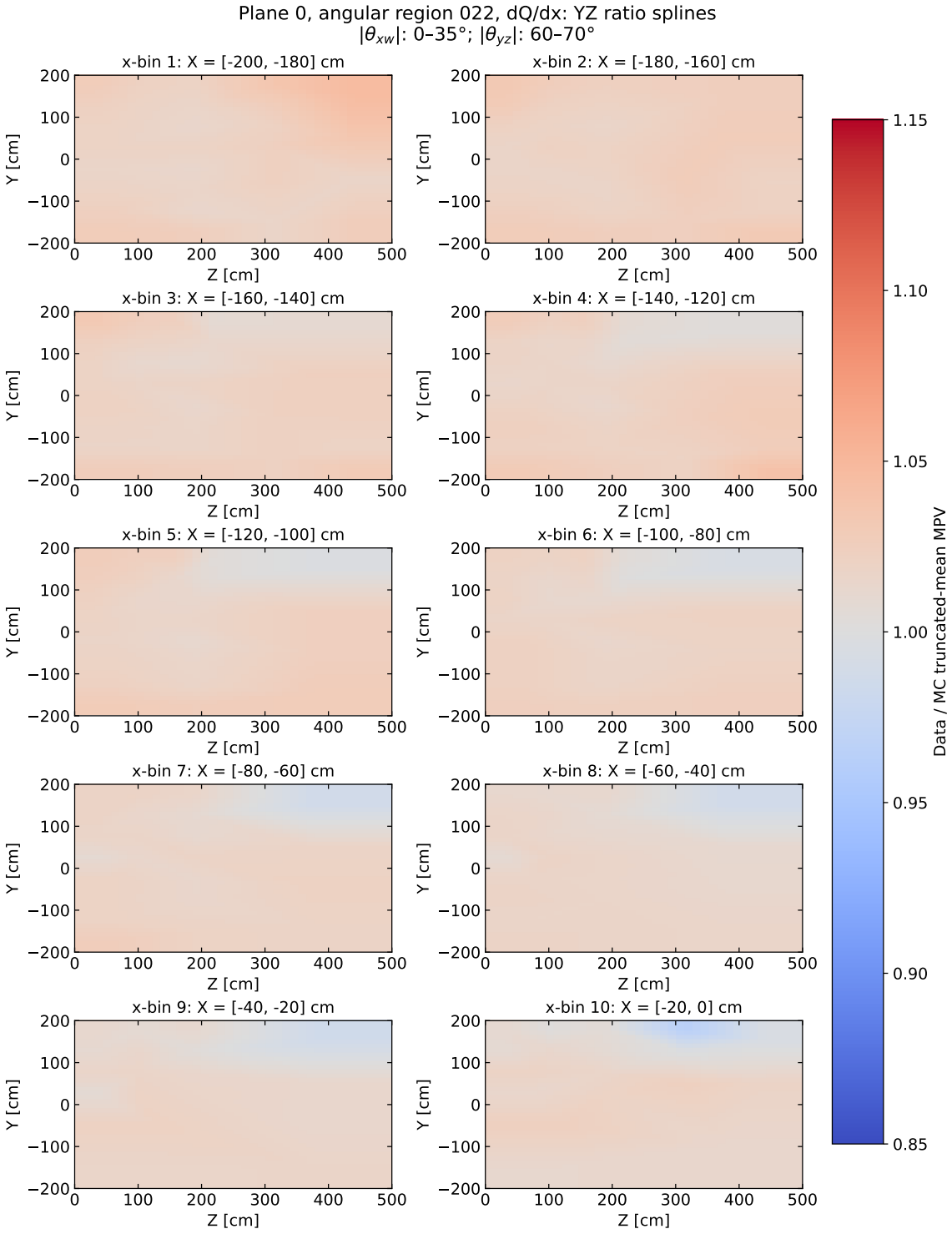


Figure 66: Plane 0, angular region 022: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 35]^\circ$ and $|\theta_{yz}| \in [60, 70]^\circ$.

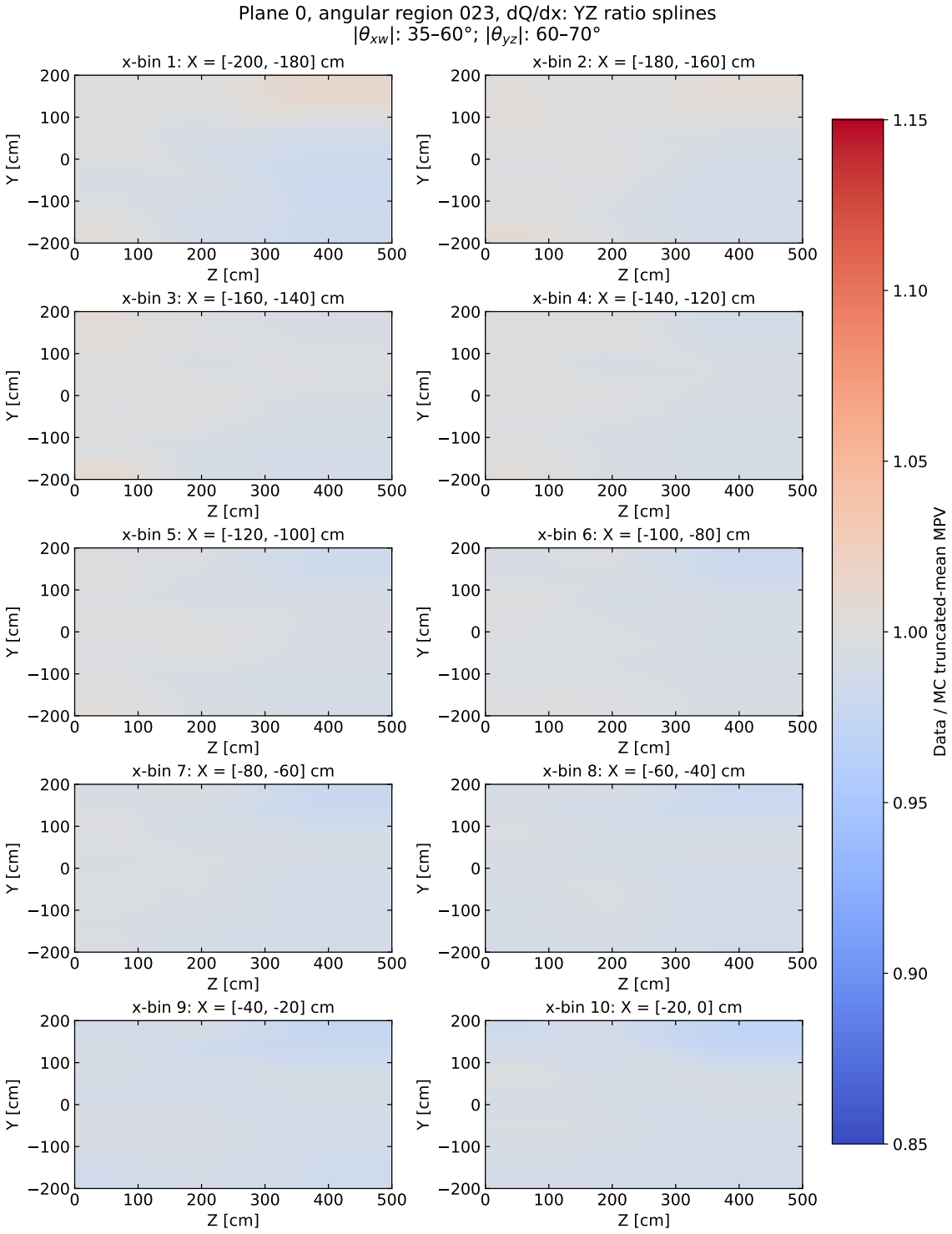


Figure 67: Plane 0, angular region 023: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [35, 60]^\circ$ and $|\theta_{yz}| \in [60, 70]^\circ$.

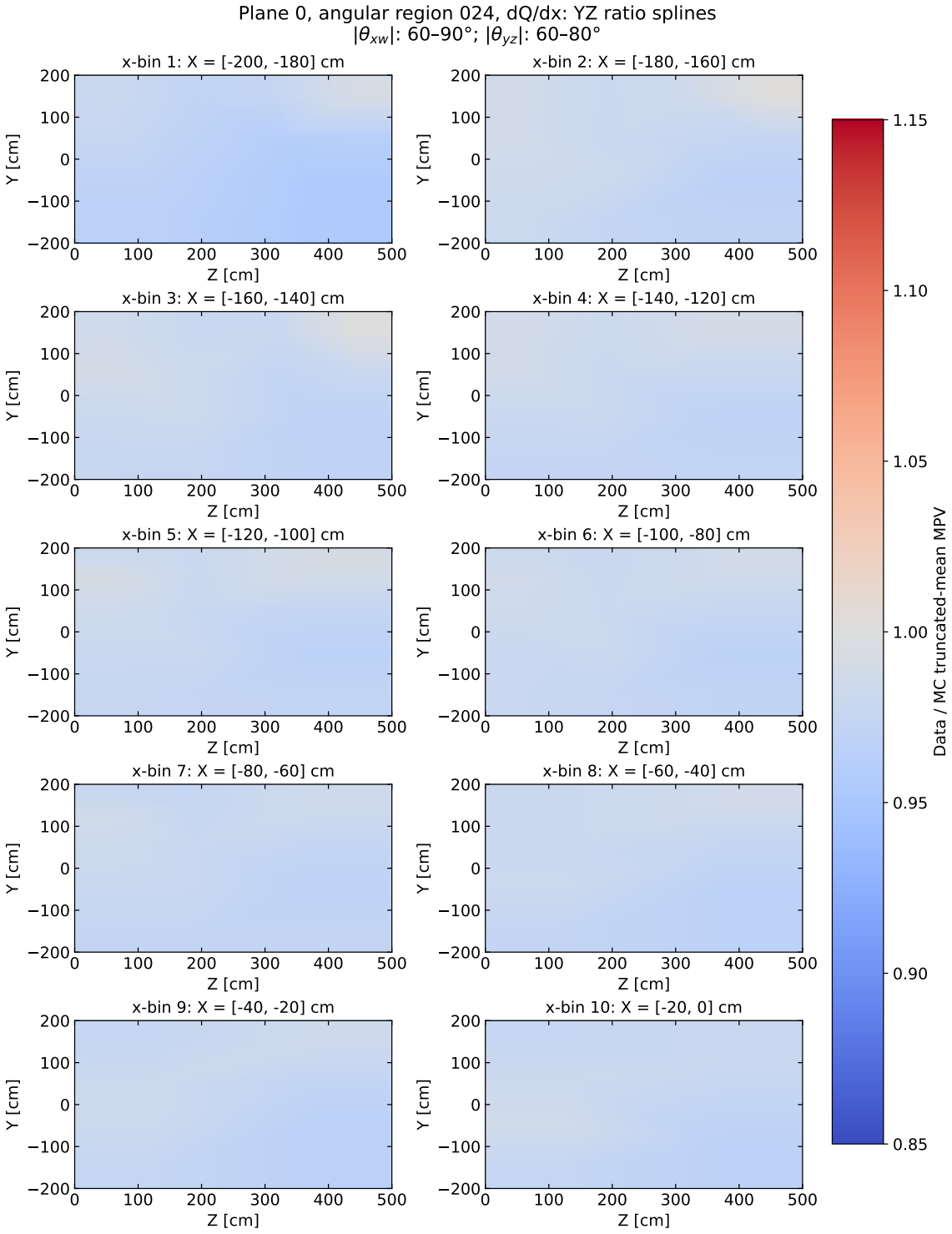


Figure 68: Plane 0, angular region 024: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [60, 80]^\circ$.

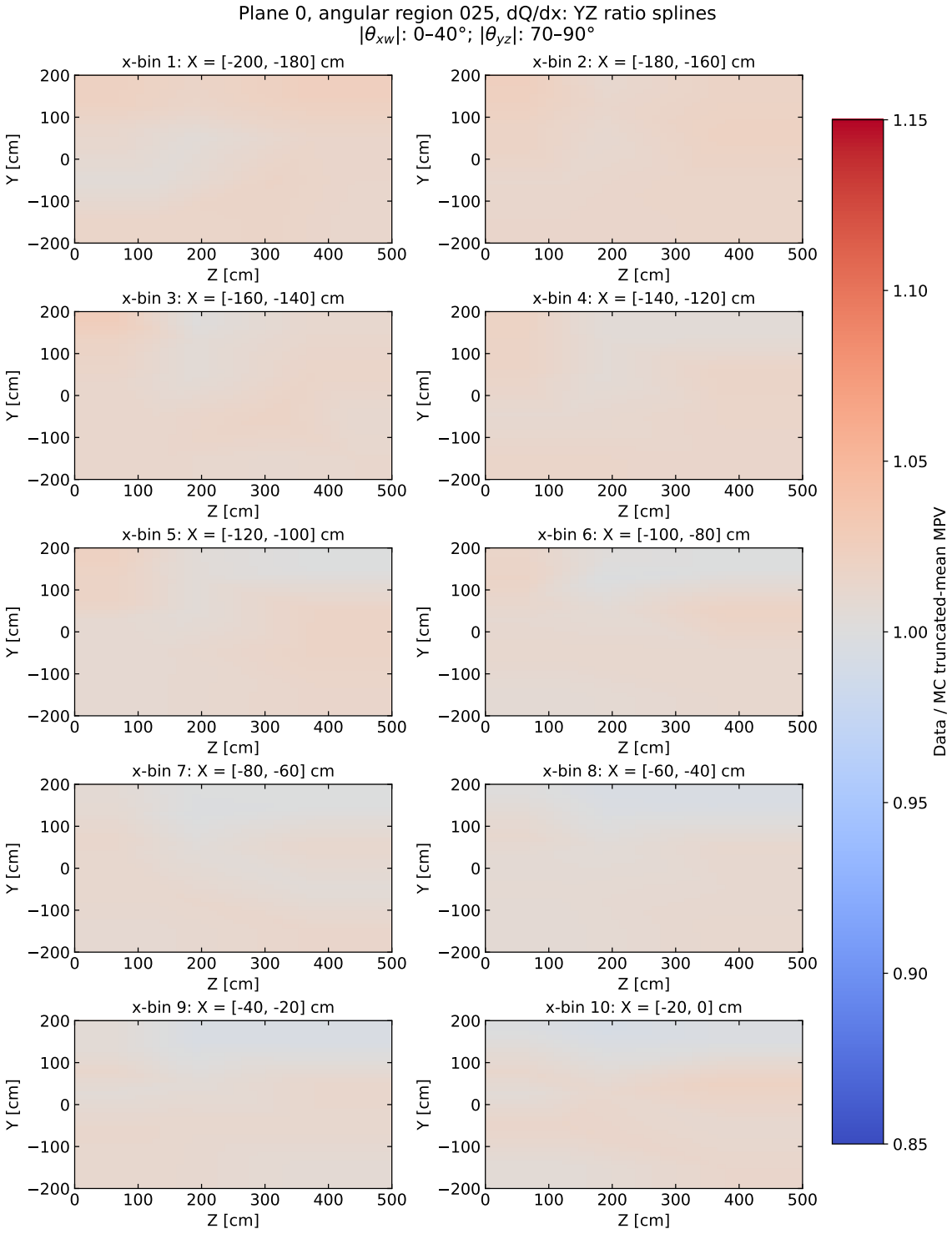


Figure 69: Plane 0, angular region 025: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 40]^\circ$ and $|\theta_{yz}| \in [70, 90]^\circ$.

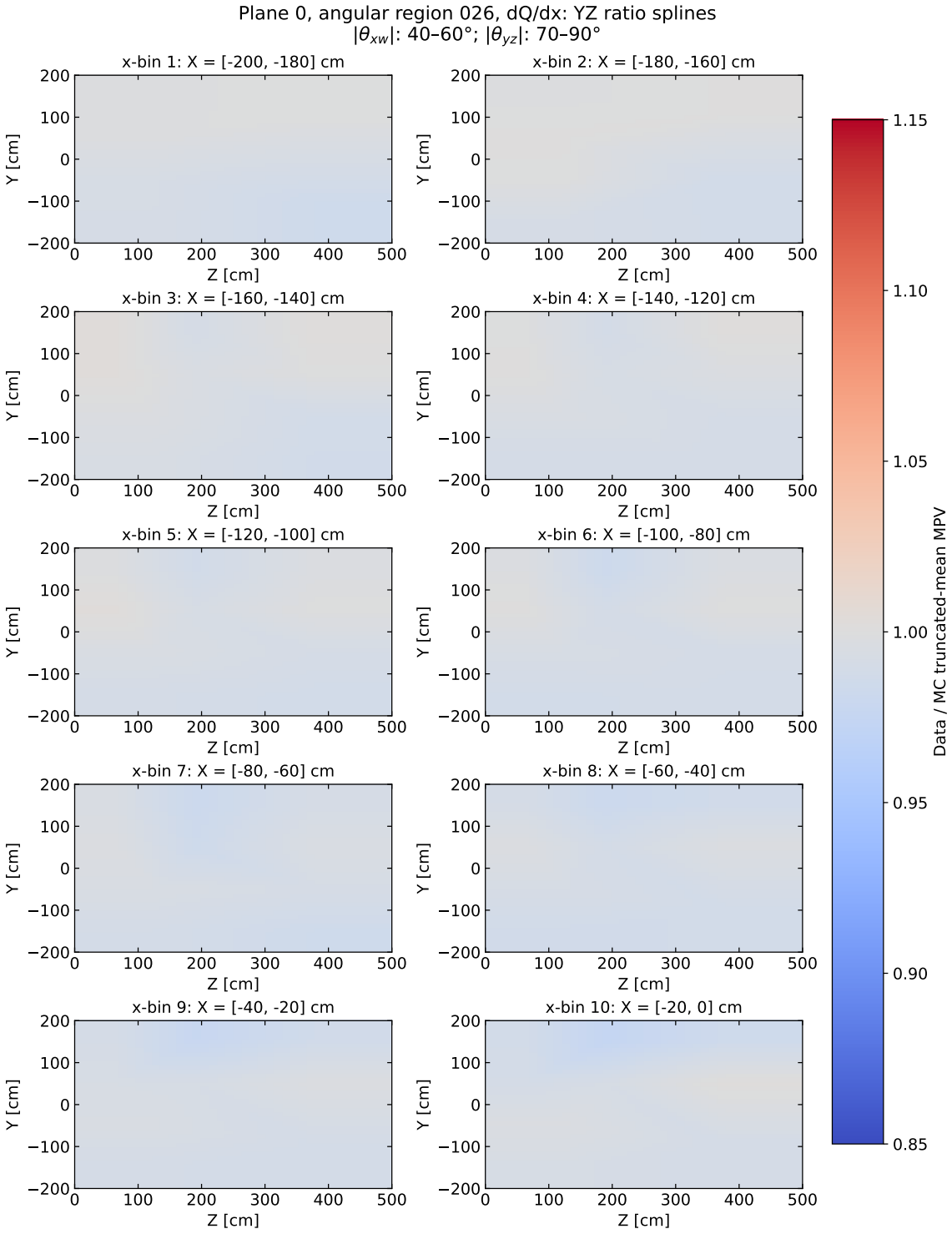


Figure 70: Plane 0, angular region 026: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [70, 90]^\circ$.

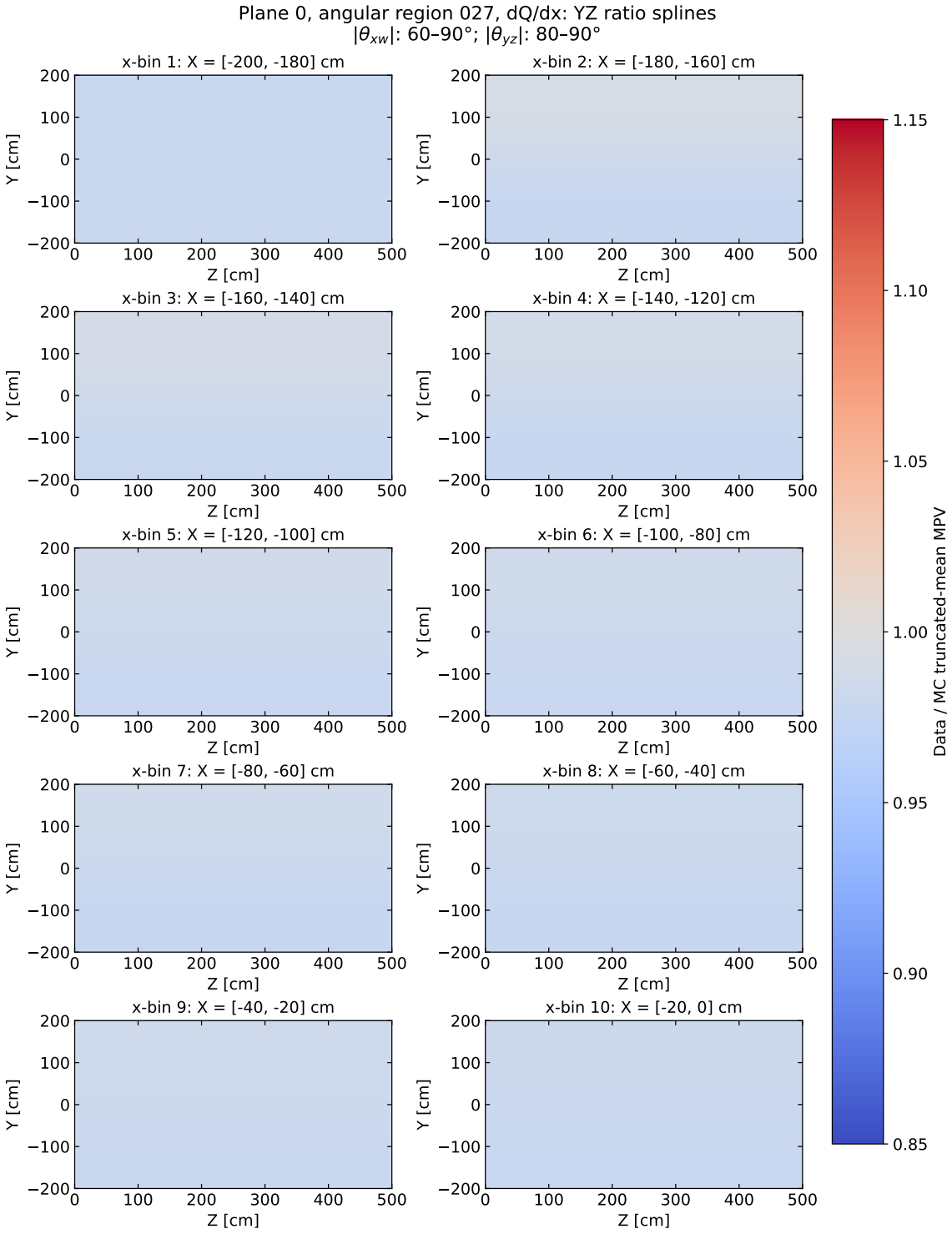


Figure 71: Plane 0, angular region 027: dQ/dx YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [80, 90]^\circ$.

B.1.3 Ratio of Q and dQ/dx Splines

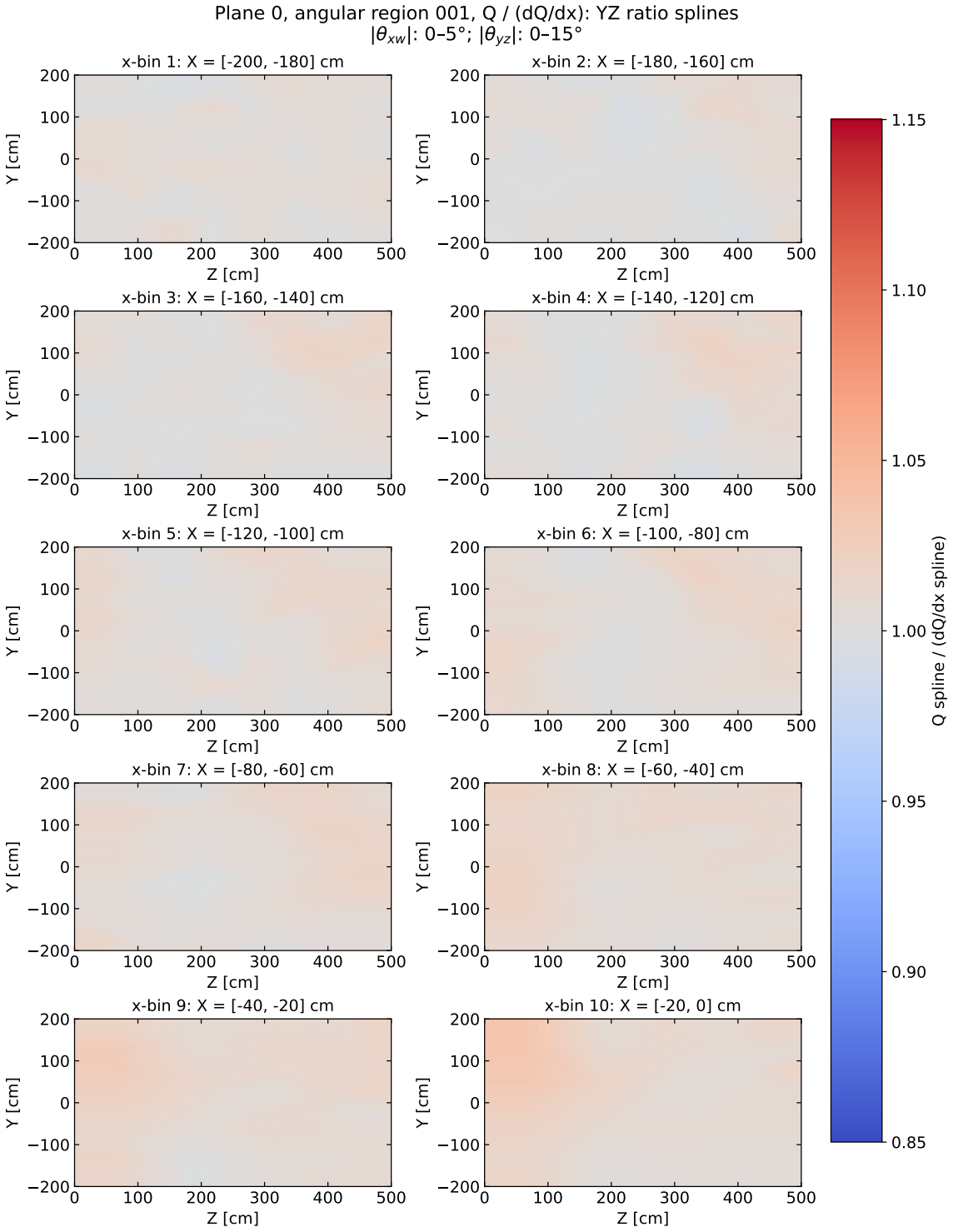


Figure 72: Plane 0, angular region 001: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 5]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

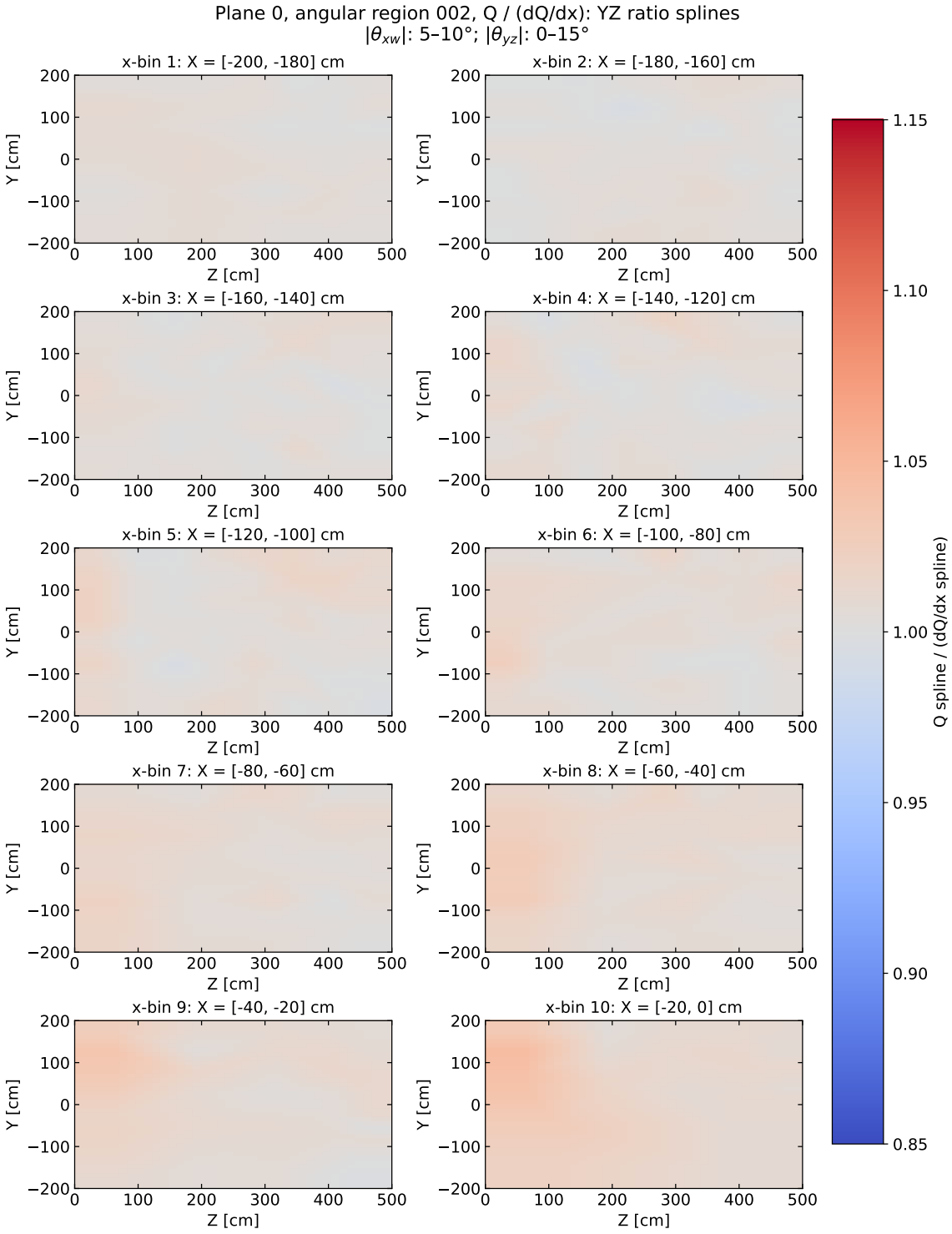


Figure 73: Plane 0, angular region 002: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [5, 10]°$ and $|\theta_{yz}| \in [0, 15]°$.

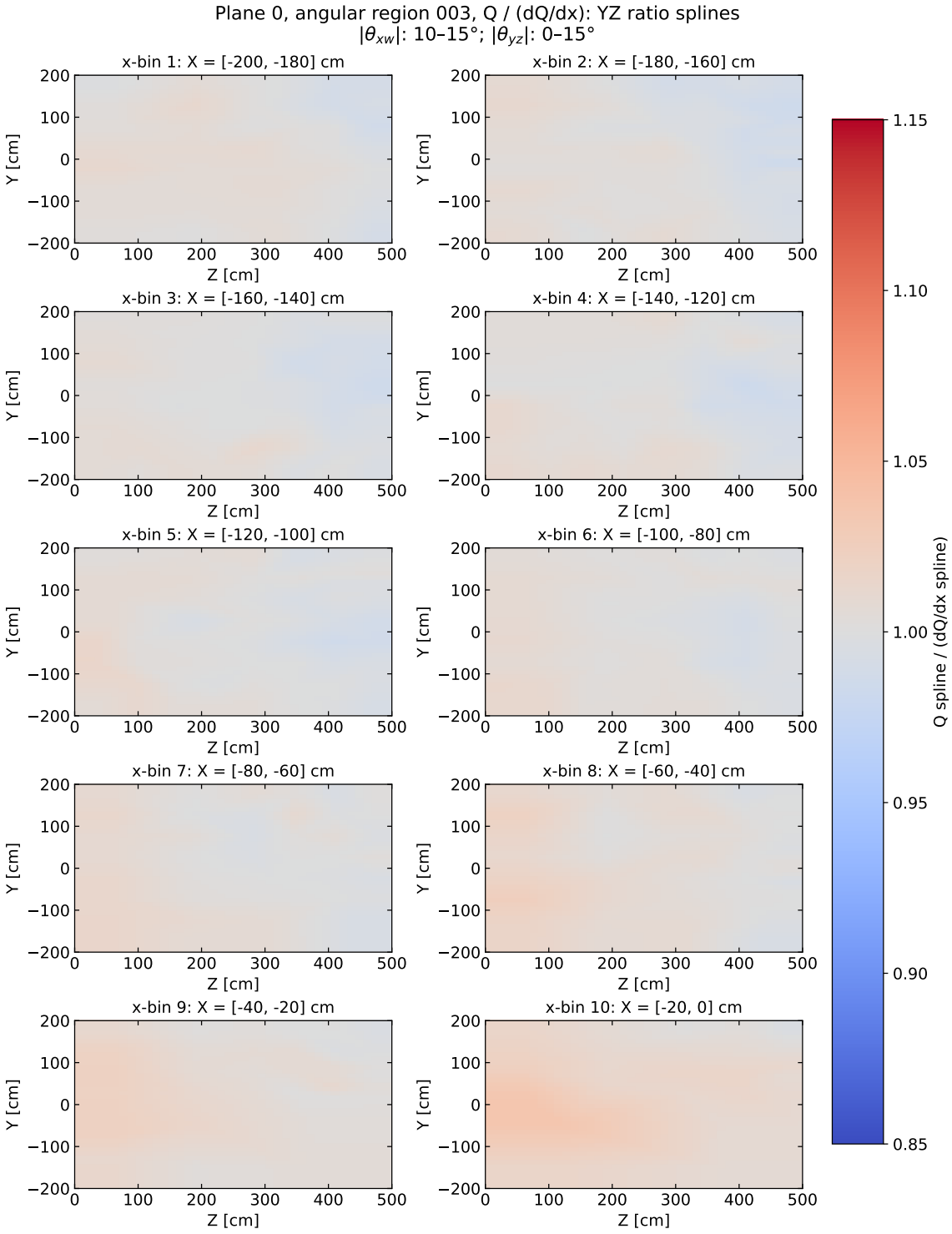


Figure 74: Plane 0, angular region 003: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [10, 15]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

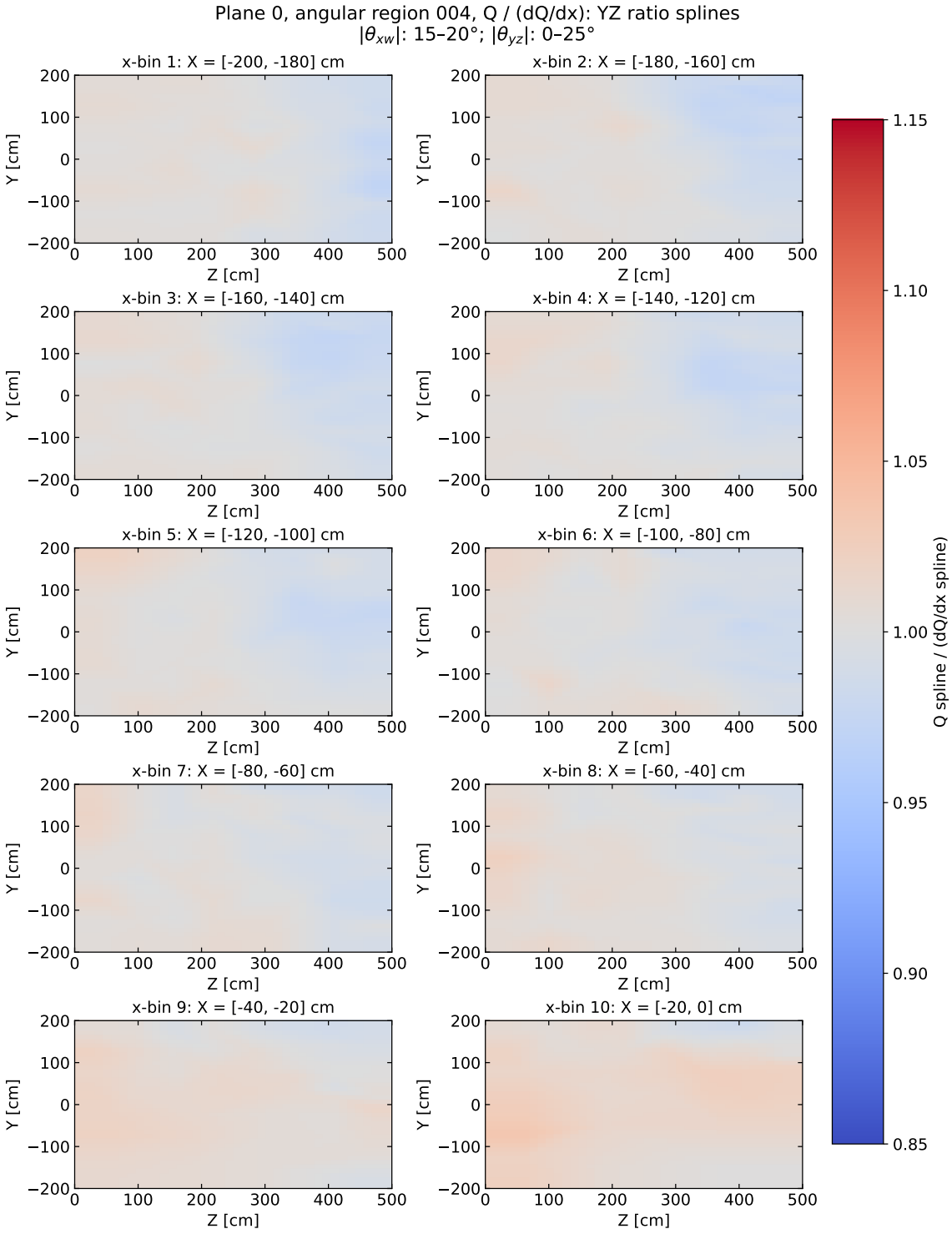


Figure 75: Plane 0, angular region 004: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 20]^\circ$ and $|\theta_{yz}| \in [0, 25]^\circ$.

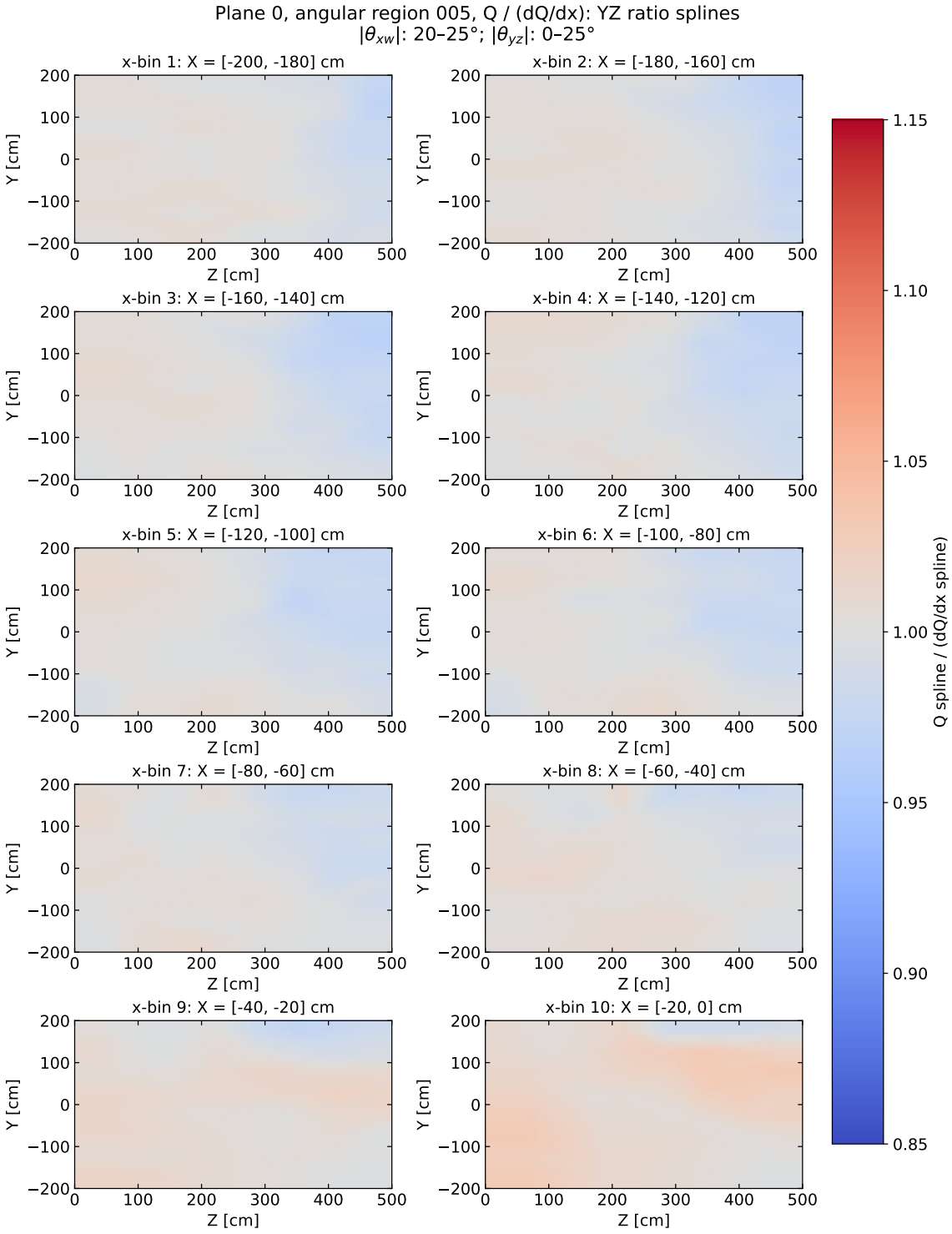


Figure 76: Plane 0, angular region 005: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [20, 25]^\circ$ and $|\theta_{yz}| \in [0, 25]^\circ$.

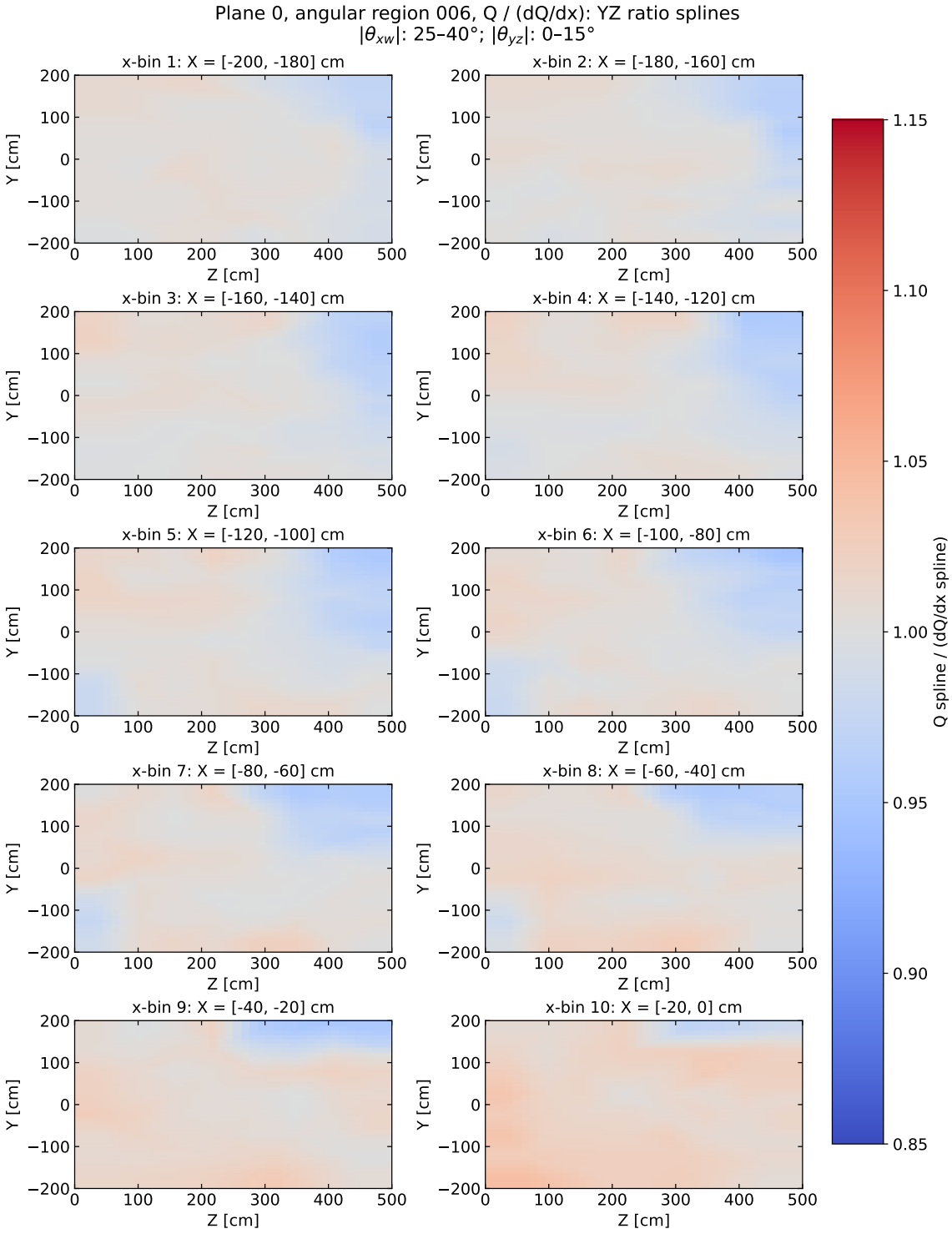


Figure 77: Plane 0, angular region 006: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

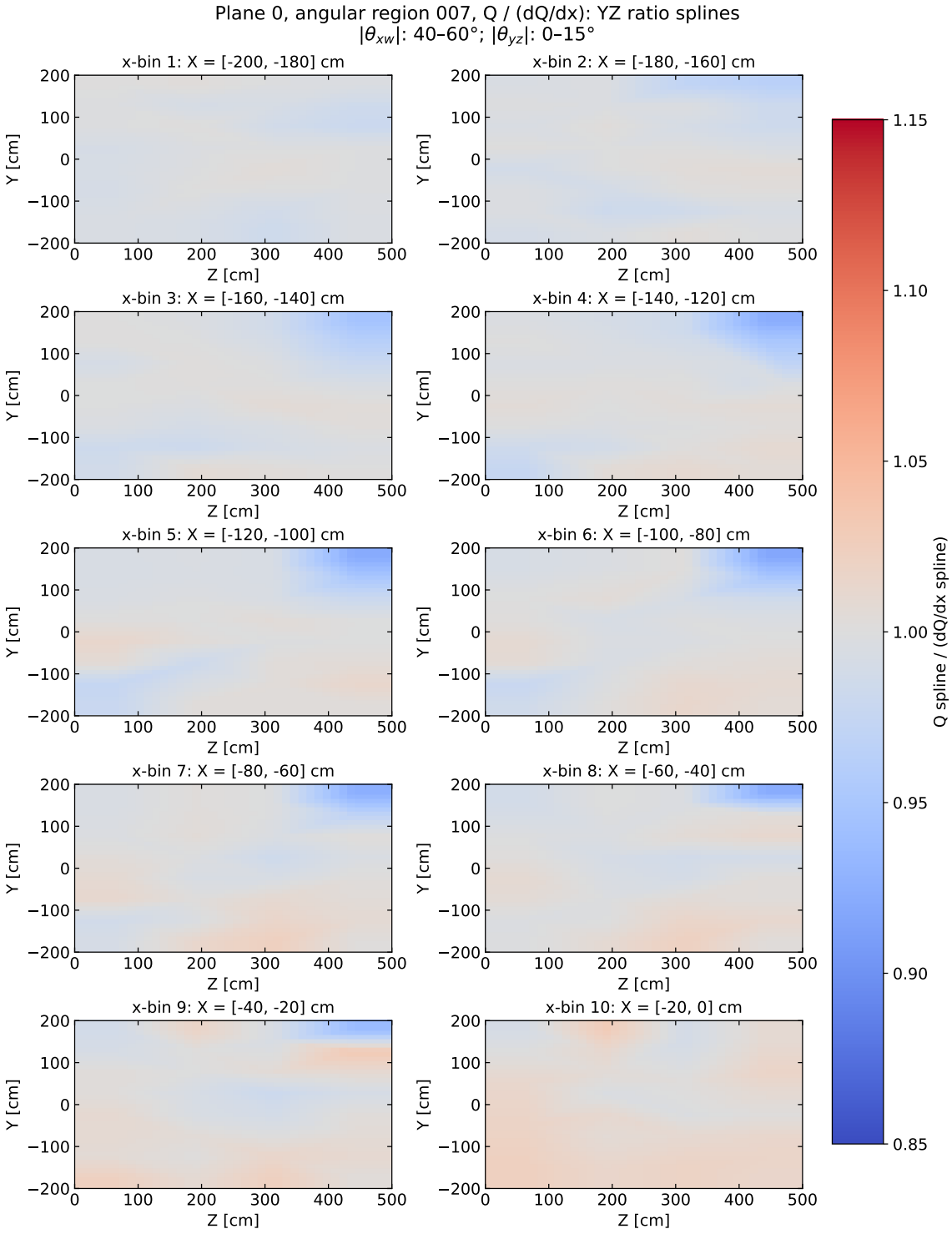


Figure 78: Plane 0, angular region 007: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [0, 15]^\circ$.

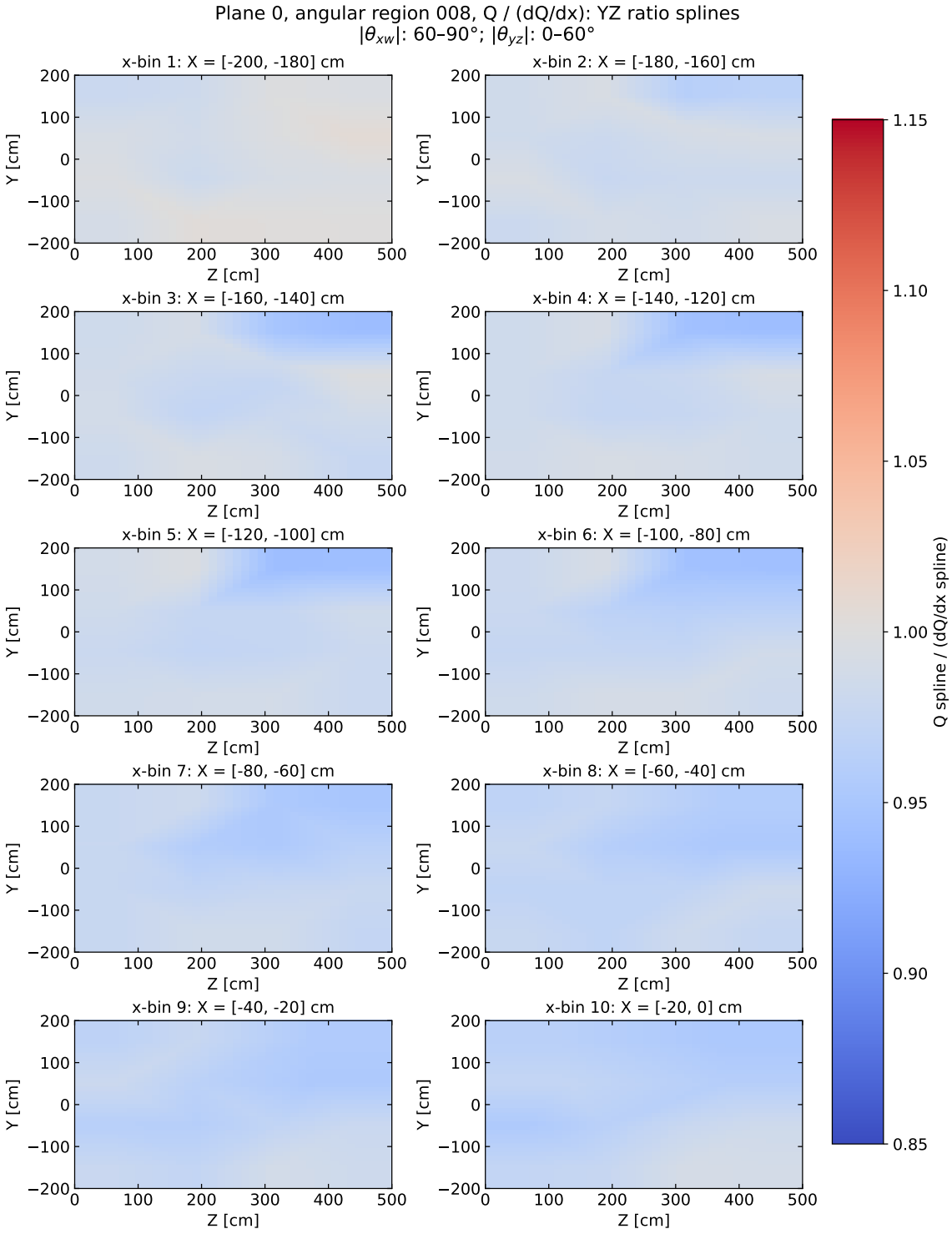


Figure 79: Plane 0, angular region 008: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [0, 60]^\circ$.

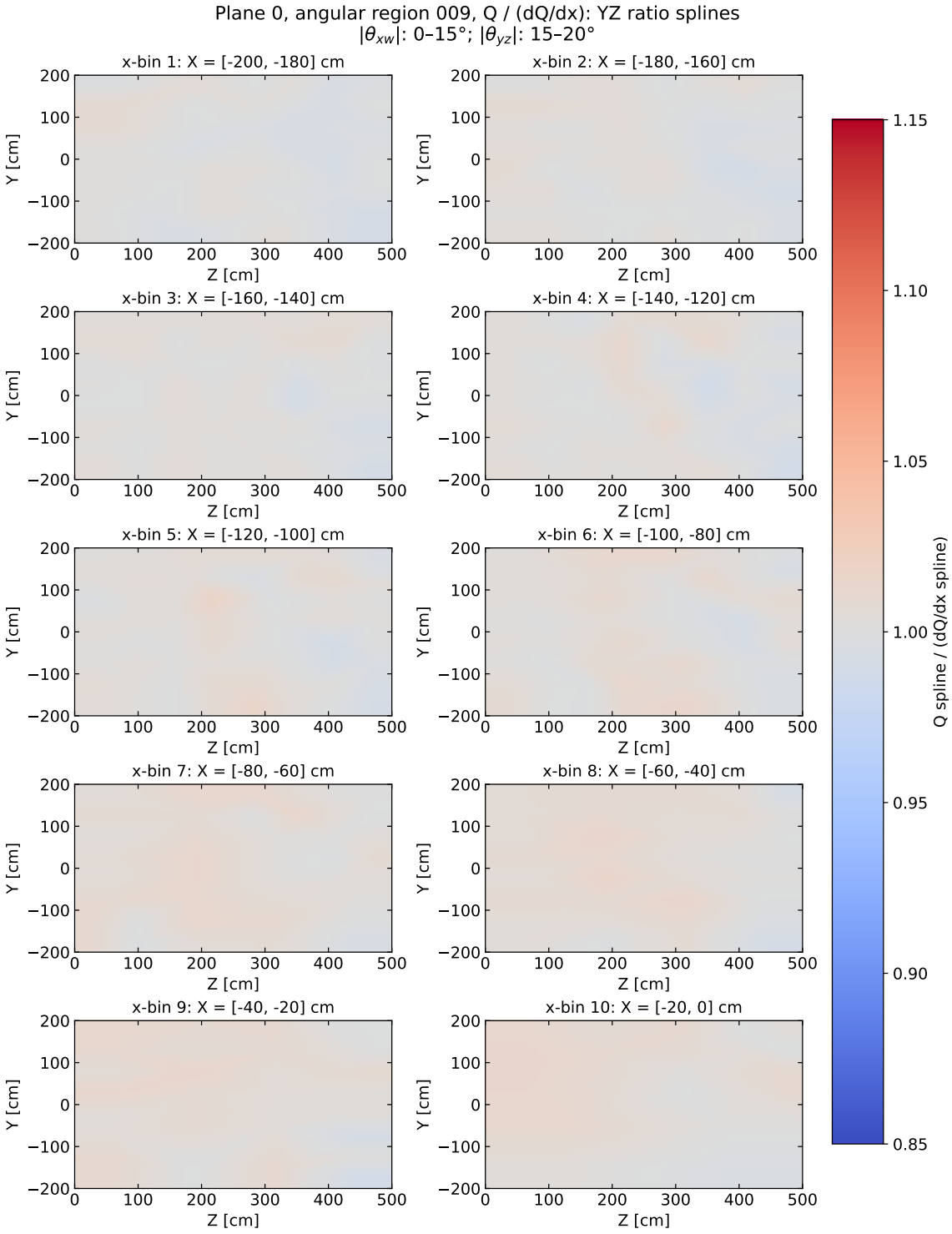


Figure 80: Plane 0, angular region 009: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [15, 20]^\circ$.

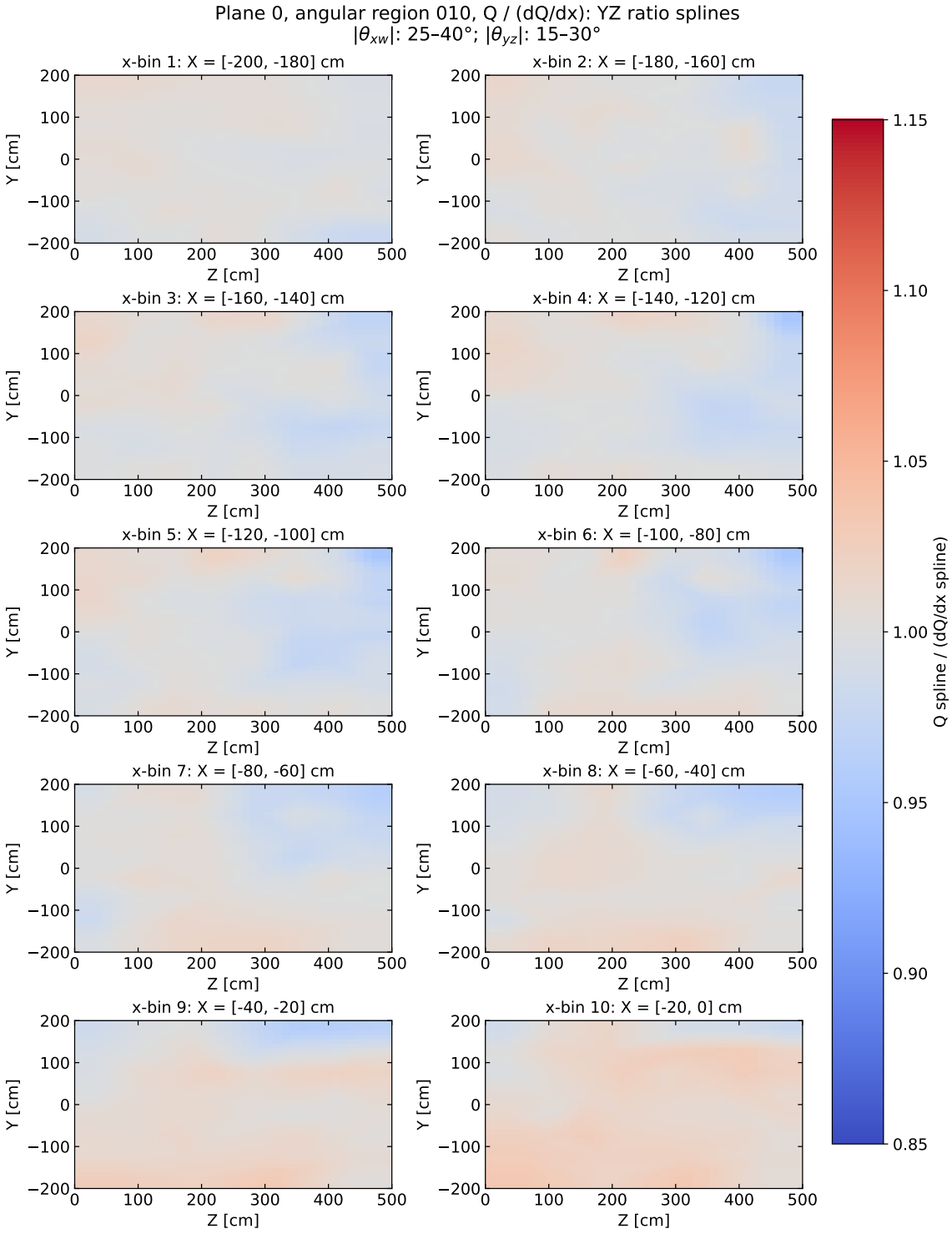


Figure 81: Plane 0, angular region 010: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [15, 30]^\circ$.

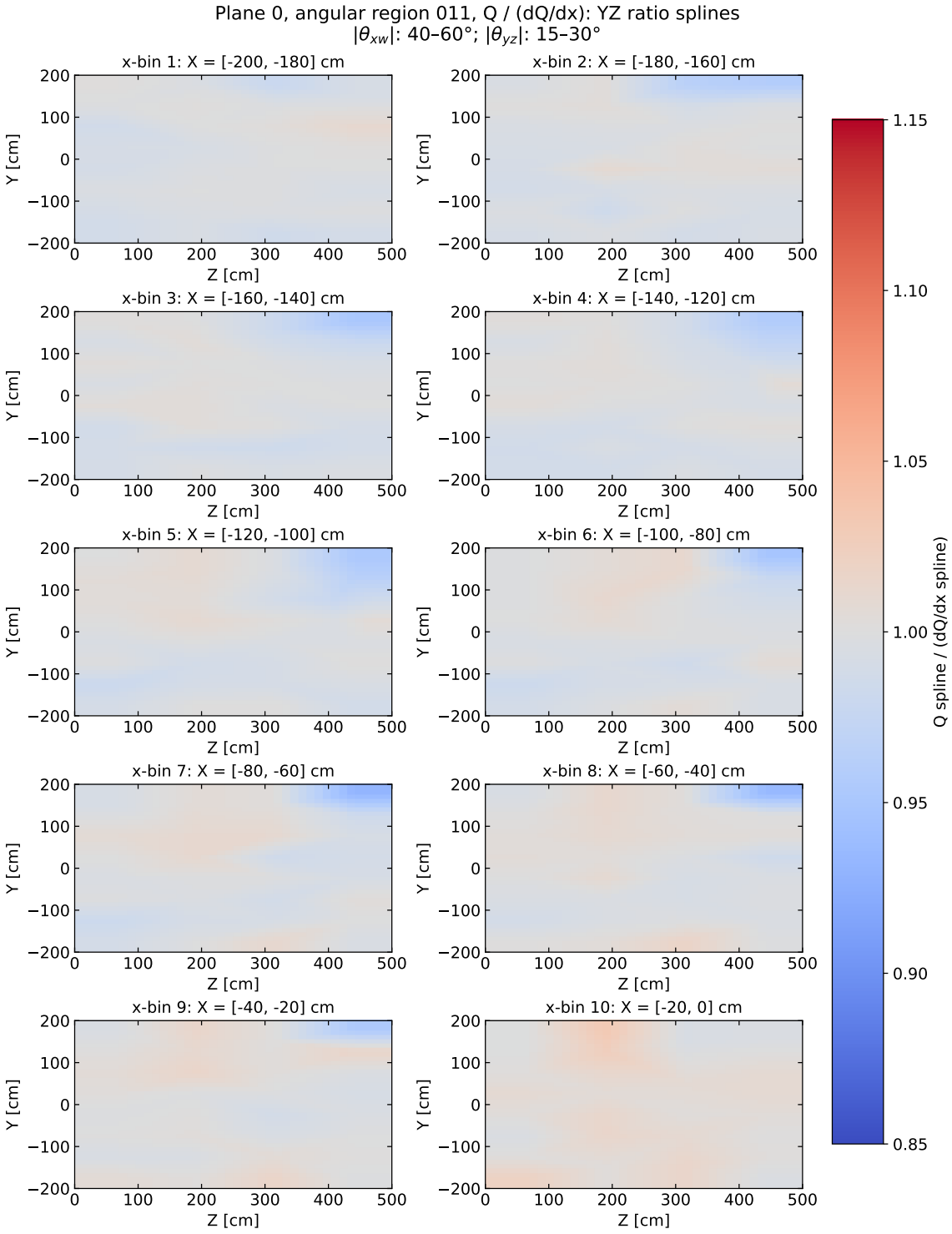


Figure 82: Plane 0, angular region 011: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [15, 30]^\circ$.

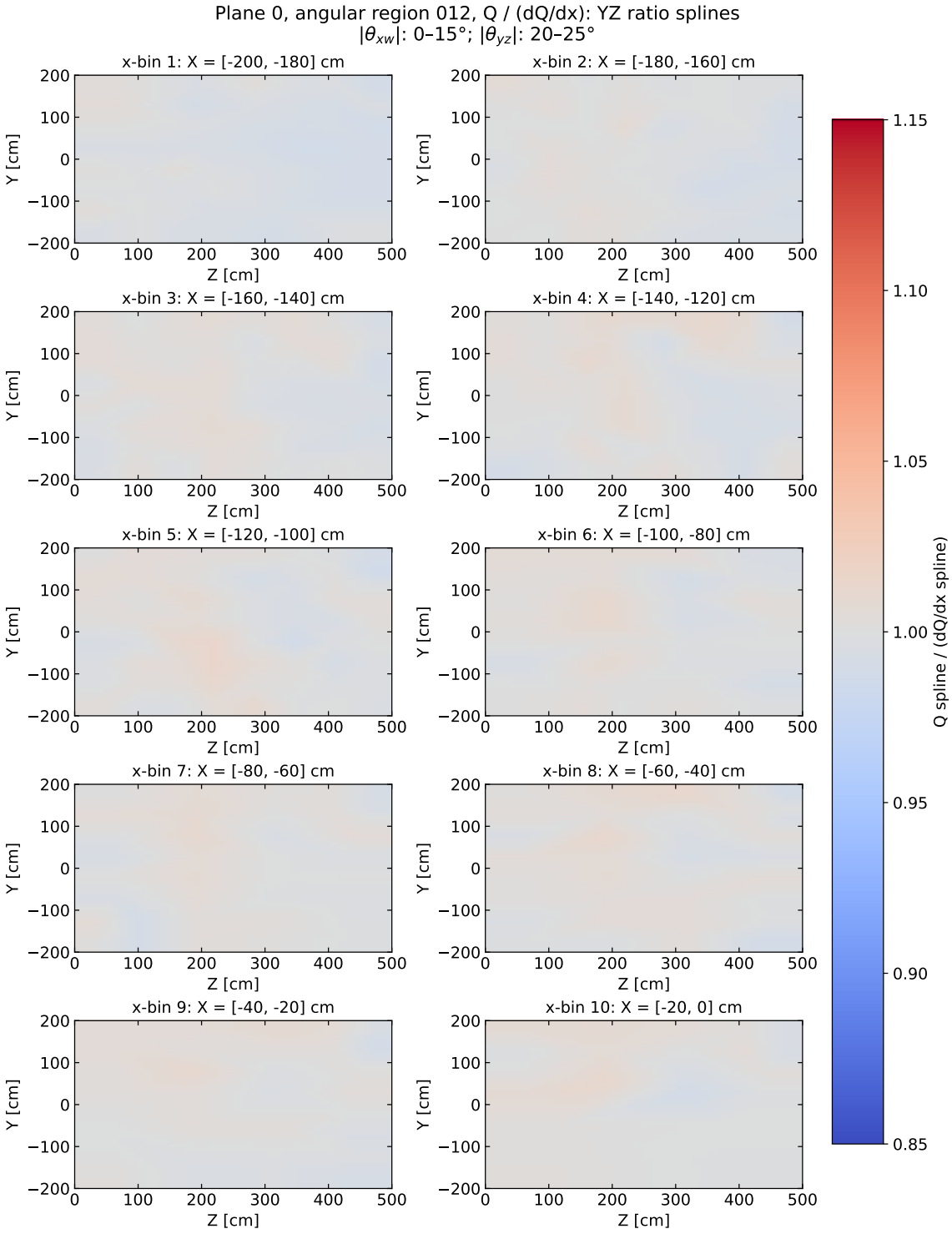


Figure 83: Plane 0, angular region 012: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [20, 25]^\circ$.

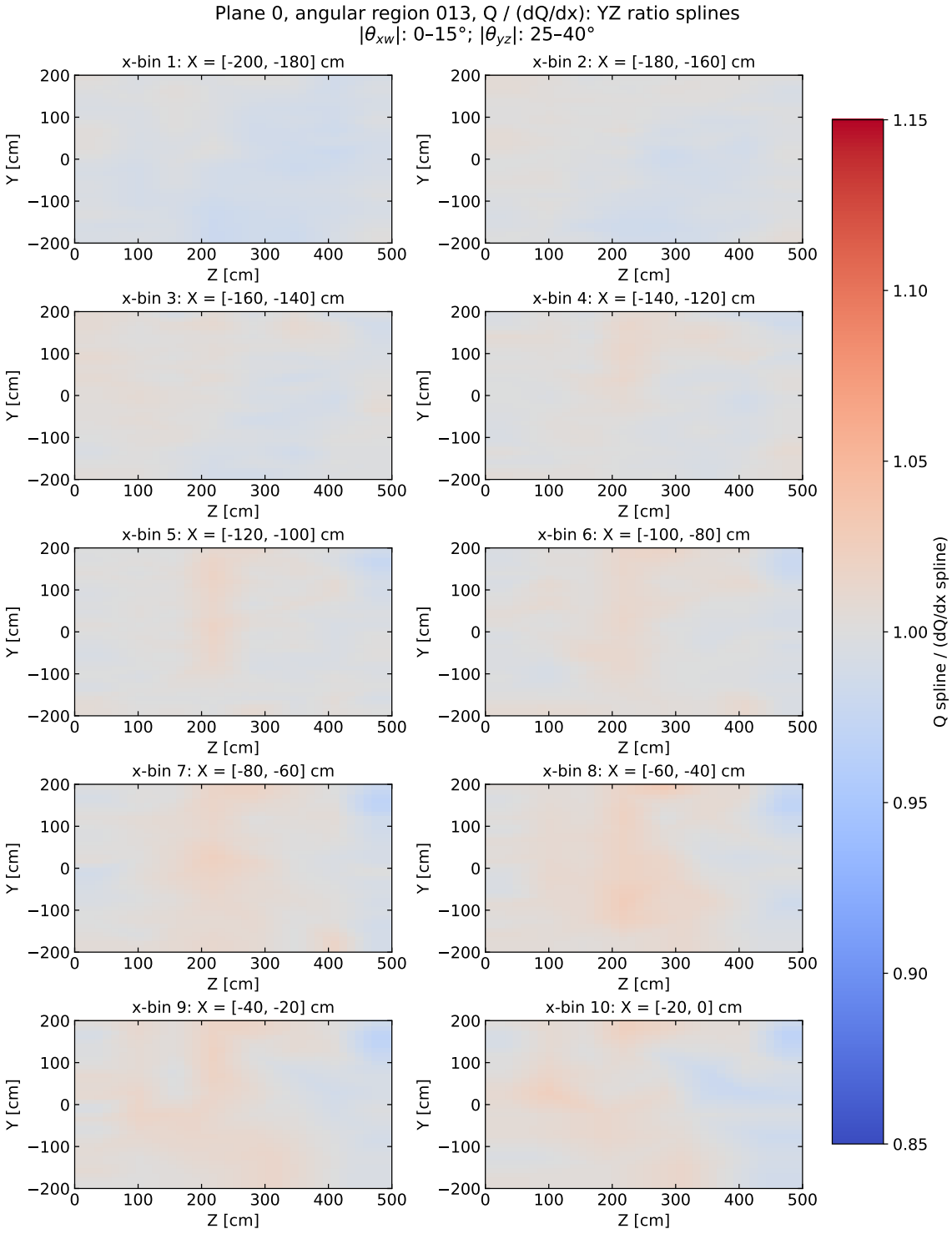


Figure 84: Plane 0, angular region 013: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [25, 40]^\circ$.

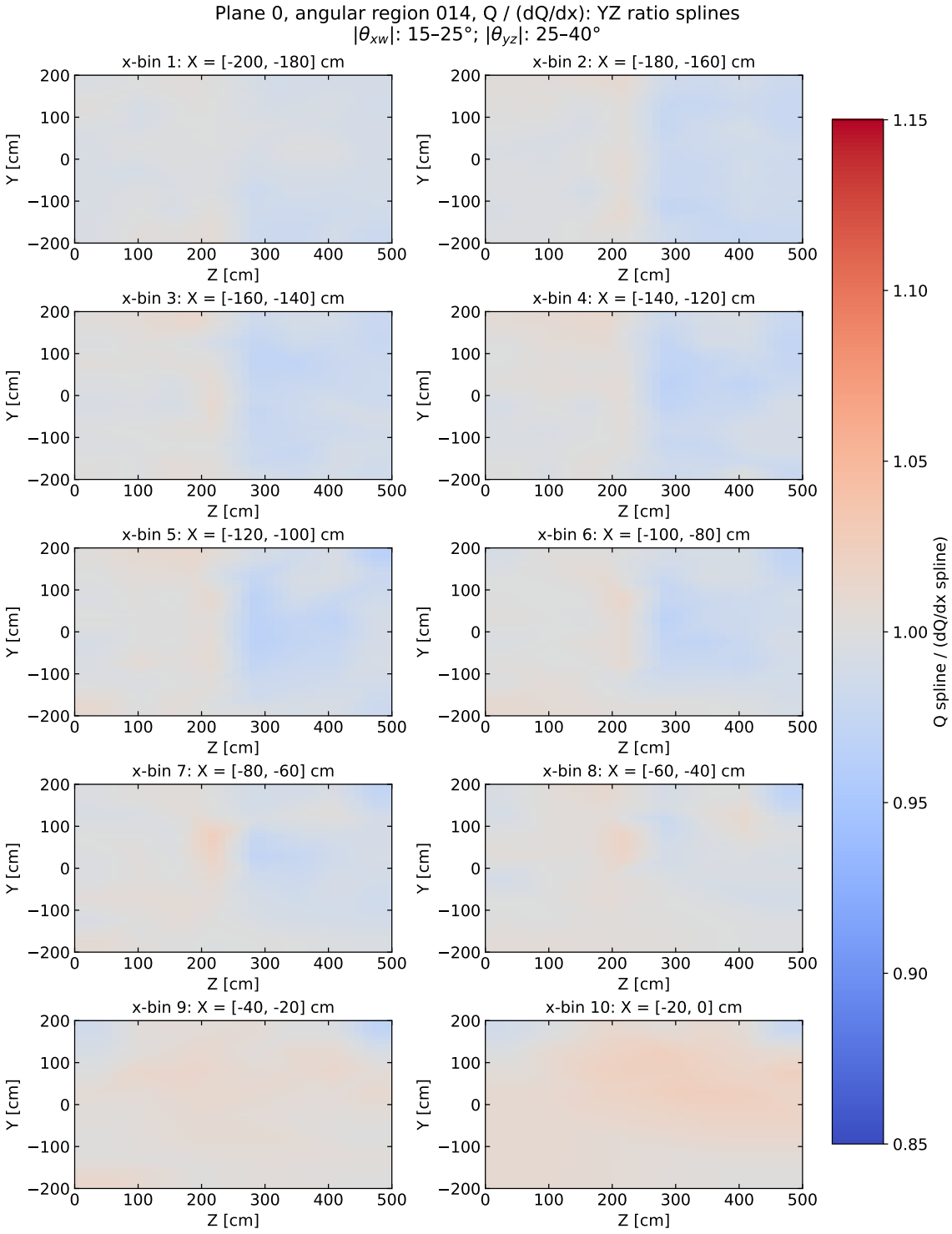


Figure 85: Plane 0, angular region 014: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 25]^\circ$ and $|\theta_{yz}| \in [25, 40]^\circ$.

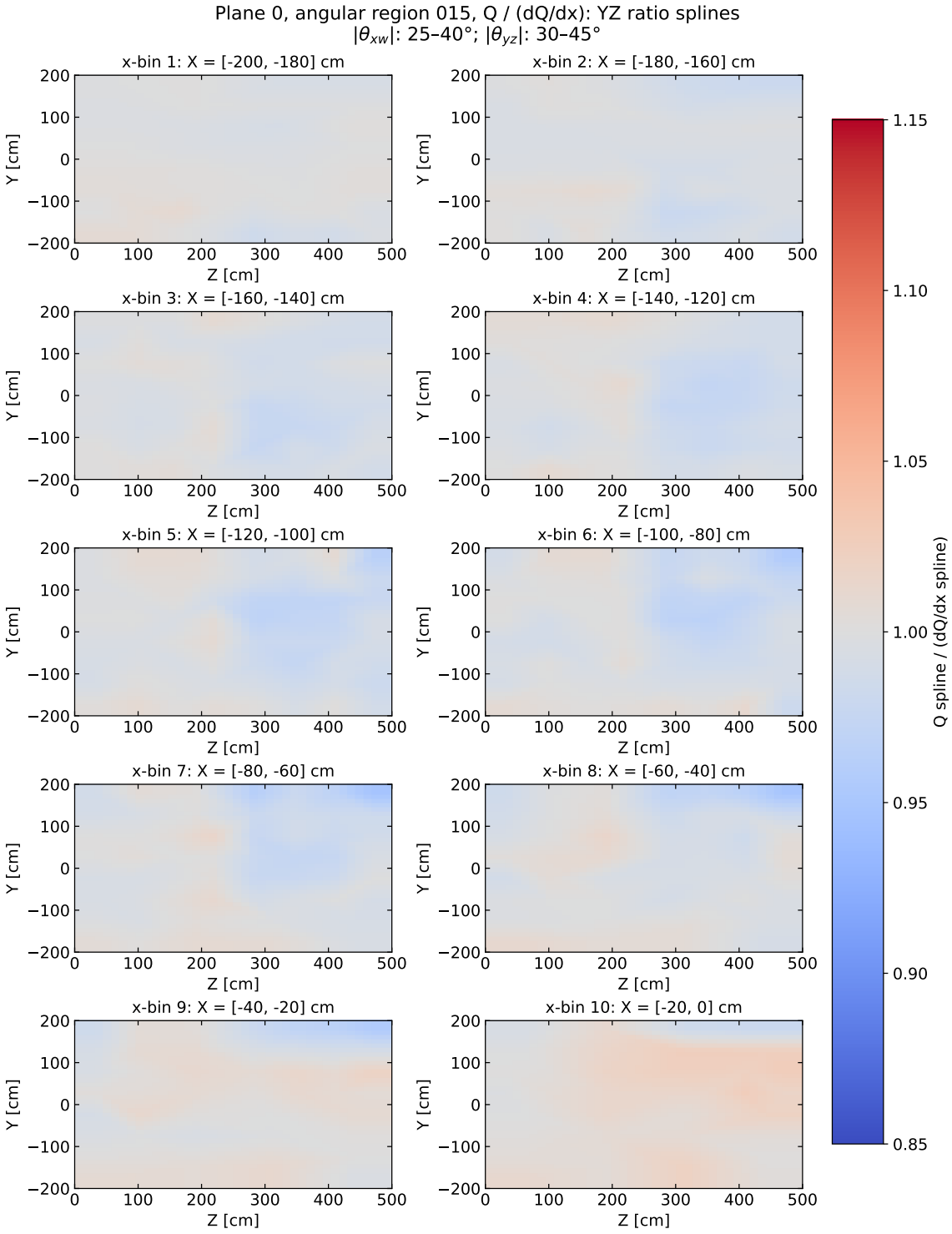


Figure 86: Plane 0, angular region 015: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [30, 45]^\circ$.

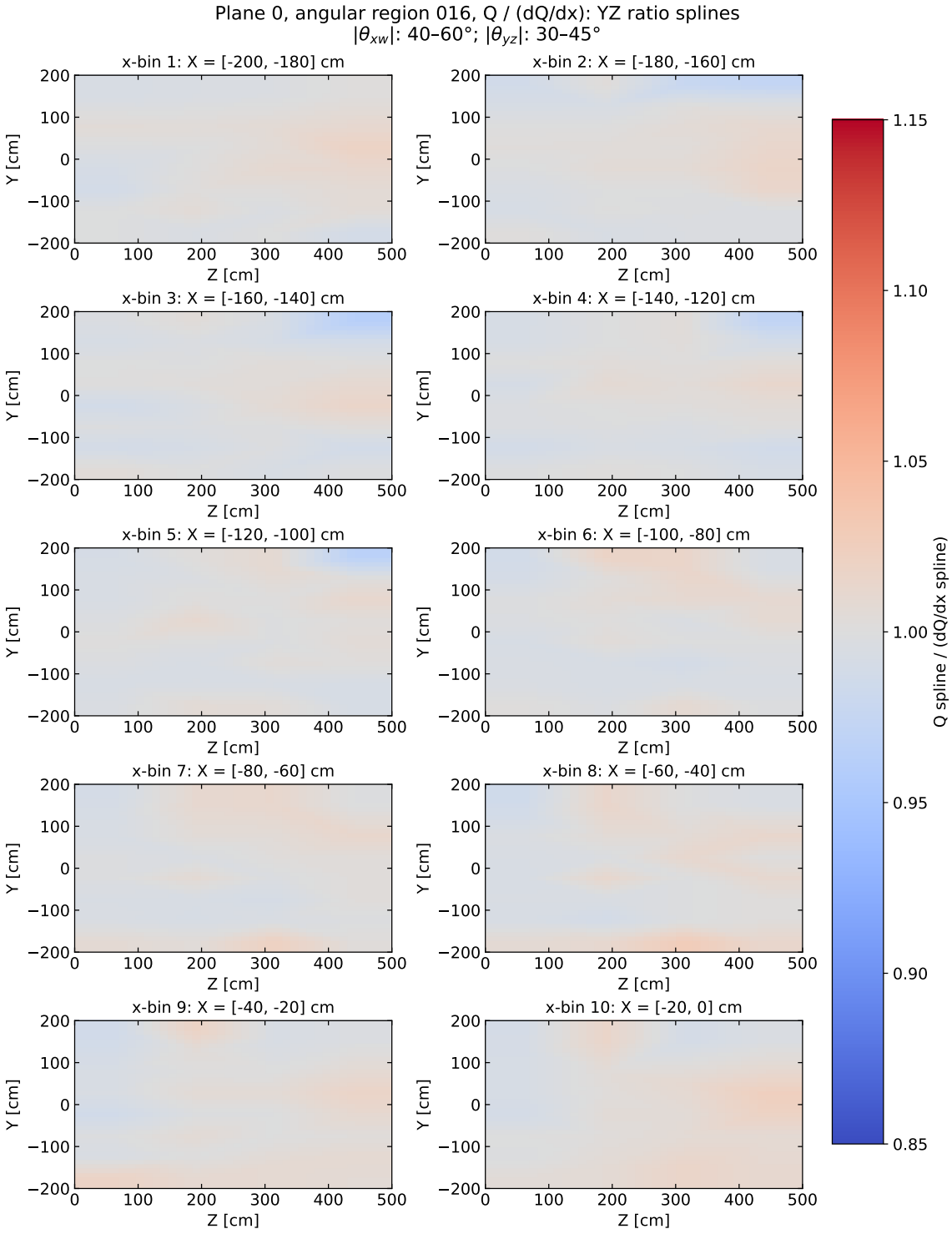


Figure 87: Plane 0, angular region 016: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [30, 45]^\circ$.

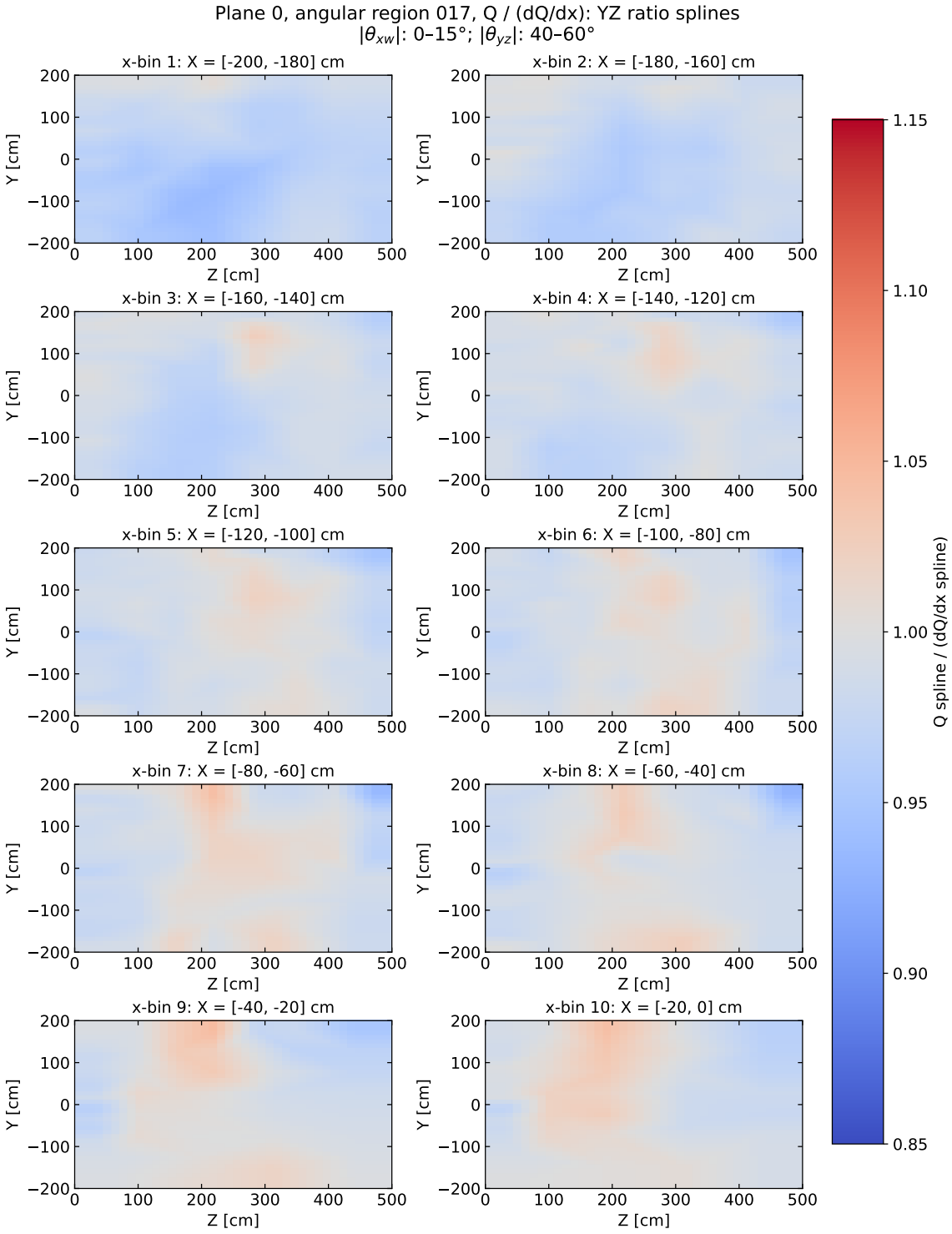


Figure 88: Plane 0, angular region 017: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [40, 60]^\circ$.

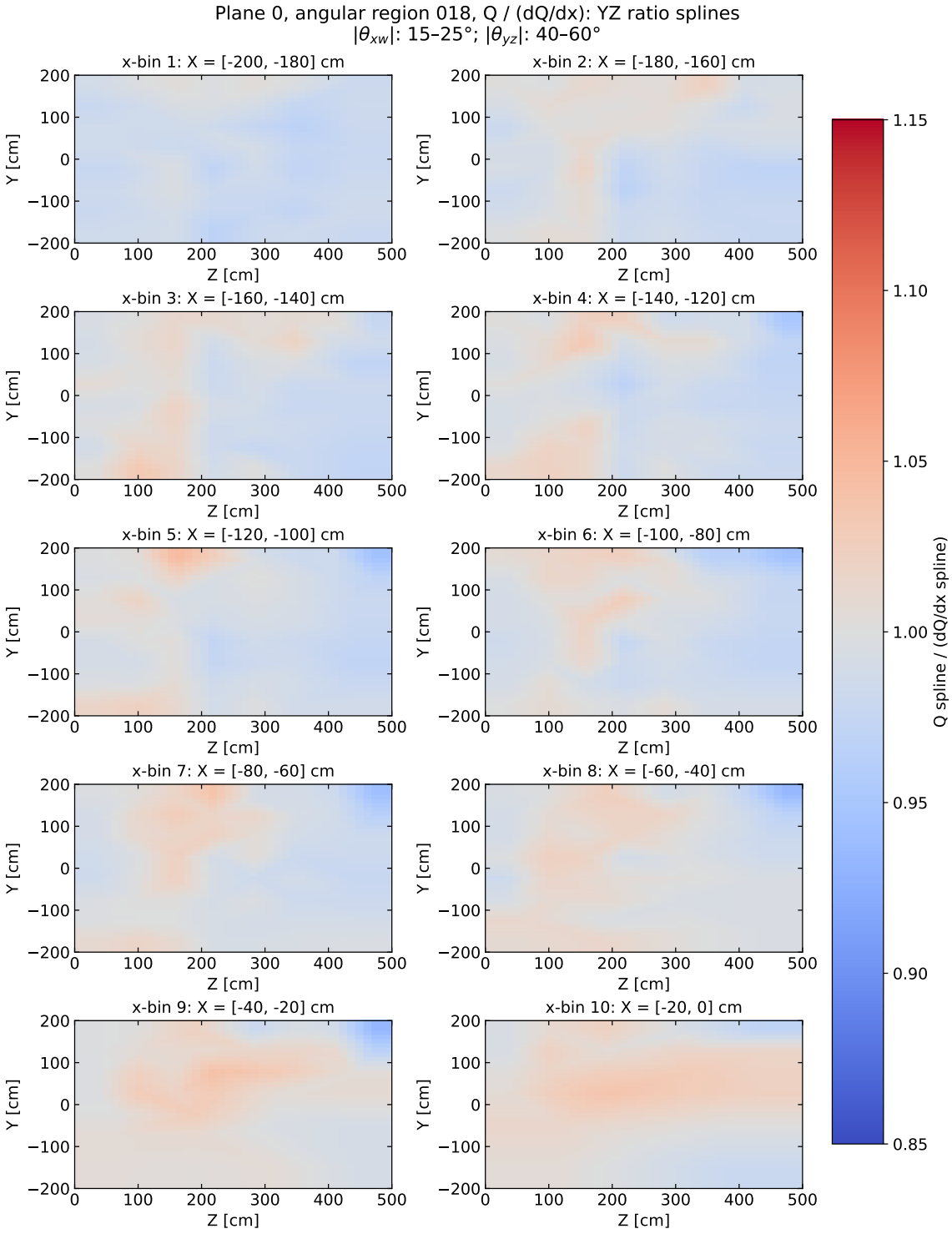


Figure 89: Plane 0, angular region 018: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 25]^\circ$ and $|\theta_{yz}| \in [40, 60]^\circ$.

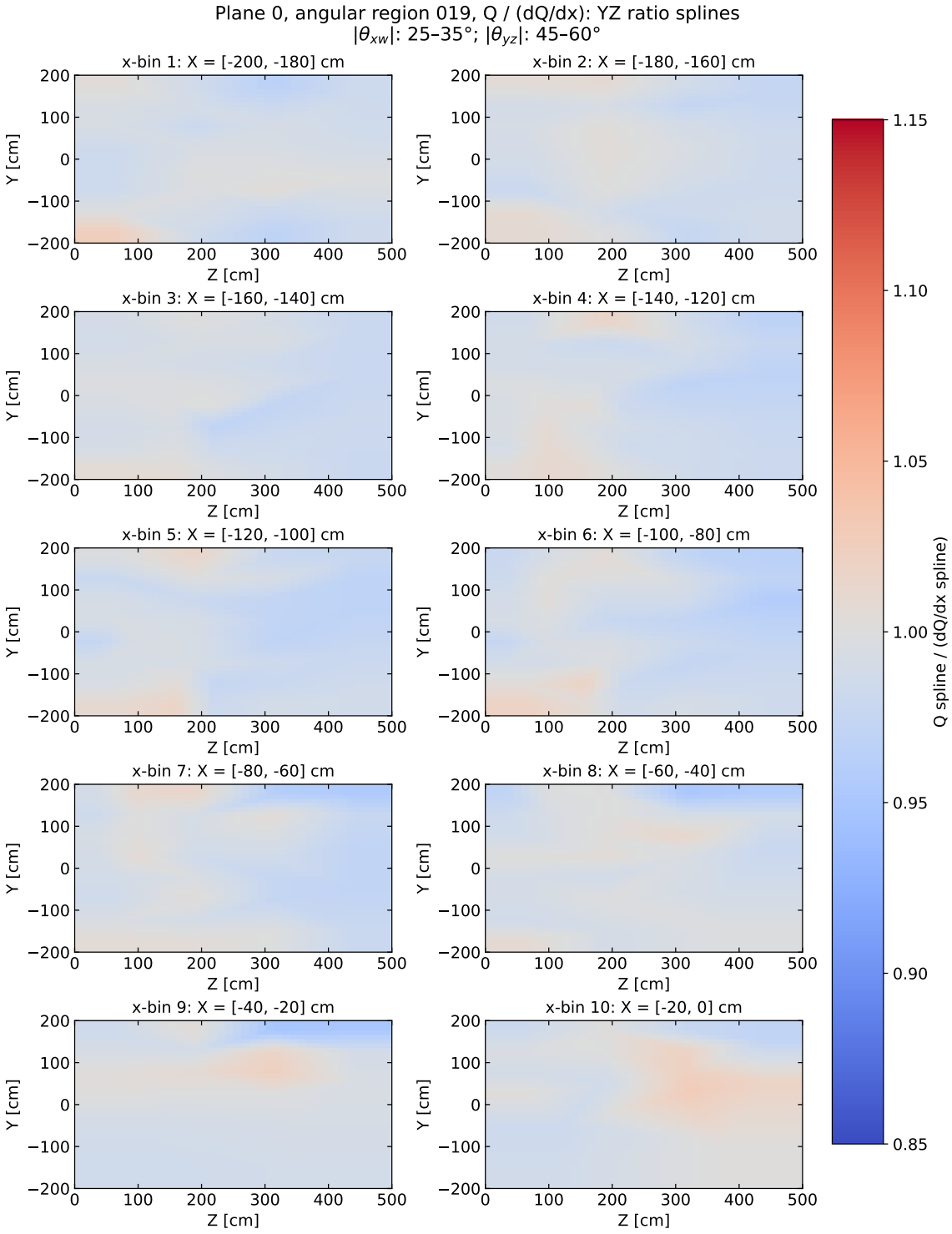


Figure 90: Plane 0, angular region 019: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 35]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

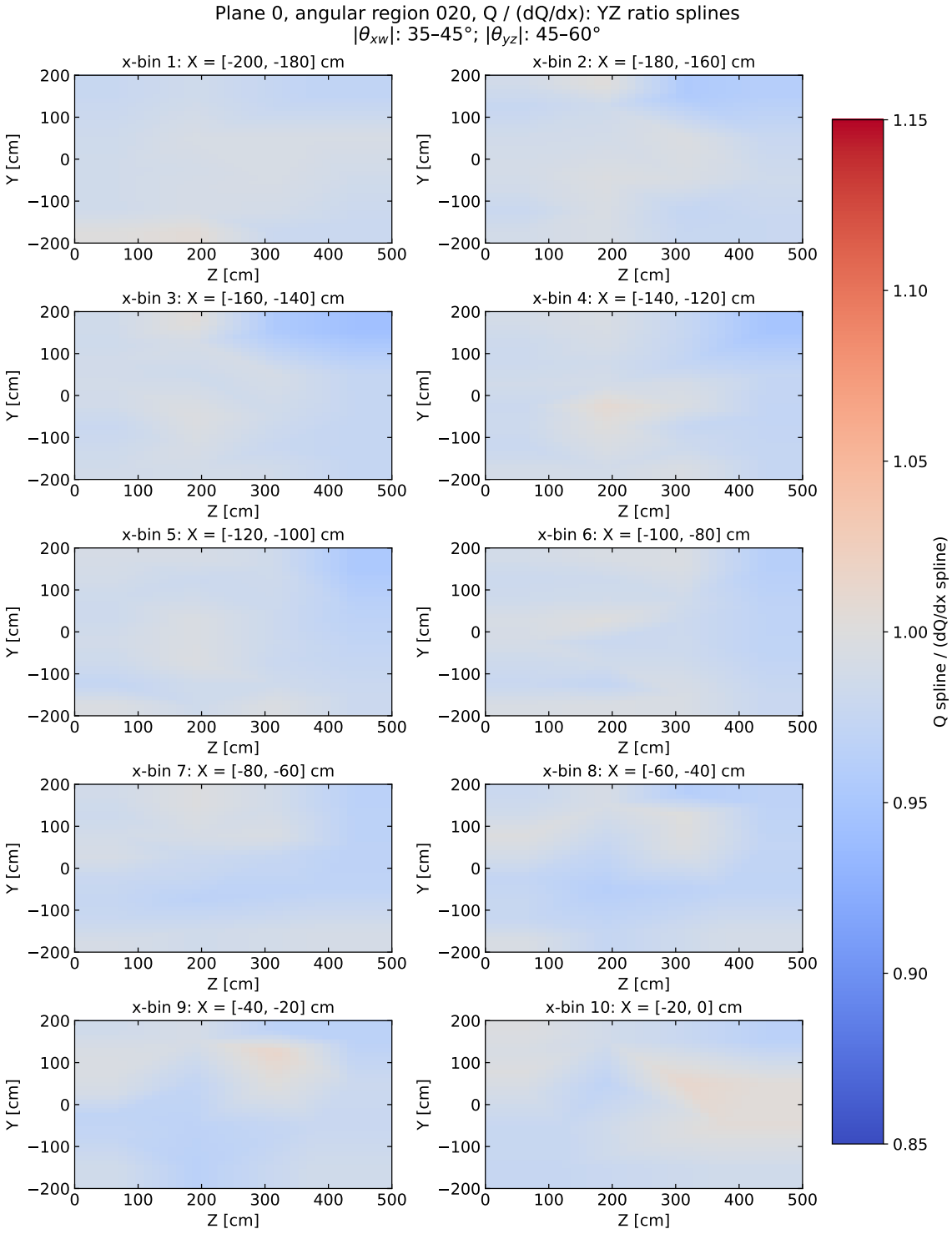


Figure 91: Plane 0, angular region 020: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [35, 45]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

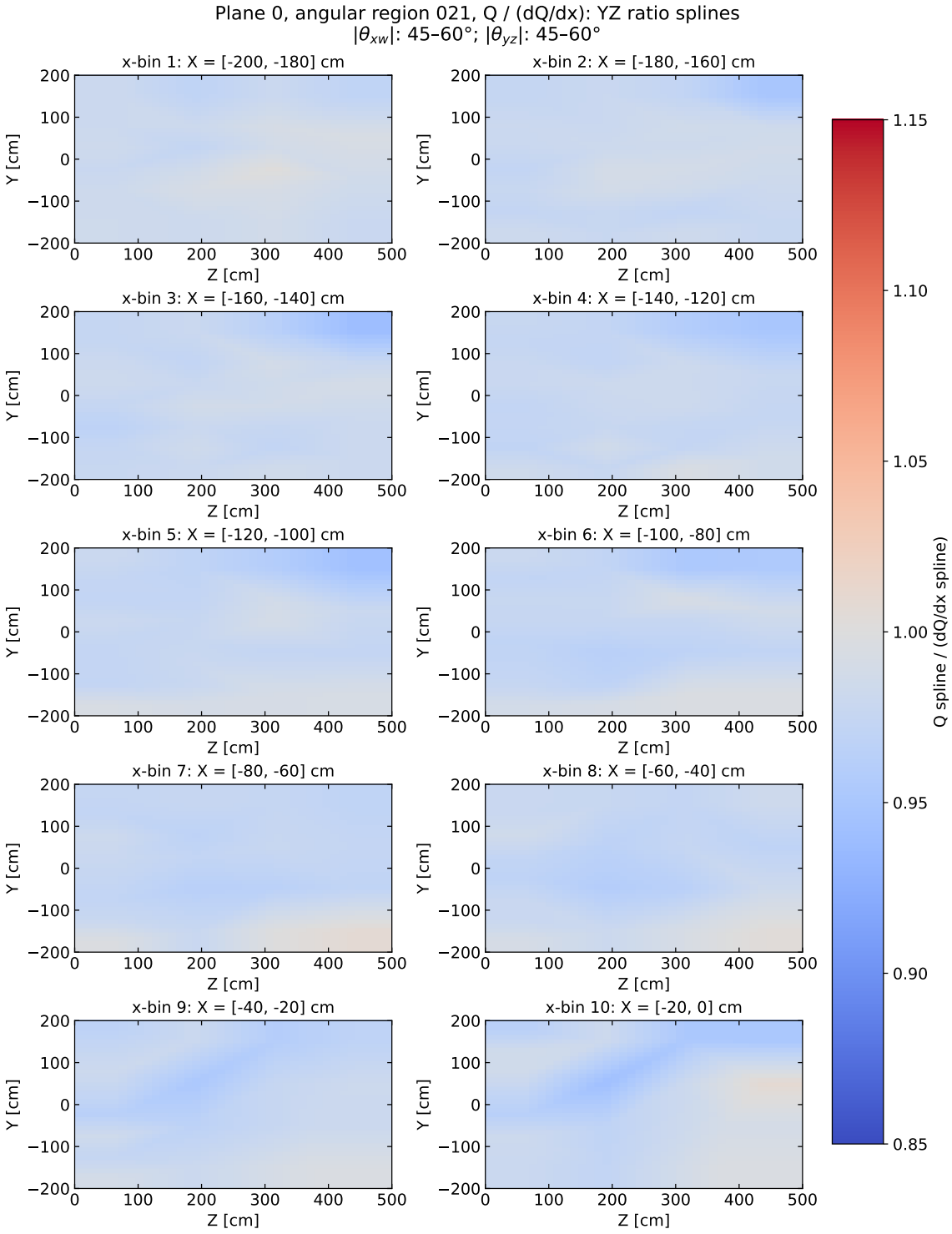


Figure 92: Plane 0, angular region 021: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [45, 60]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

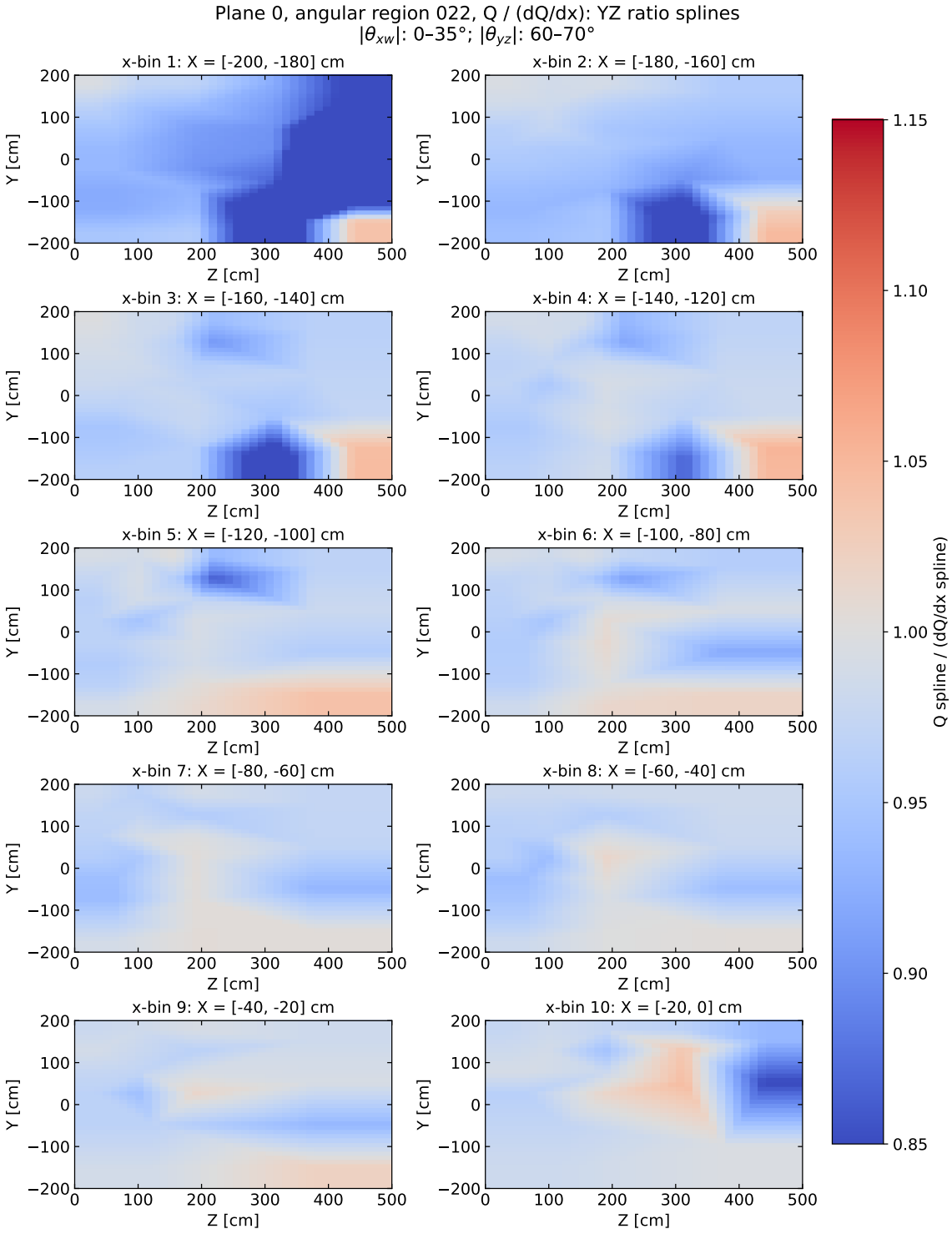


Figure 93: Plane 0, angular region 022: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 35]^\circ$ and $|\theta_{yz}| \in [60, 70]^\circ$.

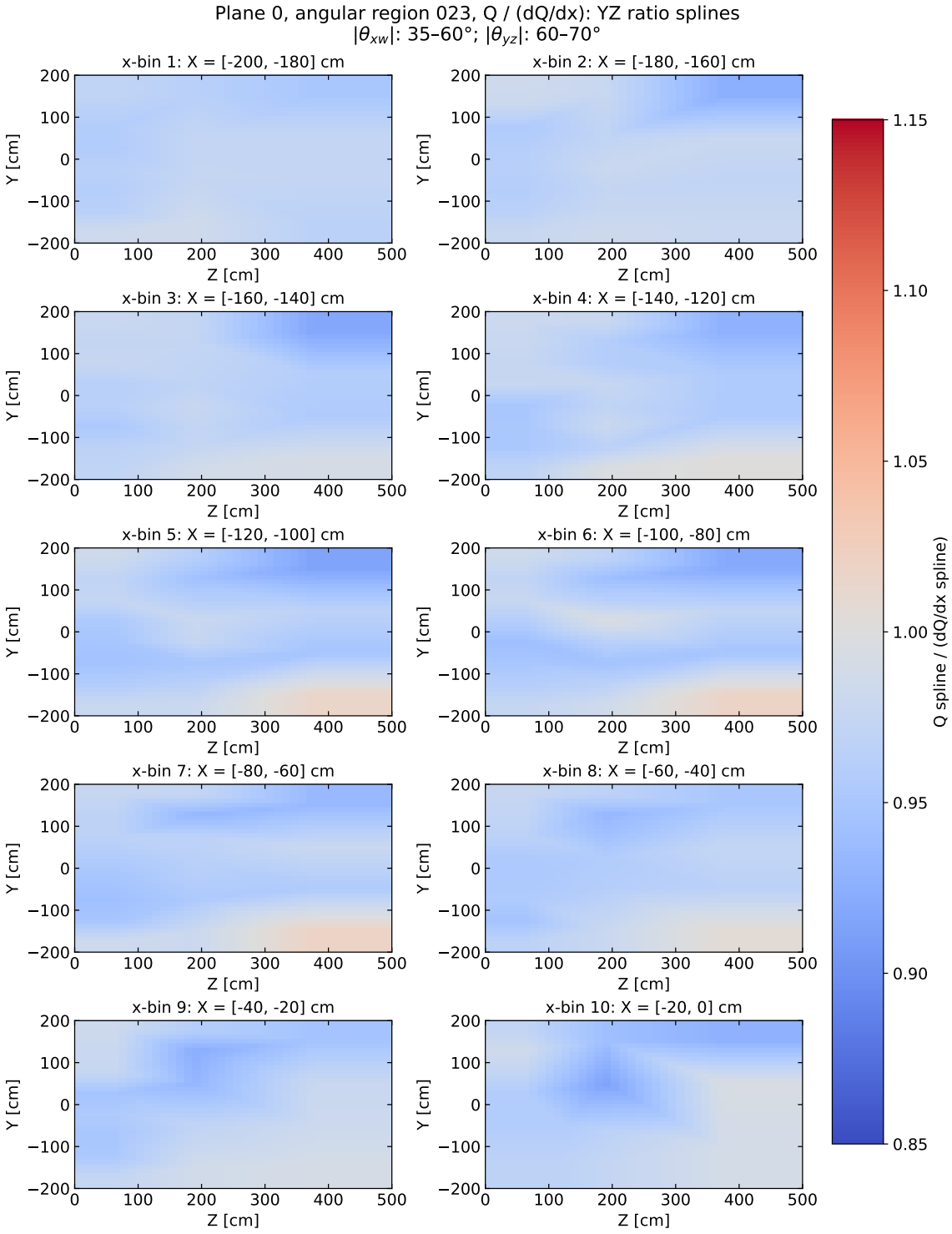


Figure 94: Plane 0, angular region 023: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [35, 60]^\circ$ and $|\theta_{yz}| \in [60, 70]^\circ$.

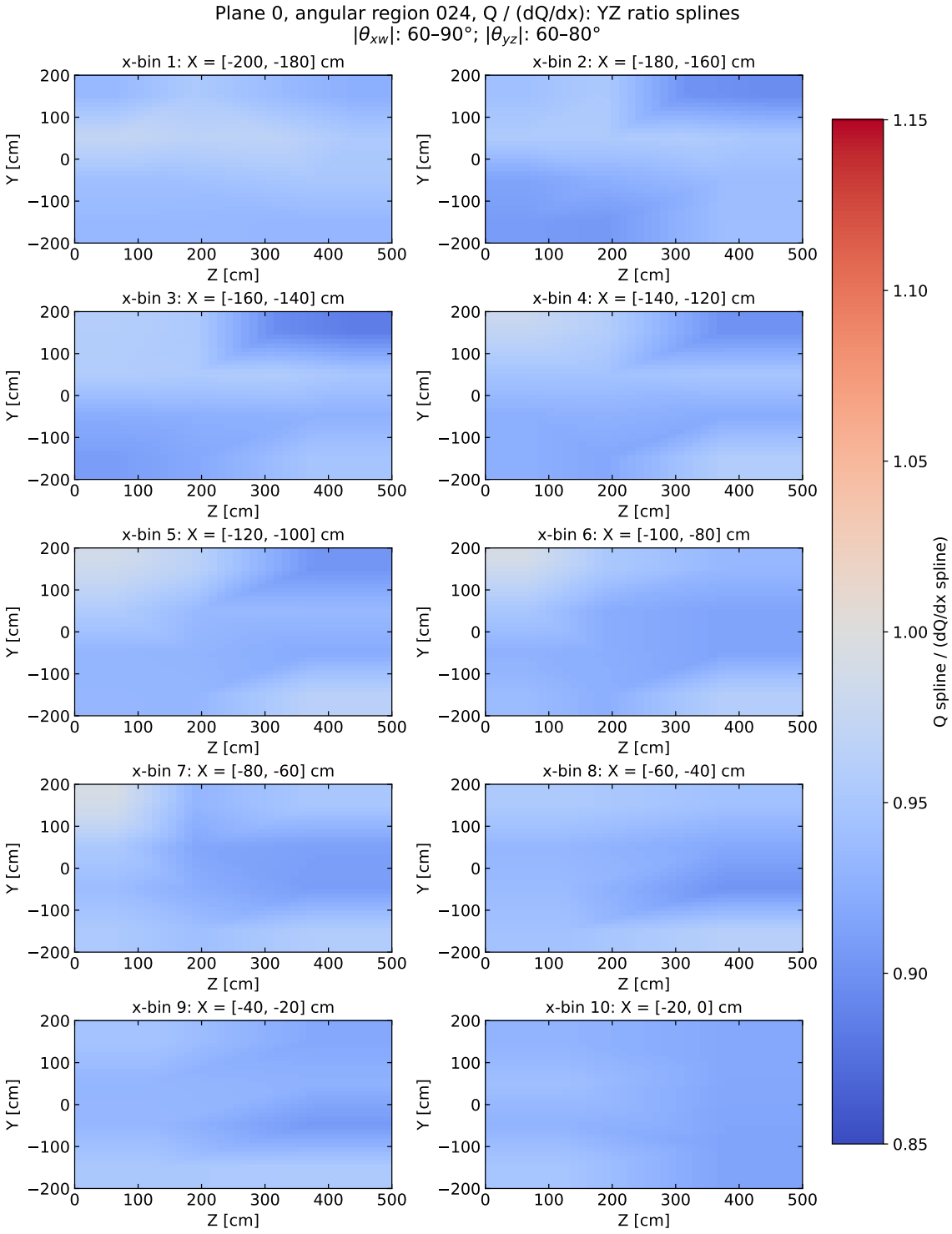


Figure 95: Plane 0, angular region 024: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [60, 80]^\circ$.

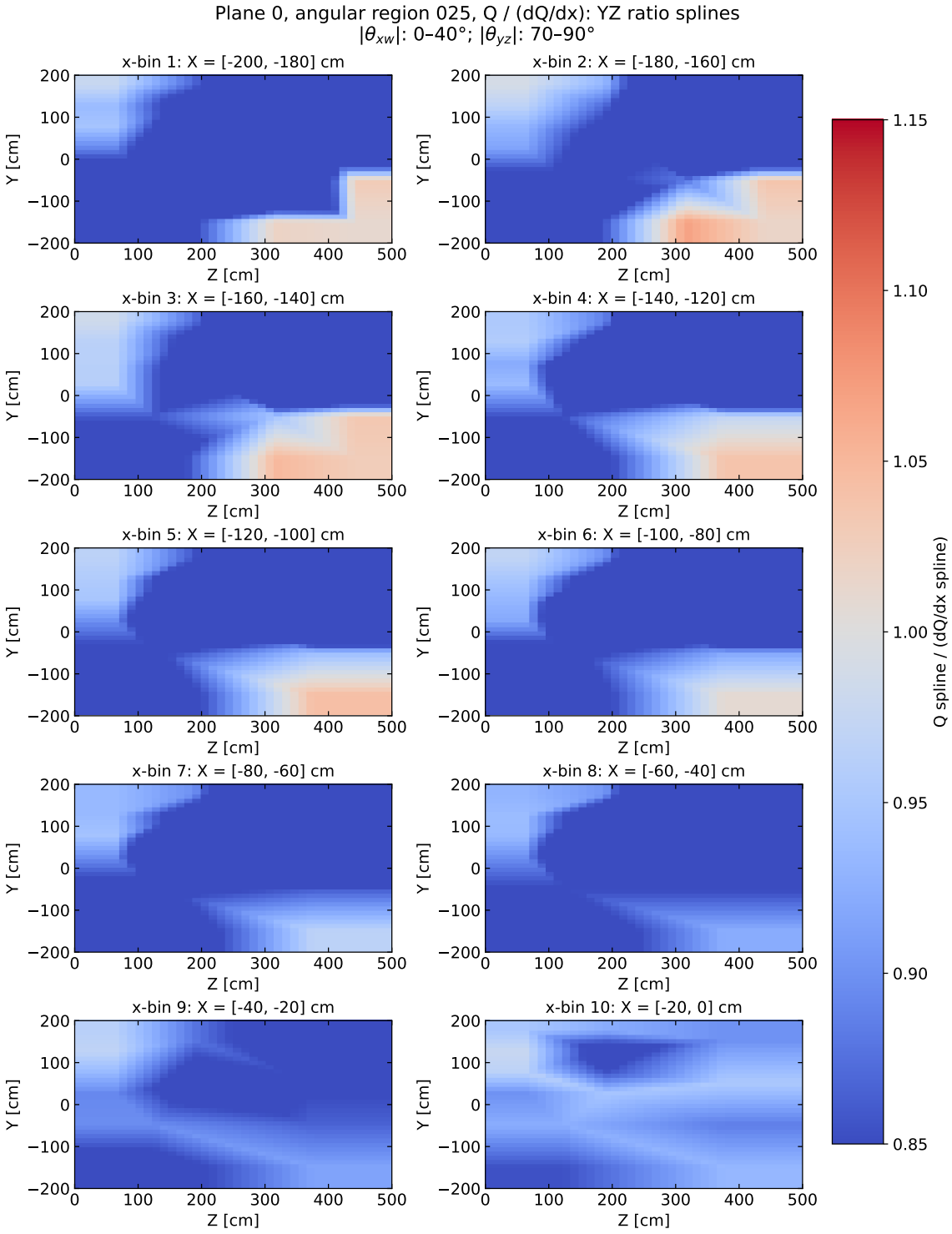


Figure 96: Plane 0, angular region 025: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 40]^\circ$ and $|\theta_{yz}| \in [70, 90]^\circ$.

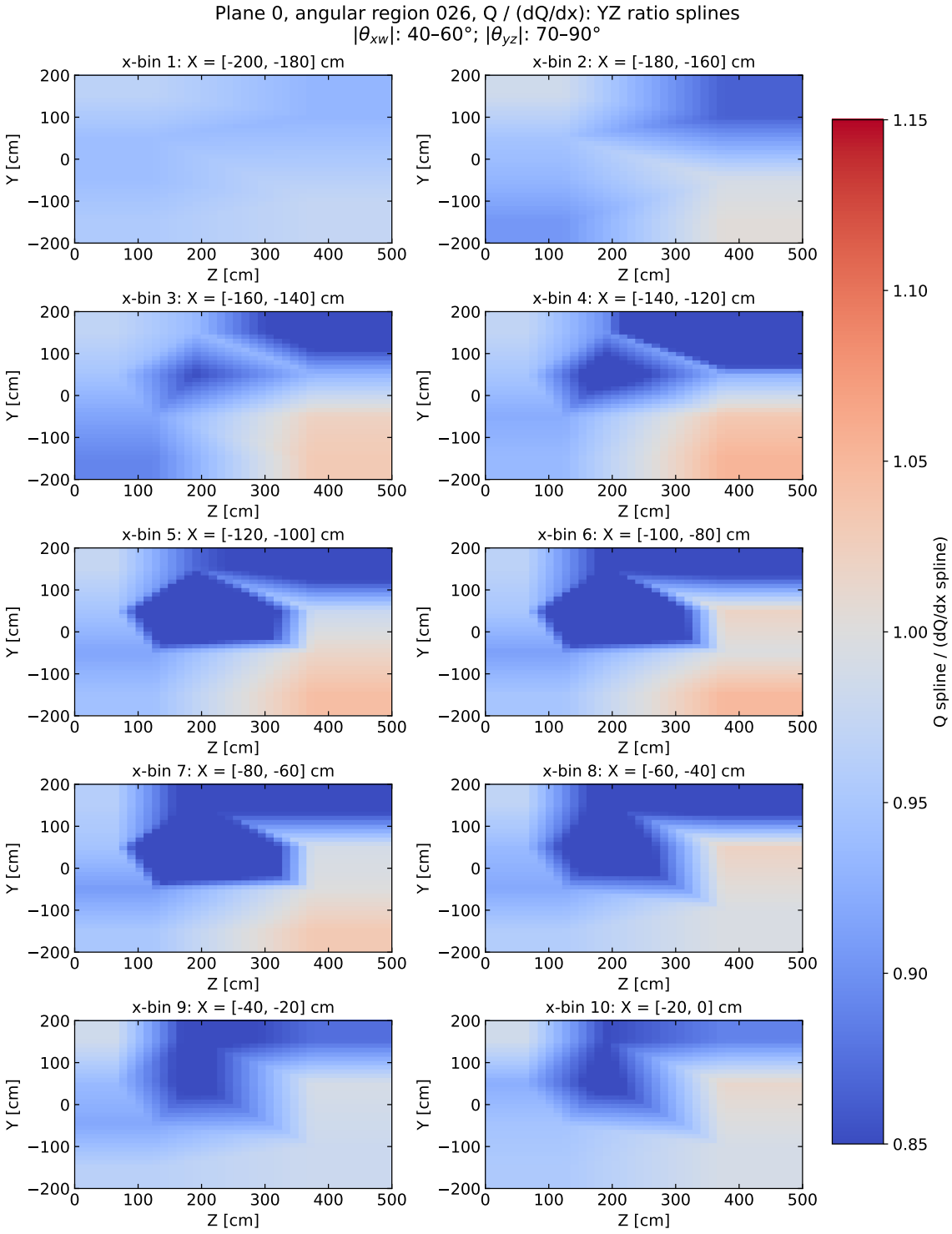


Figure 97: Plane 0, angular region 026: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [70, 90]^\circ$.

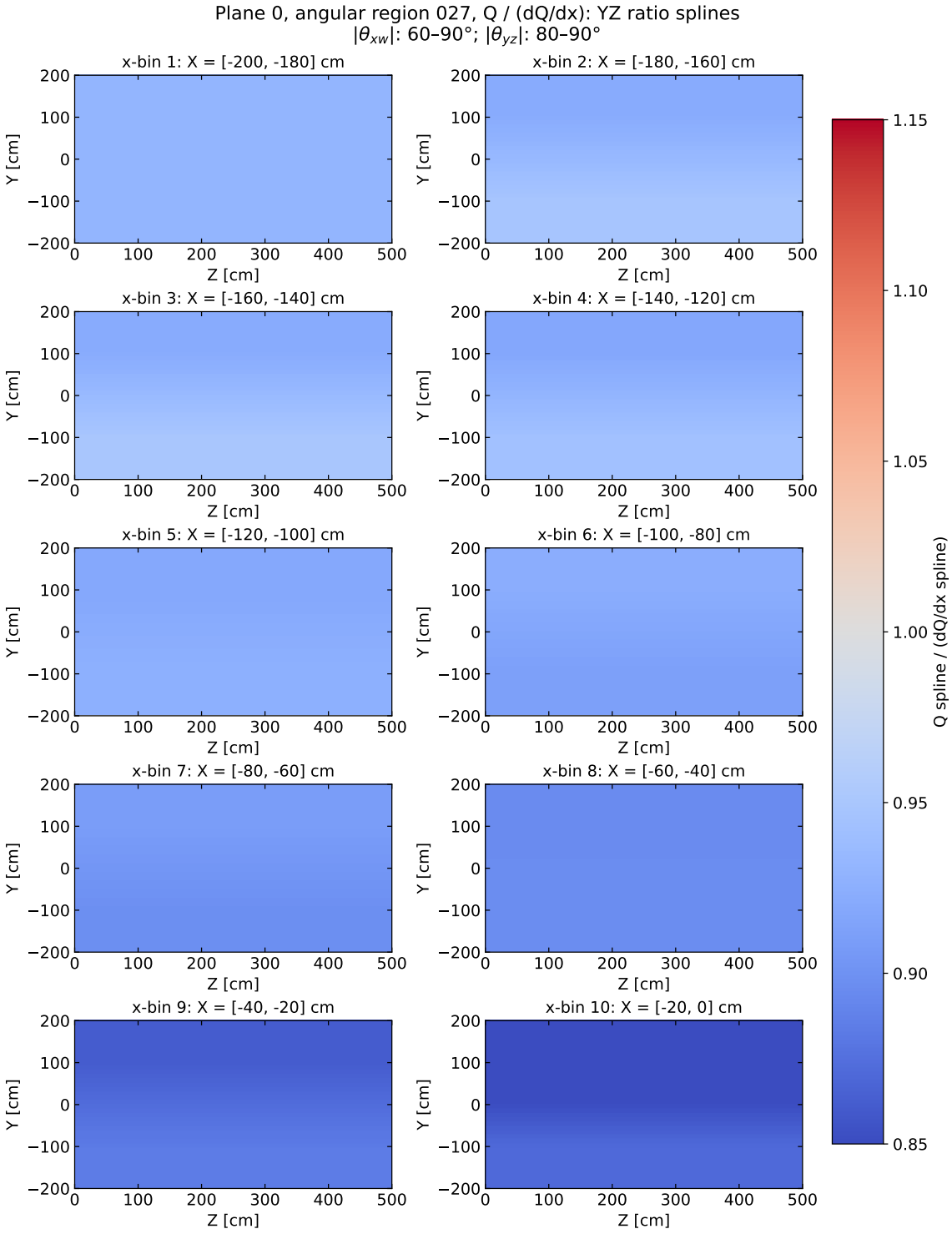


Figure 98: Plane 0, angular region 027: $R_Q/R_{dQ/dx}$ YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [80, 90]^\circ$.

B.1.4 Grouped Angular Summary of Spline Agreement

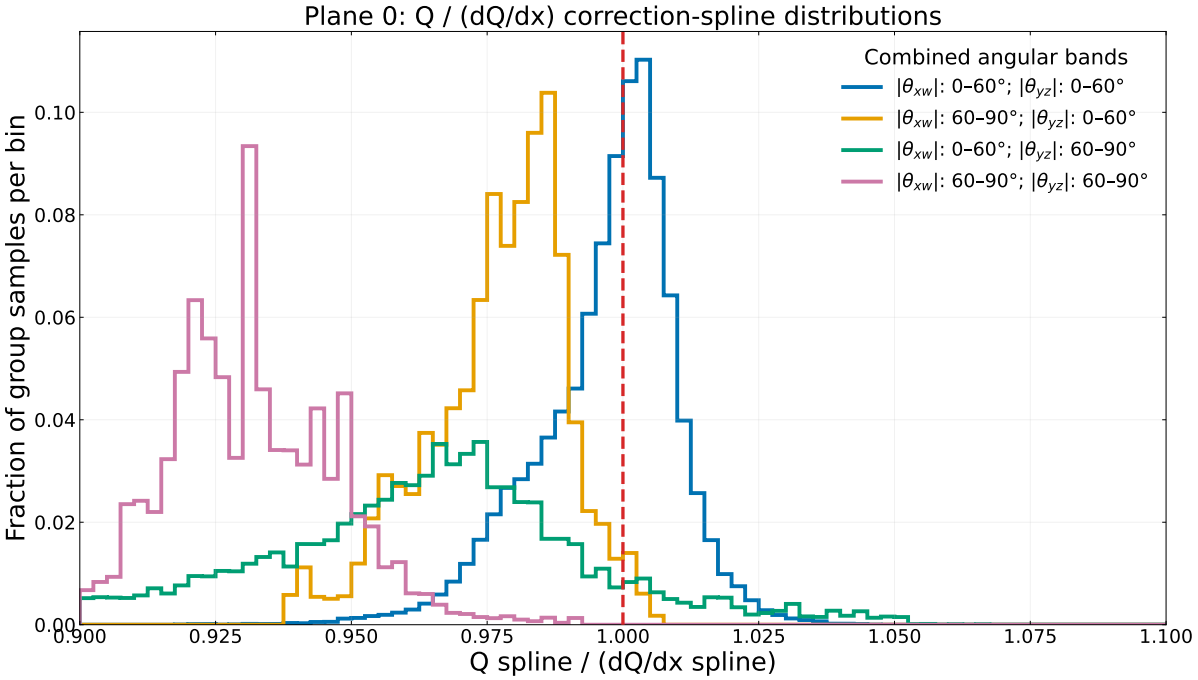


Figure 99: Plane 0 agreement of the charge and dQ/dx correction splines, expressed as $R_Q/R_{dQ/dx}$ and combined into four angular bands separated at 60° in each folded angular coordinate. The red dashed line marks unity. Each distribution pools samples from the same 40×40 interpolated YZ grids used in the preceding comparison figures over all populated x-slices. Each group is normalized by its full sample count before restricting the displayed range to 0.9–1.1; values outside this range remain included in the normalization. Samples have equal spatial-grid weight, rather than track-count weight.

B.2 Plane 2

B.2.1 Charge (Q)

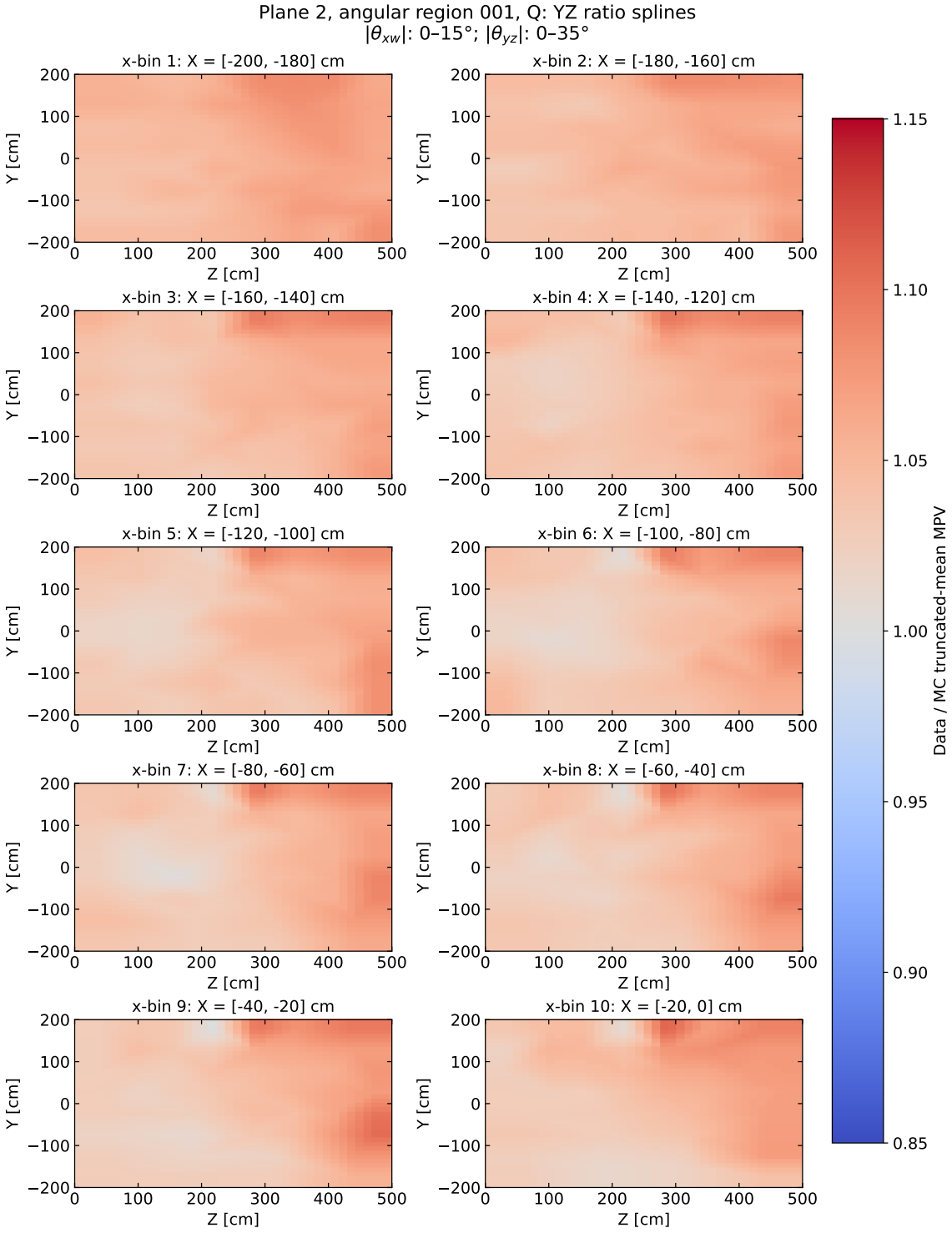


Figure 100: Plane 2, angular region 001: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [0, 35]^\circ$.

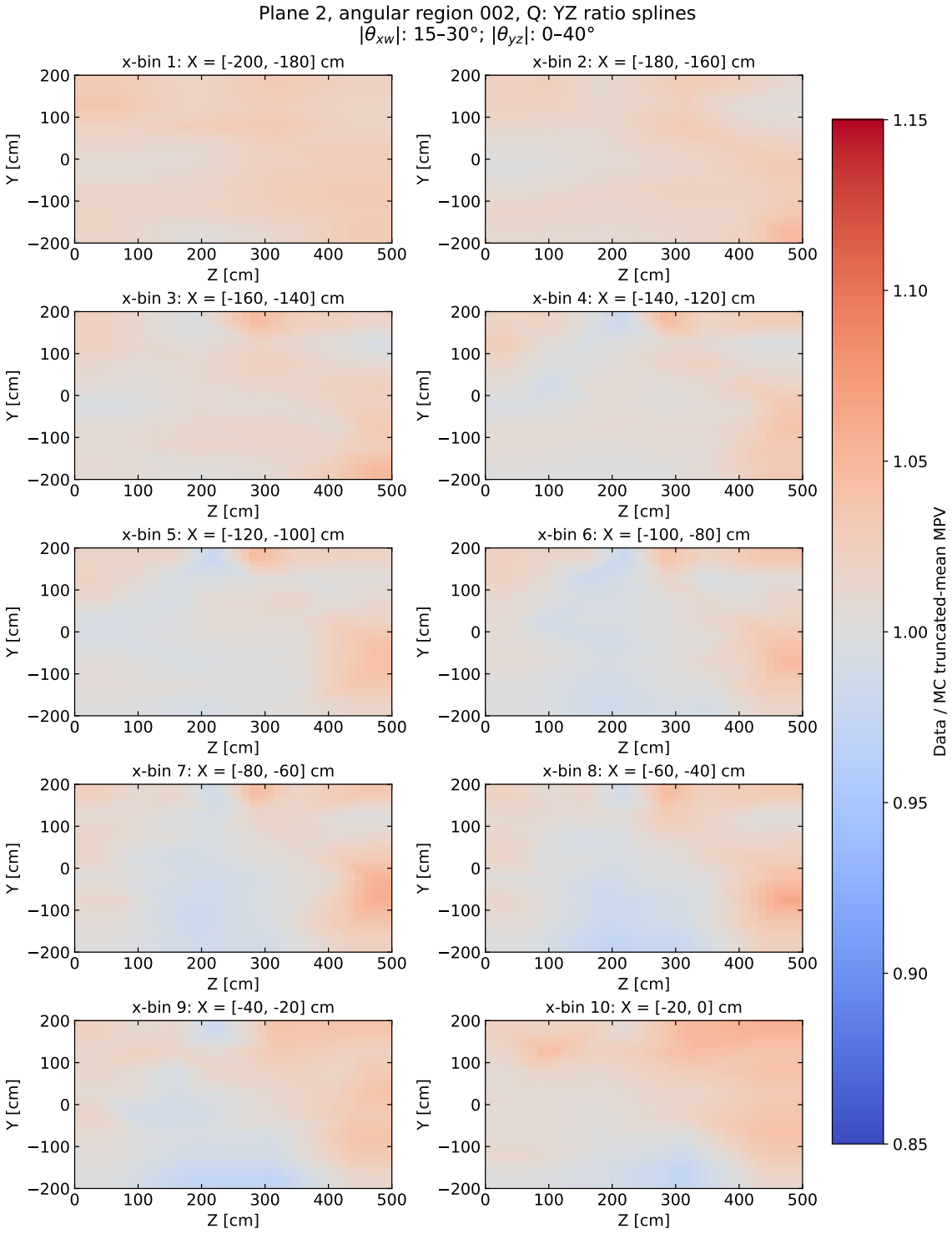


Figure 101: Plane 2, angular region 002: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 30]^\circ$ and $|\theta_{yz}| \in [0, 40]^\circ$.

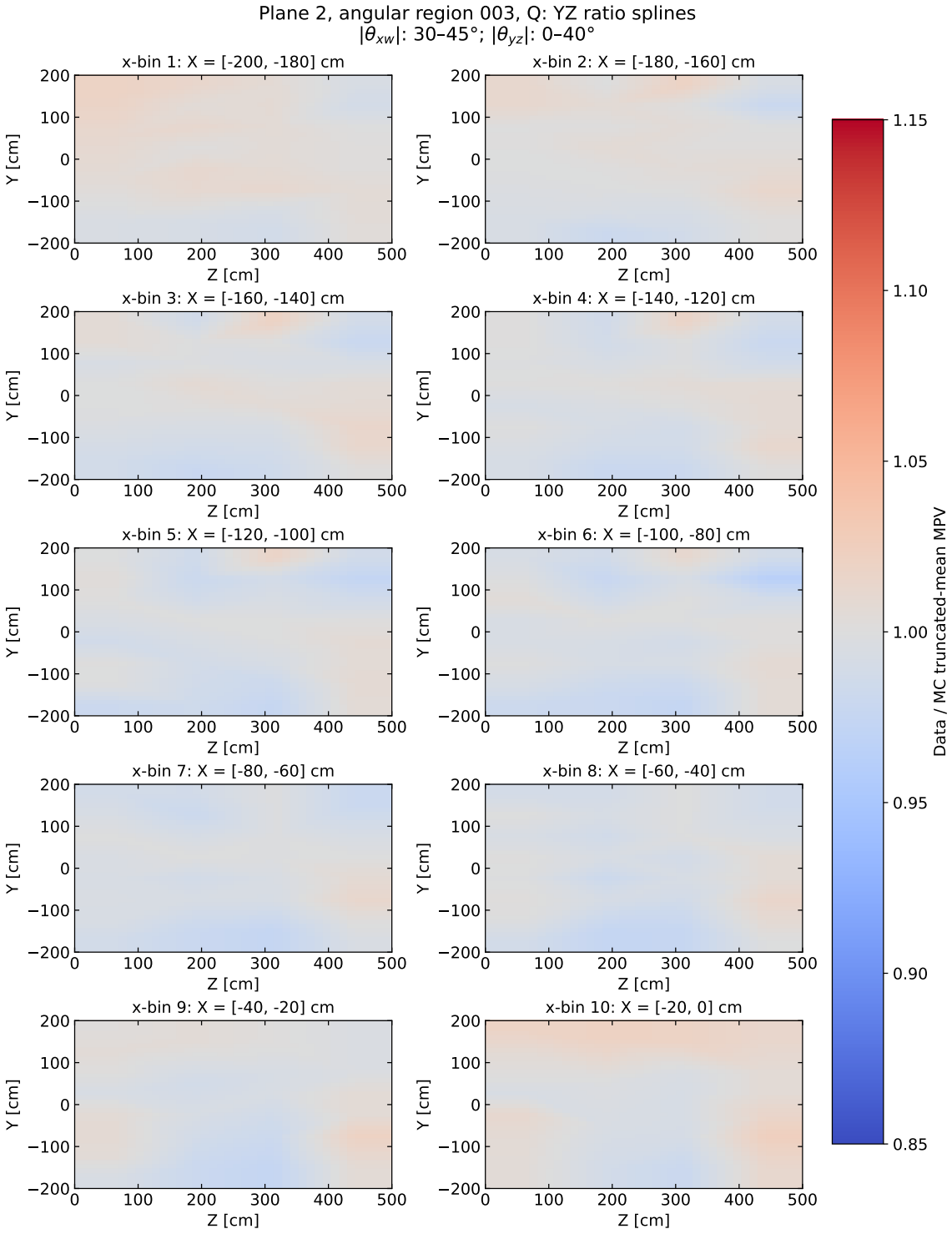


Figure 102: Plane 2, angular region 003: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [30, 45]^\circ$ and $|\theta_{yz}| \in [0, 40]^\circ$.

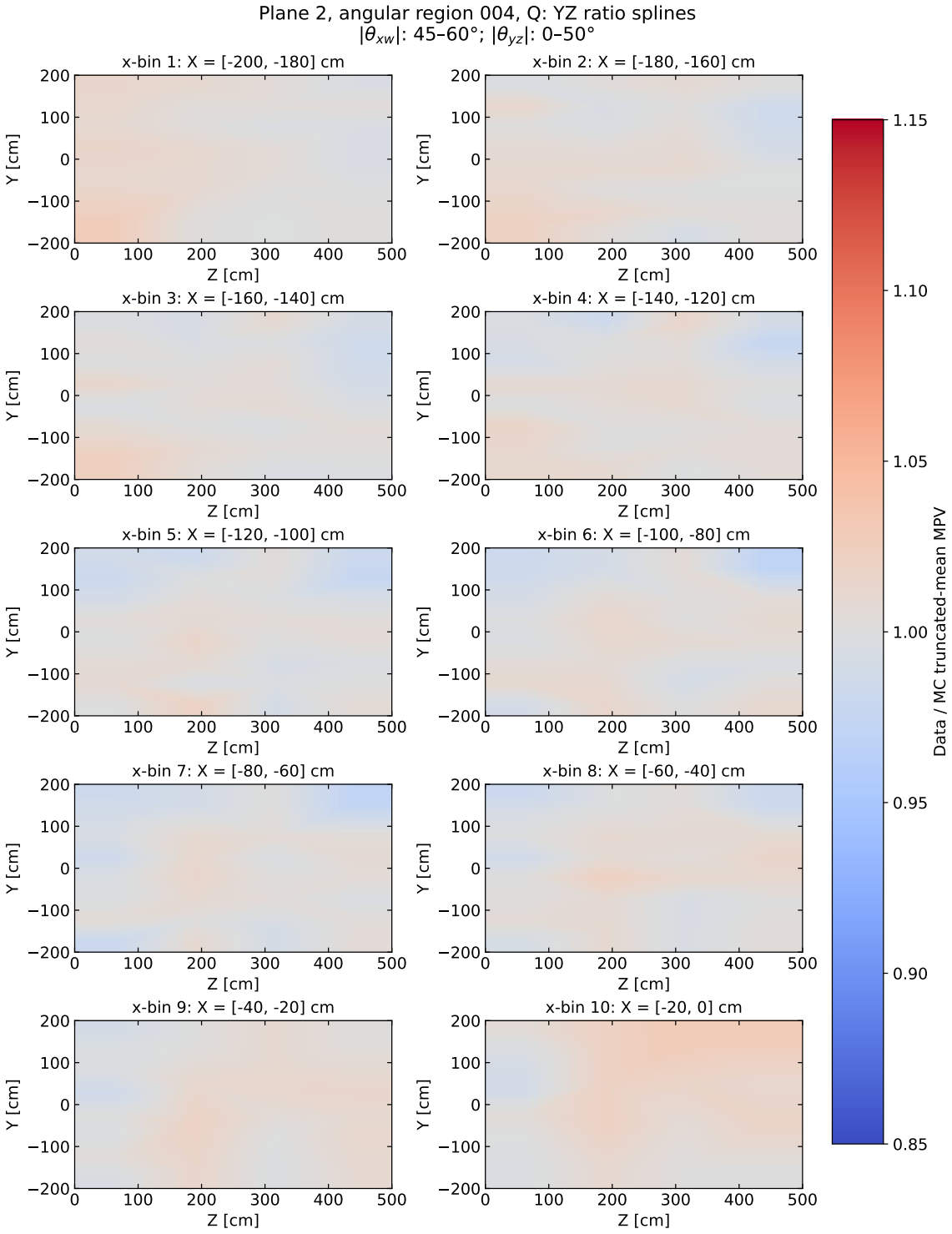


Figure 103: Plane 2, angular region 004: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [45, 60]^\circ$ and $|\theta_{yz}| \in [0, 50]^\circ$.

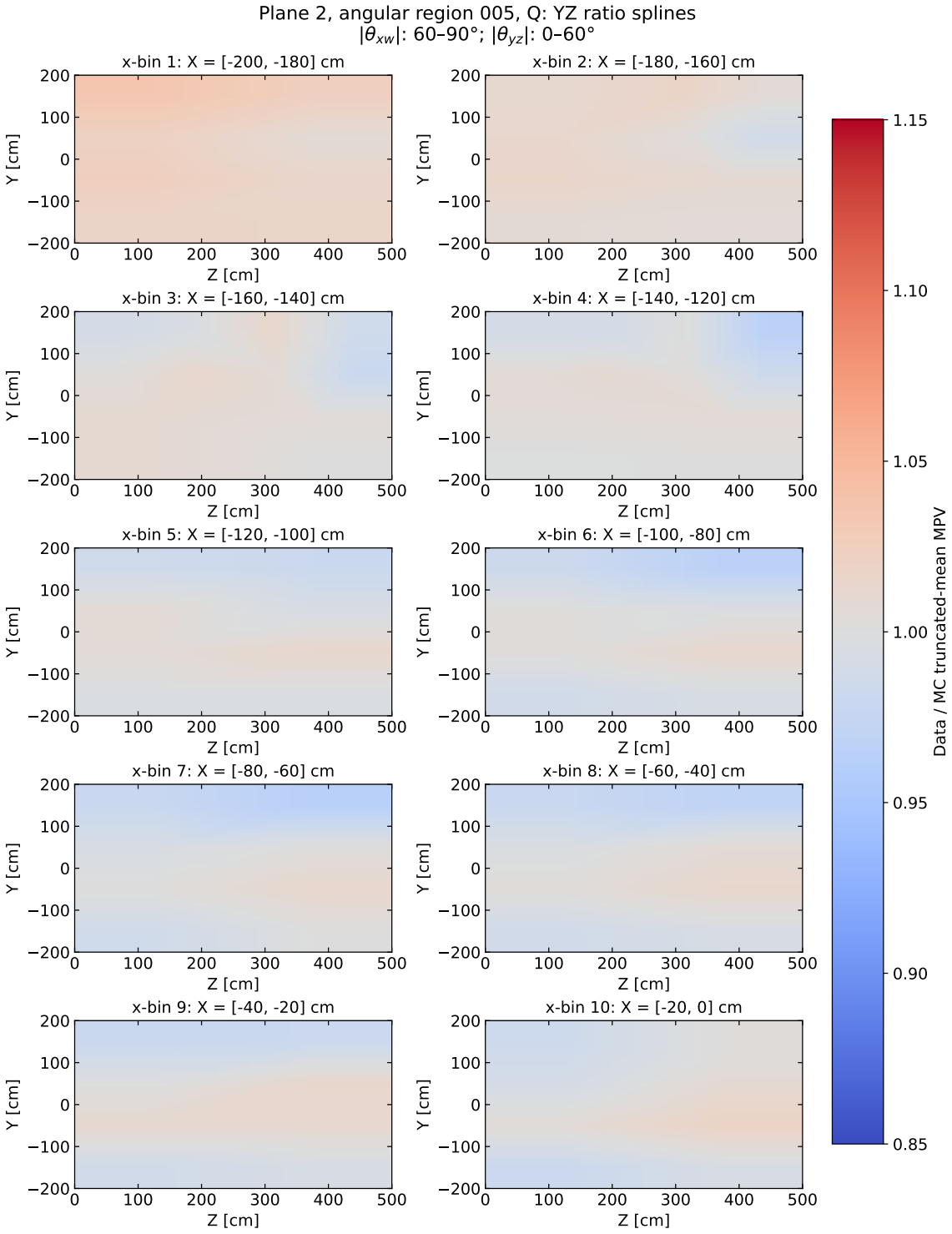


Figure 104: Plane 2, angular region 005: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 90]^\circ$ and $|\theta_{yz}| \in [0, 60]^\circ$.

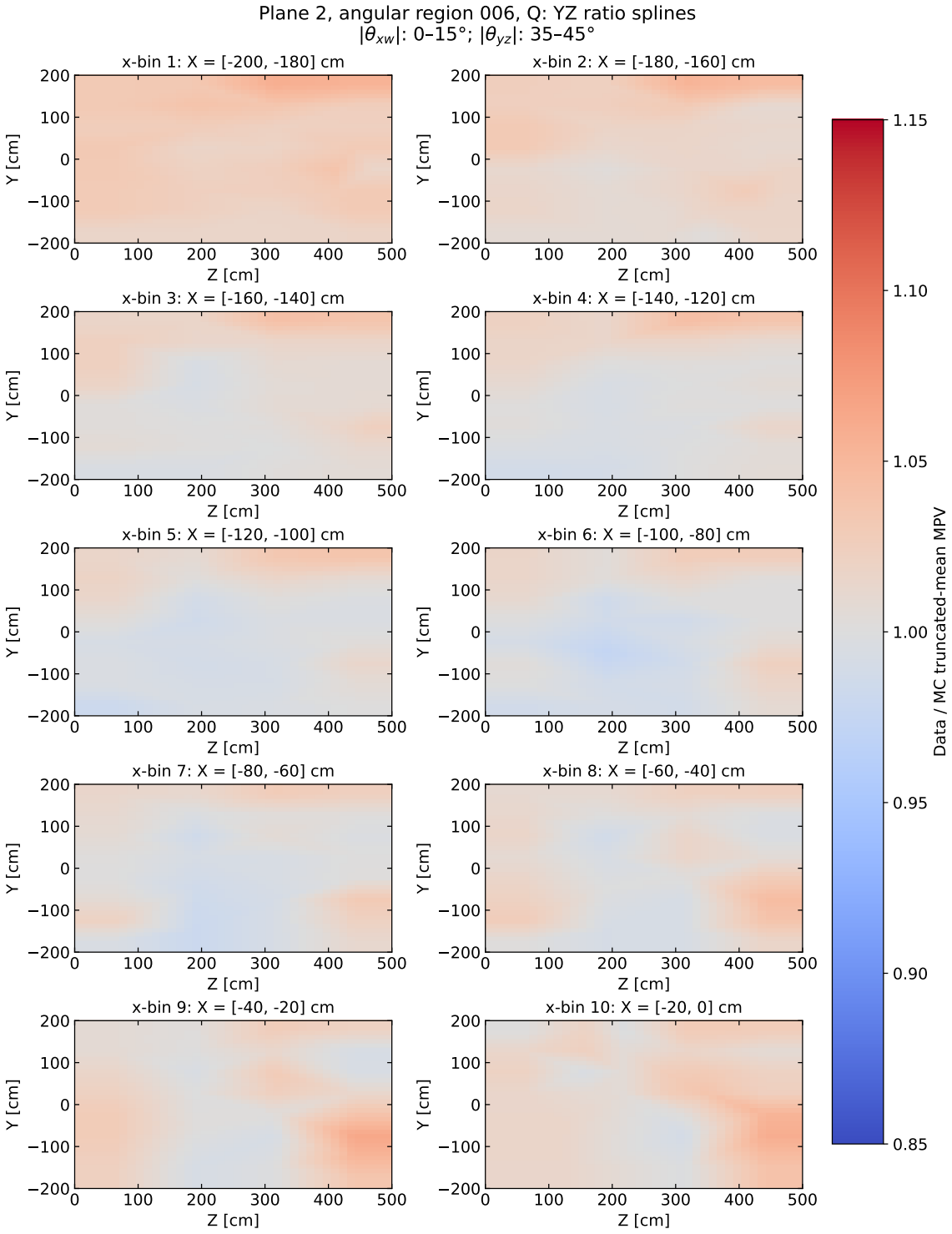


Figure 105: Plane 2, angular region 006: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [35, 45]^\circ$.

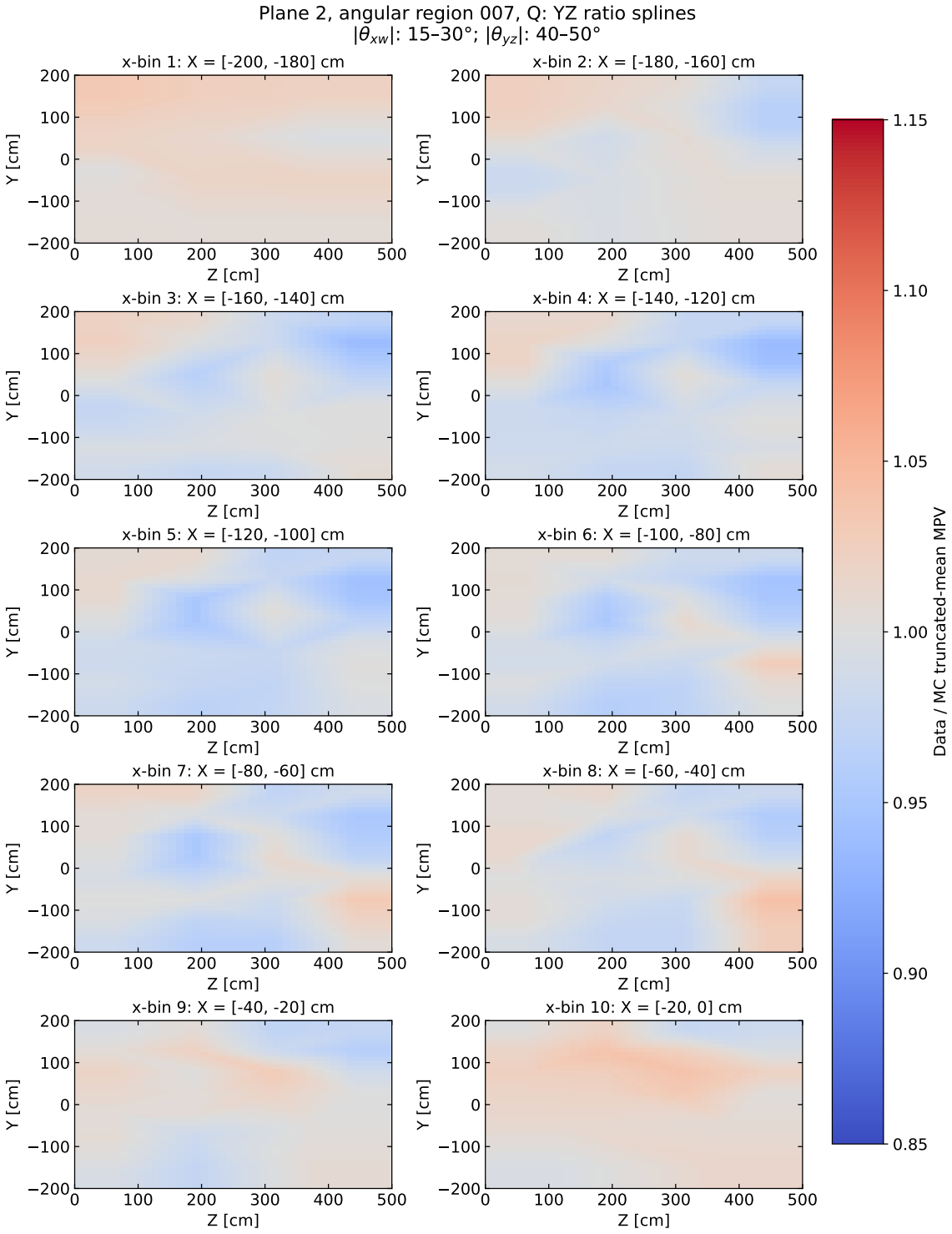


Figure 106: Plane 2, angular region 007: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 30]^\circ$ and $|\theta_{yz}| \in [40, 50]^\circ$.

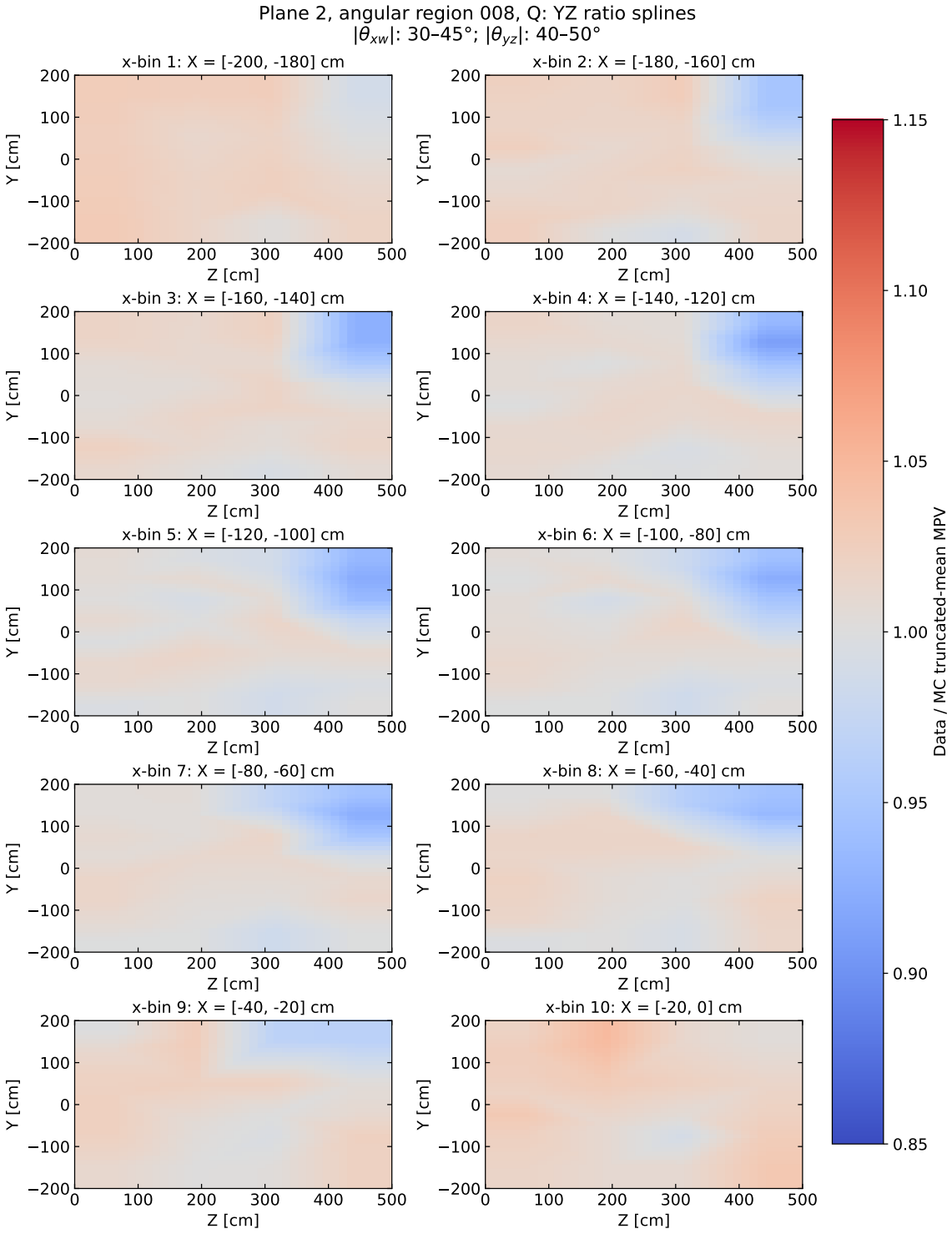


Figure 107: Plane 2, angular region 008: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [30, 45]^\circ$ and $|\theta_{yz}| \in [40, 50]^\circ$.

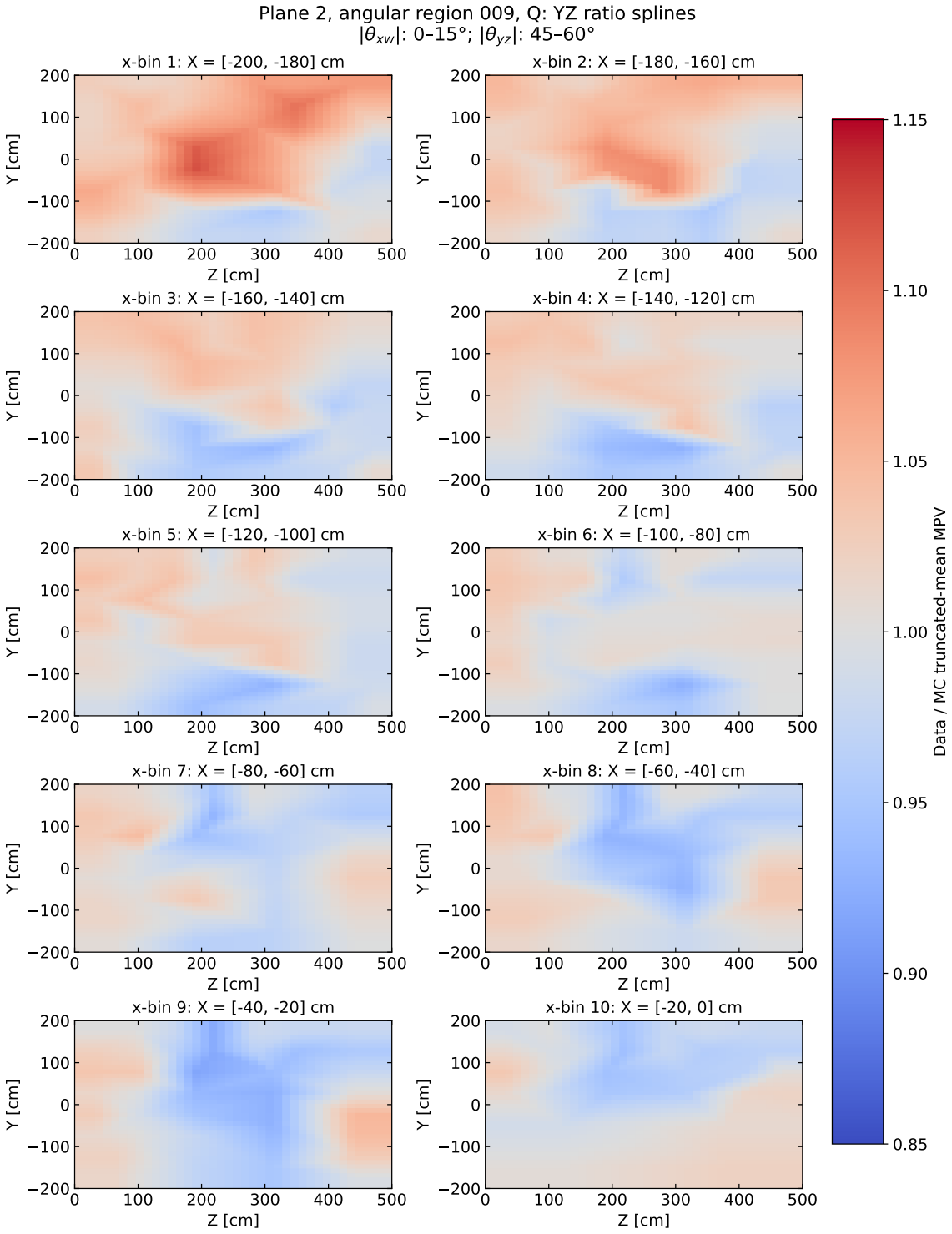


Figure 108: Plane 2, angular region 009: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 15]^\circ$ and $|\theta_{yz}| \in [45, 60]^\circ$.

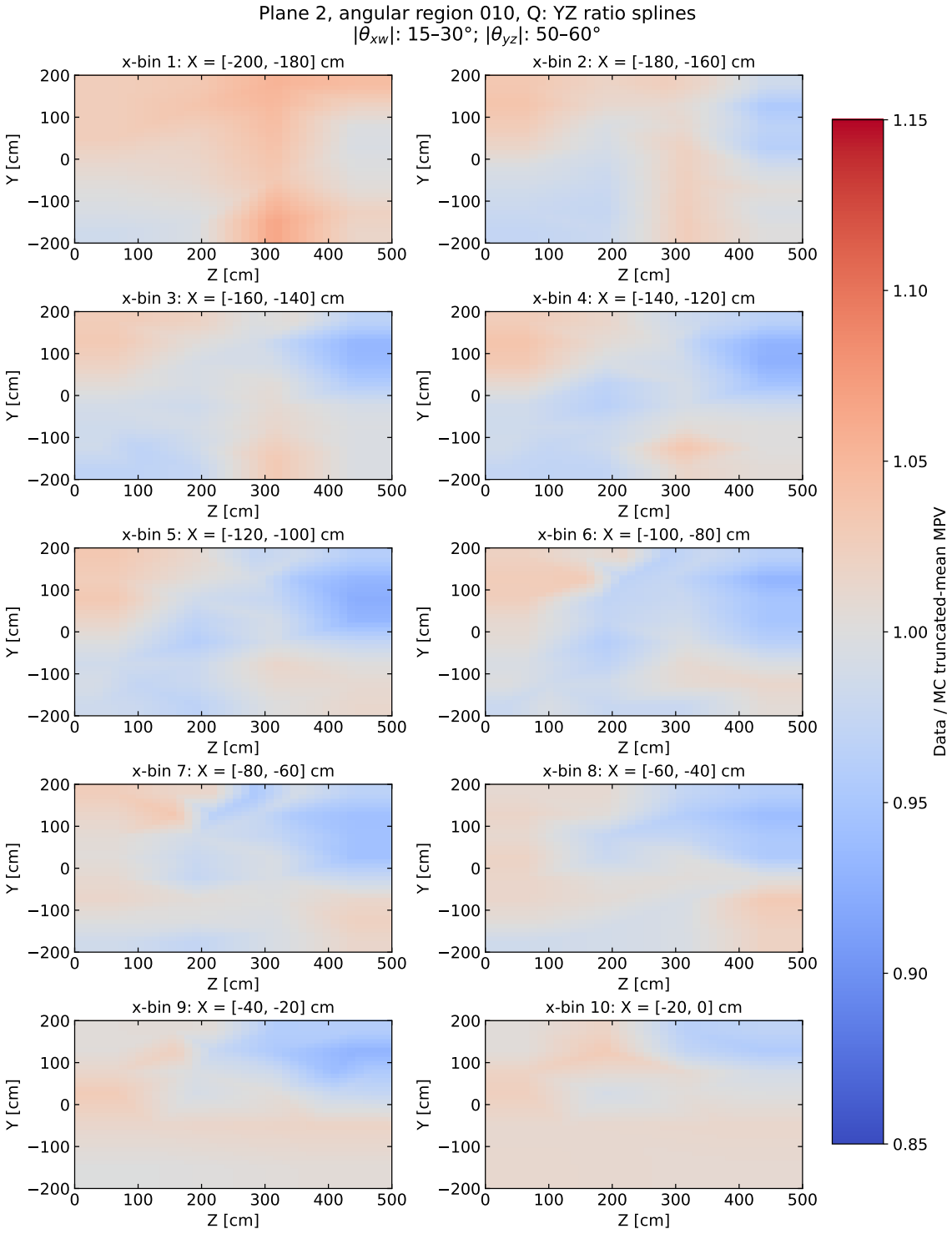


Figure 109: Plane 2, angular region 010: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [15, 30]^\circ$ and $|\theta_{yz}| \in [50, 60]^\circ$.

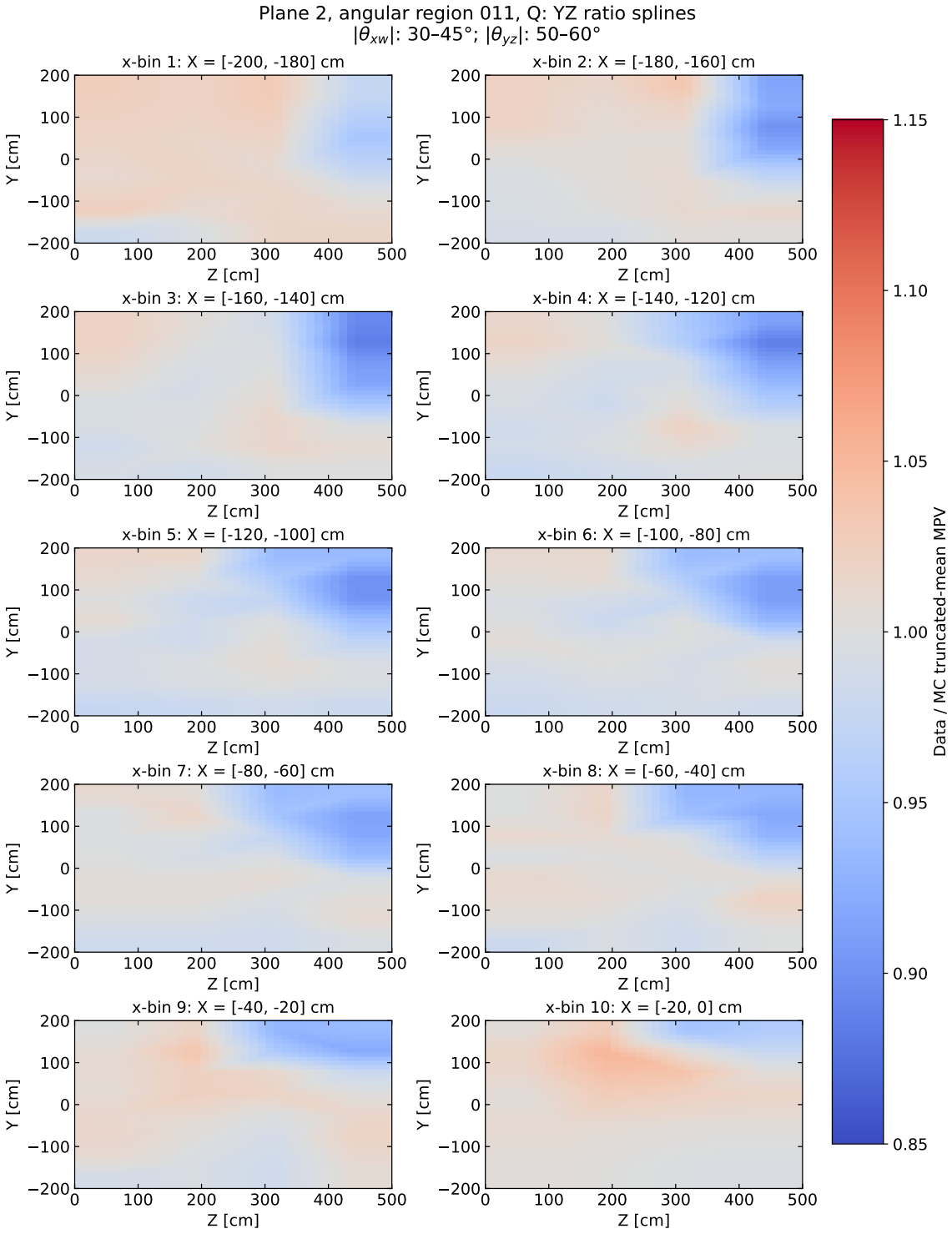


Figure 110: Plane 2, angular region 011: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [30, 45]^\circ$ and $|\theta_{yz}| \in [50, 60]^\circ$.

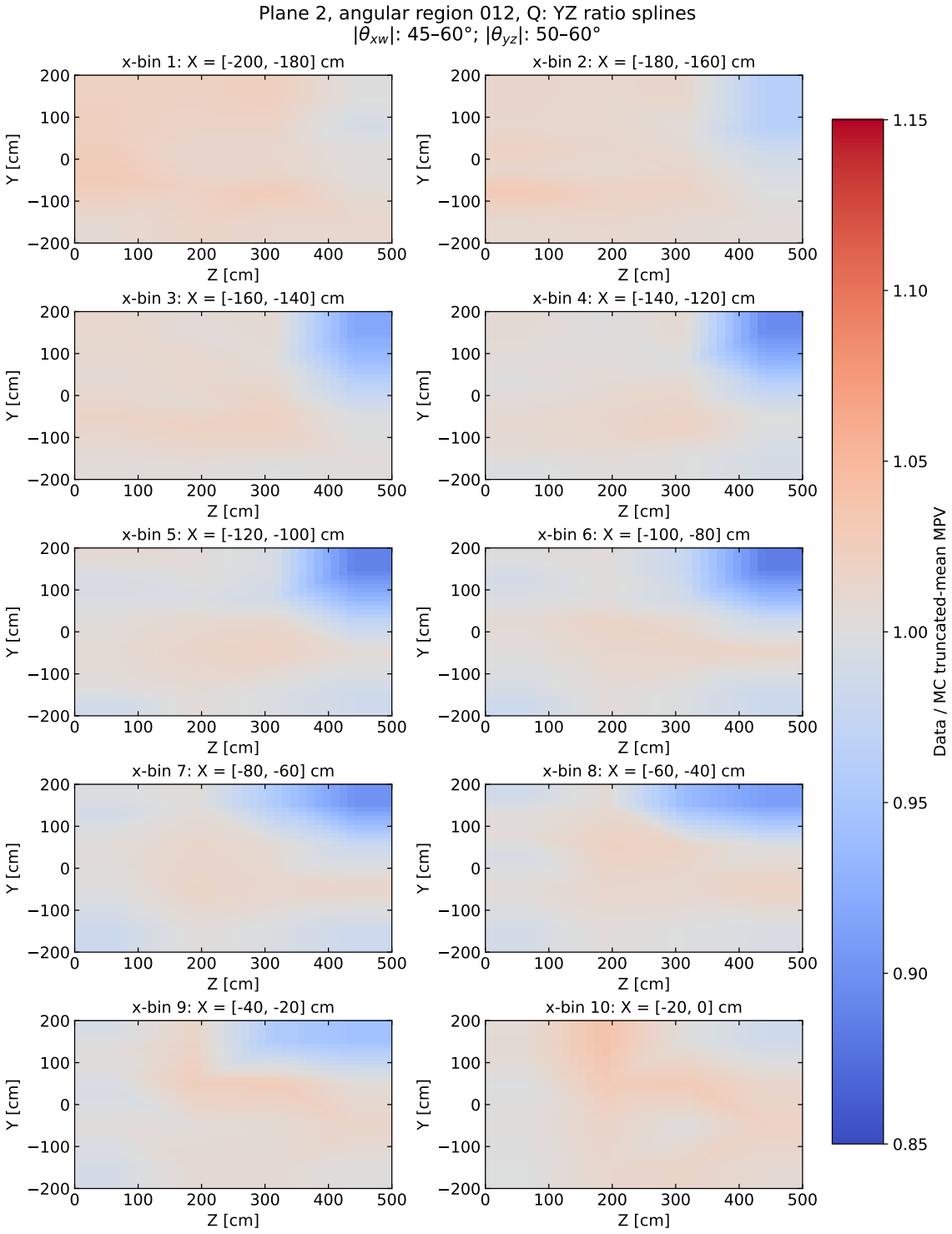


Figure 111: Plane 2, angular region 012: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [45, 60]^\circ$ and $|\theta_{yz}| \in [50, 60]^\circ$.

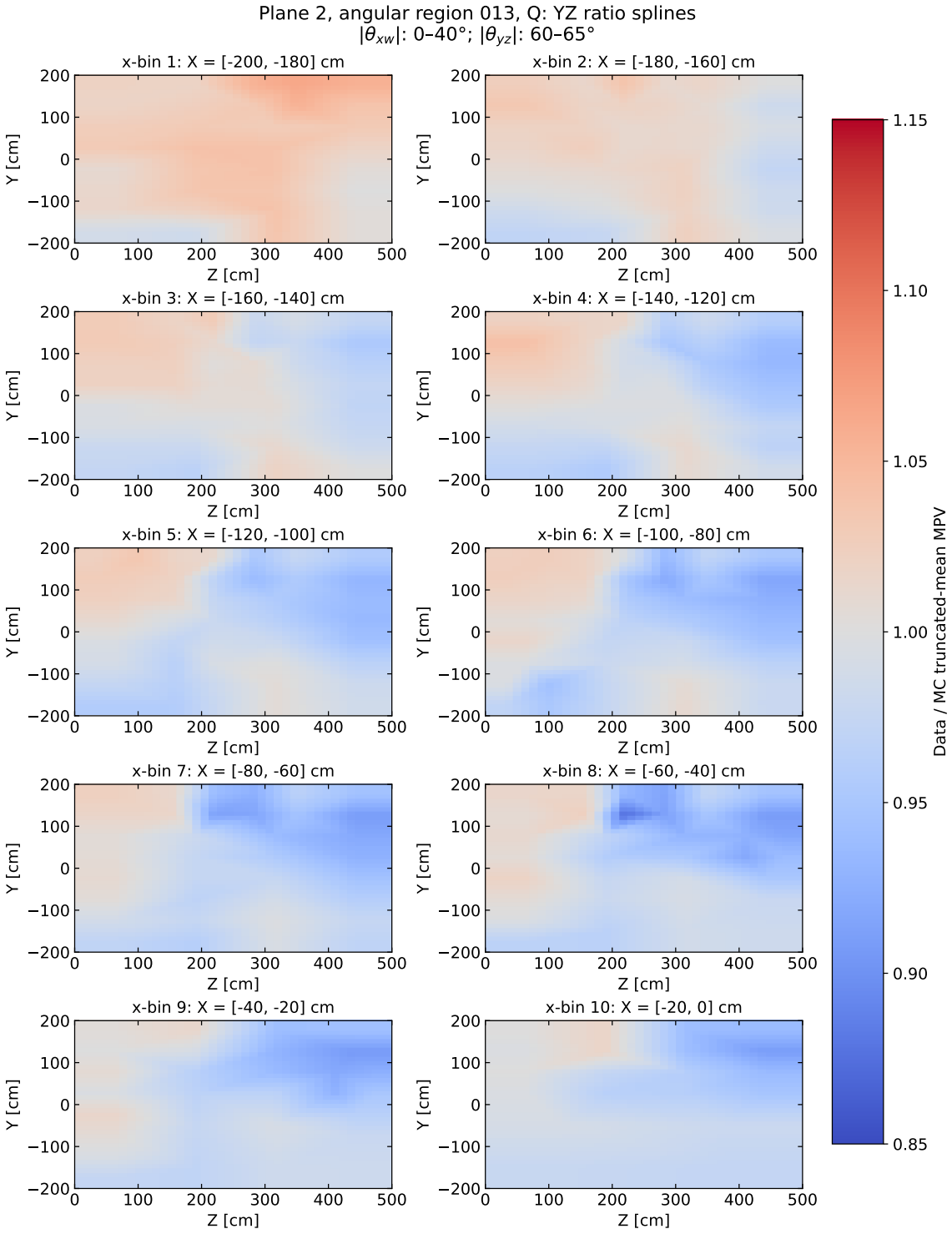


Figure 112: Plane 2, angular region 013: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 40]^\circ$ and $|\theta_{yz}| \in [60, 65]^\circ$.

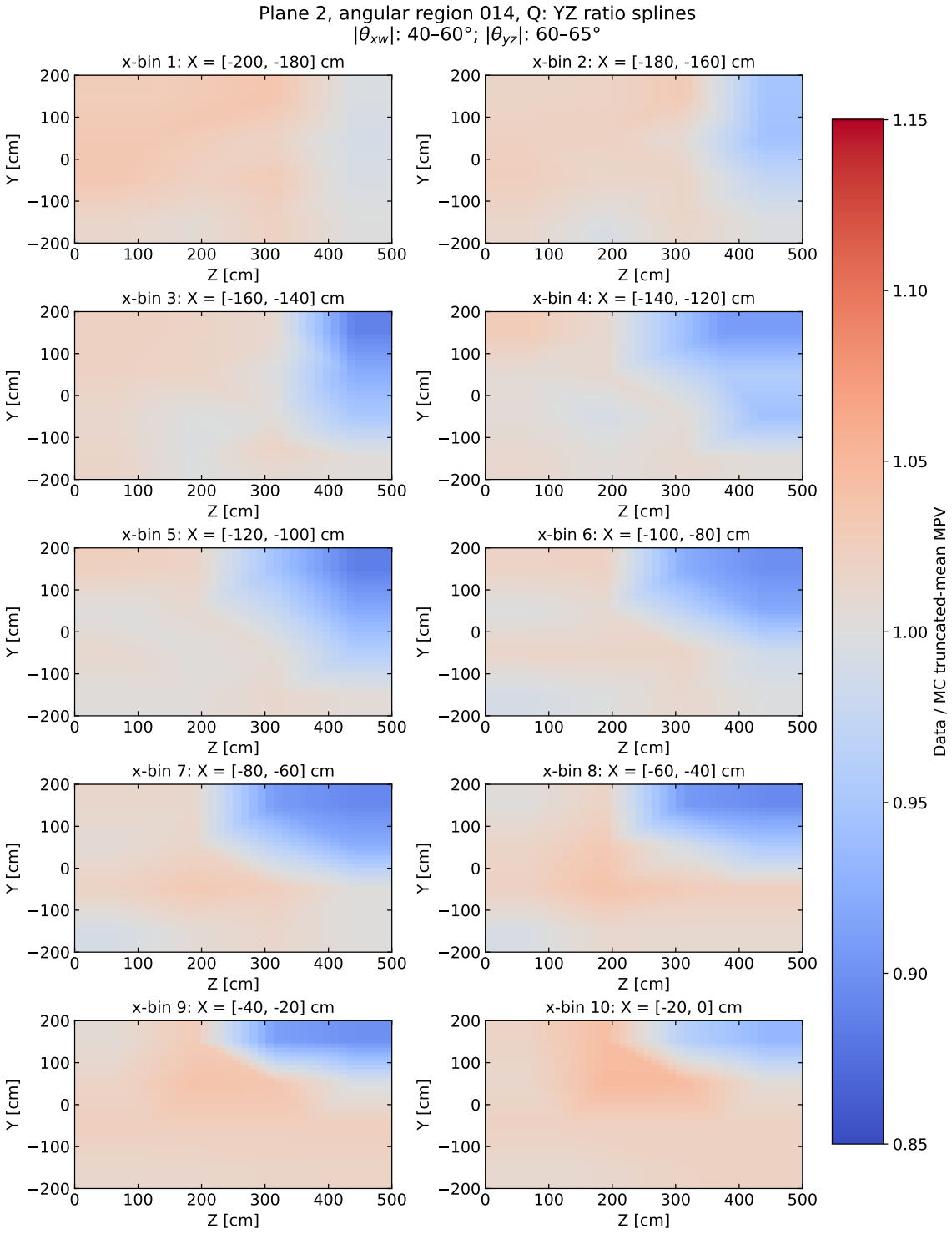


Figure 113: Plane 2, angular region 014: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [60, 65]^\circ$.

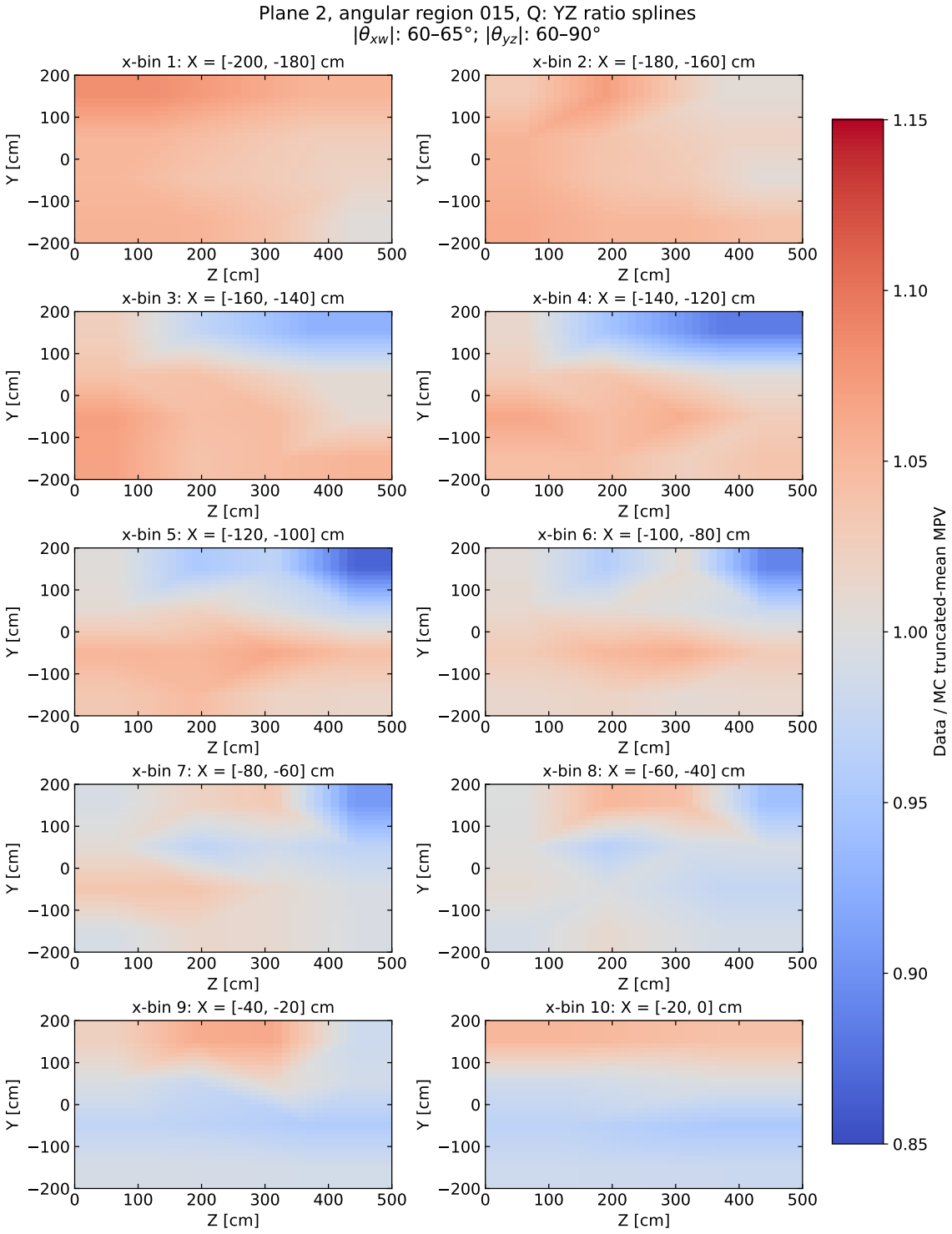


Figure 114: Plane 2, angular region 015: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [60, 65]^\circ$ and $|\theta_{yz}| \in [60, 90]^\circ$.

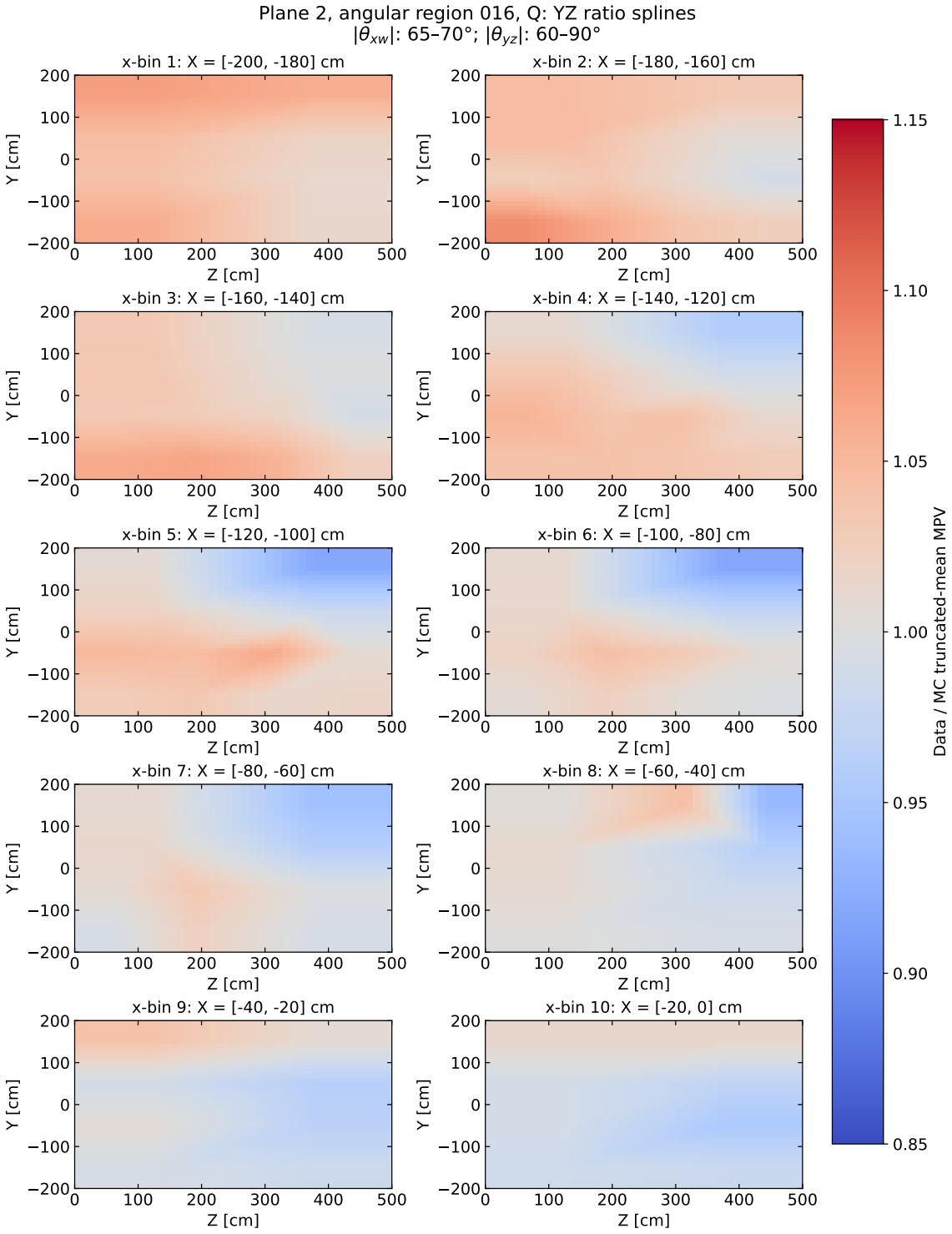


Figure 115: Plane 2, angular region 016: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [65, 70]^\circ$ and $|\theta_{yz}| \in [60, 90]^\circ$.

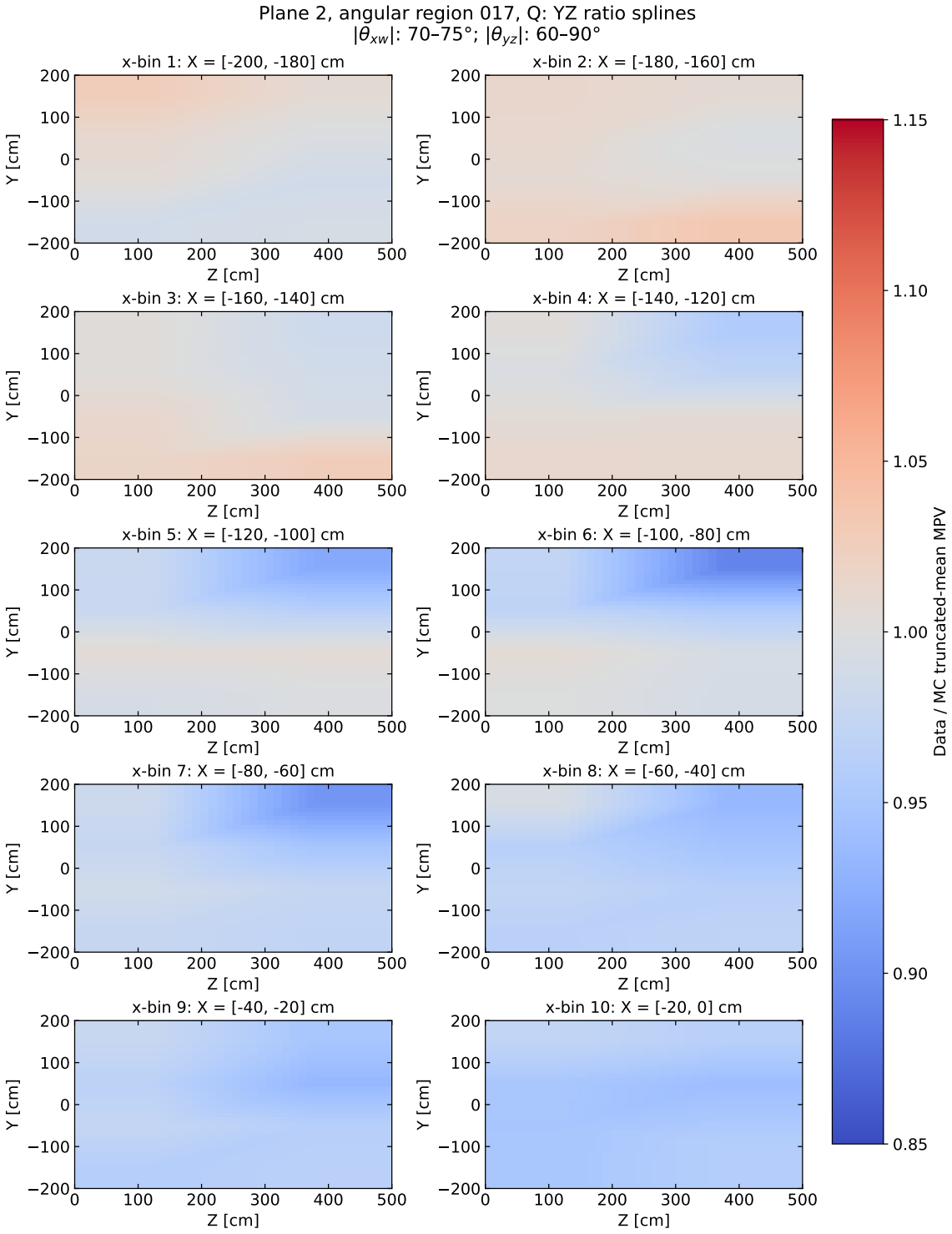


Figure 116: Plane 2, angular region 017: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [70, 75]^\circ$ and $|\theta_{yz}| \in [60, 90]^\circ$.

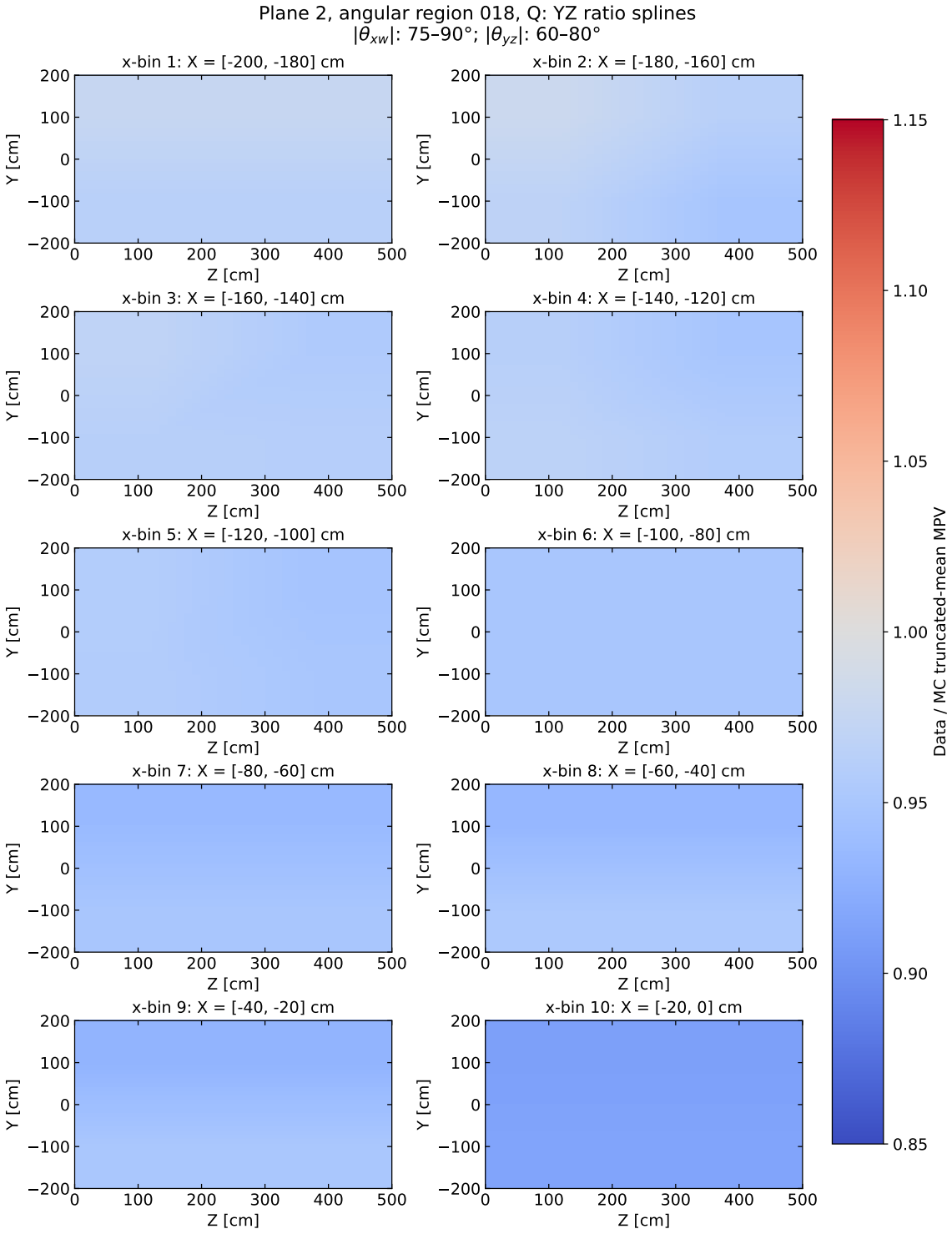


Figure 117: Plane 2, angular region 018: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [75, 90]^\circ$ and $|\theta_{yz}| \in [60, 80]^\circ$.

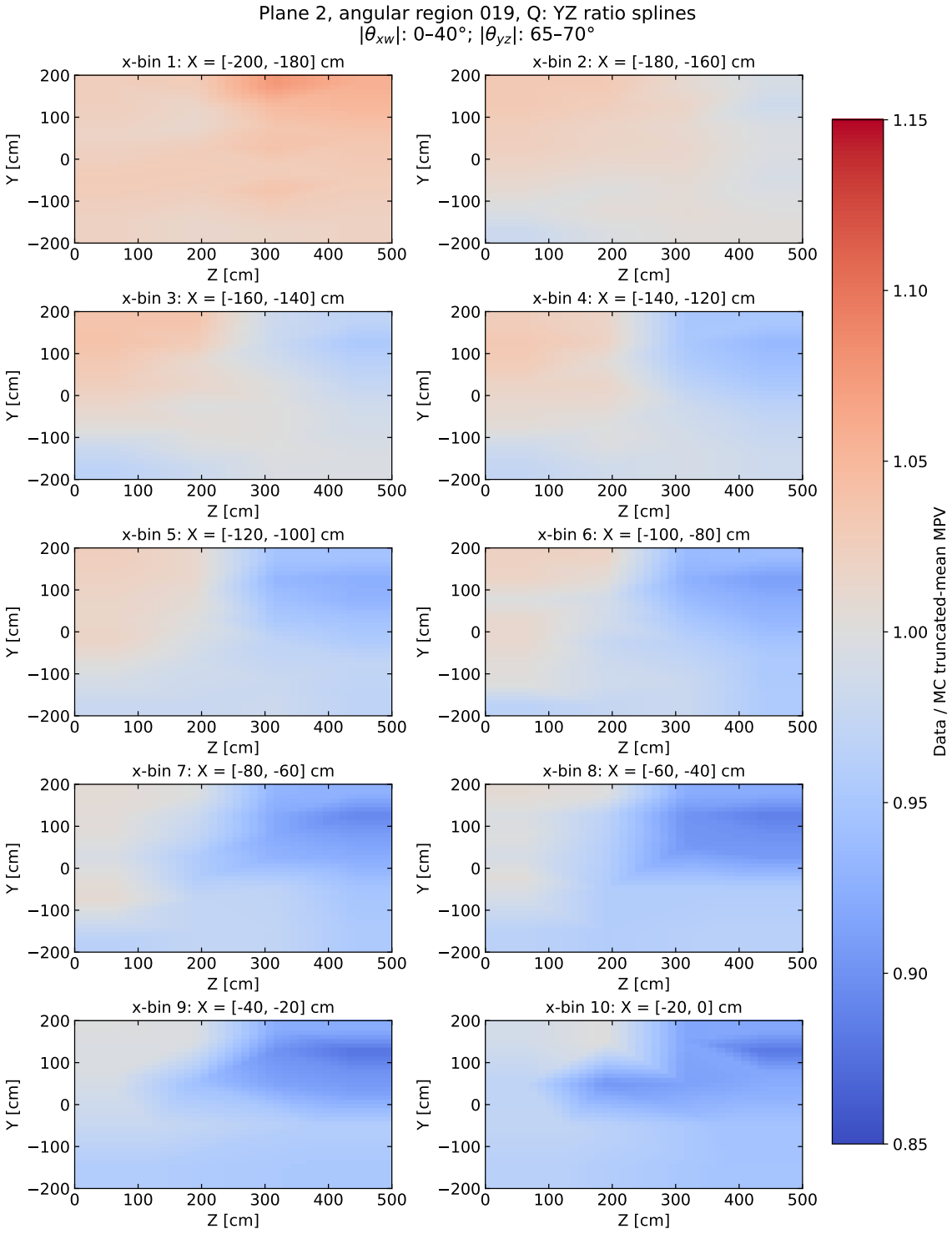


Figure 118: Plane 2, angular region 019: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 40]^\circ$ and $|\theta_{yz}| \in [65, 70]^\circ$.

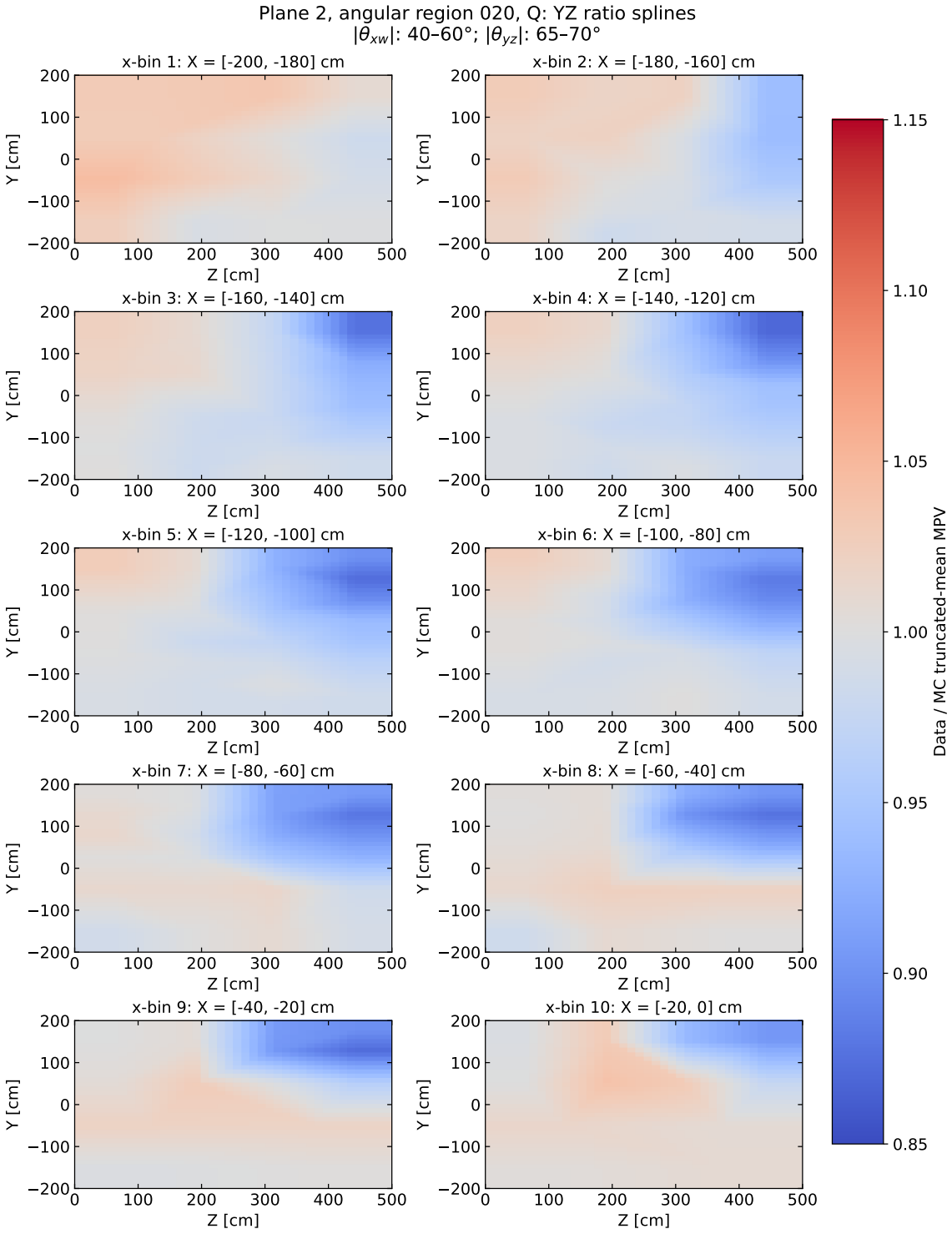


Figure 119: Plane 2, angular region 020: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [65, 70]^\circ$.

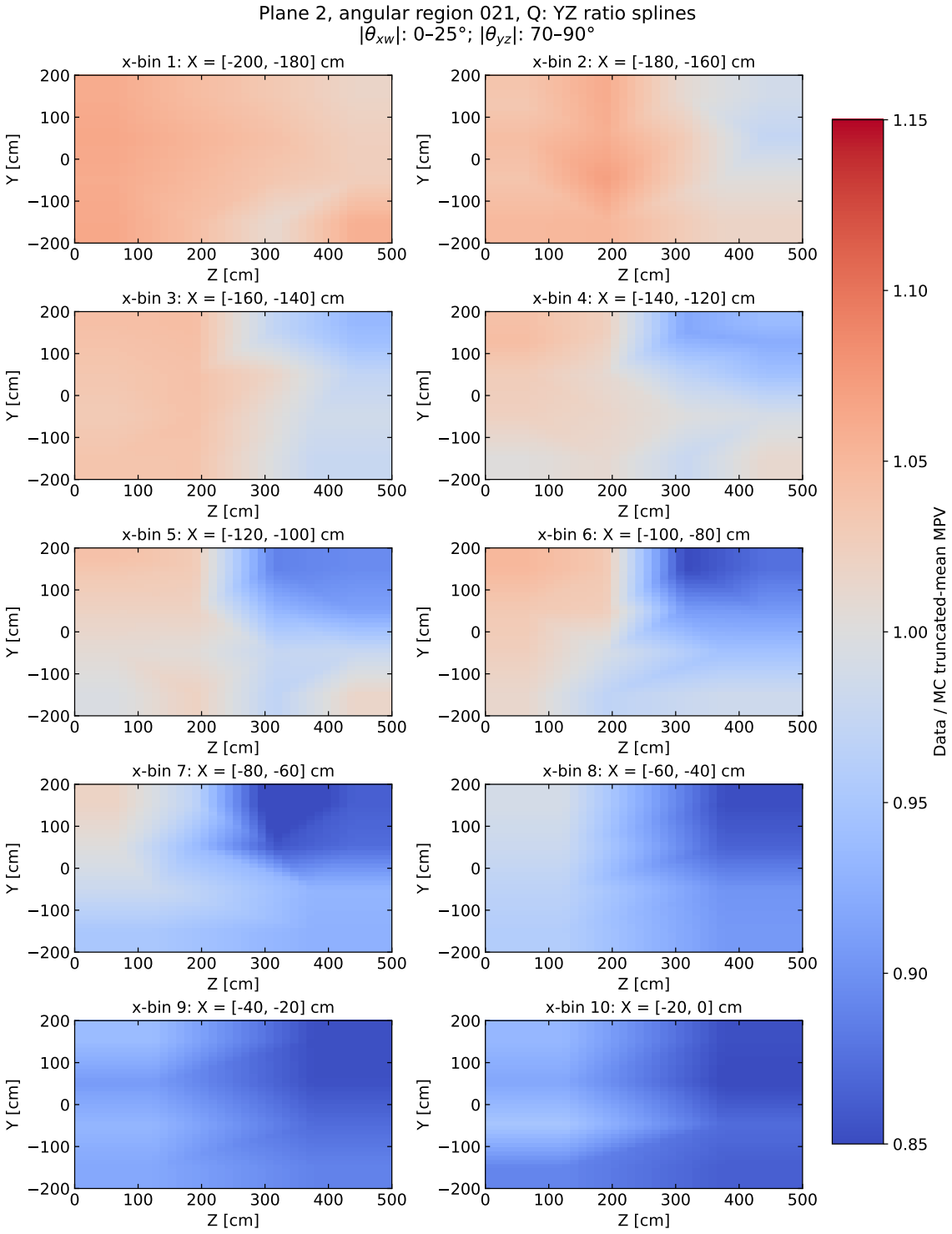


Figure 120: Plane 2, angular region 021: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [0, 25]^\circ$ and $|\theta_{yz}| \in [70, 90]^\circ$.

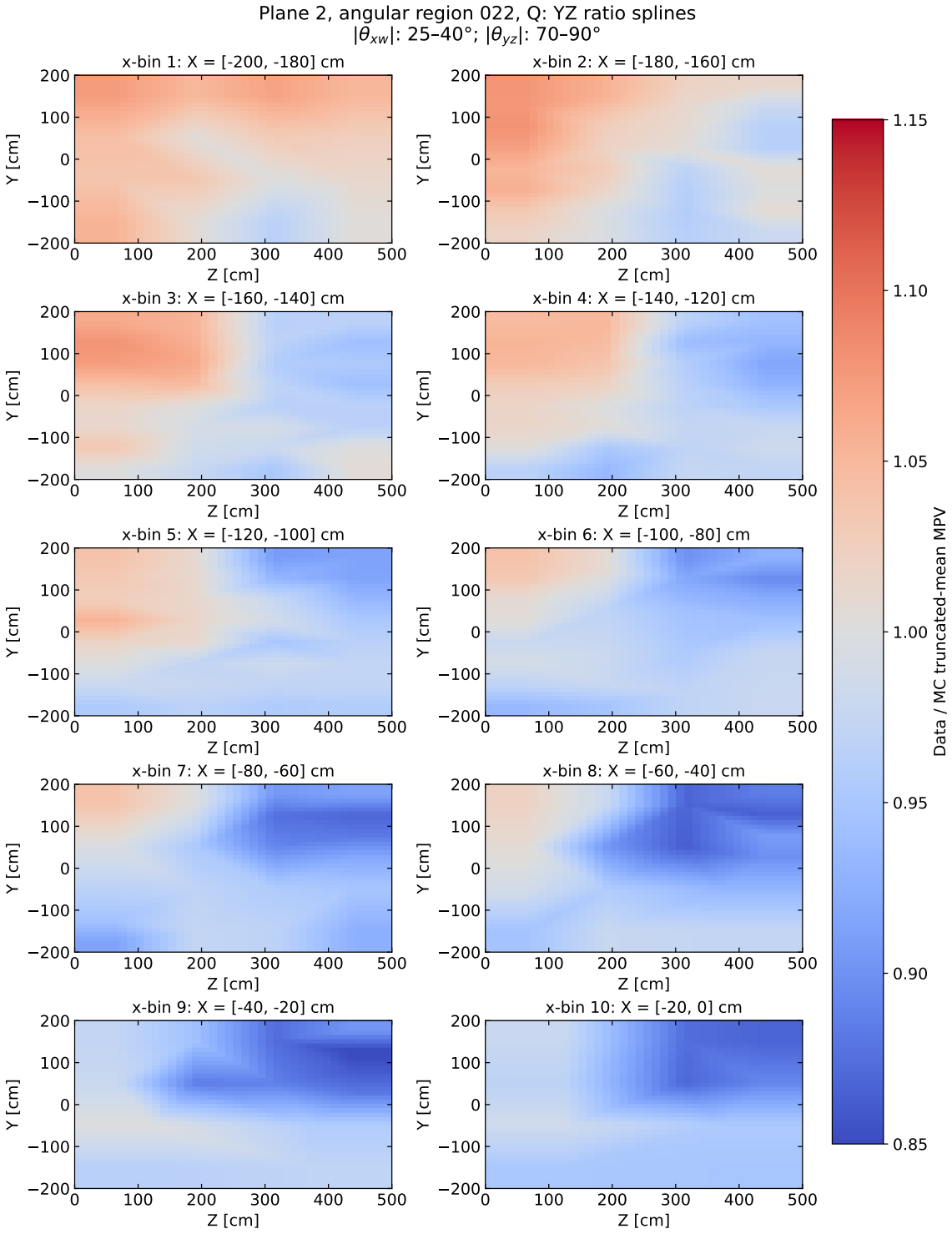


Figure 121: Plane 2, angular region 022: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [25, 40]^\circ$ and $|\theta_{yz}| \in [70, 90]^\circ$.

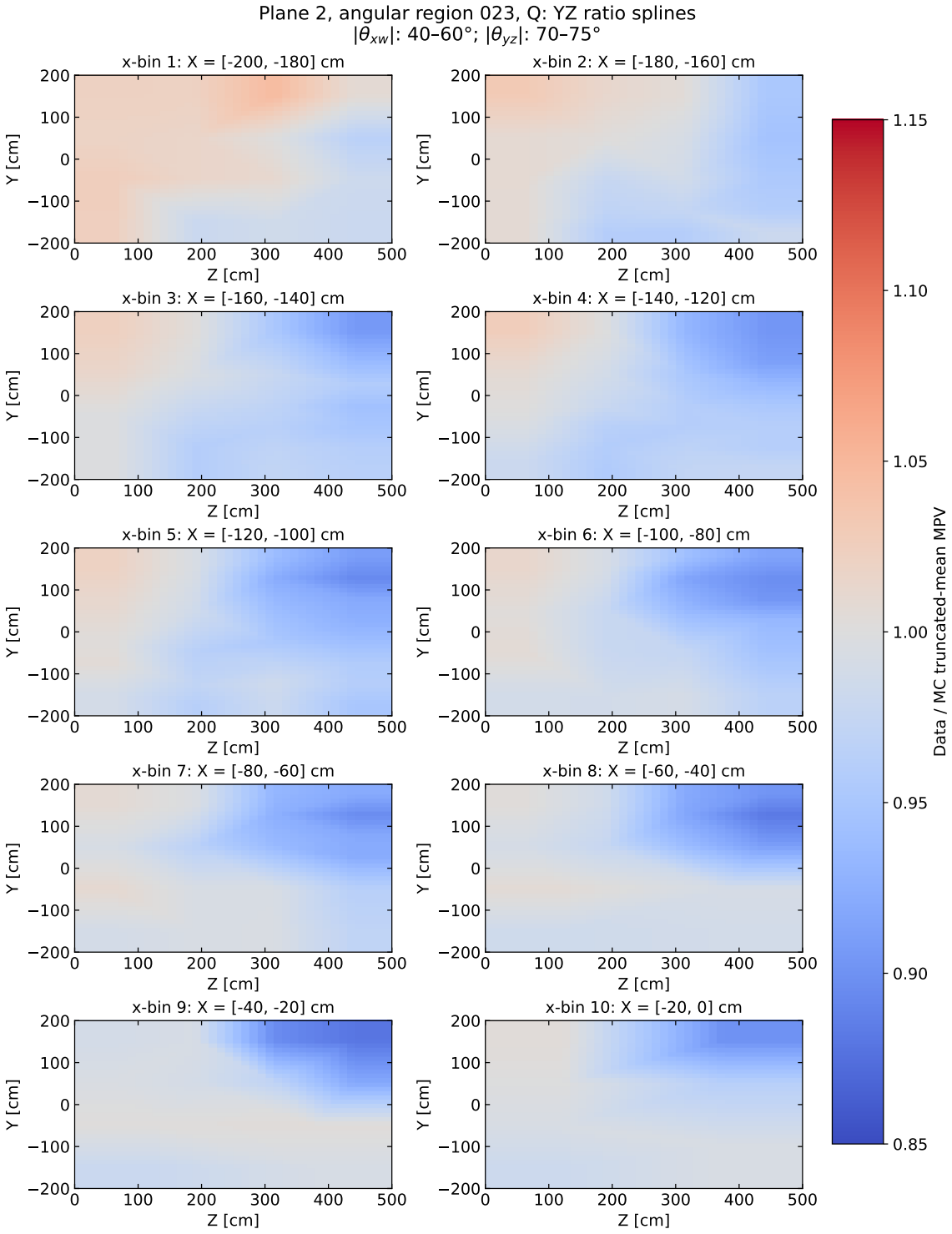


Figure 122: Plane 2, angular region 023: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [70, 75]^\circ$.

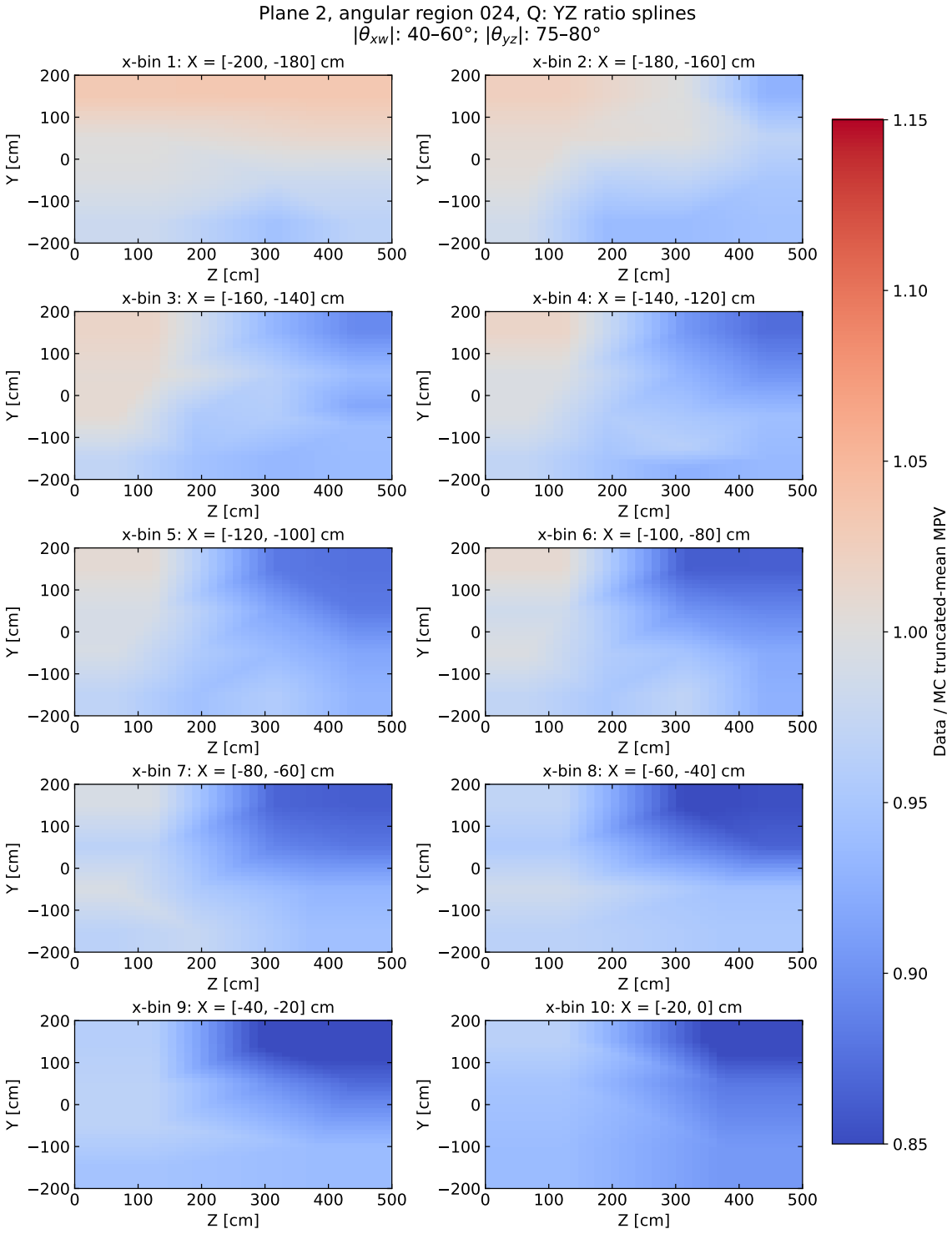


Figure 123: Plane 2, angular region 024: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [75, 80]^\circ$.

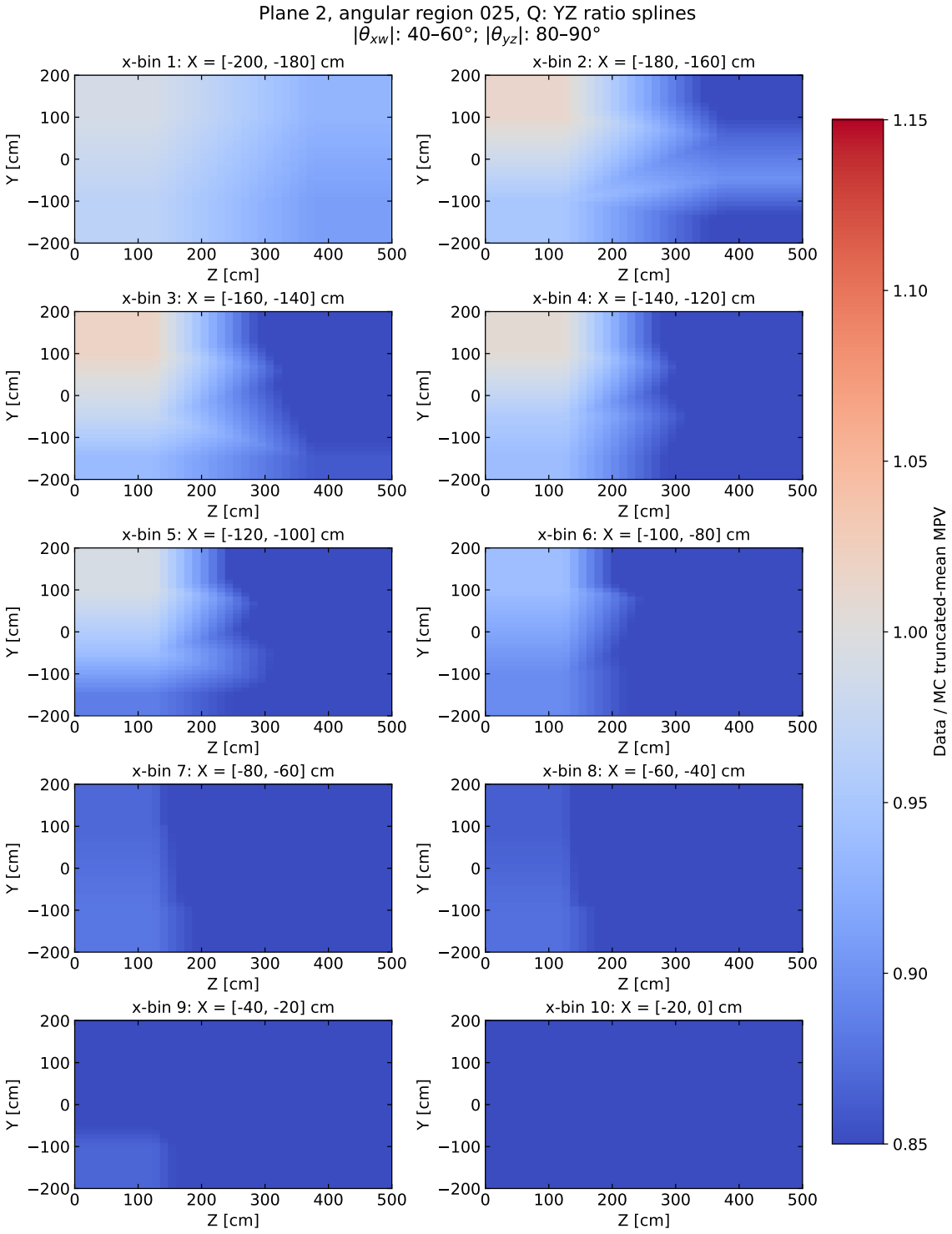


Figure 124: Plane 2, angular region 025: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [40, 60]^\circ$ and $|\theta_{yz}| \in [80, 90]^\circ$.

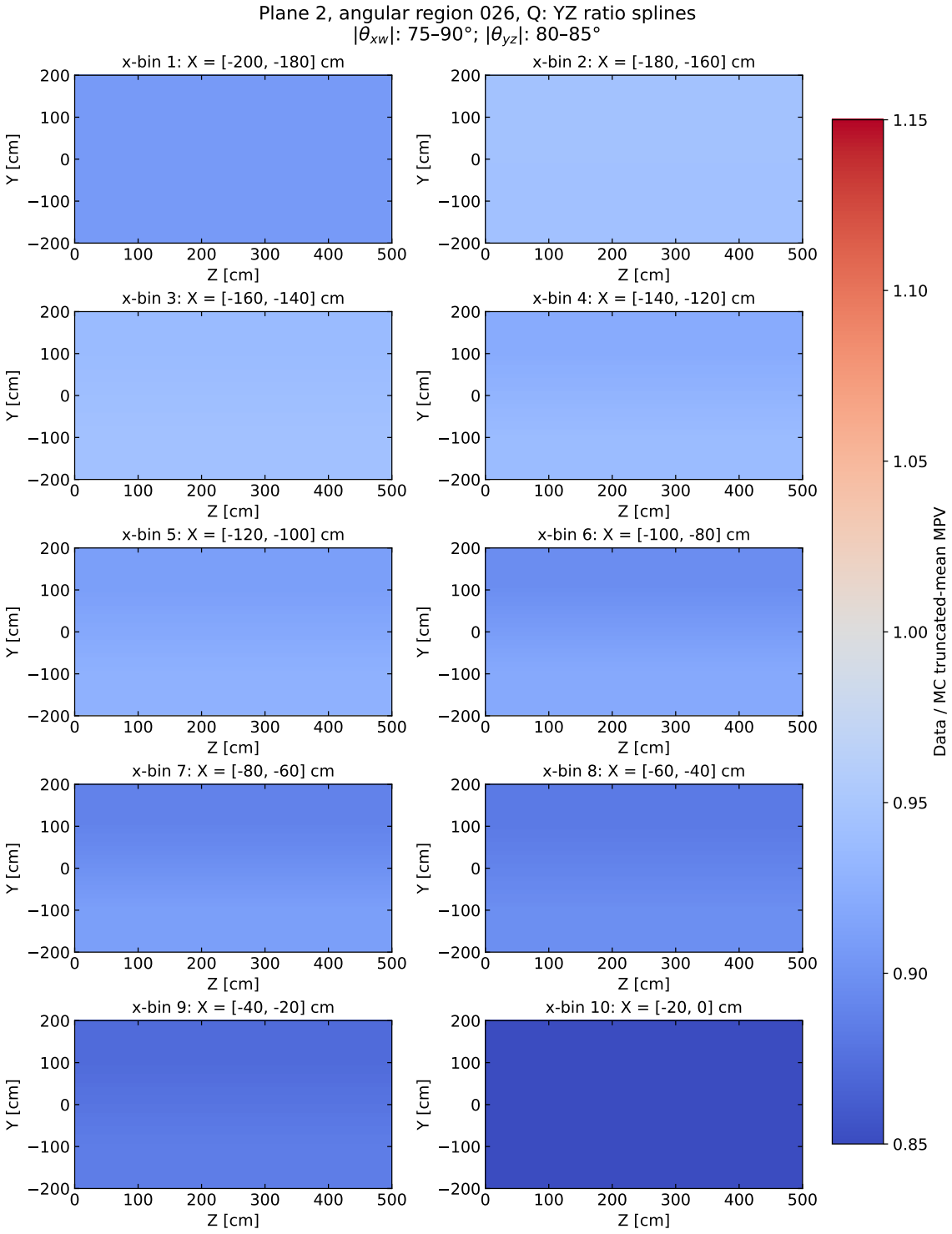


Figure 125: Plane 2, angular region 026: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [75, 90]^\circ$ and $|\theta_{yz}| \in [80, 85]^\circ$.

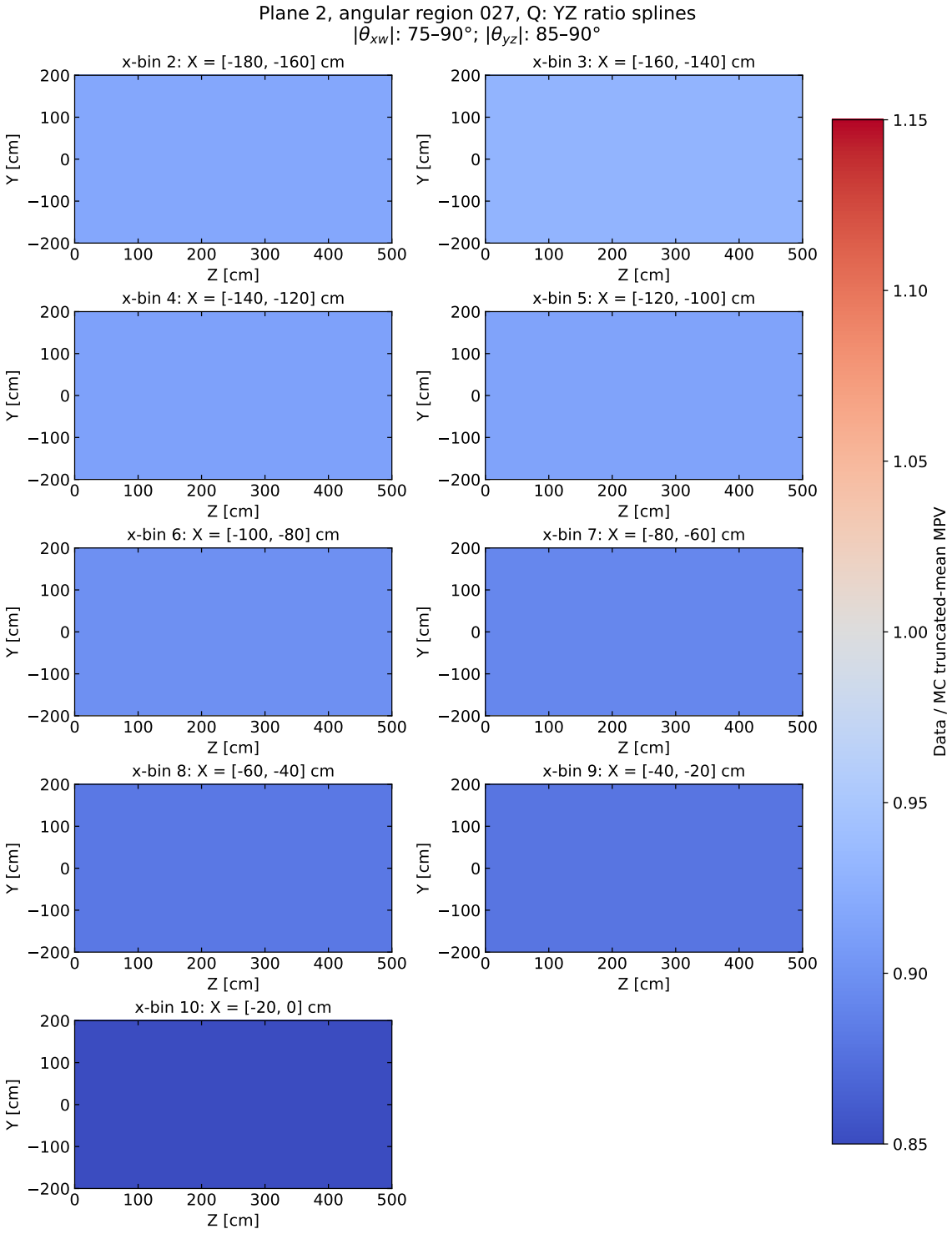


Figure 126: Plane 2, angular region 027: charge (Q) YZ data/MC ratio surfaces for all populated x-bins. The folded angular selection is $|\theta_{xw}| \in [75, 90]^\circ$ and $|\theta_{yz}| \in [85, 90]^\circ$.